

Mathematics

Student's Book

Senior 3

Version Edited in 2026

Copyright

© 2026 Rwanda Basic Education Board, Kigali.

All rights reserved.

This book is the property of the government of Rwanda.

Credit should be given to REB when the content is quoted.

FOREWORD

Dear students,

The Rwanda Basic Education Board (REB) is honoured to present the Mathematics book for Senior three (S3). This Textbook is designed to support the effective implementation of the Competence-Based Curriculum (CBC). It promotes learner-centred teaching and learning practices that align with Rwanda's vision of empowering learners to reach their full potential, actively participate in society, and pursue meaningful careers.

As Rwanda advances towards building a knowledge-based economy, education plays a pivotal role. Teachers are central to this transformation, guiding learners to connect classroom knowledge with real-life experiences, thereby fostering both individual growth and national development.

This student book serves as a comprehensive resource for the teaching and learning of Mathematics at Senior Three. Mathematics is an integral subject within the CBC, aimed at equipping learners with essential knowledge, skills, values, and attitudes necessary to become responsible and competent citizens. Through this subject, you are encouraged to develop problem-solving skills and contribute positively to your communities. In this regard, the only way to learn mathematics effectively is by doing mathematics through active problem-solving rather than passive reading. You are expected to work through problems, attempt proofs, and practise regularly, rather than merely reviewing theorems or looking up answers.

In its pursuit of quality education, the Government of Rwanda emphasizes the importance of using this kind of teaching and learning materials aligned with the curriculum.

I sincerely thank everyone who contributed to the development and refinement of this textbook. Special recognition goes to the REB team that coordinated the entire process. We also thank all partners, teachers, lecturers, and individuals who were involved from the beginning to the end. All feedback was highly valued and will contribute to improving the learning outcomes of our students.



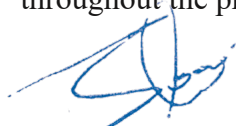
Dr. MBARUSHIMANA Nelson
Director General

ACKNOWLEDGEMENT

I would like to extend my sincere appreciation to everyone who contributed to the refinement of the Mathematics students' book for Senior Three. This achievement would not have been possible without the active involvement of various stakeholders.

Special thanks go to Rwanda Basic Education Board (REB) staff, subject experts, university lecturers and teachers for their valuable work in the writing of this textbook.

I am also grateful to all our education development partners whose support was instrumental throughout the process.

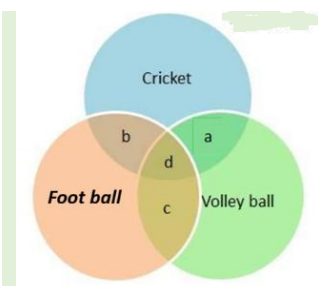


MURUNGI Joan

Head of CTLR Department

Contents

FOREWORD.....	3
ACKNOWLEDGEMENT.....	4
UNIT 1: PROBLEMS ON SETS.....	6
UNIT 2: QUADRATIC EQUATIONS.....	17
Unit 3: NUMBER BASES.....	34
UNIT 4: ALGEBRAIC FRACTIONS.....	55
UNIT 5: SIMULTANEOUS LINEAR INEQUALITIES.....	75
UNIT 6: LINEAR AND QUADRATIC FUNCTIONS.....	102
UNIT 7: COMPOUND INTEREST, REVERSE PERCENTAGE AND COMPOUND PROPORTIONAL.....	127
UNIT 8: RIGHT ANGLED TRIANGLES.....	145
UNIT 9: CIRCLE AND RELATED THEOREMS.....	174
UNIT 10: COLLINEAR POINTS AND ORTHOGONAL VECTORS IN 2D	210
UNIT 11: ENLARGEMENT AND SIMILARITY IN 2D.....	220
UNIT 12: INVERSE AND COMPOSITE TRANSFORMATIONS IN 2D....	255
UNIT 13: INTERPRETATION OF STATISTICAL GRAPHS.....	273
UNIT 14: PROBABILITY WITH VENN DIAGRAMS.....	293
BIBLIOGRAPHY.....	305



UNIT 1: PROBLEMS ON SETS

Key unit competence: Use sets notation, Venn diagrams and relations in solving related problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Explain the set operations such as union, intersection, and complement.
- Describe a real-world scenario using the language of sets (e.g., union, intersection, complement).

Skills:

- Construct Venn diagrams to represent the relationships between two or three sets.
- Translate word problems into set notations and Venn diagrams.
- Formulate and solve equations based on the cardinality of sets in a Venn diagram.
- Analyse complex Venn diagrams to extract information.

Attitudes and values:

- Collaborate effectively in a team to build and solve set models.
- Appreciate the utility of Venn diagrams for organizing and visualizing categorical data in daily life.

Introduction

In this unit, you will learn **sets** using the language of grouping and belonging that helps us organize and make sense of complex information. A **set** is simply a well-defined collection of distinct objects that can be represented in different ways including Venn diagram. In this unit, you will learn how **Venn diagrams** become our essential planning tool: circles that show relationships between groups. The operations of **union** ($\cup$) to find the total of participants, **intersection** ($\cap$) to find common members for different groups, and **complement** to find who is not in a given group, will give us precise numbers. Mastering sets and Venn diagrams will help you to visualize and solve problems involving sets with clarity and precision.



Introductory activity

In a survey about Sports Day participation, students signed up for Football (F), Athletics (A), and/or Basketball (B). Some students chose more than one sport. Assuming we get data from students in the following ways 25 say they play Football (F), 20 say they play Athletics (A); 22 say they play basketball (B). However, 10 say they play football and basketball while 8 say they play football and Athletics. In addition, 6 say they do Athletics and Basketball and there are 3 who play all three games.

How can we visually organize this information to avoid double-counting and find:

1. The total number of students who chose each sport individually.
2. The number of students in each overlap (F and B, F and A, A and B, and F and A and B)?
3. The total number of all participants.

1.1 Using Venn diagrams to represent set operations



Activity 1.1:

1) Remind your partners what a set is.

Given that A= The set of countries of the East African community, B= The set of countries bordering Rwanda, and C = The set of countries which share the Lake Victoria.

(a) With the aid of a map or an atlas, list the members of:

n(i) Set A (ii) Set B (iii) Set C

(b) Find: (i) $n(A)$ (ii) $n(B)$ (iii) $n(C)$

(c) Find: (i) $A \cap B$ (ii) $A \cap C$ (iii) $A \cap B \cap C$ (iv) $A \cup B \cap C$

2) Given that $\varepsilon = \{1, 2, 3, 4, 5, 6\}$; $A = \{2, 3, 5\}$; $B = \{3, 4, 5\}$. List the members of:

(i) A' (ii) B' (iii) $A' \cap B$ (iv) $A \cup B$ (v) $(A \cap B)'$ (vi) $A' \cap B'$ (vii) $A \Delta B$.



Content summary

A **set** is a well-defined collection of distinct objects, called elements or members, which are considered as a single conceptual entity.

The set of **common elements** which appear in two or more sets is called **the intersection of the sets**. The symbol used to denote intersection of sets is $\cap$.

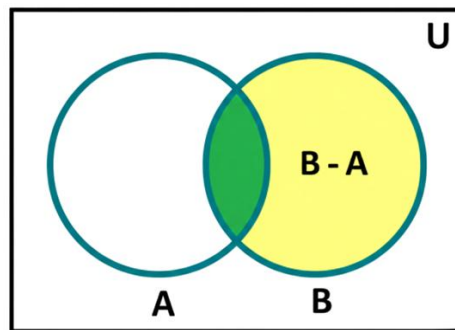
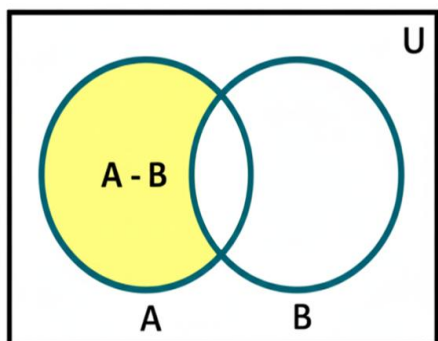
Intersection of sets is also represented by “**and**” in word statement. For example, “sets A and B” means $A \cap B$.

When the elements of two or more sets are put **together** to form a set, the set formed is known as **union of sets**. The symbol for the union of sets is $\cup$.

Union of sets is also represented by “**or**” in word statement. For example, “Sets A or B” means $A \cup B$ that is the union of sets A and B.

Complement of a set is the set of all elements in the universal set that are not members of a given set. The complement of set A is denoted by A' .

A **universal set** contains all the subsets under consideration. It is denoted by the symbol ε or U .



$A \cap B$ is in **green** color.

For two sets **A** and **B**, the **symmetric difference**, denoted by $A \Delta B$ (or sometimes $A \oplus B$), is the set of elements that belong to **exactly one** of the sets, but not both.

$A \Delta B = (A \cup B) \setminus (A \cap B)$. The number of elements in set **A** is denoted by $n(A)$ and so on.

Example 1

Given the following sets $A = \{a, b, c, d, e, f\}$ and $B = \{a, b, c, h, i\}$ find:

- (i) $A \cap B$ (ii) $A \cup B$ (iii) $A \Delta B$

Solution

- (i) $A \cap B = \{a, b, c\}$
(ii) $A \cup B = \{a, b, c, d, e, f\} \cup \{a, b, c, h, i\} = \{a, b, c, d, e, f, h, i\}$
(iii) $A \Delta B = (A \cup B) \setminus (A \cap B) = \{d, e, f, h, i\}$

Example 2

Given $\varepsilon = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 4, 6\}$ and $C = \{1, 3, 5, 7, 9\}$,

Answer the questions below about the sets **A**, **B** and **C**.

- (a) List the element of set $A \cap B$.
(b) Write down $n(A)$.
(c) List the element of set $A \cup B$.
(d) List the element of set $A \cup B \cup C$.
(e) List the element of set A' and B'

Solution

- (a) $A \cap B = \{2, 4\}$
(b) $n(A) = 5$
(c) $A \cup B = \{1, 2, 3, 4, 5, 6\}$
(d) $A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 9\}$
(e) $A' = \varepsilon - A = \{6, 7, 8, 9, 10\}$ and $B' = \varepsilon - B = \{1, 3, 5, 7, 8, 9, 10\}$

Example 3

Consider $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and

$A = \{2, 4, 6, 8, 10\}$. Then $A' = \{1, 3, 5, 7, 9\}$.



Application activity 1.1:

1) If $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3\}$, $C = \{6, 8, 10\}$ and $D = \{2, 3, 6\}$ find:

- (a) $n(A)$ (b) $n(B)$ (c) $n(C)$ (d) $n(A) + n(B)$ (e) $n(A) + n(C) - n(B)$ (f) $A \cup B \cup C$
 (g) $n(A \cup B \cup C)$ (h) $n(A \cup C \cup D)$

2) Find the union of the following sets:

$A = \{\text{positive even numbers from 0 to 20}\}$,

$B = \{\text{Integers greater than } -2 \text{ but less than } 9\}$, $C = \{\text{Prime numbers between 1 and 7}\}$

3. If $P = \{\text{counting numbers from 1 to 15}\}$ and $Q = \{\text{Even numbers from 2 to 14}\}$, find:

- (i) $(P \cup Q)'$ (ii) $(P \cap Q)'$ (iii) ε (iv) $n(\varepsilon)$

4. Given that $A = \{3, 5\}$, $B = \{7, 9, 11, 13\}$, $C = \{3, 5, 7\}$ and $\varepsilon = \{3, 5, 7, 9, 11, 13\}$, find:

- (i) A' (ii) $(A \cap B)'$ (iii) B' (iv) $(A \cap B)$ (v) $(A \cap C)'$ (vi) $(B \cap C)'$

1.2. Solving a problem involving two sets using Venn diagrams



Activity 1.2:

A survey was carried out in a shop to find the number of customers who bought bread or milk or both or neither. Out of a total of 79 customers for the day, 52 bought milk, 32 bought bread and 15 bought neither.

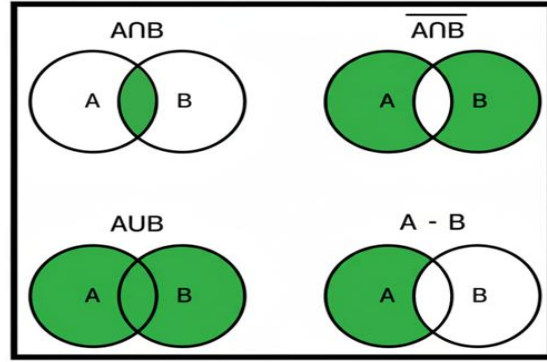
- (a) Without using a Venn diagram, find the number of customers who:
 (i) bought bread and milk
 (ii) bought bread only
 (iii) bought milk only
 (b) With the aid of a Venn diagram, work out questions (i), (ii) and (iii) in (a) above.
 (c) Which of the methods in (a) and (b) above is easier to work with? Give reasons for your answer.



Content summary

Let A and B be two sets. The Venn diagram shows all possible relationships among and between them, divided into **four regions**:

1. **Only in A:** $A \setminus B$
2. **Only in B:** $B \setminus A$
3. **In both A and B:** $A \cap B$
4. **In neither A nor B:** $\varepsilon - (A \cup B)$
5. $A \Delta B = (A \setminus B) \cup (B \setminus A) = \overline{A \cap B}$



Basing on the previous activity, we clearly see that a Venn diagram plays a very important role in analysing the set problem and helps in solving the problem very easily.

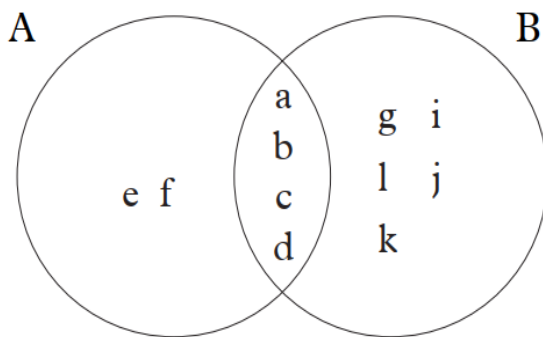
First, express the data in terms of set notations and then fill the data in the Venn diagram for easy solution.

Some important facts like “**intersection**”, “**union**”, “**complement**” and **symmetric difference** should be well considered and represented when drawing Venn diagrams.

Example 1

Consider two intersecting sets A and B such that $A = \{a, b, c, d, e, f\}$ and

$B = \{a, b, c, d, g, i, j, k, l\}$. We represent the two sets in a set diagram as shown in Fig 1.1 below.



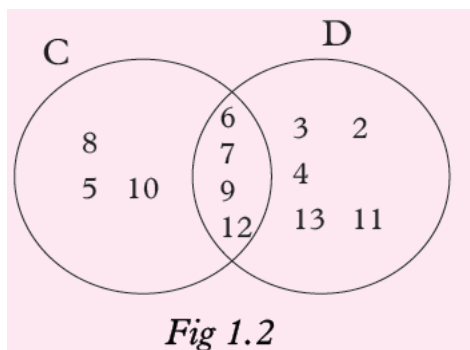
The number of elements in union of sets A and B is given by:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

In the Venn diagram, $n(A) = 6$, $n(B) = 9$ and $n(A \cap B) = 4 \Rightarrow n(A \cup B) = 6 + 9 - 4 = 11$.

Example 2

The diagram below shows the marks out of 15 scored by a number of Senior 3 students in the two groups.



Determine the total number of Senior 3 students in the two groups.

Solution:

$$n(C \cup D) = n(C) + n(D) - n(C \cap D) = 7 + 9 - 4 = 16 - 4 = 12$$

Example 3:

A survey involving 120 people about their preferred breakfast showed that 55 drink milk at breakfast, 40 drink juice at breakfast and 25 drink both milk and juice at breakfast.

(a) Represent the information on a Venn diagram.

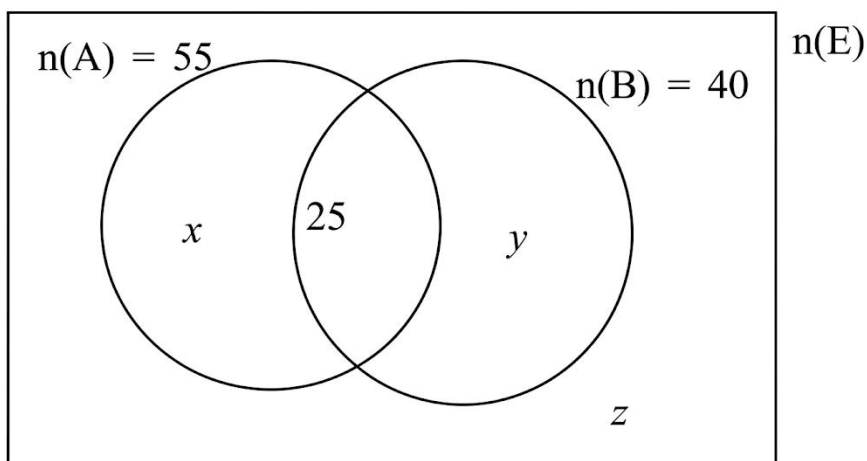
(b) Calculate the following:

- (i) Number of people who take milk only.
- (ii) Number of people who take neither milk nor juice.

Solution:

(a) Let A be the set of those who drink milk and B be the set of those who drink juice, x be the number of those who drink milk only, y be the number of those who drink juice only and z represents number of those who did not take any. By expressing data in set notation;

$$n(A) = 55, n(B) = 40, n(A \cap B) = 25, n(A' \cap B) = y, n(E) = 120$$



(b) (i) We are required to find the number of those who take milk only.

$$x = 55 - 25 = 30$$

So, 30 people take milk only.

(ii) To find the value of z ;

$$30 + 25 + 15 + z = 120.$$

$$z = 120 - (30 + 15 + 25) = 120 - 70 = 50.$$



Application activity 1.2:

1. In a certain group of children, all of them study French or German or both languages. 15 study French but not German, 12 study German of whom 5 study both languages.
 - (a) Draw a Venn diagram to show this information.
 - (b) Calculate how many children are there in the group.
2. In a group of 17 pupils, 10 offer Economics and 9 offer Mathematics. The number that offer both Economics and Mathematics is twice the number that offer none of the two subjects.
 - (a) Draw a Venn diagram to represent the information.
 - (b) Calculate the number of pupils that;
 - (i) Offer both subjects
 - (ii) Offer only one subject
 - (iii) Offer none of the subjects.

1.3 Solving a problem involving three sets using Venn diagrams



Activity 1.3:

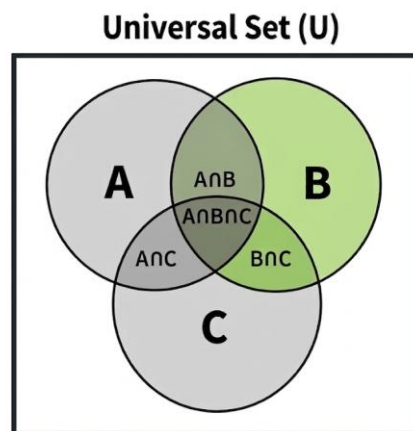
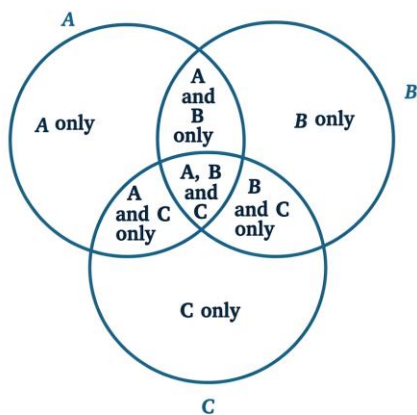
A survey was done on 50 people about which food they like among rice, sweet potatoes and posho. It was found out that 15 people like rice, 30 people like sweet potatoes, 19 people like posho. 8 people like rice and sweet potatoes, 12 rice and posho, 7 people like Sweet potatoes and posho. 5 people like all the three types of food.

- (a) Extract the data and represent it in set notation.
- (b) Without using a Venn diagram,
 - (i) Find the number of people who like none of the foods.
 - (ii) Find the number of those who like posho and rice only.
 - (iii) Find the number of those who like sweet potatoes and rice only.
- c) With the help of a Venn diagram find out the solutions for (b) (i), (ii) and (iii) above
- d) Was it easy to do (b) (i), (ii) and (iii) without a Venn diagram?



Content summary

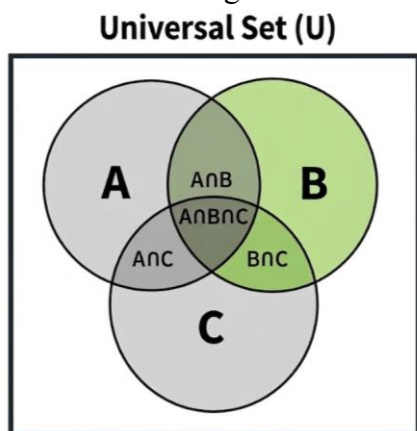
A **three-set Venn diagram** uses three intersecting circles to represent sets A, B, and C. The diagram is divided into **seven distinct regions**, each representing a different combination of membership.



From answers for activity 1.3, it is clearly seen that without a Venn diagram, the problems involving three or more sets become complicated to handle. A Venn diagram makes the problem easier because we can represent the data extracted in each region and then calculate the values required.

Steps for solving three-set problems:

- Draw a Venn diagram with three overlapping circles.



- Start with the innermost intersection $A \cap B \cap C$ and fill it first.
- Fill the two-set intersections:
 $A \cap B$ (excluding those in all three), $A \cap C$ and $B \cap C$
- Fill the regions belonging to only one set: Only in A, Only in B and Only in C
- Use totals given to determine missing values using addition or subtraction. Note that

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - [n(A \cap B) + n(A \cap C) + n(B \cap C)] + n(A \cap B \cap C)$$
- Check your total to ensure it matches the universal set.

Example 1:

Consider the Venn diagram shown below illustrating the numbers of students who take the foreign languages German (G), Spanish(S) and French(F) in a college.

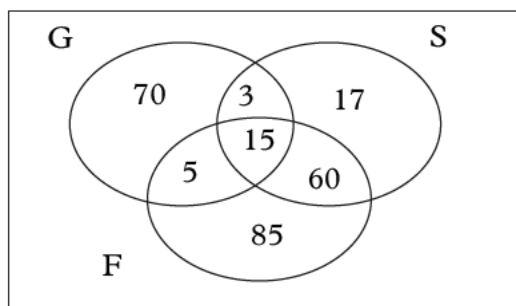


Fig. 1.6

Page 6

Find the total number of students taking languages.

Solution:

The total number of students taking languages is given by the union of the three sets as shown by the following formula:

$$n(G \cup S \cup F) = n(G) + n(S) + n(F) - [n(G \cap S) + n(G \cap F) + n(S \cap F)] + n(G \cap S \cap F)$$

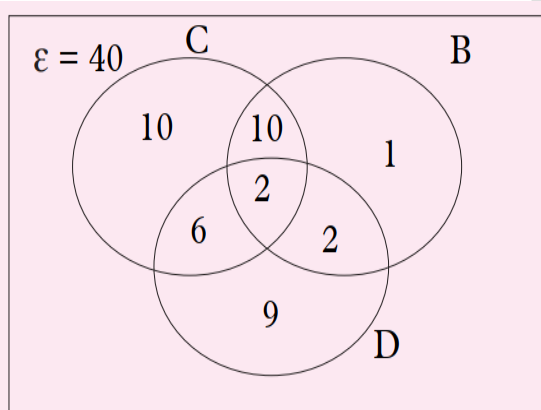
From the Venn diagram in the figure,

$$n(G \cup S \cup F) = 93 + 95 + 165 - (18 + 20 + 20 + 75) + 15 = 353 - 113 + 15 = 255$$

Example 2:

The students in Senior 3 class did a survey on their names regarding whether they contained the letters B, C and D.

The following Venn diagram shows the results of the survey in terms of the number of names in each category.



Page 6

Use the Venn diagram to determine the number of student's names that contained:

- All the three letters
- Letter D
- Letters B and D but not C
- Only two of the letters
- The total number of students

Solution:

- All the three letters $n(B \cap C \cap D) = 2$
- Letter D: $n(D) = 9 + 6 + 2 + 2 = 19$

(c) Letters B and D but not C: $n(B \cap D) - n(B \cap C \cap D) = 4 - 2 = 2$

(d) Only two of the letters = $6 + 10 + 2 = 18$

(e) The total number of students:

$$\begin{aligned} n(B \cup C \cup D) &= n(B) + n(C) + n(D) - [n(B \cap C) + n(B \cap D) + n(C \cap D)] + n(B \cap C \cap D) \\ &= 15 + 28 + 19 - \{12 + 4 + 8\} + 2 = 62 - 24 + 2 = 40 \text{ Students} \end{aligned}$$



Application activity 1.3:

1) In a class of 53 students, 30 study Chemistry, 20 study Physics, 15 study Mathematics. 6 study both Chemistry and Physics, 4 study both mathematics and Chemistry, 5 study both Physics and Mathematics. All the students study at least one of the subjects.

- Represent the information on a Venn diagram.
- Find the number of students who study all the three subjects.
- How many study;
 - Physics only.
 - Physics but not Mathematics
 - Two subjects only.

2) Out of 100 students in a school, 42 take English, 35 take Kiswahili and 10 take French. 20 take none of the subjects, 9 take French and English, 10 take French and Kiswahili, 11 take English and Kiswahili.

- Represent the information on a Venn diagram.
- Find the number of students who do not belong to debating club.
- Find the number of students who take three subjects.
- Find the number of students who take English only.
- Find the number of students who take Kiswahili and French.

3) A school has a teaching staff of 22 teachers. 8 of them teach Mathematics, 7 teach Physics and 4 teach Chemistry. 3 teach both Mathematics and Physics, none teaches Mathematics and Chemistry. No teacher teaches all the three subjects. The number of teachers who teach Physics and Chemistry is equal to the number of teachers who teach Chemistry but not Physics.

- Represent the data on a Venn diagram
- Find the number of teachers who teach;
 - Mathematics only.
 - Physics only.
 - None of the three subjects.



End of unit assessment

Assessing knowledge and understanding

1) Which of the following correctly describes the operation $A \cap B$?

- a) The set of all elements in either A or B.
- b) The set of all elements that are in the universal set but not in A or B.
- c) The set of all elements common to both A and B.
- d) The set of all elements in A that are not in B.

2) 1.2 In a survey, $n(P)=40$, $n(Q)=35$, $n(P \cap Q)=15$, and the universal set has 70 members. How many people are in $(P \cup Q)'$?

- a) 5 b) 10 c) 20 d) 25

Assessing skills

3) In a class of 45 students, 25 study History, 20 study Geography, and 15 study neither subject.

(a) Draw a clearly labeled Venn diagram to represent this information.

(b) Calculate:

- (i) The number of students who study both History and Geography.
- (ii) The number of students who study Geography only.

4) The following facts were discovered in a survey of course preferences of 110 pupils in senior six: 21 like engineering only, 63 like engineering, 55 like medicine and 34 like none of the two courses.

(a) Draw a Venn diagram representing this information.

(b) (i) How many like Engineering or Medicine?

(ii) How many like Engineering and Medicine?

(iii) How many like only Medicine?

5) A survey was done on 50 people about which hotels they eat from among H, S and L. 15 people eat at hotel H, 30 people eat at hotel S, 19 people eat at hotel L, 8 people eat at hotels H and S, 12 people eat at hotels H and L, 7 people eat at hotels S and L. 5 people eat at hotels H, S and L.

(a) How many people eat only at Hotel H?

(b) How many people eat at hotels H and S, but not at L?

(c) How many people don't eat at any of these three hotels?

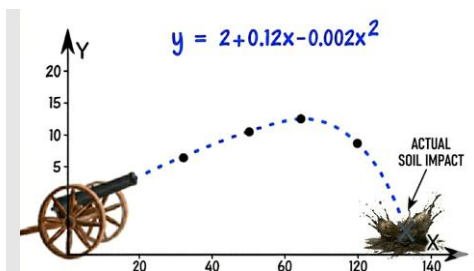
6) Translate the following word problem into set notation and solve without a diagram:

In a group of 80 tourists, 50 visited Nyungwe Forest, 40 visited Volcanoes National Park, and 30 visited Akagera Park. 20 visited both Nyungwe and Volcanoes, 15 visited both Volcanoes and Akagera, and 10 visited both Nyungwe and Akagera. If 5 visited all three, how many tourists visited at least one of the parks?

Assessing attitudes and values

7) You are working in a group to model survey data about sports preferences using a Venn diagram. One group member insists on doing all the calculations mentally without drawing the diagram, causing confusion when explaining to the class.

- (a) Explain, with a specific example, why using a Venn diagram is important for clear communication in such team tasks.
- (b) Suggest one way your team could ensure everyone's ideas are included when building the set model.



UNIT 2: QUADRATIC EQUATIONS

Key unit competence: Solve problems by formulating and manipulating quadratic equations.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Write a quadratic equation in standard form ($ax^2 + bx + c = 0$).
- Explain how to solve a quadratic equation.
- Identify the number of solutions (roots) for a given quadratic equation.

Skills:

- Solve quadratic equations accurately by factoring, completing the square, using the quadratic formula, and synthetic division.
- Solve real-world problems involving quadratic equations.

Attitudes and values:

- Appreciate the usefulness of the methods of solving quadratic equations in real life. Collaborate effectively with peers to compare solution strategies and verify results.

Introduction

This unit introduces the definition and structure of quadratic equations, explores different factorization techniques, and presents several methods for solving quadratic equations. You will learn **factorization, perfect squares, completing the squares, the quadratic formula, and synthetic division (Horner's method)**. The unit also emphasizes on interpreting and solving real-life problems that lead to quadratic equations, thereby improving both analytical and practical problem-solving abilities.



Introductory activity

Use your knowledge of solving equations to solve the following real- life problem. “A number of people planned to contribute a total of 800 000 FRW to start a small self help groups from where they could be getting small loans to boost their business. They agreed to contribute each an equal amount of money. Four of them withdrew from the venture. As a result, each of the remaining people had to contribute 10 000 FRW more to reach the target”. Determine the:

- (a) Original number (x) of people in the venture.
- (b) Amount of money each contributed after the withdrawal of the four.
1. Compare your answer with those of other classmates during the class presentations.

2.1. Definitions and examples of quadratic equations



Activity 2.1:

1. Expand the following expressions.
 - (a) $3(5x + 6)$
 - (b) $(x - 5)(x - 2)$
 - (c) $(x - 1)(x + 2)(x - 3)$
2. State the highest power of the unknown in each of the expression in (1) above.
3. Which of the expressions in question (1) above are quadratic?



Content summary

A quadratic expression has the general form $ax^2 + bx + c$ where a, b and c are real numbers with $a \neq 0$. It is a polynomial with the highest degree or power on the variable 2 or Polynomial of order 2.

Examples of quadratic expressions include $x^2 + 3x + 1$, $a^2 + 2a$, $y^2 - 2y + 1$, $c^2 - 4$ etc .

The general form of a quadratic equation is $ax^2 + bx + c = 0$, where a, b and c are real numbers and $a \neq 0$.

Examples

- 1) The following are quadratic equations: $x^2 + 3x + 1 = 0$, $x^2 - 9 = 0$, $x^2 - 5x + 6 = 0$ etc.
- 2) Expand each of the expression and state whether it is quadratic or not:
 - (a) $(x + 3)(x + 1) = x^2 + 4x + 3$. The highest power of x is x^2 , so it is quadratic.
 - (b) $(x + 2)(x - 4)(x - 1) = x^3 - 3x^2 - 6x + 8$: It is not quadratic.



Application activity 2.1:

- Expand each of the expression and state whether it is quadratic or not:
 - $(x-7)(x+3)(x-1)$
 - $x(x-4)(x-3)$
 - $(a^2+8)(a-3)$
 - $(p+2)(p-5)$
- Which of the following is not a quadratic expression? Circle the correct expression.
 - $(x^2-5)(0+4)$
 - $(x+5)(x+4)$
 - $\frac{(x+2)(5a+3)}{x-1}$
 - $(3x+2)(x-4)$
 - $(2x-7)(4x-3)$
 - $(x+3)(x+3)$
- Does the following expression give quadratic expression? Answer by **True** or **False**.
 - $(a-4)(a^2-4)$
 - $6(x+7)(x)$
 - $(5t+3)(3t+2)$
 - $(2-x)(3-x)$
 - $(3+p)(5-p)$
- A man bought a certain number of shirts for 2 000 FRW. If each shirt had cost 200 FRW less, he could have bought five more for the same money.
 - Form and simplify an expression from the above information.
 - What name can be given to the above expression?
- The perimeter of a rectangle is 42 cm. If the diagonal is 15 cm;
 - Form an expression that relates the perimeter, diagonal, length and width of the rectangle.
 - What special name is given to the expression in 5(a) above?

2.2 Factorizing Quadratic equations



Activity 2.2:

- What do you understand by the term “factor” of a product in Mathematics?
- Given that $x^2 + x = x(x+1)$ is the way of writing a quadratic expression in form of a product of factors x and $x+1$, Factorize the following expressions:
 - $4x^2 + 2x$
 - $x^2 - 4$
- Given that $x^2 + 5x + 6 = (x^2 + 2x) + (3x + 6)$, factorize $x^2 + 5x + 6$ by grouping.
- Using the fact you have learnt in 3 above, factorize $x^2 - x - 12$.



Content summary

In order to find the factors of an expression such as $ax^2 + bx$, we look for the factors

that are common in both terms.

The common factors of ax^2 and bx is x .

Thus, $ax^2 + bx = x(ax + b)$.

Therefore, the factors of $ax^2 + bx$ are x and $(ax + b)$.

This means, $ax^2 + bx$ can be factored as $x(ax + b)$.

In S2, you learnt different methods of factoring polynomials by common factor, by grouping, by using algebraic identities, using sum and product, using zeros/ roots of polynomials etc. Those methods can be used when factoring quadratic expressions.

The process of finding the factors of an expression is called **factoring** or **factorization**. This is the reverse of **expansion**.

Consider $(x + a)(x + b) = x(x + b) + a(x + b) = x^2 + bx + ax + ab = x^2 + (a + b)x + ab$

The expressions $x + a$ and $x + b$ are the factors of $x^2 + (a + b)x + ab$.

Similarly: $(x + 3)(x + 4) = x(x + 4) + 3(x + 4) = x^2 + 4x + 3x + 12 = x^2 + 7x + 12$

The last expression shows the relationship between the terms of the quadratic expression and its factors. We see that the coefficient 7 of x is $7 = 3 + 4$ (**sum**) while the independent term $12 = 3 \times 4$ (**product**).

Thus, a constant term **ab** of a quadratic expression is the **product** of the number terms a and b . Whereas, the coefficient of x is the **sum** of a and b .

We can use these facts to factorize quadratic expressions.

Example 1:

Consider the expression $x^2 + 10x + 21$.

We can factorize it as follows, compare $x^2 + 10x + 21$ and $(x + \underline{\quad})(x + \underline{\quad})$.

Considering, $x^2 + 10x + 21 = x^2 + (a + b)x + ab$, we need two numbers a and b such that $a + b = 10$ and $ab = 21$.

Clearly, a and b are factors of 21 and their sum is 10. These are 3 and 7.

Rewriting $x^2 + 10x + 21$ as $x^2 + (3 + 7)x + 21$ gives :

$$\begin{aligned} x^2 + 3x + 7x + 21 &= x(x + 3) + 7(x + 3) \text{ (factoring common terms)} \\ &= (x + 3)(x + 7). \end{aligned}$$

Therefore, $x^2 + 10x + 21 = (x + 3)(x + 7)$.

Example 2:

Factorize $x^2 + 7x + 12$.

Solution

We need two numbers a and b such that $a + b = 7$ and $ab = 12$. These numbers are 3 and 4.

Thus,

$$x^2 + 7x + 12 = x^2 + (3 + 4)x + 12 = x^2 + 3x + 4x + 12 = x(x + 3) + 4(x + 3) = (x + 3)(x + 4)$$

Example 3

Write an expression $x^2 - 6x + 8$ in terms of product of factors.

Solution

$$\text{Let, } x^2 - 6x + 8 = (x + a)(x + b).$$

$$\text{Thus } x^2 - 6x + 8 = x^2 + (a + b)x + ab.$$

$$\text{Then, } (a + b) = -6 \text{ and } ab = 8.$$

Two numbers which add up to -6 and whose product is 8 are -2 and -4 .

So,

$$\begin{aligned} x^2 - 6x + 8 &= x^2 + (-2 + (-4))x + 8 \\ &= x^2 - 2x - 4x + 8 \\ &= x(x - 2) - 4(x - 2) \\ &= (x - 2)(x - 4) \end{aligned}$$

Example 4

Factorize: $x^2 + 12x + 36$

Solution

$$\text{Let } x^2 + 12x + 36 = (x + p)(x + q) = x^2 + (p + q)x + pq$$

$$\text{Thus, } p + q = 12 \text{ and } pq = 36.$$

The factors of 36 whose sum is 12 , are 6 and 6 .

$$\text{So, } x^2 + 12x + 36 = (x + 6)(x + 6) = (x + 6)^2$$

Note: $x^2 + 12x + 36$ is perfect square because its factors are identical.

When factorizing a quadratic expression of the form $ax^2 + bx + c$ we get two factors whose sum is b and the product is ac .

Consider a quadratic expression $6x^2 + 5x - 4$. It can be factorized as follows:

Comparing $6x^2 + 5x - 4$ to $ax^2 + bx + c$, we find out that $a = 6$, $b = 5$ and $c = -4$.

We then get two values whose product is -24 and whose sum is 5 ie $ac = -24$ and $a + b = 5$. These values must be the factors of -24 and they are -3 and 8 .

$$\text{We then write } 6x^2 + 5x - 4 = 6x^2 + (8 - 3)x - 4 = 6x^2 + 8x - 3x - 4.$$

By factorizing out the common factors, We get

$$6x^2 + 5x - 4 = 2x(3x + 4) - 1(3x + 4)$$

$$\text{So, } 6x^2 + 5x - 4 = (2x - 1)(3x + 4)$$

Example 4

Write down $7 + 3x^2 - 22x$ in terms of its factors.

Solution.

Rearranging the terms, $7 + 3x^2 - 22 = 3x^2 - 22x + 7$, We have the coefficients $a = 3$, $b = -22$ and $c = 7$. Sum is -22 and product is 21 . The two factors that can add to give -22 and multiply to give 21 are -21 and -1 .

We write $3x^2 - 22x + 7 = 3x^2 - 21x - x + 7$.

By factorizing, $3x^2 - 21x - x + 7 = 3x(x - 7) - 1(x - 7) = (3x - 1)(x - 7)$.



Application activity 2.2:

1) Factorize the following expressions:

(a) $x^2 + 6x + 8$

(b) $x^2 - 49$

2) Can the following expressions be factorized? Write YES or NO for the results.

(a) $x^2 + 9x + 14$ (b) $x^2 - 11x - 12$

(c) $5u^2 + 6u + 9$ (d) $2t^2 - t - 42$

3) A man bought a certain number of shirts for 2 000 FWR . If each shirt had cost 200 FWR less, he could have bought five more for the same money .

(a) Form and simplify an expression from the above information. Is it the following:

$(n+5) \times (p-200) = 2000$?

(b) What name can be given to the above expression?

(c) Factorize the expression obtained in 3(b) above.

4) The perimeter of a rectangle is 70 cm. If the diagonal is 25 cm:

(a) Form an expression that relates the perimeter, diagonal, length and width of the rectangle.

(b) Factorize the expression obtained in 4(a) above.

2.3. Solving Quadratic Equations by factorization method



Activity 2.3:

1) In S2 we found that if $A \times B = 0$, either $A = 0$ or $B = 0$. Use this property on $(x - 2)(x + 3) = 0$.

What are the possible values of $(x - 2)$ and $(x + 3)$? Find the values of x for each expression.

2) Given that $(x + 4)(x + 6) = 0$, find two possible values of x .

3) Factorize completely $4x^2 - 2x$. Hence find the values of x given that $4x^2 - 2x = 0$.



Content summary

To solve a quadratic equation, the quadratic expression is factorized so that the equation is in the form $(x + a)(x + b) = 0$. Then either $(x + a) = 0$ or $(x + b) = 0$.

Thus $x = -a$ and $x = -b$.

Example 1

Solve: $(x-4)(x+1)=0$

Solution

If $(x-4)(x+1)=0$, then either $x-4=0$ or $x+1=0$.

Therefore, $x=4$ or $x=-1$.

Hence the roots of the equation $(x-4)(x+1)=0$ are 4 or -1.

Example 2

Solve the equation $x^2 + 7x + 6 = 0$. It is necessary to factor this expression and write it in form of $(x+a)(x+b)=0$.

Solution

Factorizing, $x^2 + 7x + 6 = 0$ gives $(x+6)(x+1)=0$.

Therefore, $x+6=0$ or $x+1=0$, Which means $x=-6$ or $x=-1$

Example 3

Solve the quadratic equation $6x^2 + 5x - 4 = 0$

Solution

$$6x^2 + 5x - 4 = 0$$

When the coefficient of x^2 , (in this case, it is 6), in the quadratic expression is numerically greater than 1, we proceed as follows when factorizing:

- Multiply the coefficient of x^2 by the constant term, i.e. $6 \times (-4) = -24$.
- Find the factors of -24 whose sum is 5, (the coefficient of x), i.e. -3 and 8.
- Rewrite the equation as:

$$6x^2 + (-3+8)x - 4 = 0 \Rightarrow 6x^2 - 3x + 8x - 4 = 0$$

$$\Rightarrow 3x(2x-1) + 4(2x-1) = 0$$

$$\Rightarrow (2x-1)(3x+4) = 0.$$

Then, either $2x-1=0$ or $3x+4=0$

$$\Rightarrow 2x=1 \text{ or } 3x=-4$$

$$x = \frac{1}{2} \text{ or } x = -\frac{4}{3}$$



Application activity 2.3:

1) Obtain the values of x from the following expressions.

(a) $(x+4)(x+2)=0$

(b) $(x-5)(x-3)=0$

(c) $(x-5)(x+7)=0$

(d) $x^2 - 9 = 0$

2) Factorize the following expressions hence find the values of unknowns.

(a) $x^2 + 9x + 14 = 0$

(b) $x^2 - 11x - 12 = 0$

(c) $u^2 + 6u + 9 = 0$

2.4 Solving Quadratic equations by use of Perfect Squares



Activity 2.4:

1) If a number squared equals **zero**, what must the number be?

If a number squared equals **25**, what could the number be?

2) You learnt in S2 the perfect squares such as $(a+b)^2$ and $(a-b)^2$. Determine whether the expression of the lefthand side of following quadratic expression is a perfect square or not. Then, solve the given equation.

(a) $x^2 + 8x + 16 = 0$

(b) $x^2 - 10x + 25 = 9$



Content summary

Recall that:

1. $(a+b)^2 = a^2 + 2ab + b^2$, and

2. $(a-b)^2 = a^2 - 2ab + b^2$

The right-hand sides of 1 and 2 are the expansions of the squares on the left hand sides. Such expressions are called perfect squares.

A quadratic equation can sometimes be expressed as:

a) Case A: Perfect Square = 0

$$(\text{expression})^2 = 0$$

This gives exactly **one solution** (one repeated root).

- The square of the sum of two numbers is equal to the sum of their squares plus twice their product. $(a+b)^2 = a^2 + 2ab + b^2$. If $a^2 + 2ab + b^2 = 0 \Rightarrow (a+b)(a+b=0)$ and $a = -b$

- The square of the difference of two numbers is equal to the sum of their squares minus twice their product. $(a-b)^2 = a^2 - 2ab + b^2$. If $a^2 - 2ab + b^2 = 0 \Rightarrow a - b = 0$ and $a = b$

b) Case B: Perfect Square = a Positive Number

$$(\text{expression})^2 = k, \quad k > 0$$

This gives two distinct real solutions.

To solve such equation,

- Move constant term to the other side.
- Write the left side as a perfect square.
- Compare to cases A or B.

- Solve for x .

Example 1

Solve: $x^2 - 6x + 9 = 0$

Solution

$$x^2 - 6x + 9 = (x-3)(x-3).$$

$$x^2 - 6x + 9 = 0 \Rightarrow (x-3)^2 = 0$$

$(x-3)(x-3) = 0 \Rightarrow x-3=0$ and $x-3=0 \Rightarrow x=3$ (Only one solution which is a repeated root).

$$S = \{3\}.$$

Example 2:

Solve: $x^2 + 4x + 4 = 36$

Solution

Factorizing $x^2 + 4x + 4 = (x+2)(x+2) = (x+2)^2$.

$$x^2 + 4x + 4 = 36 \rightarrow (x+2)^2 = 36 \rightarrow x+2 = \pm 6$$

Solve both cases:

$$x+2=6 \Rightarrow x=4$$

$$x+2=-6 \Rightarrow x=-8$$

Two solutions: $x=4$ or $x=-8$. We find **two distinct real solutions**.

$$S = \{-8, 4\}.$$

Example 3:

$$x^2 - 8x + 16 = 5$$

Solution

$$x^2 - 8x + 16 = (x-4)(x-4) = (x-4)^2$$

$$\text{Therefore } (x-4)^2 = 5 \Rightarrow x-4 = \pm 5$$

Solve both cases:

$$x-4=5 \Rightarrow x=9$$

$$x-4=-5 \Rightarrow x=-1$$

$$S = \{-1, 9\}.$$



Application activity 2.4:

1) Solve: (a) $x^2 + 2x + 1 = 0$ (b) $x^2 + 10x + 25 = 49$ (c) $x^2 - 12x + 36 = 0$

2) Solve a) $x^2 + 14x + 49 = 100$ using perfect square method.

3) For what value of m does $x^2 + mx + 36 = 0$ have a perfect square?

Solve for x in that case.

2.5 Solving quadratic equations using completing squares method



Activity 2.5:

1) Observe the following expression $x^2 + 4x + k = 0 + k$. Find the number or expression k at the that makes the Left side a perfect square: $x^2 + 4x + k$.

Then, factorize the obtained expression using the propriety of square root.

2) Use the same procedure to solve the following quadratic equations by completing squares method.

(a) $x^2 + 6x + 4 = 0$

(b) $x^2 - 8x + 10 = 0$



Content summary

Completing the square when the coefficient of x^2 is 1:

When the coefficient of x^2 is one, the quantity to be added is the square of half of the coefficient of x (or whatever letter is involved), i.e. to make $x^2 + bx$ a perfect square, we add $\left(\frac{b}{2}\right)^2$ to get

$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2, \text{ where } b \text{ is any positive or negative number.}$$

In some cases, it is not possible to solve a quadratic equation by the method of factorization because the LHS does not factorize. In such a case, we first rearrange the equation to make the LHS a perfect square, i.e. we complete the square on the LHS.

If LHS of an equation factorizes, it is better to use the method of factorization instead of completing the square.

Example 1

What must be added to $x^2 + 8x$ to make the result a perfect square?

Solution

Suppose $x^2 + 8x + k$ is a perfect square and that it is equal to $(x + a)^2$, i.e. $x^2 + 8x + k = (x + a)^2$

We know that

$$(x + a)^2 = x^2 + 2ax + a^2$$

$$\therefore x^2 + 8x + k = x^2 + 2ax + a^2.$$

The expression on the LHS and that on the RHS can only be equal if the corresponding terms are equal.

Comparing coefficients of x : $2a = 8$

$$\therefore a = 4.$$

Comparing the constant terms: $k = a^2 = 4^2 = 16$.

$\therefore 16$ must be added to the expression $x^2 + 8x$.

Then $x^2 + 8x + 16 = (x + 4)^2$.

Check by opening the bracket on the RHS!

Example 2

Solve the equation $x^2 + 8x + 9 = 0$.

Solution

The LHS does not factorize and so we use the method of completing the square.

$$x^2 + 8x = -9 \text{ (subtracting 9 from both sides)}$$

$$x^2 + 8x + \left(\frac{8}{2}\right)^2 = -9 + \left(\frac{8}{2}\right)^2 \text{ (adding } \frac{8}{2} \text{ to both sides)}$$

$$x^2 + 8x + 4^2 = 7 \Leftrightarrow (x + 4)^2 = 7 \Rightarrow x + 4 = \pm\sqrt{7} \Rightarrow x = -4 \pm \sqrt{7}$$

$$\text{Thus } x = -4 + \sqrt{7} \text{ or } x = -4 - \sqrt{7}.$$

$$S = \{-4 + \sqrt{7}, -4 - \sqrt{7}\}.$$

Completing the square when the coefficient of x^2 is different from 1:

If the quadratic equation is of the form $ax^2 + bx + c = 0, a \neq 1$ and the LHS does not factorize, first divide both sides by a to make the coefficient of x^2 one and then complete the square.

Example 3

Solve the equation $2x^2 + 14x + 9 = 0$, giving your answer correct to 3 d.p.

Solution

First make the coefficient of x^2 one by dividing both sides by 2.

$$\text{The equation becomes } x^2 + 7x + 4\frac{1}{2} = 0 \Rightarrow x^2 + 7x = -4\frac{1}{2}.$$

$$\text{Completing the square gives } x^2 + 7x + \left(\frac{7}{2}\right)^2 = -4\frac{1}{2} + \left(\frac{7}{2}\right)^2$$

$$\Rightarrow \left(x + \frac{7}{2}\right)^2 = -4\frac{1}{2} + \frac{49}{4} = -\frac{9}{2} + \frac{49}{4} = \frac{-18 + 49}{4} = \frac{31}{4}$$

$$\therefore x + \frac{7}{2} = \pm\sqrt{\frac{31}{4}} \Rightarrow x = -\frac{7}{2} + \frac{\sqrt{31}}{2} \text{ or } x = -\frac{7}{2} - \frac{\sqrt{31}}{2}$$

$$S = \left\{-\frac{7}{2} + \frac{\sqrt{31}}{2}, -\frac{7}{2} - \frac{\sqrt{31}}{2}\right\}$$



Application activity 2.5:

1) Solve the following quadratic equations by completing squares method.

(a) $x^2 - 2x - 8 = 0$

(b) $2x^2 + 6x + 1 = 0$

2) Complete the squares in the following expressions.

(a) $x^2 - 8x - 7 = 0$

(b) $3x^2 - 2x - 6 = 0$

(c) $x^2 + 12x = 0$

3) Solve the following equations by factorization, if possible, otherwise by completing the square. Give your answers to 2 d.p. where necessary.

(a) $2x^2 - 3x = 9$

(b) $2y^2 - 4y + 1 = 0$

(c) $2b^2 + b + 1 = 0$

2.6 Solving quadratic equations by formula method



Activity 2.6:

1) Use the steps that are applied in solving quadratic equations by completing squares to obtain the values of x in terms of a , b and c for which $ax^2 + bx + c = 0$

(a) First divide the whole equation by a to make the coefficient of x^2 be one.

(b) Solve for the value of x by completing the square.

(c) Make x the subject of the formula.

2) Use the expression you obtained in step 1 to solve the equation $2x^2 - 5x + 3 = 0$ by substituting $a = 2$, $b = -5$ and $c = 3$.

3) Solve the quadratic equation $2x^2 - 5x + 3 = 0$ by factorization method.

4) Compare your results for step 2 and step 3. What do you notice?



Content summary

Consider the quadratic equation $ax^2 + bx + c = 0$.

When we solve it using the completing square method, we get:

$$x = -\frac{b}{2a} \pm \sqrt{\left(\frac{b}{2a}\right)^2 - \frac{c}{a}} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When we take the discriminant $D = \sqrt{b^2 - 4ac}$ and $D \geq 0$, we find $x = \frac{-b \pm D}{2a}$.

Therefore, $x_1 = \frac{-b + D}{2a}$ and $x_2 = \frac{-b - D}{2a}$

$ax^2 + bx + c = 0$ is the same as $\left(x - \frac{-b + D}{2a}\right)\left(x + \frac{-b - D}{2a}\right) = 0$

This formula is called the **quadratic formula**, and it is used to solve any quadratic equation provided the coefficients a, b and c are all known.

Example 1

Solve the equation $2x^2 - 3x - 4 = 0$ with the aid of a quadratic formula.

Solution

In this case $a = 2$, $b = -3$, $c = -4$. The discriminant $D = \sqrt{(-4)^2 - 4 \times 2 \times (-2)} = \sqrt{41}$

$$\therefore x = \frac{3 + \sqrt{41}}{4} \text{ or } x = \frac{3 - \sqrt{41}}{4}. \quad S = \left\{ \frac{3 + \sqrt{41}}{4}, \frac{3 - \sqrt{41}}{4} \right\}.$$

Example 2

Solve the equation $3x^2 + 11x + 6 = 0$ with the aid of a quadratic formula.

Solution

In this case $a = 3$, $b = 11$, $c = 6$. Therefore, $D = \sqrt{(11)^2 - 4 \times 3 \times 6} = 7$

$$\therefore x = \frac{-11 + 7}{6} = \frac{-4}{6} = -\frac{2}{3} \text{ or } x = \frac{-11 - 7}{6} = \frac{-18}{6} = -3. \quad S = \left\{ -3, -\frac{2}{3} \right\}.$$



Application activity 2.6:

1) Solve the following with the aid of quadratic formula, giving answers to two decimal places where necessary:

(a) $2x^2 + 11x + 5 = 0$

(c) $6x^2 + 7x + 2 = 0$

(d) $3x^2 - 10x + 3 = 0$

(e) $5x^2 - 7x + 2 = 0$

2) From the questions below, circle the correct solution.

(a) When the quadratic equation $2x^2 + 5x - 3 = 0$ is solved, the results obtained are;

(i) -3 and 2

(ii) -3 and 12

(iii) 4 and -2 (iv) 6 and -1

(b) The equation $2x^2 - x - 6 = 0$ when solved gives the values of x as;

(i) 2 and -3

(ii) -1 and 3

(iii) 3 and -2

(iv) None of the above

(c) Which of the solutions below is suitable solution for $2x^2 + 3x - 5 = 0$?

(i) -2.5 and 1

- (ii) -1.5 and 3
- (iii) -3.4 and -2
- (iv) None of the above

3) Write true/false for the statements below.

- (a) 1 and 9 are the solutions to $3x^2 - 10x + 3 = 0$
- (b) $x^2 + 9x + 14 = 0$ is a perfect square
- (c) $x^2 + 9x + 14 = 0$ has no solution in the field of real numbers.
- (d) $x^2 + 5x - 500 = 0$ has two solutions one is positive and another is negative
- (e) The equation $x^2 - x - 12 = 0$ cannot be solved by factorization method.

2.7 Solving Quadratic equations by synthetic division method



Activity 2.7:

Use internet or library to recall about factorization of quadratic equations and polynomials by using Horner's rule. List the steps that are followed using your own words.



Content summary

Consider the quadratic equation $ax^2 + bx + c = 0$. To solve this quadratic equation, we proceed as follows:

Steps

1. Write the coefficients of all terms including the constant term in descending powers of the variable in the quadratic equation.

$$ax^2 + bx + c$$

↓
 a

↓
 b

↓
 c

← Coefficients

(on page 80)

2. In case there are missing powers of the variable in the equation, you should represent them with zero coefficients.

For example

$$ax^2 - bx \Rightarrow ax^2 - bx + 0x^0$$

$\downarrow \quad \downarrow \quad \downarrow$
must be put
 Coefficient $\rightarrow a \quad b \quad 0$

$$ax^2 - c \Rightarrow ax^2 + 0x - c$$

$\downarrow \quad \downarrow \quad \downarrow$
 Coefficient $\rightarrow a \quad 0 \quad c$
this must be written

$$ax^2 + 0x^1 + 0x^0$$

$\downarrow \quad \downarrow \quad \downarrow$
 $a \quad 0 \quad 0$
These must be written

3. Identify all the factors of the constant c both positive and negative. Let the factors of c be e, f, g, h etc. Choose the factor of c which can reduce the quadratic expression to zero. Let the factor of c be f. This means that f is one of the solutions of the quadratic equation provided. i.e. $x = f$ and the linear factor is $(x - f)$
4. Arrange the coefficients and the factor f as shown below.

f	a	b	c
-----	-----	-----	-----

5. Proceed to get the other factor of the quadratic equation as follows.

f	a	b	c
	$\downarrow$	$\downarrow$	$\downarrow$
	$a \times f$	$b + af$	$c + f(b + af)$

$\downarrow \quad \downarrow \quad \downarrow$
Drop the value a *This gives zero*

For vertical patterns, add terms while for the diagonal patterns, multiply by factor f.

Hence on dividing the expression $ax^2 + bx + c$ by $(x - f)$, we should get the other factor (quotient) as $ax + (b + af)$. Thus the quadratic equation

$$ax^2 + bx + c = (x - f)(ax + (b + af)) \text{ in factorized form.}$$

We therefore solve the values of x from the factors $(x - f)(ax + (b + af)) = 0$, meaning

$$x - f = 0 \text{ or } ax + (b + af) = 0$$

Example 1

Solve the quadratic equation $x^2 - x - 6 = 0$ using synthetic division method.

Solution

The factors of 6 are $\pm 1, \pm 2, \pm 3$ and -6 . The factor of -6 that can reduce the quadratic equation to zero is 3.

	1	-1	6
3	↓	↗	↗
	1	2	0

The values 1 and 2 obtained are the coefficients of the next linear factor called quotient.

Here we then get $(x-3)(x+2) = 0$. Hence $x = 3$ and $x = -2$ are the roots of the given quadratic equation.

Example 2

Solve the quadratic equation $3x^2 - 5x - 2 = 0$ by synthetic division method.

Solution

The factor of -2 that can reduce the quadratic expression to zero is $+2$.

The coefficients of the quadratic equation are 3, -5 and -2 .

	3	-5	-2
2	↓	↗	↗
	3	1	0

$$(x-2)(3x+1) = 0 \text{ and } x = 2 \text{ or } x = -\frac{1}{3}. S = \left\{ 2, -\frac{1}{3} \right\}.$$



Application activity 2.7:

Solve the following quadratic equations by synthetic division method.

1) $x^2 + x - 12 = 0$ 2) $x^2 + 9x + 14 = 0$ 3) $u^2 + 6u + 9 = 0$

4) $a^2 - 2a + 1 = 0$ 5) $x^2 + 10x = 24$



END UNIT ASSESSMENT

Assessing knowledge and understanding

1. What is a quadratic equation?
2. Which of the following is the general form of a quadratic equation?

- A) $ax+b=0$
- B) $ax^2+bx+c=0$
- C) $ax^3+bx+c=0$
- D) $(ax+b)^2$

3. Why is the equation $2x^2 - 7x + 3 = 0$ called a quadratic equation?

4. Match each term in Column A with its correct description in Column B.

Column A

Column B

- | | |
|---------------------------|------------------------------------|
| (a) Coefficient | 1. Number without a variable |
| (b) Constant term | 2. Highest power of the variable |
| (c) Degree of an equation | 3. Number multiplied by a variable |

5. Explain how a quadratic equation differs from a linear equation.

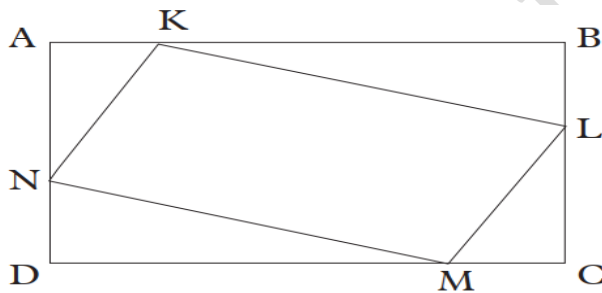
Assessing skills

6. Solve $2x^2 + 20x = 48$.

7. Solve $6x^2 - 10x - 4 = 0$

8. Solve $x^2 - 16x = 73$

9. In the following figure, ABCD is a rectangle with $AB = 12$ cm and $BC = 7$ cm. $AK = BL = CM = DN = x$ cm.



- a) What is the base of the shape KLMN?
- b) What is its height?
- c) use the result in (a) and (b) to write expression of the area of KLMN.
- d) If the area of KLMN is 54 cm^2 , find x .

Assessing attitudes and values

- 10. Why is accuracy important when solving quadratic equations?
- 11. What positive attitude should a learner show when a quadratic equation is difficult to solve?
- 12. How does learning quadratic equations help develop logical thinking?
- 13. Why is it important to check solutions of quadratic equations after solving them?
- 14. How can working in groups on quadratic equations promote good values among learners?

$$\begin{array}{r}
 \overset{1}{1} \overset{1}{1} \overset{1}{0} 1_2 \\
 1101_2 \\
 \hline
 11010_2
 \end{array}$$

Unit 3: NUMBER BASES

Key unit competence: Solve problems involving conversion, operation and equations of number bases.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Identify the digits and place values used in each base.
- Explain the process of converting between base ten and other bases using place value and grouping.

Skills:

- Convert numbers fluently between base ten and other bases and vice versa.
- Convert numbers between two non-base-ten systems (e.g., base 2 to base 5).
- Execute addition, subtraction, multiplication, and division operations on numbers in different bases.
- Solve simple equations with number bases.

Attitudes and values:

- Value the role of different number bases as foundational to modern technology (computers).
- Develop patience and systematic thinking when working with abstract numerical systems.
- Collaborate effectively to troubleshoot and verify solutions in group tasks.

Introduction

In this unit, we will learn a number of different numeration systems including the decimal (base ten) system that we are all familiar with. We will also learn how to convert between different numeration (counting) systems.



Introductory activity

In everyday life, we count or estimate quantities using groups of ten items or units. This may be so because, naturally, we have ten fingers. For example, when we count ten, i.e. we write 10 meaning one group of 10 and no units. A quantity like twenty-five, written as 25 means 2 groups of 10 and 5 units. Suppose instead we had say 6 fingers

- (i) How, in your opinion would we do our counting?
- (ii) If we had eight fingers, how would we count?
- (iv) Demonstrate symbolically how counting in groups of 3, 4, 5, 6, 7...can be done.
- (v) Do you think we could also do operations such as addition, subtraction, multiplication and division using such groups? If your answer is yes, demonstrate this with simple examples.

3.1 Meaning and examples of a number, a digit, a numeral and number base



Activity 3.1:

1) Use a dictionary or internet to define:

- (i) Number
- (ii) Numeral
- (iii) Digit

2)

- (a) $253 = 3 \times 10^0 + 5 \times 10^1 + 2 \times 10^x$. Find the value of x taken as exponent of 10 at the last term.
- (b) What the base of powers observed in this expression?
- (c) If 253 is written in base y in (b), what are other bases used to write a number?



Content summary

In mathematical numeral systems, we use basic terms such as number, numeral and digit. In order to deal with number bases, we must be able to distinguish between the three terms.

A **number** is an idea; a **numeral** is the symbol that represents the number. The number system that we use today is a **place value system**. A unique feature of this system is that the value of any of the digits in a number depends on its position. For example, the number 7 707 contains three sevens, and each of them has a particular value as shown in table 2.1.

Thousands	Hundreds	tens	Ones
7	7	0	7

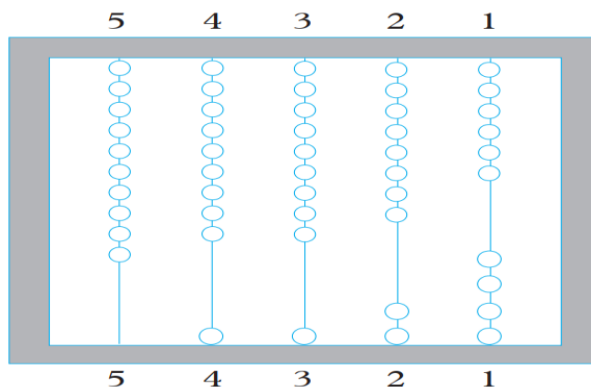
The first seven from the right represents 7 ones or units. The 2nd seven stands for 100s or 10^2 and the 3rd seven stands for 1 000s or 10^3 . The zero holds the place for the tens (10s) without which, the number would be 777 which is completely different from 7 707.

A **digit** is any numeral from 0 to 9.

A **numeral** is made of one or more digits. For example, number one hundred and thirty-five is represented by the numeral 135 which has three digits 1, 3 and 5. The number 7707 contains four digits, each of which has a specific value depending on its place value.

One device that has been used over time to study the counting in different numeration systems is the abacus.

An abacus is a calculating device consisting of beads or balls strung on wires or rods set in a frame in figure below, shows a typical abacus on which the place value concept can be developed very effectively.



On each wire, there are ten beads. Let us consider the beads at the bottom of the wire. Beginning from the right: 10 beads on wire 1 can be represented by 1 bead on wire 2. Similarly, 10 beads on wire 2 can be represented by 1 bead on wire 3 and so on.

This means:

1 bead in wire 1 represents a single bead.

1 bead in wire 2 represents 10 beads.

1 bead in wire 3 represents (10×10) beads.

1 bead in wire 4 represents $(10 \times 10 \times 10)$ beads.

So, the number shown in figure is 1 124.

If we had x beads in each wire such that $x < 10$, it would mean that:

In wire 1 we had x beads

In wire 2 we had $10x$ beads

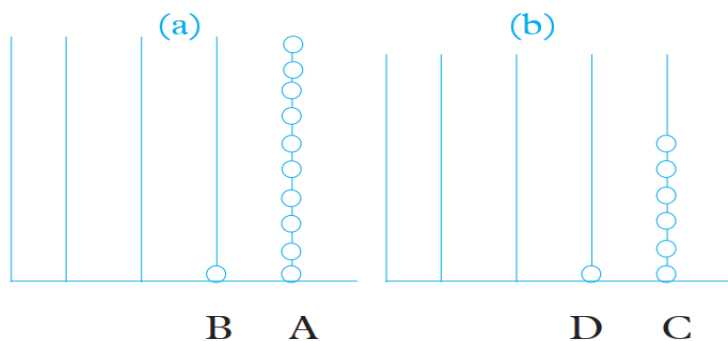
In wire 3 we had 10^2x beads

In wire 4 we had 10^3x beads and so on.

The place values from right to left are

10^0	10^1	10^2	10^3	$10^4 \dots$
Ones	10s	100s	1000s	10000s etc

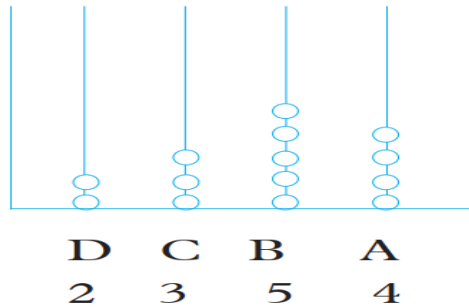
If we had 6 fingers, most probably we would count using groups of 6, if 8 fingers, groups of 8 and so on. In the system that we use, every **ten** items make **one** basic group which is represented in the next place value column to the left as shown in Fig. 2.2 below.



(a) The 1 bead in wire B represents 10 beads in wire A i.e. it represents a group of 10 beads.

- (b) The 1 bead in wire D represents 6 beads in wire C, thus making a group of 6 beads.
Counting in different groups of numbers such as 10, 6, 5, 8 etc means using different number systems. We call them base ten, base six, base five, base eight respectively etc.

Now consider Fig. 2.3.



Counting in base six, what numbers do the beads on each wire represent?

- (i) There are 4 beads in wire A. This represents 4 ones.
- (ii) There are 5 beads in wire B. This means 5 groups of 6 beads each. i.e. $5 \times 6 = 30$ beads written as 50_{six} .
- (iii) There are 3 beads in wire C. This means 3 groups of six sixes i.e. $3 \times 6 \times 6 = 108$ beads, written as 300_{six} .
- (iv) There are 2 beads in wire D. This means 2 groups of six six sixes i.e. $2 \times 6 \times 6 \times 6 = 216 \times 2 = 432$

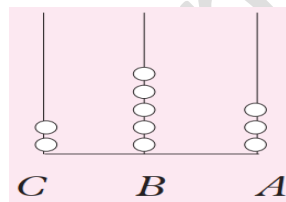
written as 2000_{six} . The whole number represented in Figure 3

is $4_{\text{six}} + 50_{\text{six}} + 300_{\text{six}} + 2000_{\text{six}} = 2354_{\text{six}}$

The answer 2354_{six} is read as; two three five four base six. The number 2 354 six has a value of 574_{ten}

Example 1

Given that the number represented here below is in base six, find the number in base 10.



Solution

Column A represents 3 ones.

Column B represents 5 sixes.

Column C represents 2 six sixes

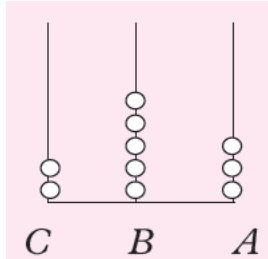
$$\therefore \text{the number} = (3 \times 6^0) + (5 \times 6^1) + (2 \times 6^2) = 3 + 30 + 72 = 105_{\text{ten}}$$

Note $253_{\text{six}} = 105_{\text{ten}}$ are two different symbols for the same number.



Application activity 3.1:

Given that the number represented in the figure below is in base five, find the number in base 10.



3.2 Converting a number from base ten to any other base and vice versa



Activity 3.2:

Consider the number 725 given in base ten.

1. Divide 725 by 8 and write down the remainder.
2. Divide the quotient obtained in (1) by 8 above and write down the remainder.
3. Repeat this process of division by 8 until the quotient is less than 8 which you should treat as a remainder and write it down.
4. Write down the number made by the successive remainders beginning with the first one on the right going left.
5. Describe the number in part (4) above in terms of a base.



Content summary

In this activity, you have just converted 725_{10} to a number in base 8. In converting any number from base ten to any other base, we use successive division of the number by the required base. The new number is obtained by writing down the remainders beginning with the first remainder on the right to the last remainder on the left.

For example, to change 425_{10} to base 6, we do successive division by 6.

$$425 \div 6 = 70 \quad \text{Rem } 5$$

$$70 \div 6 = 11 \quad \text{Rem } 4$$

$$11 \div 6 = 1 \quad \text{Rem } 5$$

$$1 \div 6 = 0 \quad \text{Rem } 1$$

The successive remainders read upwards form the number 1545

$$\therefore 425_{10} = 1545_6$$

Example 1

Convert 92_{10} to base 6.

Solution

To convert 92_{10} to base 6, we perform successive divisions by 6 until the remainder is less than 6.

6	92		
6	15	Rem 2	↑
6	2	Rem 3	↑
	0	Rem 2	↑

$92 \div 6 = 15 \text{ rem } 2$
 $15 \div 6 = 2 \text{ rem } 3$
 $2 \div 6 = 0 \text{ rem } 2$

$92_{10} = 232_6$

Example 2

Convert 194_{10} to base 8

Solution

194 is divided by 8 successively until the remainder is less than 8.

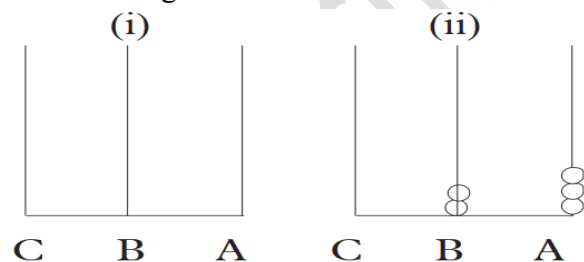
8	194		
8	24	Rem 2	↑
8	3	Rem 0	↑
	0	Rem 3	↑

$194_{10} = 302_8$

To convert a number from one base to base ten, we use number place values. For example, to convert 253_9 to base 10, we say: 253_9 means 5 ones + 3 nines + 2 nine nines.

$$\begin{aligned} \therefore 253_9 &= (5 \times 9^0) + (3 \times 9^1) + (2 \times 9^2) \\ &= (5 \times 1) + (3 \times 9) + (2 \times 9^2) = 5 + 27 + 162 \\ &= 194 \\ \therefore 253_9 &= 194_{10} \end{aligned}$$

Consider Figure 1 below.



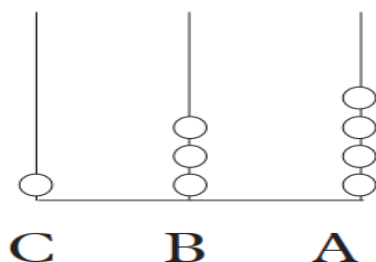
Suppose in the figure here above, each spike is designed to hold six beads, and that each bead in spike B represents six beads in spike A.

Thus, in figure (ii) there are two beads in B and three beads spike A. The 2 beads mean 2 groups of six i.e 2×6 or 12 beads.

The 3 beads are said to represent 3 ones. Thus, the number represented in figure (ii) is written as 23_{six} .

Therefore, $23_{\text{six}} = 15_{\text{ten}}$. This is read as **two three** base six equal **one five** base ten: 23_{six} and 15_{ten} are different numerals for the same number.

Now consider the figure below:



The number shown in Fig 2.6 can be written as 134_{six} . What does the single bead in spike C represent? It is the same as six beads in spike B which is equal to $\text{six} \times \text{six}$ (or thirty-six) beads in spike A. Hence 134_{six} means:

The 1 stands for 1 six sixes or 36_{ten}

The 3 stands for 3 six or $3 \times 6_{\text{ten}}$

The 4 stands for 4 ones or 4_{ten}

So, we would write

$$134_{\text{six}} \text{ as } (36 + 18 + 4)_{\text{ten}} = 58_{\text{ten}}$$

$$\text{i.e } 134_{\text{six}} = 58_{\text{ten}}$$



Application activity 3.2:

1) Convert the following numbers from specified base to base 10.

(a) 85_9 (b) 1001_2 (c) 2343_5 (d) 12_3 (e) 615_7 (f) 142_5 (g) 1232_4

2) Are the following valid or invalid?

(a) 123_2 (b) 234_5 (c) 1002_2 (d) 3467_6

3.3. Converting a number from one base to another



Activity 3.3:

Consider the number 46_7 .

(a) Convert 46_7 to a number in base 10 as you did in activity 3.2

(b) Use your answer to part (a) above and convert it to a number in base 5.

(c) Describe the procedure of converting a number from a number in base x to a number in base y where $x \neq y$.



Content summary

Suppose we wish to change from base m to base n where $m \neq n \neq 10$ and m and n are positive numbers.

To convert a number from a base other than 10 to another base, we first convert from the given base to base 10. Then from base 10 to the required base.

Example 1

Convert 514_8 to base 9.

Solution

To convert from base 8 to base 9,

(i) First convert to base 10

(ii) Then convert result (i) to base 9

$$514_8 = (4 \times 1 + 1 \times 8 + 5 \times 8^2)_{10} = (4 + 8 + 320)_{10} = 332_{10}$$

To convert 332_{10} to base 9, we do successive division by 9, noting the remainder at each step.

Thus:

Thus:

9	332	
9	36	Rem 8
9	4	Rem 0
	0	Rem 4

From down upwards the remainders form number 408.

This means 8 ones

0 nines

4 nine-nines

Thus $514_8 = 408_9$.

Numbers in other bases can be expressed in the same way as we have done.

The following are some other bases and the numerals used.

Base Numerals

Nine: 0 1 2 3 4 5 6 7 8

Eight: 0 1 2 3 4 5 6 7

Seven: 0 1 2 3 4 5 6

Six: 0 1 2 3 4 5

Five: 0 1 2 3 4

Four: 0 1 2 3

Three: 0 1 2

Two: 0 1

and so on.



Application activity 3.3:

1) Convert the following to base 7.

(a) 411_5 (b) 321_6 (c) 15_6 (d) 302_4

2) Express 637 to base 5

3) Given that $85_{10} = 221_x$. Find the value of x.

4) Convert the number 703_8 to:

(a) Base 6 (b) Base 10 (c) Base 9 (d) Base 2 (e) Base 4

5) Write the first twenty numerals of:

(a) Base six (b) Base seven (c) Base eight

3.4 Addition and subtraction on number bases



Activity 3.4:

Table below shows part of the addition table for numerals in a certain base

(i) State the base. (ii) Copy and complete the table.

(iii) Use your table to evaluate.

$15 - 10$; $13 - 5$; $12 - 3$;

+	0	1	2	3	4	5	10
0	0	1	4	10			
1	4	10					
2	3	10					
3	3	5	12				
4	11	14					
5	11						
10	10						



Content summary

To add or subtract, numbers must be in the same base. In performing addition or subtraction, whatever the base, the digits to be added or subtracted must be in the same place value. For example, in $65_{\text{ten}} + 18_{\text{ten}}$, 5 and 8 have the same place value while 6 and 1 have another place value.

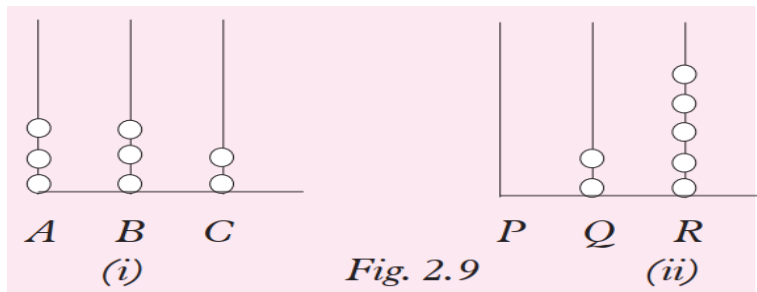
Example 1

Evaluate: $332_{\text{six}} + 25_{\text{six}}$

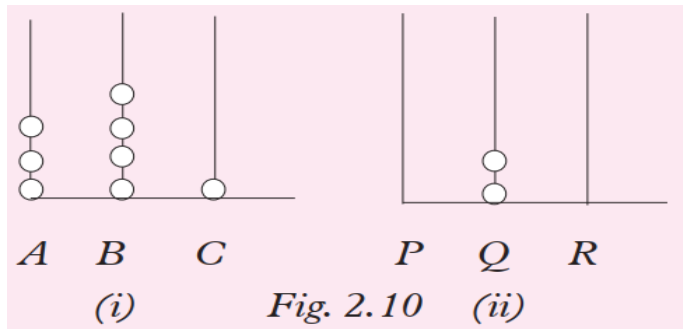
Solution

It is best to set work vertically so that the place values correspond. Follow the step:

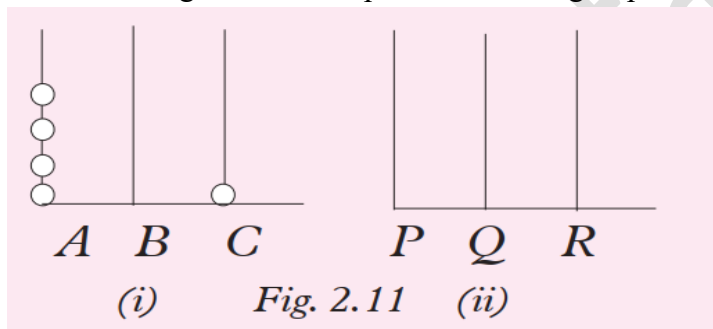
1) Illustrate the two numbers on different abacus



2) Remove all the 5 beads from R and place them in C to make 7 beads. One bead remains at C another goes to B to represent another group of six



3) Remove the two beads from Q and place them on B to make 6 beads. No bead remains at B, but one bead goes to A to represent another group of six sixes.



4) The result of the addition is 401_{six} Alternatively,

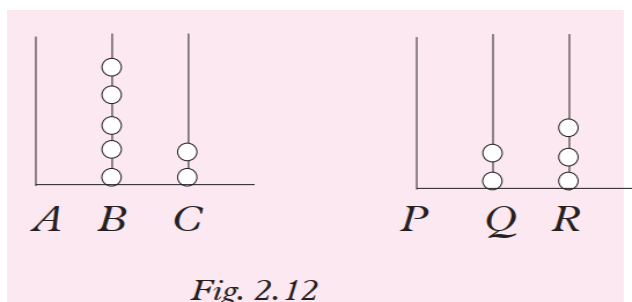
$$\begin{array}{rcl}
 332_{\text{six}} & \rightarrow & 330 + 2 \\
 25_{\text{six}} & \rightarrow & 20 + 5 \\
 & & 350 + 11 \\
 = & & 350 \\
 & & + 11 \\
 \hline
 & & 401_{\text{six}}
 \end{array}$$

Example 2

Use abacus to evaluate: $52_{\text{eight}} - 23_{\text{eight}}$

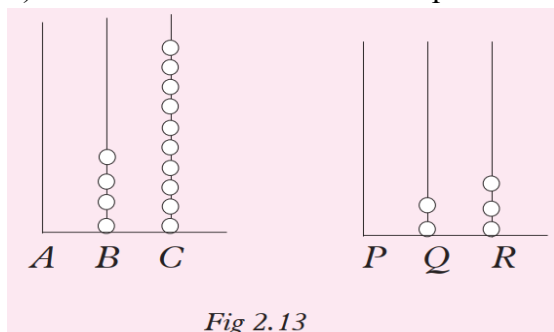
Solution

Fig. 2.12 shows the two numbers on different abaci.

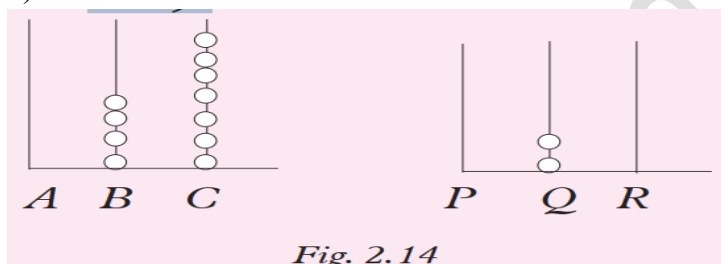


Since we cannot subtract beads in R from beads in C,

1) Remove one bead from B and place it on wire C so that there is a total of 10 in C, (Fig. 2.13)

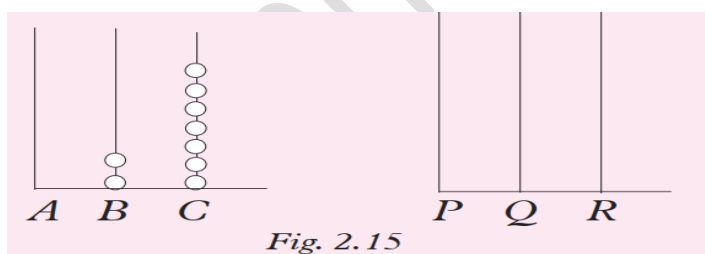


2) Remove 3 beads from C and R .



3) Remove 2 beads from B and Q so that the result is as represented in figure below

$$52_8 - 23_8 = 27_8$$



Alternatively,

$$\begin{array}{rcl} 52_8 & \rightarrow & 50 + 2 \rightarrow 40 + 10 \\ - 23_8 & \rightarrow & 20 + 3 \rightarrow 20 + 3 \\ & & 20 + 7 = 27_{\text{eight}} \end{array}$$



Application activity 3.4:

1. Work out the following in base eight:

(a) $17 + 211$ (b) $106 + 12$ (c) $257 + 462$

2. Evaluate the following in base six:

(a) $31 - 25$ (b) $145 - 51$ (c) $55 - 43$ (d) $403 - 54$

3. Evaluate the following in base nine:

(a) $122 + 85$ (b) $103 - 86$ (c) $17 - 8$ (d) $66 + 35$

4. The following calculations are correct.

State the base used in each case.

(a)
$$\begin{array}{r} 36 \\ + 26 \\ \hline 64 \end{array}$$
 (b)
$$\begin{array}{r} 53 \\ + 36 \\ \hline 111 \end{array}$$
 (c)
$$\begin{array}{r} 3 \\ + 23 \\ \hline 31 \end{array}$$

(d)
$$\begin{array}{r} 172 \\ - 169 \\ \hline 3 \end{array}$$
 (e)
$$\begin{array}{r} 82 \\ - 53 \\ \hline 28 \end{array}$$
 (f)
$$\begin{array}{r} 65 \\ - 36 \\ \hline 26 \end{array}$$

5. Each of the following calculations were done using a certain base. Three of them are correct. Identify:

(a) the base

(b) the incorrect ones and explain why.

(i)
$$\begin{array}{r} 22 \\ - 16 \\ \hline 6 \end{array}$$
 (ii)
$$\begin{array}{r} 68 \\ + 15 \\ \hline 84 \end{array}$$

(iii)
$$\begin{array}{r} 100 \\ - 64 \\ \hline 25 \end{array}$$
 (iv)
$$\begin{array}{r} 177 \\ + 19 \\ \hline 207 \end{array}$$

3.5 Multiplication and division on number bases



Activity 3.5:

1) Table below shows part of the multiplication table for numerals in a certain base.

x	0	1	2	3	4	5	6	7	8
0									0
1					4				8
2									17
3		3		10				23	26
4									35
5							33		44
6			13	20			40		53

7									62
8	0	8	17	26	35	44	53	62	71

- (a) Identify the base.
 (b) Copy and complete the table.
 (c) Given that x is a numeral, use your table to find the value of x if:
 (i) $6x = 13$ (ii) $3x = 23$ (iii) $7x = 46$ iv) $5x = 27$ v) $4x = 31$ vi) $6x = 46$

Use your table to formulate three equations using a variable of your choice.

- 2) (a) Given that a , b and c are numerals in base ten such that $ab = c$, express:

- (i) a in terms of b and c .
 (ii) b in terms of a and c .
 (iii) Describe the operation used to obtain the results above.

- (b) Given that $2_{\text{six}} \times 5_{\text{six}} = 14_{\text{six}}$, express:

- (i) 2_{six} in terms of 5_{six} and 14_{six} .
 (ii) 5_{six} in terms of 2_{six} and 14_{six} .

What operation have you used to obtain your results?

- (c) Make a multiplication table for base six and use it to confirm your findings in part (b) above.
 (d) Use the table in (c) above to create more examples of division.



Content summary

Note that for any base;

- (i) The highest numeral is always one less than the base and
 (ii) The least is zero (0).

Consider the product:

$$2_{\text{six}} \times 3_{\text{six}}$$

Whether in base ten or base six, 2 by 3 remains the same

$$2 \times 3 = 6_{\text{ten}} = 10_6$$

Multiplication table in base 9:

$\times$	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	4	6	8	11	13	15	17
3	3	6	10	13	16	20	23	26
4	4	8	13	17	22	26	31	35
5	5	11	16	22	27	33	38	44
6	6	13	20	26	33	40	46	53
7	7	15	23	31	38	46	54	62
8	8	17	26	35	44	53	62	71

Example 1

Use long multiplication to evaluate

$$45_{\text{six}} \times 23_{\text{six}}.$$

Solution

$$\begin{array}{r} \times 45 \\ 23 \\ \hline 1340 \\ + 223 \\ \hline 2003 \end{array}$$

(i) 1st row products

$$5 \times 2 = 10_{\text{ten}} = (14_{\text{six}}, \text{we write 4 and carry 1})$$

$$2 \times 4 = 8_{\text{ten}} (12_{\text{six}} \text{ plus the 1 we carried})$$

$$= 12_{\text{six}} + 1$$

$$= 13_{\text{six}} (\text{we write } 13_{\text{six}})$$

$$45 \times 2 = 134_{\text{six}}$$

(ii) 2nd row products

$$3 \times 5 = 15_{\text{ten}} = 23_{\text{six}} (\text{We write 3 and carry 2})$$

$$3 \times 4 = 12_{\text{ten}} = 20_{\text{six}} (20 \text{ plus 2 we carried})$$

$$\Rightarrow 20_{\text{six}} + 2_{\text{six}} = 22_{\text{six}}$$

$$\therefore 45 \times 3 = 223_{\text{six}}.$$

Add the products in the 1st and 2nd rows to get 2003_{six}

Now consider the example $23_{\text{six}} \div 5_{\text{six}}$.

To do this, you ask yourself, 'by what can I multiply 5_{six} to obtain 23_{six} ?'

This question can be answered using the multiplication table.

Example 2

Evaluate: $15_{\text{six}} \div 2_{\text{six}}$

Solution

$$\begin{array}{r} 5 \\ 2 \overline{)15} \\ \underline{14} \\ 1 \end{array}$$

$$2 \times 5 = 10_{\text{ten}} = 14_{\text{six}}$$

$$\therefore 15_{\text{six}} \div 2_{\text{six}} = 5 \text{ Rem } 1$$

We could also divide by first changing the number to base 10, then change back to base 6.

$$15_{\text{six}} = (1 \times 6) + 5 = 11$$

$$2_{\text{six}} = 2_{\text{ten}}$$

$$\therefore 15_{\text{six}} \div 2_{\text{six}} = 11_{\text{ten}} \div 2_{\text{ten}} = 5 \text{ Rem } 1$$

$$5_{\text{six}} = 5_{\text{ten}} \text{ and } 1_{\text{six}} = 1_{\text{ten}}$$

$$\therefore 15_{\text{six}} \div 2_{\text{six}} = 5_{\text{six}} \text{ Rem } 1$$

But this is a long and an unnecessary process.

A binary system

A **binary** system is a number system that uses only two digits 0 and 1. Numbers are expressed as powers of 2 instead of powers of 10 as in the decimal system. Computers use binary notation, the two digits corresponding to two switching position, on and off, in the individual electronic devices in the logic circuits. Remember; in any base there is no numeral equal to the base.

Example 1

Calculate in binary.

$$(a) 101 + 10101 \quad (b) 1001 - 101 \quad (c) 1101 \times 11 \quad (d) 1010101 \div 101$$

$$\begin{array}{rcl}
 (a) & 101 & (b) \quad 1001 \\
 + & \underline{10101} & - \underline{101} \\
 & 11010 & 100
 \end{array}$$

$$\begin{array}{r}
 (c) \quad 1101 \\
 \times \quad 11 \\
 \hline
 1101 \\
 1101 \\
 \hline
 100111
 \end{array}$$

$$\begin{array}{r}
 (d) \quad \begin{array}{r} 10001 \\ 101 \overline{) 1010101} \\ \underline{101} \\ 00101 \\ \underline{101} \\ 000 \end{array}
 \end{array}$$

Note: Just as in division in decimal system, remember to put a zero in the answer any time the divisor fails to divide.

A duodecimal system

It is a system of numbers whose base is twelve. When buying or selling in bulk, often, items are counted in groups of twelve i.e. dozens.

Earlier in the chapter, we saw that the numeral equivalent to the base is always represented by 10. Therefore, in the case of base twelve, we have to define two different variables to use in place of 10 and 11 to avoid confusion. Such substitutions are necessary when working with any base greater than 10, i.e. base eleven, thirteen etc. To be able to list the digits used in base twelve, let letter A represent 10, and B represent 11.

Thus, the digits in base twelve are:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B.

Example 1

Evaluate the following in base 12.

$$(a) 12A + 4AB \quad (b) 23B \times A \quad (c) 789 - AB$$

Solution

$$\begin{array}{r} (a) \quad 12A \\ + 4AB \\ \hline 619_{12} \end{array}$$

$A + B = 10 + 11 = 21_{ten} = 19_{12}$
Write 9 carry 1.
 $1 + 2 + A = 3 + 10 = 13_{ten} = 11_{12}$
Write 1 carry 1: $1 + 4 + 1 = 6_{12}$

$$\begin{array}{r} (b) \quad 23B \\ \times A \\ \hline 1B32 \end{array}$$

$A \times B = 10 \times 11 = 110_{ten} = 92_{12}$
Write 2 carry 9.
 $3A + 9 = 39_{ten} = 33_{12}$
Write 3 carry 3: $2A + 3 = 23_{ten} = 1B_{12}$

$$\begin{array}{r} (c) \quad 589 \\ - AB \\ \hline 49A \end{array}$$

$9 - B = 9 - 11$ (not possible)
Borrow 1 ($\equiv 12$) from 8.
 $(9+12)-11 = 21-11 = 10_{ten} = A$ in base 12
 $7 - A = 7 - 10$ (not possible)
Borrow 1 ($\equiv 12$) from 5
 $(7+12)-10 = 19-10 = 9$
Drop down 4



Application activity 3.5:

1) Copy and complete the multiplication table in base eight and use it to answer the questions below.

$\times$	0	1	2	3	4	5	6	7	10
0	0	0	0	0	0	0	0	0	0
1	0								
2	0				10				
3	0		6			17			
4	0								
5	0				24				
6	0	6		22					60
7	0							61	
10	0								

(a) $52_8 \div 7_8$ (b) $43_8 \div 5_8$ (c) $34_8 \div 7_8$ (d) $20_8 \div 4_8$

2) Evaluate the following:

(a) $15_{\text{six}} \times 11_{\text{six}}$ (b) $21_6 \times 12_6$ (c) $5_6 \times 5_6$ (d) $1333_6 \div 35_6$

3) (a) $2122_4 \div 23_4$ (b) $100122_4 \div 203_4$

4) (a) $1216_8 \div 3_8$ (b) $1032_6 \div 4_6$

3.7 Solving equations involving number bases



Activity 3.7:

(1) Solve the equations: (ii) $x - (-3) = -5$ (iii) $4x + 3 = 5$ given that you are working with base 10 system.

(2) Solve the equations in (a) above using base 6.



Content summary

Solving equations that involve number bases requires understanding how numbers behave in bases other than ten. The main idea is to **convert numbers to a common base (usually base 10)** or **operate directly using place values in the given base**.

Now, consider the equations below.

(i) $x + 5 = 12$ (ii) $x - (-2) = 4$ (iii) $\frac{2}{3}x + 3 = 5$

Working with base ten system,

(i) $x + 5 = 12 \Rightarrow x = 7$

(ii) $x + 2 = 4 \Rightarrow x = 2$

(iii) $23x + 3 = 5 \Rightarrow x = 3$

Now, working in base six,

(i) $x + 5 = 12$ can be written as $x + 5 = 20$ since $12_{10} = 20_6$

$\Rightarrow x + 5 = 20$ $x = 20 - 5 = 11_{\text{six}}$

Alternatively, we can assume that $x + 5 = 12$ is already in base six

$x + 5 = 12$

$x = 12 - 5 = (6 + 2) - 5$

$x = 3_{\text{six}}$

This means that it is necessary to follow instructions.

(ii) $x - (-2) = +4$

$x + 2 = 4$

$x = -2 + 4 = 2_{\text{six}}$

$$(iii) \frac{2x}{3} + 3 = 5 \dots 3 \times 3 = 9_{10} = 13_{six} \text{ and } 5 \times 3 = 15_{10} = 23_{six}$$

Thus $\frac{2x}{3} + 3 = 5$ becomes $2x + 13 = 23$ (base six).....multiplying each term by 3

$$2x = 23 - 13 \dots \dots \dots \text{subtracting } 13_{six} \text{ from both sides} = 10_{six}$$

$$x = 3_{six} \dots \dots \dots \text{after dividing both sides by } 2_{six}$$

Example 1

Solve for the unknown in:

$$(a) 2x + 15 = 17 \text{ (base 8)}$$

$$(b) 12x - 3x = 20 \text{ (base 6)}$$

$$(c) \frac{3}{4}x + \frac{3}{2}x = 9 \text{ (base 10)}$$

Solution

$$(a) 2x + 15 = 27$$

$$2x = 27 - 15$$

$$2x = 12 \Rightarrow x = 6$$

$$x = 6_8$$

$$(b) \frac{1}{2}x - 3x = 20_6$$

$$x - (2 \times 3)_6 x = 20_6 \times 2_6$$

$$x - 10_6 x = 40_6$$

$$5_6 x = -40_6$$

$$x = \frac{-40_6}{5_6} = \left(-4\frac{4}{5}\right)_6$$

$$(c) \frac{3}{4}x + \frac{3}{2}x = 9_{10}$$

$$3x + 6x = 36$$

$$9x = 36$$

$$x = 4$$

Example 2

Solve for x in the equation $36_x + 26_x = 64_x$ given that x is a number other than base ten. Verify your answer.

Solution

$$36_x + 26_x = 64_x \text{ can be expressed as } 3 \times x + 6 + 2 \times x + 6 = 6 \times x + 4$$

(convert the equation to base ten then solve for x)

$$3x + 6 + 2x + 6 = 6x + 4 \quad 5x + 12 = 6x + 4$$

$$12 - 4 = 6x - 5x$$

$$8 = x$$

Thus $x = 8$.

To verify the answer, substitute $8 = x$ in the given equation $36_x + 26_x = 64_x$

$$\text{In } 36_x + 26_x = 64_x, \text{ LHS } 36_x + 26_x = 3x + 6 + 2x + 6$$

$$= 3 \times 8 + 6 + 2 \times 8 + 6$$

$$= 24 + 6 + 16 + 6 = 52$$

$$\text{RHS } 64_x = 6x + 4 = 6 \times 8 + 4 = 48 + 4 = 52$$

Thus $\text{LHS} = \text{RHS} = 52$

x Represents base 8 in the given equation.



Application activity 3.7:

1. Solve for x in:

$$(a) 2102_3 = 72_x$$

$$(b) 110011_{two} = 23_x$$

2. Solve for x if $110101_2 = x_8$

3. Given that A and B represent ten and eleven respectively in a certain base x , solve for x in:

$$(a) A7_x + 5B_x = 198_{10}$$

$$(b) BA1_x = 1705_{10}$$

4. Find x if $10011_{two} = 23_x$.

5. Given that x is the base, solve the equation:

$$(a) 25_x + 13_x = 42_x \quad (d) 12_x - 6_x = 5_x$$

$$(b) 32_x + 24_x = 100_x \quad (e) 31_x - 16_x = 12_x$$

$$(c) 142_x + 33_x = 215_x \quad (f) 32_x - 24_x = 6_x$$

6. Given that A and B represent 10 and 11 respectively in base twelve, solve for x in;

$$(a) 12A_{12} + 4AB_{12} = x_{ten}$$

$$(b) 789_{12} - AB_{12} = x_{nine}$$



END OF UNIT ASSESSMENT

Assessing knowledge and understanding

- What is a number base?
- Which of the following correctly represents a binary number?

A. 128

- B. 204
- C. 1011
- D. 789

3. What is meant by the place value of a digit in a number base?
4. Match each number base in Column A with its correct description in Column B.

Column A	Column B
(a) Base 2	1. Uses digits 0–7
(b) Base 8	2. Uses digits 0 and 1
(c) Base 10	3. Uses digits 0–9

5. Explain why computers use the binary number system.

Assessing skills

1. Add 3554_{six} to 44_{six} giving your answer in the same base.
2. (a) Express 101_{eight} in base 2.
(b) Calculate $110_{\text{two}} \times 1010_{\text{two}}$ giving your answer in base two and also in base ten.
3. Write 230_n as an algebraic expression in terms of n .
4. Given that $10022_{\text{three}} = 155_n$, find the value of n .
5. Write each of the numbers as a mixed number in the base number
(a) 101.11_{ten}
(b) 21.01_{five}
6. If $13 \times 21 = 303$ find the base of the multiplication.
7. Carry out the following in base six
(a) $115 + 251 + 251$
(b) $53412 - 34125$
(c) 123×54
- 8) A factory prints serial numbers in **base seven**. If it prints serial numbers from 0_7 to 156_7 , how many items does it produce (in decimal)?

Assessing attitudes and values:

1. Why is accuracy important when converting numbers between different bases?
2. What attitude should a learner show when they find it difficult to understand a new number base?
3. The unit mentions that computers use the binary system (base 2) because it corresponds to "on" and "off" states in electronic circuits.
In your own words, explain why it is valuable to learn number bases beyond base ten in today's technology-driven world.

$$\frac{7a}{3} = \frac{14a}{6}$$

The diagram illustrates the process of simplifying or expanding an algebraic fraction. It shows the fraction $\frac{7a}{3}$ on the left and $\frac{14a}{6}$ on the right, with an equals sign between them. Two curved arrows, one above and one below the equals sign, point from the first fraction to the second. Each arrow is labeled with 'x2', indicating that both the numerator and the denominator of the first fraction were multiplied by 2 to obtain the second fraction.

UNIT 4: ALGEBRAIC FRACTIONS

Key Unit Competence: Use operation properties and simplification of rational expressions in solving real life problems.

Learning objectives:

At the end of this unit, you should be able to:

Knowledge and understanding:

- Explain a rational algebraic expression.
- Determine the restrictions on the variable (conditions for existence) for a given algebraic fraction.
- Explain the rationale behind each step when simplifying or operating with algebraic fractions.

Skills:

- Simplify algebraic fractions by factoring and cancelling common factors.
- Perform addition, subtraction, multiplication, and division on algebraic fractions with linear and quadratic denominators.
- Solve rational equations and check for extraneous solutions.
- Model real-world situations (e.g., work, motion, sharing cost, flow rates) using rational expressions and solve them.

Attitudes and values:

- Value logical, step-by-step reasoning and the importance of checking solutions.
- Demonstrate precision and attention to detail, particularly in identifying domain restrictions.
- Collaborate constructively in teams, showing respect for different problem-solving approaches during peer review.

Introduction

Algebraic fractions are simply fractions that contain unknown variables, like x or a , in their numerator, denominator, or both. Examples include $\frac{x}{3}$ or $\frac{2}{y+1}$. They follow all the same rules as ordinary numerical fractions for operations like simplification, addition, and multiplication. Learning to manipulate algebraic fractions is a key skill in algebra because they act as a bridge

between basic arithmetic and more complex mathematical formulas where quantities are not yet known.

The properties of algebraic fractions are applied in many real-life situations that involve rates, mixing, or comparing quantities. For instance, they are used to calculate average speed, work out unit prices, determine the concentration of a mixture, or share costs proportionally.



Introductory activity

1) Students are making a special drink using cups. The procedure says the perfect sweetness is achieved when the ratio of fruit juice (in x cups) to yogurt (in cups) is $\frac{x+1}{2}$.

a) If you use 3 cups of juice ($x=3$), how many cups of yogurt should you add to match the procedure's ratio? Show your working.

b) In your own words, what do you think the x in this fraction represents in a real kitchen?

What does the whole fraction $\frac{x+1}{2}$ help a cook, or a business owner figure out?

2) In a game, the amount of money F in USD to be received or to be paid is calculated using $F = \frac{2}{2x-4}$ where x is the whole number from written on a card to be picked from the basket and the numbers vary from 0 to 10. During the game, the organizer removed the card with the number 2.

(i) Find the amount of money to be received when $x = 0, 1, 2, 3, 4, 5, 6, 7, 8, 10$.

(ii) Is there any value of x for which the amount of money to be received is not known in (i) above? If your answer is yes, explain why the organizer did not avail the card with number $x=2$.

4.1 Meaning and identification of algebraic fractions



Activity 4.1

Consider the fractions: $\frac{3y}{1-x}$, y , $\frac{5}{y+4}$, $\frac{5x-6}{x+6}$, $\frac{2}{7}$, $\frac{3}{4}$, x^2 . Identify the algebraic fractions from them.



Content summary

An **algebraic fraction** is a fraction where the numerator and/or the denominator contain **algebraic expressions** (terms with variables like x , y , a , etc).

Example: 1

- a) $\frac{3}{x}$: However, $\frac{3}{x}$ exist in the set of real numbers for the variable x different from zero. The fraction is valid for all real numbers **except** when $x = 0$.
- b) $\frac{x+1}{2}$ the variable x is on the numerator.
- c) $\frac{a^2 - b^2}{a + b}$ is an algebraic fraction with variable a and b but exist in the set of real numbers for a different from $-b$.
- d) $\frac{5}{7}$ is not an algebraic fraction, it has no variable.
- e) x^2 this is not an algebraic fraction because its denominator is 1 and is a simple polynomial.
- f) Select all algebraic fractions from the list:

i) $\frac{a+b}{c}$

ii) $\frac{5}{t}$

iii) $y - \frac{1}{y}$ entire expression.

iv) $\frac{\pi}{r^2}$

v) $\frac{x-3}{2}$

vi) $\frac{1}{x^{-1}}$ (simplified to x as given)

...



Application activity 4.1:

Question 1: Which of the following are algebraic fractions?

- (a) $\frac{3}{x+2}$ (b) $\frac{\sqrt{x}}{5}$ (c) $\frac{7}{4}$ (d) $x^2 + 3x - 1$ (e) $\frac{2x+1}{x^2-4}$

Question 2: True or False — The following expressions are algebraic fractions:

1. $\frac{1}{\sqrt{x+1}}$ 2. $3x^2$ 3. $\frac{(x-1)(x+2)}{(x-3)}$ 4. $\frac{2+\pi}{y}$ 5. $\frac{0}{x+5}$

4.2 Restriction on variables or conditions of existence



Activity 4.2

Consider the fractions: $\frac{3y}{1-x}$, y , $\frac{5}{y+4}$, $\frac{5x-6}{x+3}$, $\frac{2}{7}$, $\frac{3}{4}$, x^2

- (1) Find the value of the variable that makes the denominator
 - (i) $x + 3$ (ii) $y + 4$ (iii) $1 - x$ of given fraction zero.
- (2) Is it acceptable to divide by zero in the set of real numbers?
- (3) Use your answer in (1) above and discuss what conditions to have so that the following expressions exist in the set of real numbers: $\frac{5x-6}{x+3}$, $\frac{5}{y+4}$, $\frac{3y}{1-x}$.



Content summary

In general, an algebraic fraction exists in the set of real numbers only if the denominator is not equal to zero. The values of the variable that make the denominator zero is called a restriction on the variable(s). An algebraic fraction can have more than one restriction.

Examples :1

(i) The fraction $\frac{2}{x}$, is valid for all real numbers **except** when $x = 0$. If $x = 0$, it means dividing 2

by zero is not defined. Therefore, $\frac{2}{x} \in \mathbb{R}$ only if $x \neq 0$.

(ii) $\frac{x+3}{x-1}$ exists **only** if $x - 1 \neq 0$ and $x - 1 \neq 0$ if $x \neq 1$. If $x = 1$, the denominator or divisor becomes zero which is not defined/which does not exist. The fraction is not defined when $x = 1$.

(iii) The fraction $\frac{y-4}{2y-6}$ is defined (exists) **only if the denominator is not equal to zero**.

Thus if $2y - 6 = 0$, then $2y = 6$, and $y = 3$. Therefore, $\frac{y-4}{2y-6}$ exists for all real values of y except when $y = 3$.

(iv) Identify the restriction on the variable in the fraction $\frac{3xy}{(x+3)(x-2)}$

Solution:

There are two factors in the denominator i.e. $(x + 3)$ and $(x - 2)$.

The denominator equals zero if; $x + 3 = 0$ or $x - 2 = 0$

Therefore, $x = -3$ or $x = 2$

If $x = -3$, $(x + 3)(x - 2) = 0$ and if $x = 2$, $(x + 3)(x - 2) = 0$.

As we do not divide by zero in the set of real numbers, $\frac{3xy}{(x+3)(x-2)}$ exist for the restrictions

$x \neq -3$ or $x \neq 2$.



Application activity 4.2:

1. Identify the restrictions on the variables of each of the following fraction.

(a) $\frac{a+2}{a}$ (b) $\frac{2x}{y-2}$ (c) $\frac{7}{3m+6}$ (d) $\frac{3w-9}{w+3}$

2. Find the restrictions on the variables in the following fractions:

(a) $\frac{x^2-2x-15}{4(x-5)}$ (b) $\frac{2y-2}{(y-2)(y-1)}$ (c) $\frac{2y-7}{(2y-7)(y+3)}$ (d) $\frac{6x^2-x-1}{(2x-1)(x+1)}$

4.3 Simplification of the algebraic fractions



Activity 4.3

Write the following fractions in the simplest form:

a) $\frac{8}{14}$; $\frac{30}{45}$; $\frac{9}{15}$

b) $\frac{12a}{15a^2}$; $\frac{3ab}{4a^2b}$; $\frac{12x^2y^4}{3xy^2}$

What is the technique do you use to simplify expressions of (b)?



Content summary

A fraction $\frac{a}{b}$ where a and b are real numbers, is in its simplest form if its numerator and denominator do not have common factors. To simplify means to divide both numerator and denominator by the common factors.

A numerator and a denominator can be divided by the same factor without altering the value of the fraction.

For example, in $\frac{8}{12}$, the numerator and denominator have a **common factor 4**.

$$\therefore \frac{8}{12} = \frac{8 \div 4}{12 \div 4} = \frac{2}{3}$$

We say that $\frac{8}{12}$ and $\frac{2}{3}$ are **equivalent fractions**.

Similarly, $\frac{30}{45} = \frac{30 \div 15}{45 \div 15} = \frac{2}{3}$

$\therefore \frac{30}{45}$ is equivalent to $\frac{2}{3}$.

Also, $\frac{9}{12} = \frac{9 \div 3}{12 \div 3} = \frac{3}{4}$.

$\therefore \frac{9}{12}$ is equivalent to $\frac{3}{4}$.

In $\frac{3ab}{4a^2b}$, a and b are common factors.

$$\therefore \frac{3ab}{4a^2b} = \frac{3ab \div ab}{4a^2b \div ab} = \frac{3}{4a}$$

In $\frac{12x^2y^4}{3xy^2}$, 3 , x , y^2 are common factors.

$$\begin{aligned}\text{Therefore, } \frac{12x^2y^4}{3xy^2} &= \frac{12x^2y^4 \div 3xy^2}{3xy^2 \div 3xy^2} \\ &= \frac{4xy^2}{1} = 4xy^2.\end{aligned}$$

If both the numerator and denominator of a fraction have more than one term, we simplify the fraction by:

- (i) Factorizing both numerator and denominator where necessary.
- (ii) Cancelling by the common factor.

Remember: In Senior 2, you learnt to factorize algebraic expressions. To simplify the algebraic fraction, it is recommended to:

- **Factor everything possible:** Factor the numerator and the denominator completely into: Numbers (numerical factors), Variables (with exponents), Polynomials (using methods like GCF, grouping, trinomials, special formulas).

- **Identify restrictions (domain):** The denominator cannot be zero.

Set each factor of the **original denominator** equal to zero to find excluded values.

- **Cancel common factors:** Cancel factors that appear in both numerator and denominator (only factors multiplied across the whole numerator/denominator).

- **Rewrite the simplified form:** After cancelling, write the final simplified fraction, keeping the domain restriction.

Example 1:

Simplify the following fractions:

(a) $\frac{8x^2y^3}{2x^3y}$ (b) $\frac{2x^2 + 5x^3}{2x^2 + 4x^3}$

Solution

(a) $\frac{8x^2y^3}{2x^3y}$

$8x^2y^3$ is the numerator and

$2x^3y$ is the denominator

The common factors are 2 and x^2y

$$\therefore \frac{8x^2y^3}{2x^3y} = \frac{8x^2y^3 \div 2x^2y}{2x^3y \div 2x^2y}$$

$$= \frac{4y^2}{x}; x \neq 0$$

(b) $\frac{2x^2 + 5x^3}{2x^2 + 4x^3}$

Both the numerator and the denominator contain two terms each.

$$2x^2 + 5x^3 = x^2(2 + 5x)$$

$$2x^2 + 4x^3 = 2x^2(1 + 2x)$$

$$\frac{2x^2 + 5x^3}{2x^2 + 4x^3} = \frac{x^2(2 + 5x)}{2x^2(1 + 2x)}$$

(divide both numerator and denominator by x^2)

$$\frac{2 + 5x}{2(1 + 2x)} = \frac{1}{2} \frac{(2 + 5x)}{1 + 2x}$$

Example 2:

Simplify: $\frac{6x^2 - 54}{3x^2 + 12x + 9}$

Solution:

Step 1 – Factor:

Numerator: $6x^2 - 54 = 6(x-3)(x+3)$

Denominator: $3x^2 + 12x + 9 = 3(x+1)(x+3)$

Step 2 – Restrictions:

$\frac{6x^2 - 54}{3x^2 + 12x + 9}$ exists for $3x^2 + 12x + 9 \neq 0$ $3x^2 + 12x + 9$

$3(x+1)(x+3) \neq 0 \Rightarrow x+1 \neq 0$ and $x+3 \neq 0$

$\Rightarrow x \neq -1$ and $x \neq -3$

Step 3 – Cancel:

$\frac{6x^2 - 54}{3x^2 + 12x + 9} = \frac{6(x-3)(x+3)}{3(x+1)(x+3)}, x \neq -1, x \neq -3$. The common factors is $3(x+3)$.

$\frac{6(x-3)(x+3)}{3(x+1)(x+3)} = \frac{2(x-3)}{x+1}, x \neq -1 \text{ and } x \neq -3$

Even though $x \neq -3$ is no longer in the denominator after simplifying, it is still a restriction from the original expression.



Application activity 4.3:

For each of the following fractions in questions below:

(i) write the expression in factor form

(ii) note the restrictions on the variables

(iii) simplify the fractions.

$$(a) \frac{x^2 - 2x - 15}{4x - 20}$$

$$(b) \frac{2y - 7}{2y^2 - y - 21}$$

$$(c) \frac{6x^2 - x - 1}{4x^2 - 1}$$

$$(d) \frac{x^2 - 5x + 6}{x - 3}$$

$$(e) \frac{y^2 - 7y + 6}{y - 6}$$

$$(f) \frac{x^2 - 9x + 20}{x - 5}$$

$$(a) \frac{x^2 - 10xy + 25y^2}{x + 5y}$$

$$(b) \frac{6x^2 - 5xy + y^2}{6x^2 - xy - y^2}$$

$$(c) \frac{x^2 - y^2}{3x^2 - 2xy - y^2}$$

$$(d) \frac{2x^2 + xy - y^2}{4x^2 - y^2}$$

4.4 Addition and subtraction of algebraic fraction with linear denominator



Activity 4.4

1. Find the LCM of 2, 6 and 7.

2. Express the fractions $\frac{1}{2}$, $\frac{1}{6}$ and $\frac{1}{7}$ as equivalent fractions with the calculated LCM as the common denominator. Hence find the value of $\frac{1}{2} + \frac{1}{6} + \frac{1}{7} =$

3. Find the value of $\frac{1}{2a} + \frac{1}{6a^2} + \frac{1}{7a^2}$ in the simplest form.



Content summary

To add or to subtract simple fractions, first, find the LCM of the denominator then convert each fraction into a fraction having this LCM as denominator which becomes LCD (**Least Common Denominator**). Add or subtract the numerators and simplify your answer.

- **Factor denominators completely:** Factor each denominator into its prime/polynomial factors. This helps find the LCD.

Example: add $\frac{2}{x^2 - 4} + \frac{3}{x + 2} = \frac{2}{(x - 2)(x + 2)} + \frac{3}{x + 2}$

- **Find the Least Common Denominator (LCD):** List all **distinct factors** from all denominators. For each factor, take the **highest power** that appears in any denominator.

In our example; Denominators: $(x - 2)(x + 2)$ and $(x + 2)$

Factors: $x - 2$ (max power 1), $x + 2$ (max power 1)

LCD = $(x - 2)(x + 2)$.

- **Rewrite each fraction with the LCD:** Multiply numerator and denominator of each fraction by whatever factor is missing from its denominator to match the LCD.

Example: First fraction: already has LCD.

Second fraction: needs $(x - 2)$ factor in denominator: $\frac{3}{x + 2} = \frac{3}{x + 2} \times \frac{x - 2}{x - 2} = \frac{3(x - 2)}{(x + 2)(x - 2)}$

So, $\frac{2}{(x-2)(x+2)} + \frac{3}{x+2} = \frac{2}{(x-2)(x+2)} + \frac{3(x-2)}{(x+2)(x-2)}$

- **Combine numerators:** Add/subtract the numerators, keep the LCD.

$$\frac{2}{(x-2)(x+2)} + \frac{3(x-2)}{(x+2)(x-2)} = \frac{2+3(x-2)}{(x+2)(x-2)}$$

-**Simplify the resulting fraction**

- Simplify the numerator (expand, combine like terms).
- Factor numerator if possible.
- Cancel common factors with the denominator.

Example: Numerator: $2+3x-6=3x-4$

Result: $\frac{2+3(x-2)}{(x+2)(x-2)} = \frac{3x-4}{(x+2)(x-2)}$

- **State restrictions**

The expression $\frac{2}{x^2-4} + \frac{3}{x+2}$ exists if we exclude values that make **any original denominator** zero.

From example: $x^2 - 4 \neq 0 \Rightarrow x \neq 2$, and $x \neq -2$

$x+2 \neq 0 \Rightarrow x \neq -2$

So restrictions: $x \neq 2$, and $x \neq -2$

Final answer: $\frac{2}{x^2-4} + \frac{3}{x+2} = \frac{3x-4}{(x+2)(x-2)}$ with restrictions $x \neq 2$, and $x \neq -2$.

Note:

1. Addition of algebraic fractions is performed in the same way as in numerical fractions as seen in the previous activity.
2. You can only add fractions if their denominators are the same.
3. The basic rule governing fractions is that numerator and denominator can be multiplied by the same factor without altering the value of the fraction.
4. Skills in arithmetic should extend to skills in algebra. If the denominators do not have common factors, work as in the following: Let us add $\frac{1}{x} + \frac{1}{y}$. The LCD is $x \times y$.

Therefore, $\frac{1}{x} + \frac{1}{y} = \frac{1 \times y + 1 \times x}{xy} = \frac{y+x}{xy}$

Example 1:

Example 2:

Simplify: $\frac{5}{4ab} + \frac{-2}{3a}$

Solution

Restrictions in those fractions are $a \neq 0$ and $b \neq 0$.

$$\frac{5}{4ab} + \frac{-2}{3a} = \frac{5}{4ab} - \frac{2}{3a}$$

LCM of $4ab$ and $3a$ is $12ab$.

$$\begin{aligned} \frac{5}{4ab} - \frac{2}{3a} &= \frac{5 \times 3}{4ab \times 3} - \frac{2 \times 4b}{3a \times 4b} \\ &= \frac{15}{12ab} - \frac{8b}{12ab} \\ &= \frac{15 - 8b}{12ab} \quad (\text{Subtract the numerators.}) \end{aligned}$$

This fraction cannot be simplified further.

Work out: $\frac{1}{x+1} - \frac{1}{2x+2}$

Solution

The denominators are $(x+1)$ and $(2x+2)$.

To find the LCM, we factorise the denominators.

Restrictions in these fractions are

$$x \neq -1$$

$(x+1)$ is a prime expression.

$(2x+2)$ has two terms with common factors 2.

$$\therefore 2x+2 = 2(x+1)$$

The factors in the denominator are 2, and $(x+1)$

$$\therefore \text{LCM} = 2(x+1)$$

$$\begin{aligned} \therefore \frac{1}{x+1} - \frac{1}{2x+2} &= \frac{1}{x+1} - \frac{1}{2(x+1)} \\ &= \frac{2 \times 1}{2(x+1)} - \frac{1 \times 1}{2(x+1)} \\ &= \frac{2}{2(x+1)} - \frac{1}{2(x+1)} \\ &= \frac{2-1}{2(x+1)} \quad \text{adding the numerators} \\ &= \frac{1}{2(x+1)} \end{aligned}$$



Application activity 4.4:

1) Add or subtract and simplify the fractions in the following questions:

(a) $\frac{3x}{2} + \frac{5x}{4}$

(b) $\frac{3}{2a} + \frac{5}{4a}$

(c) $\frac{3x}{2a} + \frac{5x}{3a}$

(d) $\frac{2x}{m} - \frac{3x}{n}$

(e) $\frac{6}{x} + \frac{2}{xy}$

(f) $\frac{8}{x^2y} + \frac{6}{xy^2}$

(a) $\frac{6x+1}{3} - \frac{2x-3}{4}$

(b) $\frac{8x-2}{9} - \frac{x-3}{6}$

(c) $\frac{2x-1}{6} - \frac{3x+1}{4} - \frac{2x-5}{2}$

2) Add or subtract:

3) You are given $f(x) = \frac{x(2x+3) - (2x+3)(x-3)}{2x+3}$ for $x \neq \frac{-3}{2}$. Simplify completely $f(x)$ and

circle its simplified form here below:

A: $f(x) = 2x+3$

B: $f(x) = 2x-3$

C: $f(x) = 3$

D: $f(x) = -3$

4) Add the following algebraic fractions and simplify:

a) $\frac{2}{a^2-4a+4} + \frac{5}{a-2}$

b) $\frac{x^2-4x+4}{x^2-4}$

5) Work out and simplify:

Questions for Group A

Question for Group B:

$$(a) \frac{3}{x-2} - \frac{x}{x-1}$$

$$(b) \frac{3}{x-3} - \frac{8}{x+3}$$

$$(c) \frac{3}{x-1} - \frac{x}{1-x}$$

$$(d) \frac{1}{b} - \frac{1}{b-c}$$

$$(e) \frac{1}{2} - \frac{b}{a-2b}$$

$$(f) \frac{1}{a+2} - \frac{1}{a-2}$$

$$(a) \frac{3}{a-b} - \frac{4}{a+b}$$

$$(b) \frac{2}{y-1} - \frac{3y}{y+5}$$

$$(c) \frac{1}{x+3} - \frac{1}{x-3}$$

$$(d) \frac{3}{x+1} - \frac{2}{x-1}$$

$$(e) 1 - \frac{x}{x-y}$$

$$(f) \frac{c}{c-d} - \frac{c}{c+d}$$

$$(g) r - \frac{r^2}{r+s}$$

4.5 Multiplication of algebraic fractions



Activity 4.5

1) Multiply the following fraction: $\frac{4}{5} \times \frac{a}{3}$ and explain how you do it. We think that you multiply numerators by numerators and denominators by denominator.

2) Extend this skill to multiply $\left(\frac{a-4}{a}\right) \times \left(\frac{a+5}{a+2}\right)$



Content summary

In order to multiply fractions, Multiply numerators by numerators, and multiply denominators by denominators. Then, identify common factors, or possible factors of given expression and divide both the numerator and denominator by the common factors.

If it is possible to factor the expressions in both numerators and denominators, factor them and simplify where possible. Follow the steps:

- Factor everything possible: Factor **all** numerators and denominators completely

Example 1 : Multiply $\frac{x^2-4}{x+3} \times \frac{x+1}{x^2+2x}$

Factor each:

$$x^2-4=(x-2)(x+2).$$

$x+3$ (already factored)

$x+1$ (already factored)

$$x^2+2x=x(x+2)$$

$$\text{So: } \frac{x^2-4}{x+3} \times \frac{x+1}{x^2+2x} = \frac{(x-2)(x+2)}{x+3} \times \frac{x+1}{x(x+2)}$$

- **State domain restrictions:** Set each **original denominator** equal to zero to find excluded values:

- First denominator: $x+3 \neq 0 \Rightarrow x \neq -3$
- Second denominator: $x^2 + 2x \neq 0 \Rightarrow x(x+2) \neq 0 \Rightarrow x \neq 0, x \neq -2$

Restrictions: $x \neq -3, 0, -2$.

Multiply numerators and denominators

Multiply numerators together; multiply denominators together.

$$\frac{(x-2)(x+2)}{x+3} \times \frac{x+1}{x(x+2)} = \frac{(x-2)(x+2) \times (x+1)}{(x+3) \times x(x+2)}$$

- **Cancel common factors:** Cancel any factor that appears in both numerator and denominator (across the entire expression). $(x+2)$ appears in numerator and denominator, cancel.

Result:

$$\frac{(x-2)(x+2) \times (x+1)}{(x+3) \times x(x+2)} = \frac{(x-2)(x+1)}{x(x+3)}$$

- **Write final answer with restrictions**

$$\frac{x^2 - 4}{x+3} \times \frac{x+1}{x^2 + 2x} = \frac{(x-2)(x+1)}{x(x+3)}, x \neq -3, x \neq 0, x \neq -2$$

Example 2:

Multiply: $\frac{y^2 + y}{y^2 - 4y} \times \frac{y^2 - 4y - 21}{y + 3}$

Solution

Restrictions on the variable are $y \neq 4$, or $y \neq -3$ or $y \neq 0$

There are no visible factors, but we can factorise the expression in both numerator and denominator.

$y^2 + y$ is a binomial with a common factor y

So $y^2 + y = y(y + 1)$ factor out common factor y

$y^2 - 4y = y(y - 4)$ factor out common factor y

$y^2 - 4y - 21$ is a quadratic expression which can be expressed in the form $y^2 - 7y + 3y - 21$ (split the middle term,

$y(y - 7) + 3(y - 7)$ factorise by grouping

$$= (y - 7)(y + 3).$$

$y + 3$ is a prime expression.

$$\begin{aligned} \therefore \frac{y^2 - y}{y^2 - 4y} \times \frac{y^2 + 4y - 21}{y + 3} \\ &= \frac{y(y + 1)}{y(y - 4)} \times \frac{(y - 7)(y + 3)}{y + 3} \\ &= \frac{(y + 1)(y - 7)}{y - 4} \end{aligned}$$

Example 3:

Multiply and simplify: $\frac{2x+6}{x^2-1} \times \frac{x(x+1)}{4}$

Solution:

Factor: $\frac{2x+6}{x^2-1} \times \frac{x(x+1)}{4} = \frac{2(x+3)}{(x+1)(x-1)} \times \frac{x(x+1)}{4}$

Restriction: $x \neq 1, x \neq -1$.

Cancel: $\frac{2(x+3)}{(x+1)(x-1)} \times \frac{x(x+1)}{4} = \frac{x(x+3)}{2(x-1)}$

Therefore, $\frac{2x+6}{x^2-1} \times \frac{x(x+1)}{4} = \frac{x(x+3)}{2(x-1)}, x \neq -1, x \neq 1$



Application activity 4.5:

Workout each of the following expressions and simplify the result:

Group A:

- (a) $\frac{m}{n} \times \frac{b}{a}$
- (b) $\frac{-6}{5} \times \frac{-x}{3}$
- (c) $-3m^2 \times \frac{-5}{6m}$
- (d) $\frac{-6x}{3} \times \frac{-3}{-6x}$
- (e) $\frac{3a^2}{5b^2} \times \frac{10b}{4b^3}$
- (f) $\frac{6}{x^2y} \times \frac{x^3y^2}{x}$
- (g) $\frac{x}{3m} \times \frac{-3m}{-y}$
- (h) $\frac{3x^2}{2y} \times \frac{-2y}{3}$

Group B:

- (a) $\frac{36}{y^2+2y} \times \frac{y+2}{9}$
- (b) $\frac{a^2-b^2}{a^2-16} \times \frac{a-4}{a-b}$
- (c) $\frac{3x-12}{4x+20} \times \frac{x^2+5x}{x^2-4x}$
- (d) $\frac{x^2-x}{2x+2} \times \frac{6x}{x^2-1}$
- (e) $\frac{x^2-y^2}{x^2-2xy-y^2} \times \frac{1}{xy+y^2}$
- (f) $\frac{y-3}{2-y} \times \frac{y-2}{3-y}$

4.6 Division and simplification of algebraic fractions



Activity 4.6

You know that the division of fractions means multiply the fraction by the reciprocal of the second fraction.

1) Apply it to calculate $\frac{2}{3} \div \frac{4}{5}$ and complete: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{-}{-} = \frac{5}{6}$

2) Given that $\frac{\frac{A}{B}}{\frac{C}{D}} = \frac{A}{B} \times \frac{D}{C}$ where A, B, C, and D can be algebraic expressions (but B, C, D $\neq$ 0).

Use this rule to divide and simplify: $\frac{x}{x+1} \div \frac{2}{x^2-1}$ where $x \neq -1$ and $x \neq 1$.



Content summary

To divide algebraic fractions, multiply the first fraction by the reciprocal of the second fraction.

$$\frac{\frac{A}{B}}{\frac{C}{D}} = \frac{A}{B} \times \frac{D}{C} \text{ where } A, B, C, \text{ and } D \text{ are algebraic expressions, with } B \neq 0, C \neq 0, \text{ and } D \neq 0.$$

To divide, use these steps:

- **Flip the Second Fraction:** Keep the first fraction as it is, Change the division sign ($\div$) to multiplication ($\times$), **Flip** (take the reciprocal of) the second fraction.
- **Update Domain Restrictions: Original restrictions** come from: All denominators in original expression $\neq 0$. **Plus** any numerator of a divisor that becomes a denominator after flipping.
- **Factor Everything:** Factor all numerators and denominators completely before canceling.
- Cancel any factor that appears in both a numerator and denominator (can be across the multiplied fractions).
- **Multiply Straight Across:** Multiply remaining numerators together and denominators together.
- **State Final Answer with Restrictions:** Write the simplified expression and explicitly list all domain restrictions.

Case 1: Division by a Whole Number or a Monomial:

$$\frac{\frac{A}{B}}{c} = \frac{A}{B} \times \frac{1}{c}$$

Example: $\frac{\frac{x+1}{x-2}}{3} = \frac{x+1}{x-2} \times \frac{1}{3} = \frac{x+1}{3(x-2)}$ with restriction: $x \neq 2$

Case 2: Division of a Whole Number by a Fraction:

$$\frac{\frac{C}{A}}{\frac{B}{A}} = \frac{C}{A} \times \frac{A}{B} = \frac{CB}{A}$$

Example: $\frac{\frac{2}{x-1}}{x+3} = \frac{2}{x-1} \times \frac{x+3}{x-1} = \frac{2(x+3)}{(x-1)^2}$ with restrictions $x \neq 1$, and $x \neq -3$

Note: The restrictions you state at the beginning **must stay with the answer**, even if the variable causing the restriction cancels out during simplification.

Example 1:

Example 2:

Simplify: $\frac{3x^2y}{4a^2} \div \frac{9xy}{5a}$

Solution

In $\frac{3x^2y}{4a^2} \div \frac{9xy}{5a}$, $\frac{9xy}{5a}$ is the divisor.

The reciprocal of $\frac{9xy}{5a} = \frac{1 \times 5a}{9xy}$
 $= \frac{5a}{9xy}$.

$$\begin{aligned} \therefore \frac{3x^2y}{4a^2} \div \frac{9xy}{5a} &= \frac{3x^2y}{4a^2} \times \frac{5a}{9xy} \\ &\text{(Cancel by } x \text{ and } y.) \\ &= \frac{3x \times 5}{4a \times 9} \text{ (Cancel by 3.)} \\ &= \frac{5x}{12a} \end{aligned}$$

Solve: $\frac{x^2 - 4y^2}{2x^2 - 5xy + 2y^2} \div \frac{y^2 - xy}{2x - y}$

Solution

Start by fractionising the individual expression in the given fractions.

$x^2 - 4y^2$ is a difference of two squares

$$\therefore x^2 - 4y^2 = (x - 2y)(x + 2y)$$

$2x^2 - 5xy + 2y^2$ is a quadratic homogeneous expression.

$$\begin{aligned} \therefore 2x^2 - 5xy + 2y^2 \\ = 2x^2 - 4xy - xy + 2y^2 \end{aligned}$$

$$\begin{aligned} &= 2x(x - 2y) - y(x - 2y) \\ &\text{factorise by grouping} \\ &= (x - 2y)(2x - y) \end{aligned}$$

$y^2 - xy$ is a binomial,

$$y^2 - xy = y(y - x)$$

$$\begin{aligned} \therefore \frac{x^2 - 4y^2}{2x^2 - 5xy + 2y^2} \div \frac{y^2 - xy}{2x - y} \\ &= \frac{(x - 2y)(x + 2y)}{(x - 2y)(2x - y)} \div \frac{y(y - x)}{2x - y} \\ &= \frac{(x - 2y)(x + 2y)}{(x - 2y)(2x - y)} \times \frac{2x - y}{y(y - x)} \\ &\text{(Cancel common factors.)} \\ &= \frac{(x + 2y)}{y(y - x)} \end{aligned}$$



Application activity 4.6:

Simplify each of the following:

Question 1:

(a) $\frac{x}{y} \div \frac{a}{b}$

(c) $\frac{-2}{7} \div \frac{3}{-x}$

(b) $\frac{-x}{y} \div \frac{a}{b}$

(d) $-6 \div \frac{3}{y}$

Question 2:

(a) $\frac{x}{-4} \div \frac{-5}{16y}$

(c) $\frac{x^2}{3} \div \frac{20x^2y}{-9}$

(b) $\frac{x}{ab} \div \frac{-x}{ab}$

(d) $-9 \div \frac{21}{ab}$

Question 3:

(a) $\left(\frac{-5}{2x} + \frac{-2}{3x}\right) \div \left(\frac{-3}{x} + \frac{5}{-4x}\right)$

(b) $\left(\frac{-2}{3x} + \frac{8}{-2y}\right) \div \frac{-3}{x^2}$

(c) $\left(\frac{3}{2a} + \frac{-2}{3b}\right) \div \left(\frac{2}{b} + \frac{3}{a}\right)$

Question 4:

(a) $\left(\frac{-5}{2x} + \frac{-2}{3x}\right) \div \left(\frac{-3}{x} + \frac{5}{4x}\right)$

(b) $\left(\frac{3}{2a} + \frac{2}{3b}\right) \div \left(\frac{2}{b} + \frac{3}{a}\right)$

4.7 Solving rational equations



Activity 4.7

If I have x friends, and I know that each gets $\frac{12}{x}$ cookies. Given that each friend gets 3 cookies, we need to solve the equation $\frac{12}{x} = 3$ to find the number of friends I have. Such an equation is called a rational equation. Solve: a) $\frac{12}{x} = 3$ b) $\frac{x-3}{2} = \frac{x}{5}$



Content summary

In order to be able to solve rational equations;

- (i) Start by finding the LCM of the denominators in each equation.
- (ii) Use the LCD to eliminate the denominators by multiplying each fraction or term by the LCD.
- (iii) Then solve the resulting equation.

Example 1:

$$\frac{3}{x+2} + \frac{1}{x} = \frac{5}{x+2}$$

- Identify Restrictions (Domain)

Set each denominator $\neq 0$:

$$x+2 \neq 0 \rightarrow x \neq -2 \text{ and } x \neq 0$$

Restrictions: $x \neq -2, 0$.

- Find the Least Common Denominator (LCD)

Denominators are $x+2$, x , and $x+2$. $\text{LCD} = \text{LCM}(x+2, x, x+2) = x(x+2)$.

- **Multiply Every Term by the LCD:** Multiply both sides of the equation by $x(x+2)$:

$$x(x+2) \cdot \frac{3}{x+2} + x(x+2) \cdot \frac{1}{x} = x(x+2) \cdot \frac{5}{x+2}$$

- Simplify Each Term (Cancel Denominators)

$$x \cdot 3 + (x+2) \cdot 1 = x \cdot 5$$

$$3x + (x+2) = 5x$$

- Solve the Resulting Equation

$$3x + x + 2 = 5x$$

$$x = 2$$

- Check Against Restrictions

$x=2$ is **not** -2 Correct. $x=2$ is **not** 0 correct

- Verify the Solution (Optional but Recommended)

Substitute $x=2$ into original equation: $\frac{3}{2+2} + \frac{1}{2} = \frac{5}{2+2}$ Correct.

Final Answer: $x=2$.

Example 2:

Solve the equation $\frac{x+3}{2} - \frac{x+4}{3} = \frac{x-2}{4}$

Solution

There are three fractions whose denominators are 2, 3, 4.

2 and 3 are prime numbers,

$$4 = 2 \times 2 = 2^2.$$

Therefore, LCM of 2, 3 and 4 is $2^2 \times 3 = 12$.

To eliminate the denominators, we multiply each fraction by 12 and cancel by the common factors.

$$12\left(\frac{x+3}{2}\right) - 12\left(\frac{x+4}{3}\right) = 12\left(\frac{x-2}{4}\right)$$

This step may be done mentally.

$$6(x+3) - 4(x+4) = 3(x-2)$$

$$6x + 18 - 4x - 16 = 3x - 6$$

$$6x - 4x + 18 - 16 = 3x - 6$$

$$2x + 2 = 3x - 6$$

$$3x - 2x = 2 + 6$$

$$x = 8$$

By substituting 8 for x in the given equation, we can verify that $LHS = RHS$.

Example 3:

Solve the equation $\frac{x+5}{x-3} = \frac{x+3}{x-4}$

Solution

Restrictions are $x \neq 3$ and $x \neq 4$.

In $\frac{x+5}{x-3} = \frac{x+3}{x-4}$, multiply the equation by the least common multiple (LCM).

$$LCM = (x-3)(x-4)$$

$$(x-3)(x-4)\frac{x+5}{x-3} = (x-3)(x-4)\frac{x+3}{x-4}$$

This step may be done mentally.

$$(x-4)(x+5) = (x-3)(x+3)$$

$$x(x+5) - 4(x+5) = x^2 - 9 \text{ (expansion of binomials)}$$

$$x^2 + 5x - 4x - 20 = x^2 - 9$$

$$x^2 - x^2 + 5x - 4x = -9 + 20$$

$$x = -9 + 20$$

(Collect like terms and simplify)

$$\therefore x = 11$$

**Application activity 4.7:**

1. Solve for x and in each case, state the restrictions on the variable.

(a) $\frac{2}{3x} + \frac{3}{5} = \frac{1}{4x}$

(b) $\frac{x+18}{4} + x = \frac{x-5}{2} + 2$

(c) $\frac{3}{x} + 2 = \frac{2x+3}{x-1}$

(d) $\frac{x+1}{x} - 1 = \frac{x-2}{2-x} + 2$

2. Solve and verify your answers and state the restrictions in each case:

(a) $\frac{x+8}{x+3} = \frac{x}{x-3}$

(b) $\frac{3x-1}{x} = \frac{5}{2}$

(c) $\frac{x-1}{x} = \frac{x+1}{x+3}$

(d) $\frac{y+1}{y-2} = \frac{y+3}{y-4}$

4.8 Solving simple problems involving algebraic fractions



Activity 4.8

A fruit punch recipe requires mixing **orange concentrate** and **cranberry juice**. The amount of orange concentrate needed is $\frac{x+1}{x}$ litres, and the amount of cranberry juice is $\frac{3}{x+2}$ litres.

What is the **total volume** of the punch mixture? Write your answer as a single algebraic fraction.

Solution:

$$\text{Total volume} = \text{Orange concentrate} + \text{Cranberry juice} = \frac{x+1}{x} + \frac{3}{x+2}$$

The LCD is $x(x+2)$.

$$\text{Total volume} = \frac{x+1}{x} + \frac{3}{x+2} = \frac{(x+1)(x+2) + 3x}{x(x+2)} = \frac{x^2 + 6x + 2}{x(x+2)}$$

Restrictions on x : $x \neq 0$ and $x \neq -2$ (these values would make the original denominators zero, which is undefined). In the real-world context, x should be positive since we're measuring volumes.



Content summary

To solve real-world problems with algebraic fractions:

- **Identify the quantities** and the operation needed (add, subtract, multiply, divide).
- **For multiplication/division** (e.g., area, work rates, scaling):
 - o Multiply straight across, then factor and cancel.
 - o For division, multiply by reciprocal.
- **For addition/subtraction** (e.g., total volume, combined speeds, total costs):
 - o Find the least common denominator (LCD).
 - o Rewrite fractions with LCD, combine numerators, simplify.
- **Simplify** the final expression and state **restrictions** (values that make any denominator zero, considering the real-world context).

Example 1:

1) A rectangular field has a length given by $\frac{x+2}{x-1}$ cm, and its width is $\frac{3}{x+1}$ cm where x is the number of trees planted in that field.

Write a simplified expression for the area of the rectangle. What do you need to do first when the expressions involve fractions?

Solution:

Area of a rectangle = length $\times$ width

$$\text{Area} = \frac{x+2}{x-1} \times \frac{3}{x+1} = \frac{3(x+2)}{(x-1)(x+1)} = \frac{3(x+2)}{x^2-1}$$

Restrictions: The original denominators $x-1$ and $x+1$ cannot be zero.

So $x \neq 1$ and $x \neq -1$.

Also, since this is a real-world measurement, $x > 1$ in order to have positive number of trees.

Example 2:

Tom can paint a room in $\frac{2x}{x-1}$ hours where x is the number of tiles for the room's floor. His

friend can paint the same room in $\frac{x}{2}$ hours.

If they work together, their combined work rate is the sum of their individual rates.

Write a simplified expression for their **combined rate** (in rooms per hour).

Solution:

$$\text{Tom's rate: } \frac{1}{\frac{2x}{x-1}} = \frac{x-1}{2x} \text{ rooms/hour}$$

$$\text{Friend's rate: } \frac{1}{\frac{x}{2}} = \frac{2}{x} \text{ rooms/hour}$$

$$\text{Combined rate: } \frac{x-1}{2x} + \frac{2}{x}. \text{ LCD} = 2x$$

$$\text{Combined rate: } \frac{x-1}{2x} + \frac{2}{x} = \frac{(x-1) + 2 \times 2}{2x} = \frac{x+3}{2x}$$

Restrictions: $x \neq 0$ (and realistically $x > 1$ since the time and the number of tiles must be positive).

Example 3:

A car travels $\frac{100}{x}$ km per litre of petrol. Another car travels $\frac{150}{x+5}$ km per litre.

If you have 3 litres of petrol for the first car and 2 litres for the second car, what is the **total distance** both cars can travel?

Solution:

$$\text{First car distance: } 3 \times \frac{100}{x} \text{ km} = \frac{300}{x} \text{ km}$$

$$\text{Second car distance: } 2 \times \frac{150}{x+5} \text{ km} = \frac{300}{x+5} \text{ km}$$

$$\text{Total distance: } \frac{300}{x} \text{ km} + \frac{300}{x+5} \text{ km. LCD} = x+5$$

Total distance: $\frac{300}{x} + \frac{300}{x+5} = \frac{300(x+5) + 300x}{x(x+5)} = \frac{600x + 1500}{x(x+5)}$ km

Restrictions: $x \neq 0$ and $x \neq -5$ (and $x > 0$ for real-world meaning).



Application activity 4.8:

1) A taxi price is $\frac{50}{x-2}$ dollars for the first x km, plus $\frac{30}{x}$ dollars for each additional km.

If you travel $(x+1)$ km, find the **total price**. Then, if 4 people share the price equally, how much does each pay? (Express as a single simplified fraction.)

2) A **school trip** involves renting buses. Each bus can hold $\frac{x+10}{2}$ students where x is the number

of benches in the bus. There are $\frac{3x}{x-3}$ buses available.

- Write an expression for the **total number of students** that can be transported.
- If $x=8$, how many students can be transported?
- State the restrictions on x and explain their meaning in this real-world context.



END UNIT ASSESSMENT

Assessing knowledge and understanding:

1. Define an algebraic expression and explain why the fraction $\frac{a+5}{(a-2)(a+2)}$ has restrictions on the variable a .

2. Explain the steps of how to solve the equation $\frac{3}{x-1} + 2 = \frac{5}{x-1}$

Assessing skills:

3. Given the expression: $\frac{x+1}{x^2-1} \div \frac{3x}{x-1}$

- State restrictions.
- divide and simplify.

4. Perform the operation and simplify:

- $\frac{x}{3} + \frac{2x-1}{4}$
- $\frac{2}{x-1} - \frac{1}{x}$

5. Two pipes fill a tank. Pipe A can fill it in $\frac{h+2}{3}$ hours. Pipe B can fill it in $\frac{h}{2}$ hours.

- Find their combined rate (fraction of tank/hour) as a single simplified expression.

- b) If together they fill the tank in 2 hours, their combined rate is $\frac{1}{2}$ tank/hour, find the value of h .

Assessing Attitudes and Values

6. A student solved $\frac{x}{x-3} = \frac{3}{x-3}$ and got $x=3$ as the solution.

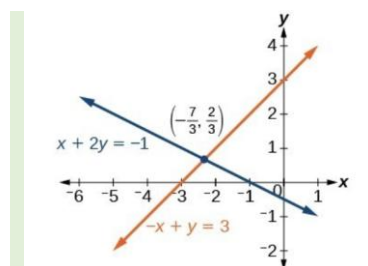
- Identify the error in reasoning.
- Demonstrate the correct solution process with attention to restrictions.
- Explain why checking solutions is crucial when solving rational equations.

7) In a group work, students were simplifying $\frac{x^2 - 4x + 4}{x^2 - 4} \times \frac{x+2}{x-2}$

Student A says: "We can cancel $(x-2)$ immediately."

Student B says: "We should factor first, then state restrictions, then cancel."

- Whose approach demonstrates better mathematical practice? Why?
- Show the complete simplification using the better approach.
- Explain how respectful collaboration improves problem-solving.



UNIT 5: SIMULTANEOUS LINEAR INEQUALITIES

Key unit competence: Solve real-world problems using systems of linear inequalities.

Learning objectives:

By the end of this unit, students should be able to

Knowledge and understanding:

- State methods for solving various inequalities, including single-variable inequalities, inequalities involving the product and ratio of two binomials, and the table of signs approach.
- Explain system of simultaneous linear inequalities in two unknowns.
- Identify the solution set of a system from its graphical representation.
- Explain the meaning of the point of intersection and the shaded region in the context of a problem

Skills:

- Solve various inequalities, including single-variable inequalities, inequalities involving the product and ratio of two binomials, and those using the table of signs.
- Graph the solution set for a system of linear inequalities.
- Translate word problems into systems of inequalities.
- Interpret the solution of a system in the context of the original real-world problem

Attitudes and values:

- Demonstrate persistence and accuracy when solving various inequalities, including single-variable inequalities.
- Appreciate the power of graphical and algebraic models for making informed decisions.
- Develop logical reasoning and systematic problem-solving strategies.
- Collaborate effectively to analyze problems and justify solutions.

Introduction

Simultaneous linear equations and inequalities introduce students to solving **two or more** algebraic conditions that must be true at the same time. Students learn how linear equations represent straight lines and how their solutions show where these lines intersect. They also explore linear inequalities, which describe regions on the coordinate plane, and determine the common region that satisfies all given conditions. This unit strengthens students' algebraic skills, improves logical thinking, and prepares them to apply mathematics to real-life problems involving comparisons, constraints, and decision-making.



Introductory activity

1) (a) On the same axes of the Cartesian plane draw the graph of the lines whose equations are:

$$\begin{cases} -2x + y = -1 & \dots\dots\dots(i) \\ x + 2y = 4 & \dots\dots\dots(ii) \end{cases}$$

(b) State the coordinates of the point where the two lines intersect. What meaning do you attach to this answer?

2) Consider the inequalities:

$$(i) y > x + 1 \quad (ii) 2x + y \leq 5$$

a) On a Cartesian plane, draw the line $y = x + 1$. Pick a point clearly not on the line and substitute the x and y values in $y = x + 1$ to determine the region of $y > x + 1$. On the same axis draw the line $y + 2x = 5$. By substitution identify the region $y + 2x \leq 5$

b) Discuss with your classmates how you can find the region that satisfies both the inequalities $y > x + 1$ and $y + 2x \leq 5$.

5.1 Introduction on equations and inequalities



Activity 5.1:

Using the equations

(i) $y = 2x + 1$ (ii) $2y = x - 4$

1) Complete the table of values for each equation for values of x : $-3 \leq x \leq 3$ as given in the following table:

X	-3	-2	-1	0	1	2	3
$y = 2x + 1$							
$y = (x - 4)/2$							

2) On the same axes, draw the two lines.

3) Do the lines intersect? If yes state, the coordinates of the point where they intersect.

4) Describe the significance or the meaning of the point of intersection of the two lines.



Content summary

To form simultaneous equations from a given situation, we must define the two variables, say x and y .

- Relate the two variables using the given information i.e. form two distinct equations in x and y .
- Simultaneous equations can be solved either algebraically or graphically.
- To solve simultaneous equations, graphically, we draw the lines representing the two equations on the same Cartesian Plane .
- If the equations have a solution, the lines will intersect at a point. The x and y values at the point of intersection represent the solution of the equations.
- If no solution, the lines will be parallel.
- If the lines are coincident, it means the equations have an infinite solution.

Example 1

Solve graphically the simultaneous equations

$$\begin{cases} 2y + 3x = 12, \\ 3y = 5x - 1 \end{cases}$$

Solution

Step 1: Make a table of values for each equation. Remember three pairs of points are sufficient for each equation.

Table 4.1 and 4.2 represent the relations $2y + 3x = 12$ and $3y = 5x - 1$ respectively.

x	0	4	2
y	6	0	3

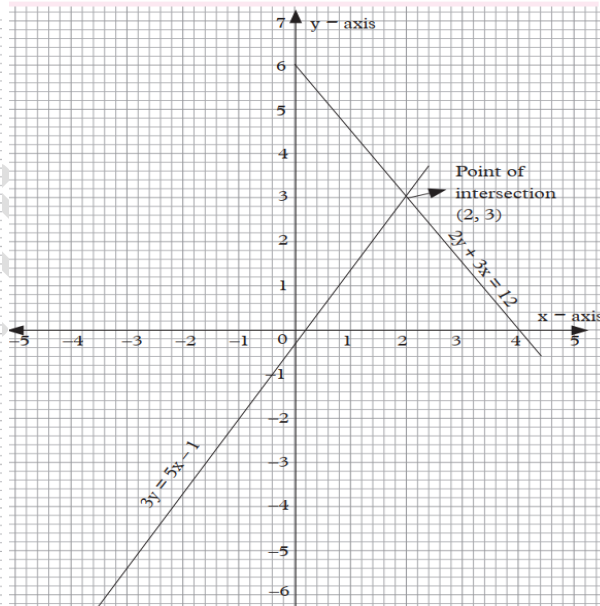
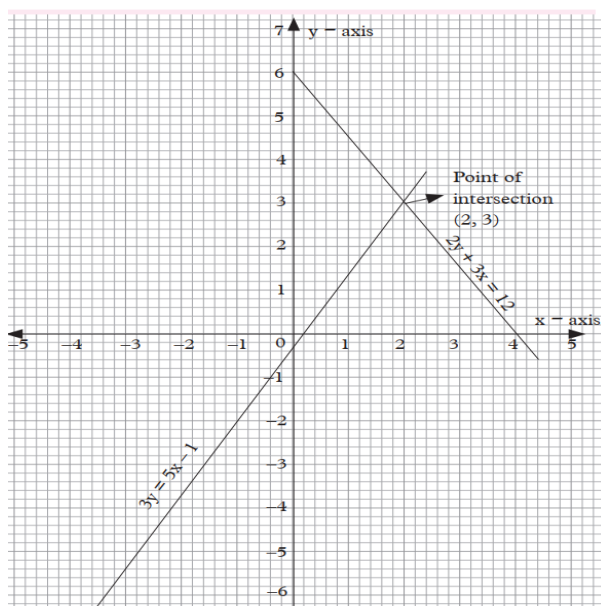
Table 4.1

x	-1	2	-4
y	-2	3	-7

Table 4.2

Step 2: Choose a suitable scale and plot the points on the same axes

- Draw both lines on the same graph
- If the lines do not intersect within the chosen interval extend them until they meet, unless the lines are parallel.



Step 3: Point of intersection has coordinates (2, 3) as read from the graph.

The solution of the simultaneous equations $2y + 3x = 12$ and $3y = 5x - 1$ is $x = 2$ and $y = 3$.

Now, check that the solutions satisfy the two equations.

Substitute 2 for x and 3 for y in the equations (1) and (2) in turns.

(i) In $2y + 3x = 12$ use 2 for x and 3 for y .

$$LHS = (2 \times 3) + (3 \times 2) = 6 + 6 = 12$$

$$LHS = RHS = 12$$

(ii) In $3y - 5x = -1$

use $x = 2$ and $y = 3$

$$LHS = (3 \times 3) - (5 \times 2) = 9 - 10 = -1$$

$$LHS = RHS = -1$$

Note:

We can also solve the equations by another method to verify that we obtain the same Solution. Now; solve the two equations using (i) elimination (ii) substitution and comparison methods.



Application activity 5.1:

1) Use graphical method to solve the following simultaneous equations. Verify your answers by substituting your solutions in the given equations

1. $x + y = 3$

$$4x - 3y = 5$$

3. $x + y = 0$

$$2y - 3x = 10$$

5. $4x - 3y = 1$

$$x - 4 = 2y$$

2. $x - 2y = 5$

$$2x + y = 5$$

4. $x + 2y = 5$

$$4x + 2y = 1$$

6. $6x - y = -1$

$$4x + 2y = -6$$

2) Solve:

- i) questions 1 to 2 using Elimination method
- ii) 3 to 4 questions using substitution method
- iii) 5 to 6 questions using comparison method.

5.2 Compound statements and inequalities



Activity 5.2

You know that 2, 4 and 7 are real numbers such that $2 < 4$ and $4 < 7$.

- 1) Write the inequalities $2 < 4$ and $4 < 7$ as a single inequality
- 2) Write a numerical example of the inequality in (1) above.
- 3) Draw a number line, and on it mark the relative positions of 2, 4 and 7. And show the region that satisfies the inequality found in (1) here above.

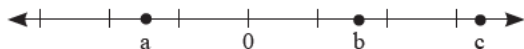


Content Summary

Some examples of such numbers are 3, 4, 5, - 5, -2, 3 etc.

if $a < b$, and $b < c$, we can say that a is the smallest, c is the largest and b is the middle number and we can write $a < b < c$.

similarly in 3, 4, 5 for example, $3 < 4 < 5$



a, b and c are such that a is to the left of b and b is to the left of c.

Any number to the left of another on a number line is less than it. $a < b < c$.

Sometimes, two simple inequalities may be combined into one **compound statement** such as $a < x < b$. This statement is the compound inequality of $a < x$ and $x < b$ or $x > a$ and $x < b$.

i.e x lies between a and b.

Example 1

Write the following pairs of simple inequality statements as compound statements and illustrate them on number lines.

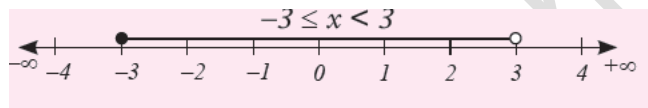
(a) $x \geq -3, x < 3$

(b) $x < 1, x \geq -2$

Solution

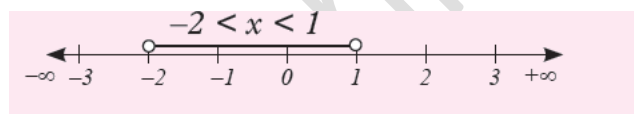
(a) $x \geq -3, x < 3$ becomes $-3 \leq x < 3$

$\therefore x$ lies between -3 and 3, and -3 is included.



(b) $x < 1, x \geq -2$ becomes $-2 < x < 1$

$\therefore x$ lies between -2 and 1.



Solving compound inequalities

It is important to remember that if three numbers are such that $a < b < c$, then $a < b$, $b < c$ and $a < c$.

Let us use the inequality $(3x - 2) < 10 + x < 2 + 5x$ to make three simple inequalities.

• The required inequalities are:

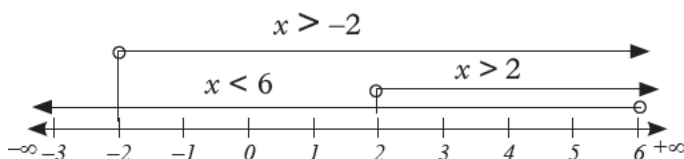
(i) $3x - 2 < 10 + x$

(ii) $10 + x < 2 + 5x$

(iii) $3x - 2 < 2 + 5x$

• Solutions include the region that satisfy the 3 at the same time:

(i) $x < 6$ (ii) $x > 2$ (iii) $x > -2$ respectively



- From the number line, the solution set is between 2 and 6 exclusive.

Any value of x chosen between 2 and 6 satisfies the inequality and this interval is common to all the three inequalities.

Example 2:

Find the range of values of x which satisfy the inequality

$$\frac{1}{4}(2x - 1) < \frac{1}{4}(x + 3) < 3(x + 4)$$

Solution

We begin by first splitting the compound inequalities to create three simple inequalities.

$$\frac{1}{4}(2x - 1) < \frac{1}{4}(x + 3) < 3(x + 4)$$

Separating and solving

$$\frac{1}{4}(2x - 1) < \frac{1}{4}(x + 3)$$

(Multiplying through by 4)

$$2x - 1 < x + 3$$

$$x < 4$$

and

$$\frac{1}{4}(x + 3) < 3(x + 4)$$

(Multiplying through by 4)

$$x + 3 < 12x + 48$$

$$-11x < 45$$

$$x > -4\frac{1}{11} \text{ (on dividing by } -11)$$

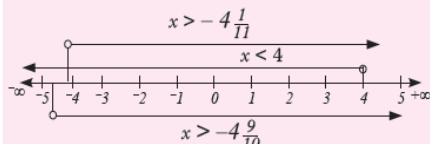
Also,

$$\frac{1}{4}(2x - 1) < 3(x + 4)$$

$$2x - 1 < 12x + 48 \text{ (Multiplying by 4)}$$

$$-49 < 10x$$

$$-4\frac{9}{10} < x \Rightarrow x > -4\frac{9}{10}$$



The range for x : $-4\frac{1}{11} < x < 4$

Example 3:

Find all the integral values of x which satisfy the inequality

$$2(2 - x) < 4x - 9 < x + 11$$

Solution

$$2(2 - x) < 4x - 9 < x + 11$$

Separating into three inequalities

$$2(2 - x) < 4x - 9 \dots (i)$$

$$4x - 9 < x + 11 \dots (ii)$$

$$\text{and } 2(2 - x) < x + 11 \dots (iii)$$

Solving each equation

$$2(2 - x) < 4x - 9 \dots (i)$$

$$4 - 2x < 4x - 9$$

$$-6x < -13$$

$$x > 2\frac{1}{6} \text{ (Dividing by } -6)$$

$$4x - 9 < x + 11 \dots (ii)$$

$$4x - x < 20$$

$$3x < 20$$

$$x < 6\frac{2}{3}$$

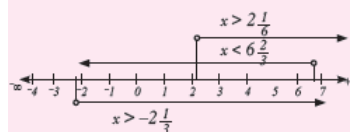
$$2(2 - x) < x + 11 \dots (iii)$$

$$4 - 2x < x + 11$$

$$-3x < 7$$

$$x > -\frac{7}{3} \text{ (Dividing by } -3)$$

$$x > -2\frac{1}{3}$$



$\therefore 2\frac{1}{6} < x < 6\frac{2}{3}$ i.e. solution lies between $2\frac{1}{6}$ and $6\frac{2}{3}$

$\therefore$ The solution is $\{3, 4, 5, 6\}$.



Application activity 5.2

1) Write each of the following pairs of simple inequality as a compound inequality and illustrate the answer on a number line.

(a) $x < -2, x > -4$ (b) $x > -3, x \leq 0$ (c) $x \leq 5, x > -2$

(d) $x > -\frac{1}{4}, x \leq 0$ (e) $x > -4\frac{1}{2}, x < \frac{1}{2}$

2) Solve the inequalities and represent your answer on a number line.

(a) $x - 4 < 3x + 2 < 2(x + 5)$ (b) $-5 \leq 2x + 1 < 5$

(c) $2(3x + 1) \geq 4(x - 1) < 12$.

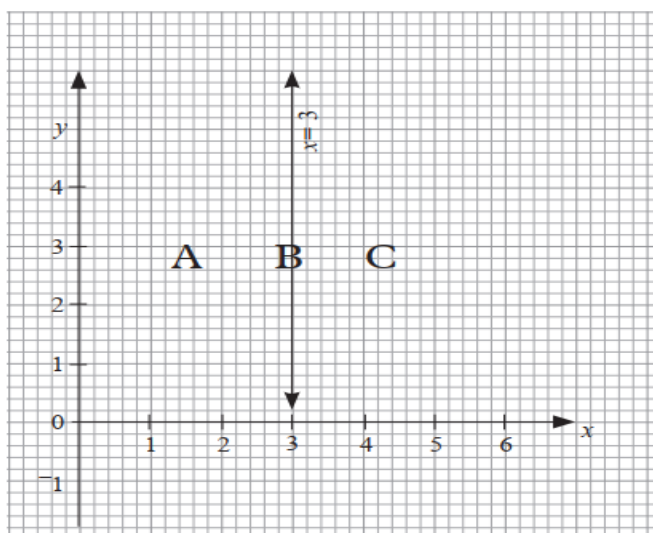
5.3 Graphical representation of linear inequalities



Activity 5.3:

Imagine that a Cartesian plane extends indefinitely in all directions and that a line also extends indefinitely in two directions.

- 1) Into how many sets of points does the line $x = 3$ divide the Cartesian plane?
2. Identify and describe each set of points with reference to the line.
3. Into how many regions (areas) does the line divide the plane?



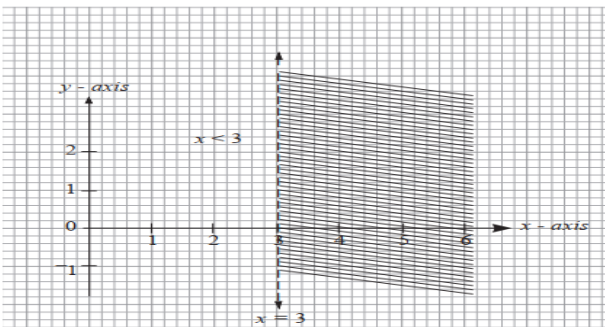
Content summary

Any line divides a plane into three sets of points i.e.

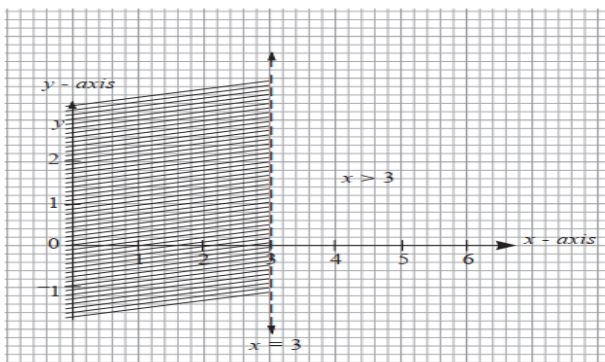
- (i) Points on the line.
- (ii) Points on either side of the line

In a Cartesian plane, each set of points can be defined with reference to the line. For example, points on the line are defined by the equation of the line. Points on either side of the line can be described using the inequalities notation(s) $>, \geq, <, \leq$ with reference to the equation of the line.

As shown in the graphs below



The region represented by $x < 3$ is at the left side of the line with equation $x = 3$ (it is not shaded).

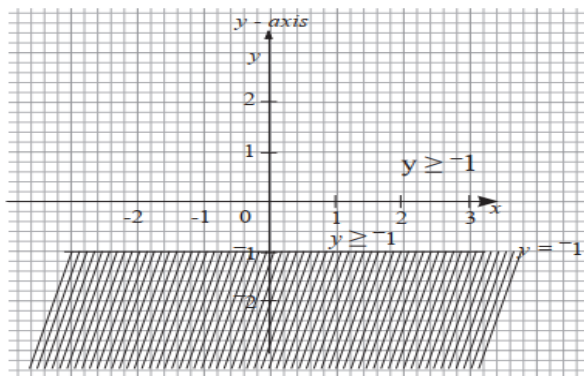


The region represented by $x > 3$ is at the right side of the line with equation $x = 3$ (it is not shaded).

Note:

In both cases, the **unwanted region** (i.e. the region in which the inequality is not satisfied) is shaded. Also, all points on the line $x = 3$ are not wanted, so the line is 'dotted'.

The following graph shows the region for which $y \geq -1$.



Note that the line $y = -1$ is continuous. This means that the points on the line are included in the required region. To represent an inequality on a graph, we use the equation corresponding with that inequality as the equation of the boundary line; e.g. $y > -1$: Boundary line is $y = -1$. If points on the line are included in the required region, the line is continuous (solid). If not, the line is dotted (broken). The normal convention is to shade the unwanted region.



Application activity 5.3:

Show each of the following regions on a graph in the Cartesian plane.

- 1.(a) $x > 1$ (b) $x < 5$ 2.(a) $x < -2$ (b) $x \geq -1$

$$3.(a)y \leq 2 \quad (b)y > 0 \quad 4.(a)y < 1 \quad 2 \quad (b)y \geq -1.5$$

5.4 Solving simultaneous inequalities with one unknown



Activity 5.4

Consider the inequalities

(i) $1 - 3x > 10$

(ii) $3 - 2x < 15$

On the same number line, represent the two solutions. Write down the range of the values of x which satisfy both the inequalities.



Content Summary

Solve the inequalities

(i) $1 - 3x > 7 - 5$

(ii) $3 - 2x < 9$

Now consider

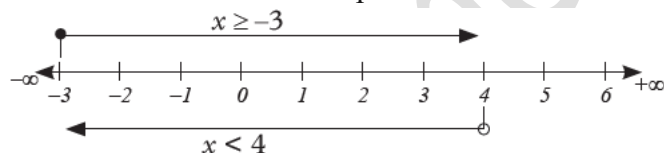
From activity 3.14 you should have observed an argument similar to the following:

(i) $x - 2 \leq 3x + 4 \rightarrow 3x - x \geq -2 - 4 \rightarrow 2x \geq -6 \rightarrow x \geq -3$

(ii) $3x + 4 < 16$

$3x < 12 \rightarrow x < 4$

The two solutions can be represented on a number as:



Any value of x between -3 inclusive and $+4$ satisfy the two inequalities i.e. $-3 \leq x < 4$.

Thus the inequalities $x - 2 < 3x + 4$ and $3x + 4 < 16$ are simultaneously satisfied by x : $-3 \leq x < 4$ they are simultaneous inequalities in one unknown. Inequalities that must be satisfied **at the same time** are called simultaneous inequalities.

Example 1:

Solve the following pair of simultaneous inequalities.

$$2(3 - x) < 10, 3(2x - 5) < 21$$

Solution

$$6 - 2x < 10$$

$$\Rightarrow -2x < 4$$

$$\Rightarrow x > -2 \quad \dots\dots\dots(i)$$

$$\text{Also } 6x - 15 < 21$$

$$\Rightarrow 6x < 36$$

$$\Rightarrow x < 6 \quad \dots\dots\dots(ii)$$

Combining (i) and (ii), we have $-2 < x < 6$.

Thus, x lies between -2 and 6 .

$$-2 < x < 6$$

This is represented on a number line as in Fig. 3.9.

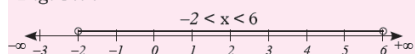


Fig. 3.9

$$s = (-2, 6) \text{ or } s =]-2, 6[$$

Example 2:

Solve the inequality:

$$x - 22 < -10 - x < -12 + x$$

Solution

$$x - 22 < -10 - x < -12 + x$$

Split the inequality into two simultaneous inequalities as:

$$x - 22 < -10 - x \quad \dots\dots\dots(i)$$

$$\text{and } -10 - x < -12 + x \quad \dots\dots\dots(ii)$$

Solving each inequality separately.

$$(i) \quad x - 22 < -10 - x$$

$$\Rightarrow 2x < 12$$

$$\Rightarrow x < 6 \quad (\text{Dividing by } 2)$$

$$(ii) \quad -10 - x < -12 + x$$

$$\Rightarrow 2 < 2x$$

$$\Rightarrow 1 < x \quad (\text{dividing by } 2)$$

Combine (iii) and (iv) to get $1 < x < 6$

This is represented on a number line as in Fig. 3.10.

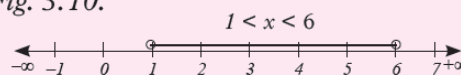


Fig. 3.10



Application activity 5.4

Solve the following simultaneous inequalities and represent each solution on a number line.

1) (a) $2x < 10$, $5x \geq 15$

(b) $3x \leq 9$, $2x > 0$

2) (a) $x + 7 < 0$, $x - 2 > -10$

(b) $x \geq 3$, $2x - 1 \leq 13$

3) (a) $4x - 33 < -1$, $-2 < 3x + 1$

(b) $2x - 5 < 2$, $2 \leq 5x - 6$.

5.5 Table of signs of a binomial



Activity 5.5

Imagine a thermometer showing the temperature in a classroom. At 20°C , the teacher says:

“When the temperature is **below** 20°C , the room feels cold.

When the temperature is **above** 20°C , the room feels warm.

And **at 20°C**, it feels neutral.” Consider the temperature x different from 20. Think of **20°C** as the **boundary point** in the thermometer story.

1) What happens when $x < 20$?

2) What happens when $x = 20$?

3) What happens when $x > 20$?



Content Summary

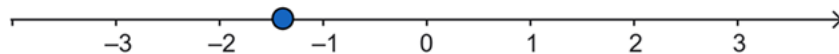
Consider a binomial $ax + b$.

To create a sign table, we first find the **zero of that binomial** (the value where the expression becomes 0):

$$ax + b = 0 \Rightarrow x = \frac{-b}{a}$$

This is the **critical point** that divides the number line into two intervals.

$$x = \frac{-b}{a}$$



General sign table on a number line

$-\infty$ and $x < \frac{-b}{a}$	$x = \frac{-b}{a}$	$x > \frac{-b}{a}$ and $+\infty$
Opposite sign of a	0	Same sign of a

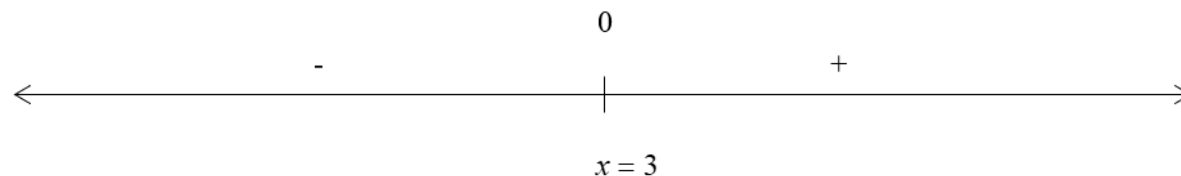
Example 1: The table of sign for the binomial $2x - 6$

Finding the zero: $2x - 6 = 0$

$$x = \frac{6}{2} = 3. \text{ The zero is } x = 3$$

	$-\infty$ and $x < 3$	$x = 3$	$x > 3$ and $+\infty$
	Opposite sign of 2	0	Same sign of 2
Sign of $2x-6$	-	0	+

Number line representation:



Example 2:

Consider the binomial $x + 4$ complete the table of sign

Solution

The root for $x + 4$ is when $x + 4 = 0 \Rightarrow x = -4$

This value divides the number line into two intervals. Thus, three cases arise

(i) $x < -4$ (ii) $x = -4$ and (iii) $x > -4$

Thus, the table of sign is

Interval/point	$x < -4$	$x = -4$	$x > -4$
Sign of $x + 4$	–	0	+

Example 3:

Evaluate the sign of the binomial $5 - 2x$

Solution

Sometimes it is important to write the binomial: $5 - 2x = -2x + 5$

The coefficient of x is **negative**

To find the critical value (root): we solve: $5 - 2x = 0$. We get $x = \frac{5}{2}$

The critical value or root is $x = \frac{5}{2}$ or $x = 2.5$

The table of signs:

Interval/point	$x < 2.5$	$x = 2.5$	$x > 2.5$
Sign of $5 - 2x$	+	0	–

Note: When $x < 2.5$, $5 - 2x$ is positive because the subtraction is smaller than 5.

When $x > 2.5$, $5 - 2x$ is **negative** because $2x$ becomes larger than 5.



Application activity 5.5

Evaluate the sign of the following binomials

a) $3x - 9$ b) $5 - 2x$ c) $x + 7$ d) $4x + 1$ e) $6 - 3x$

5.6 Solving inequalities involving the product and ratio of two binomials using analytical method



Activity 5.6

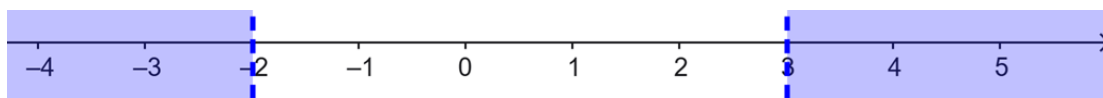
Imagine a **factory machine** controlled by two sensors:

- Sensor A: reads $x - 3$
- Sensor B: reads $2x + 4$

The machine operates safely only when both sensors indicate the same condition (both positive or both negative).

Mathematically, this corresponds to the inequality $(x - 3)(2x + 4) > 0$

1) Find the critical values



- 2) Divide the number line and identify intervals
- 3) Evaluate each factor and the product without using a table.
- 4) Share your work with other colleagues.



Content Summary

The general representation of inequalities involving the product of two binomials is given by

$$(ax + b)(cx + d) > 0,$$

$$(ax + b)(cx + d) < 0,$$

$$(ax + b)(cx + d) \geq 0,$$

$$(ax + b)(cx + d) \leq 0$$

To solve the above inequalities analytically,

- 1) Solve $ax + b = 0$ and $cx + d = 0$ to find **critical points**. The critical points

$$x = -\frac{b}{a}, \text{ and } x = -\frac{d}{c}, \text{ respectively.}$$

- 2) Locate the critical points on the number line and find the main intervals of that number line.
- 3) To identify the sign of the region of each interval, select one point of the interval (for example one $x < -\frac{b}{a}$,) substitute it in is in the product $(ax + b)(cx + d)$ and note the sign of the value.

Notation of intervals:

Notation	Inequality Form	Meaning	Endpoints Included?
(a, b)	$a < x < b$	Open interval	No
$[a, b]$	$a \leq x \leq b$	Closed interval	Yes
$[a, b)$	$a \leq x < b$	Half-open (left closed)	Only a
$(a, b]$	$a < x \leq b$	Half-open (right closed)	Only b
$[a, \infty)$	$x \geq a$	From a to infinity	a included

Example 1

Solve $(x - 2)(x + 3) > 0$

- a) Finding critical values for each factor gives,

$$x - 2 = 0 \Rightarrow x = 2, \quad x + 3 = 0 \Rightarrow x = -3$$

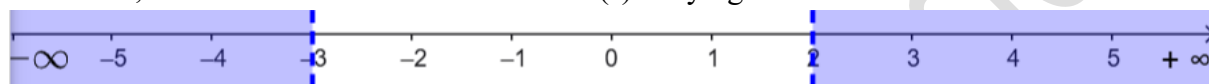
- b) On a number line, they are arranged in the order: $-\infty, -3, 2, +\infty$. Therefore, there are 3 main intervals: $(-\infty, -3); (-3, 2); (2, +\infty)$

c) Let's study the sign for each interval

i) For the interval $(-\infty, -3)$: Let us take $x = -4$: $(x-2)(x+3) = (-4-2)(-4+3) = +6 > 0$: The product is positive (which obeys the condition for $(x-2)(x+3) > 0$). This interval is a part of the solution.

ii) For the interval $(-3, 2)$: Let us take $x=1$, $(x-2)(x+3) = (1-2)(1+3) = -4 < 0$: the product is negative(**does not** obey the condition for $(x-2)(x+3) > 0$). This interval is not a part of the solution.

iii) For the interval $(2, +\infty)$: Let us take $x = 3$; $(x-2)(x+3) = (3-2)(3+3) = 4 > 0$: the product is positive (which obeys the condition for $(x-2)(x+3) > 0$). This interval is a part of the solution. Therefore, the solution is found in the interval(s) obeying the condition for > 0 .



$\therefore$ The solution is the set of all $x \in (-\infty, -3) \cup (2, +\infty)$.

The general representation of inequalities involving the ratio of two binomials is given by

$$\frac{ax+b}{cx+d} > 0, \quad \frac{ax+b}{cx+d} < 0,$$

$$\frac{ax+b}{cx+d} \geq 0, \quad \frac{ax+b}{cx+d} \leq 0,$$

To solve the inequalities, we solve numerator = 0 and denominator = 0. Note that value which make the denominator zeros are **excluded**. The obtained values are referred as critical values.

We use the critical points to separate the number line into intervals.

By studying the sign of the products of the numerator and denominator, we can identify the region that satisfies to the expected sign.

Example 2

Solve $\frac{x-1}{x+2} \leq 0$

Solution

a) Find critical values:

Numerator $x-1 = 0 \Rightarrow x = 1$

Denominator $x+2 = 0 \Rightarrow x = -2$ (*cannot include*)

It is shown that intervals on number line are $(-\infty, -2)$, $(-2, 1)$, $(1, +\infty)$

i) For the interval $(-\infty, -2)$: Let us take $x = -3$; $\frac{x-1}{x+2} = \frac{-3-1}{-3+2} = 4 > 0$: The ratio is positive (which

does not obey the condition). The interval is not part of the solution.

ii) For the interval $(-2, 1)$: Let us take $x=0$; The ratio is negative (which **obeys** the condition).

The interval is part of the solution.

iii) For the interval $(1, +\infty)$: let us take $x=3$; $\frac{x-1}{x+2} = \frac{3-1}{3+2} = \frac{2}{5} > 0$: The ratio is positive (which does not **obey** the condition). This interval is not part of the solution.

Note: We will include numerator $= 0$ (i.e $x = 1$) because of the sign ≤ 0 . We exclude $x = -2$ (because denominator cannot be zero).

The solution set is the set of $x \in (-2, 1]$.



Application activity 5.6

Solve the following inequalities without using a table of signs

1. $(x+4)(2x-1) \geq 0$
2. $\frac{3x-5}{x+1} < 0$
3. $(5-x)(x+2) < 0$
4. $\frac{2x-2}{x-1} \geq 0$

5.7 Solving inequalities involving the product two binomials using table of signs



Activity 5.7

Imagine a **school greenhouse** with two sensors that control heating: Each sensor gives a reading depending on the temperature x .

Sensor A reading $x - 4$

Sensor B reading $2x + 6$

The system works properly **only when both sensors indicate the same condition** (both readings positive and both negative).

In mathematical terms, the system works when:

$$(x-4)(2x+6) > 0$$

- 1) Find the critical points
- 2) Draw and divide the number line using critical points
- 3) Predict the sign of each factor $(x-4)$ and $(2x+6)$ on the number line.
- 4) Basing on the sign for each factor, predict the sign of the product $(x-4)(2x+6)$. Deduce The interval that satisfy $(x-4)(2x+6) > 0$.
- 4) Discuss with colleagues why the product becomes positive when both factors have the same sign.



Content Summary

Inequalities involving the **product** or **ratio** of two binomials appear in many real-life contexts, such as comparing costs, planning resources, or controlling systems.

The **analytical method** uses **critical values** and intervals to determine the solution systematically, to track where expressions are positive, negative, or zero.

The general representation of inequalities involving the product of two binomials is given by

$$(ax + b)(cx + d) > 0, (ax + b)(cx + d) < 0,$$

$$(ax + b)(cx + d) \geq 0, (ax + b)(cx + d) \leq 0$$

To solve the inequalities, it is possible to proceed as follows:

1) Find the roots of each binomial:

$$\text{i.e. } ax + b = 0 \Rightarrow x = -\frac{b}{a}, \quad cx + d = 0 \Rightarrow x = -\frac{d}{c}$$

2) Divide the number line into intervals using the roots.

3) Determine the sign of each binomial in each interval

4) Apply the inequality condition.

i.e. Product > 0 : factors have the same signs.

Product < 0 : factors have opposite signs.

Example 1: Solve $(x - 3)(2x + 4) > 0$

Solution:

The roots of factors:

$$\text{For } x - 3 = 0 \Rightarrow x = 3$$

$$\text{For } 2x + 4 = 0 \Rightarrow x = -2$$

Thus, the intervals on number line are: $(-\infty, -2), (-2, 3), (3, +\infty)$

Table of signs:

Interval/point	$x < -2$	$x = -2$	$-2 < x < 3$	$x = 3$	$x > 3$
Sign of $x - 3$	-		-	0	+
Sign of $2x + 4$	-	0	+	+	+
Sign of $(x - 3)(2x + 4)$	+	0	-	0	+

Applying condition > 0 , The solution is the set of all $x \in (-\infty, -2) \cup (3, +\infty)$

The general representation of inequalities involving the ratio of two binomials is given by

$$\frac{ax+b}{cx+d} > 0, \frac{ax+b}{cx+d} < 0,$$

$$\frac{ax+b}{cx+d} \geq 0, \frac{ax+b}{cx+d} \leq 0,$$

To solve inequalities:

- 1) Find critical values (roots of numerator and denominator)
- 2) Divide the number line into intervals.
- 3) Determine the sign of the numerator and denominator.
- 4) Apply the inequality condition.

i.e Ratio > 0 : numerator and denominator have the same sign

Ratio < 0 : numerator and denominator have opposite sign.

- 5) Exclude points where the denominator = 0

Example 2: Solve $\frac{x+1}{3x-9} \leq 0$

Solution:

The root of the numerator: $x+1=0 \Rightarrow x=-1$

The root of the denominator: $3x-9=0 \Rightarrow x=3$ (exclude)

Thus, the intervals on number line are: $(-\infty, -1), (-1, 3), (3, +\infty)$

Table of signs

Interval/point	$x < -1$	$x = -1$	$-1 < x < 3$	$x = 3$	$x > 3$
Sign of $x+1$	-	0	+	+	+
Sign of $3x-9$	-	-	-	0	+
Sign of $\frac{x+1}{3x-9}$	+	0	-		+

Applying the condition ≤ 0 , The solution is the set of all $x \in [-1, 3)$



Application activity 5.7

- 1) Given inequality

$$(x-3)(3x+6) > 0$$

- (a) Find the roots of each binomial.
- (b) Divide the number line into intervals using the roots.
- (c) Construct a sign table.
- (d) Determine the solution set for x .

- 2) Solve the inequalities

(a) $(4-x)(x+1) < 0$ (b) $\frac{2x-5}{x+3} \leq 0$ (c) $(2x-3)(x+5) \geq 0$

(d) $\frac{x-4}{2x+1} < 0$ (e) $(3-x)(x-1) < 0$ (f) $\frac{2x+5}{x-1} \geq 0$

5.8 Solving simultaneous linear inequalities with one unknown graphically



Activity 5.8:

A school sports team is selecting players for an inter-class competition.

The coach says:

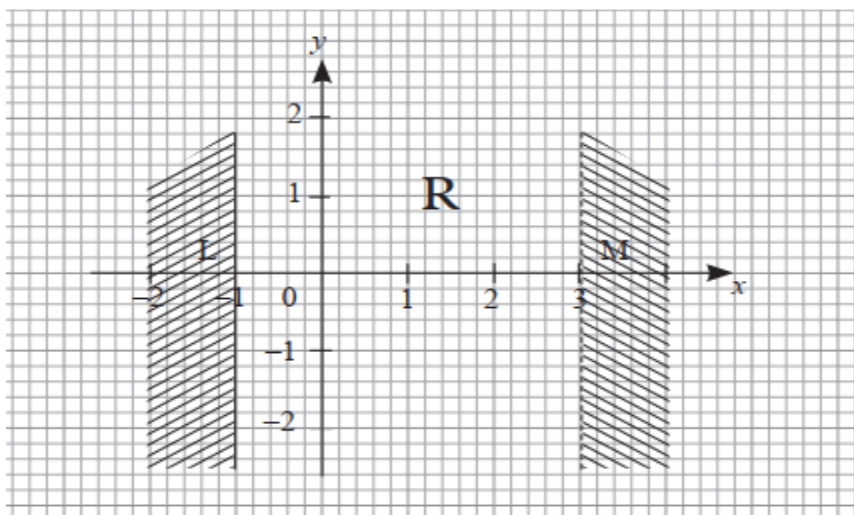
- “A player must be **older than 12 years.**”
- “A player must also be **13 years or younger** to join the junior team.”

Find the possible ages



Content summary

When regions are defined by two or more inequalities, those inequalities are referred to as simultaneous inequalities.



In graph. 4.16, the unshaded region R lies between two inequalities. The two boundary lines are $x = -1$ and $x = 3$. The region R satisfies both the inequalities $x < 3$ and $x \geq -1$ simultaneously.

$\therefore$ We can say $-1 \leq x$ and $x < 3$. In one statement, we write $-1 \leq x < 3$.

In summary “To Solve simultaneous linear inequalities with one unknown graphically, it consists of graphing each inequality, identifying the common region, and writing the solution set clearly by inequality form and interval notation.”



Application activity 5.8:

By shading the unwanted regions, show the regions which satisfy the given inequalities in questions 1 to 4.

1. (a) $3 < x < 4$ (b) $1 < x \leq 5$

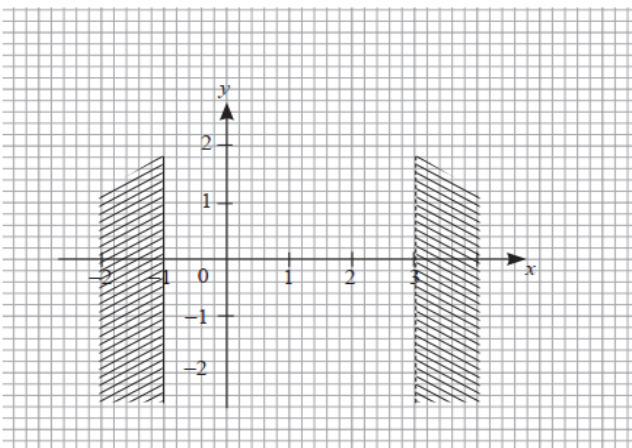
2. (a) $-2 \leq y < 2$ (b) $-1 \leq y \leq 1$

3. (a) $1\frac{1}{2} \leq x \leq 4$ (b) $2x > 7$ and $3x \leq 18$

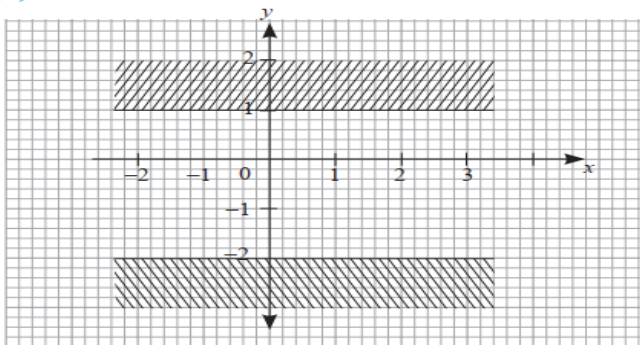
4. $y + 5 \leq 4y < 2y + 14$

5. Find the inequalities represented by the following unshaded regions. Write your answer in a single statement, that is $a \leq x \leq b$ or $a \leq y \leq b$

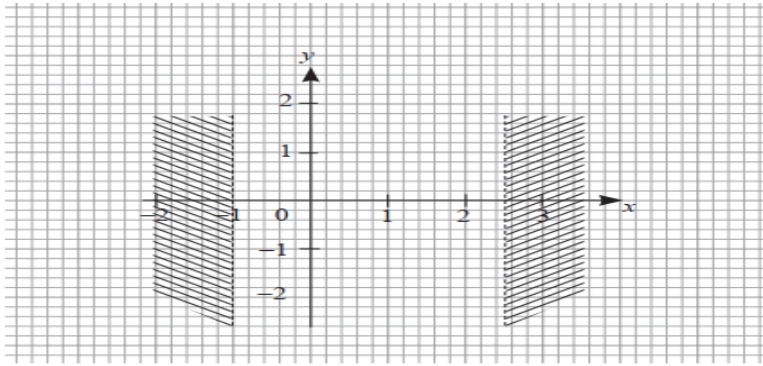
(a)



(b)



(c)



5.9 Meaning of simultaneous linear inequalities in two unknowns



Activity 5.9:

A drone is flying over a rectangular farm. The pilot sets two rules for safety:

1. The drone must fly **below the line** $y = 6$.
2. The drone must also fly **to the right of the line** $x = 2$.

Represent unshaded region for all possible safe positions (x, y) where the drone can fly.

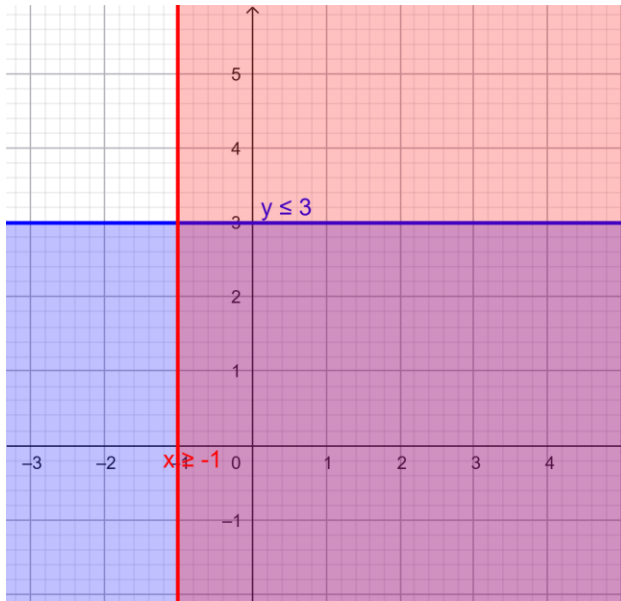


Content summary

Simultaneous linear inequalities in two unknowns are **two or more inequalities** involving **two variables**, usually x and y , that must be **true at the same time**. Each inequality represents a **region (a half-plane)** on the coordinate grid. To find the solution to the simultaneous inequalities, we determine the **overlapping region** that satisfies **all** the inequalities.

Example 1

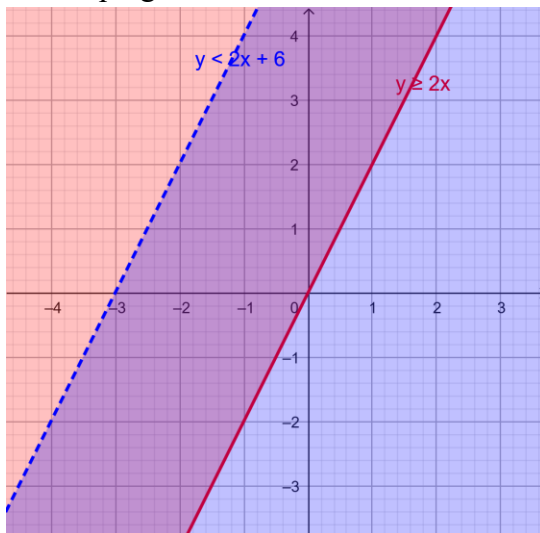
The meaning of $y \leq 3$ and $x \geq -1$ is that the points must be **below or on** the line $y = 3$ **and to the right of** $x = -1$.



The required region is shade twice.

Example 2

The meaning of $y > 2x$ and $y < 2x + 6$ is that the points lie in the **band or strip** between the two sloping lines.



The required region is shade twice.



Application activity 5.9:

- 1) Which statement describes “simultaneous inequalities”?
 - A. One inequality only
 - B. Two inequalities solved separately
 - C. Two inequalities that must be true at the same time
 - D. Two inequalities that contradict each other

- 2) What do the inequalities $x > 1$ and $y > 2$ mean?
- Points above the line $y = 2$
 - Points to the right of $x = 1$
 - Both A and B
 - Neither A nor B
- 3) What does the pair of inequalities $x \geq 0$ and $y < 4$ mean about where the point (x, y) can lie?
- Left of the y-axis and above the line $y = 4$
 - Right of the y-axis and below the line $y = 4$
 - Anywhere in the plane
 - Only on the line $y = 4$
- 4) The inequalities $y > x$ and $y < x + 5$ mean that the point (x, y) for the solution must lie:
- Between the two lines
 - Above both lines
 - Below both lines
 - On the line $y = x$ only

5.10 Solving simultaneous linear inequalities in two unknowns



Activity 5.10:

- 1) Using a graph paper, draw a line represented by the equation $x + y > 3$.
- Into how many regions does the line divide the Cartesian plane?
 - Identify one point (x, y) on one side of the line and substitute for x and y in the given equation. What do you notice? Describe the region containing the point satisfying using inequality $x + y > 3$
 - Do a similar substitution as in (b) above with a point from the other side of the line. Comment on your answer.
- 2) Using the same axis show the regions satisfying the following inequalities:
- $x + y > 3$
 - $3x + 2y < 12$
- In each case, find the equation of the boundary line.
 - Draw the line, one at a time.
 - In each case show the required region by shading the unwanted region.
 - Denote the unshaded region with letter R.
 - What can you say about this region?

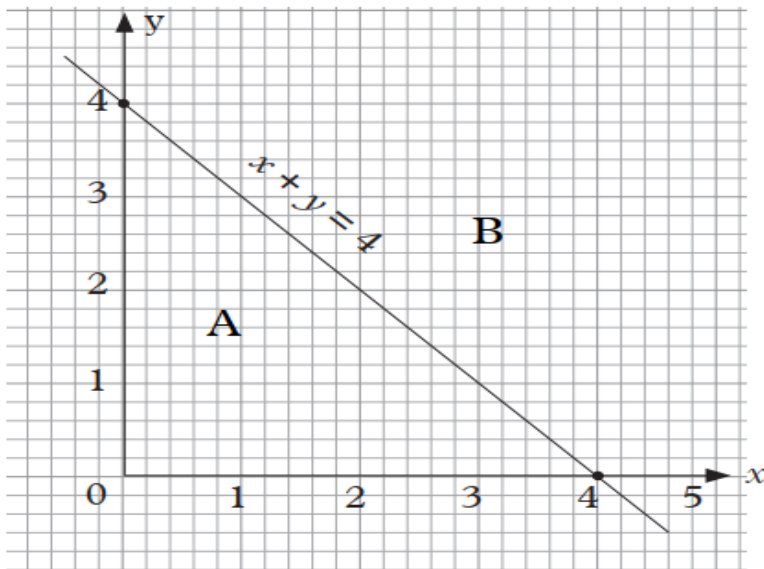


Content summary

Linear inequalities in two unknowns

Simultaneous linear inequalities in two unknowns are two or more inequalities involving the same variables, usually x and y , that must be satisfied at the same time. To solve them, we first draw each boundary line and shade the unwanted region (Region which does not correspond to the inequality). The solution of the simultaneous inequalities is the unshaded region, known as the feasible region, where all conditions are satisfied together. This method is commonly used to model real-life situations, such as planning areas, resource allocation, or movement constraints, where multiple conditions must be fulfilled simultaneously.

Now repeat Activity 5.7 above using the equation $x + y = 4$. The graph representing the equation $x + y = 4$ is the following:



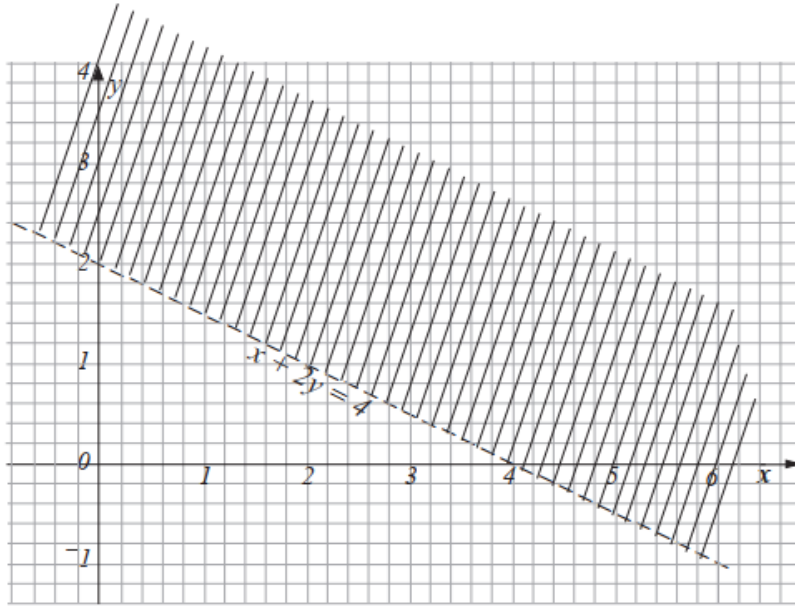
- (a) The line $x + y = 4$ divides the plane into two regions A and B
- (b) Using a point such as $(2, 1)$ substitute in the equation, $x + y = 4$.

$$\text{LHS} = 2 + 1 = 3$$

Therefore $(2, 1)$ satisfies $x + y < 4$. Thus $x + y$ is less than 4. The point $(2, 1)$ in region A satisfies the inequality $x + y < 4$. But since the line is drawn solid, we can denote the region as $x + y \leq 4$.

Similarly, points on the other side satisfy the inequality $x + y \geq 4$.

Now consider the graph in the graph.



The boundary line in the **graph** is broken. This means unshaded (required) region does not include points on the line in order to define the required region, we use coordinates of any point not on the line. For example, use the point (1,1) substitute for x and y in the equation $x + 2y = 4$ as follows: in the equation $x + 2y = 4$, LHS: $x + 2y = 1 + 2 \times 1 = 1 + 2 = 3$

$$\text{RHS} = 4$$

The value on the LHS is less than the value on the RHS, thus, $3 < 4$.

This means $x + 2y < 4$.

Therefore, the unshaded region contains the points that satisfies the inequality $x + 2y < 4$.

Note: If the boundary line does not pass through the origin, it is more convenient and faster to determine the required region using the origin.

Graphical solution of simultaneous linear inequalities with two unknowns

When solving linear simultaneous equations, we look for values of the two unknowns that make the two equations true at the same time. Similarly, linear inequalities with two unknowns are solved to find a range of values of the two unknowns which make the inequalities true at the same time. The solution is represented graphically by a region.

To identify the required region, we deal with one inequality at a time to avoid confusion.

- (i) Draw one line.
- (ii) Identify the required region by shading the unwanted inequalities.
- (iii) Repeat parts (ii) and (i) for each of the given inequalities.
- (iv) The unshaded region represents the set of the given inequalities.

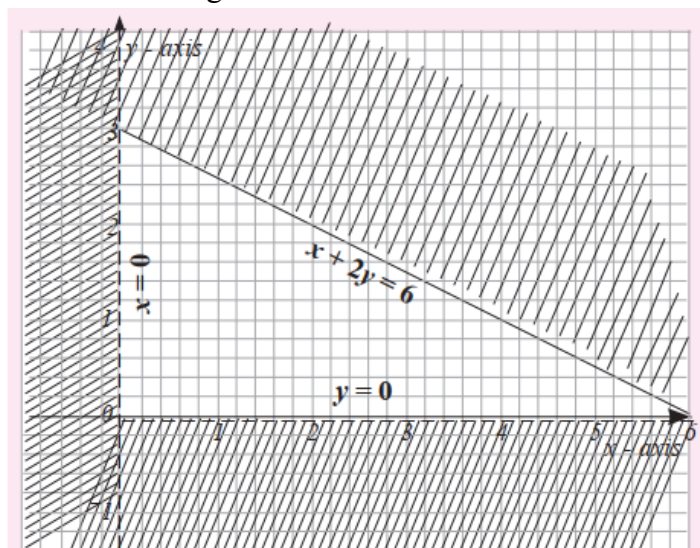
Example 1

Draw the region which satisfies the following inequalities simultaneously:

$x > 0$, $y > 0$, $x + 2y \leq 6$. State the integral values of x and y that satisfy the inequalities.

Solution

In the figure below, $x + 2y = 6$ (solid line), $x = 0$ (broken line), $y = 0$ (broken line) are the boundary lines. All the ordered pairs of values (x, y) that satisfy the three inequalities lie within the unshaded region.



The integral pairs of x and y values that satisfy the inequalities are $(1, 1)$, $(1, 2)$, $(2, 1)$, $(2, 2)$, $(3, 1)$, $(4, 1)$.



Application activity 5.10

I) Show the region which contain the set of points represented by each of the following inequalities in questions 1 to 4.

1) (a) $x + y < 4$ (b) $y - x < 4$ (c) $2x + 3y \geq 6$

2) (a) $x - y \leq 0$ (b) $4x + 5y \leq 10$ (c) $y + 3x > -6$

3) (a) $3x < y + 6$ (b) $\frac{1}{2}x - 2y > 2$ (c) $y + \frac{5}{2}x \leq -5$

4) (a) $x + y \geq 3$ (b) $4y - 3x > 0$ (c) $5y - x < 15$

II) For questions 1 to 5, show the region defined by all inequalities in each question.

1) $x + y \geq 0$, $x < 1$, $y > 1$

2) $2x + 3y > 6$, $y > 0$, $x > 0$

3) $y - x < 0$, $x < 4$, $y \geq 0$

4) $3x + 5y > 15$, $5x + 3y < 30$, $x > 0$, $y > 0$

5) $y \geq 0$, $y < 4$, $4x + 3y > 0$, $5x + 2y < 15$



END OF UNIT ASSESSMENT

Assessing knowledge and understanding

1) Which of the following best describes the solution to a system of two linear equations in two unknowns?

A) A single point where the lines are parallel

B) The point where the two lines intersect on a graph

C) The shaded region above both lines

D) The set of all points that satisfy one of the equations.

2) When graphing the inequality $y > 2x - 1$, how should the boundary line be drawn if we shade the unwanted region?

A) Solid line, shade above the line

B) Dashed line, shade above the line

C) Solid line, shade below the line

D) Dashed line, shade below the line

3) What does the unshaded region (as solution) in a graph of simultaneous linear inequalities represent?

A) The area where none of the inequalities are satisfied

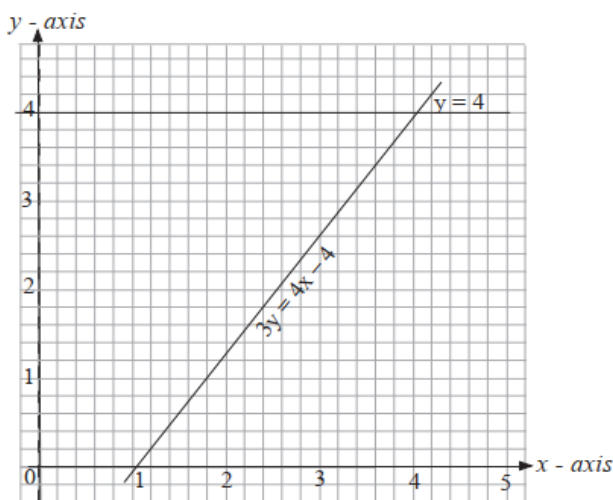
B) The area where at least one inequality is satisfied

C) The area where all inequalities are satisfied simultaneously

D) The area where the equations intersect

Assessing skills

4) Find the solution of the simultaneous inequalities $y < 4$ and $3y > 4x - 4$. The graph to be used is here below:



5) Solve

$$a) \begin{cases} x + y < 1 \\ x - y \leq 5 \end{cases} \quad b) \begin{cases} 2x + y > 4 \\ 2y - x \leq -6 \end{cases}$$

6) A school is buying sports equipment. Each football costs f FRW and each basketball costs b FRW.

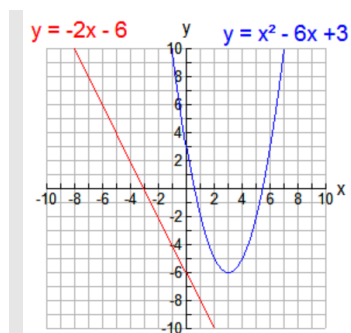
If 4 footballs and 3 basketballs cost 8,600 FRW, and 2 footballs and 5 basketballs cost 8,400 FRW:

- Write down two simultaneous equations to represent this situation.
- Solve the equations to find the cost of one football and one basketball.

Assessing attitudes and values

7) When working in a group to solve a real-world problem using inequalities, what is the most important practice?

- Letting one person do all the work to save time
- Ensuring each step is justified and everyone contributes ideas
- Only using the method the teacher prefers
- Ignoring constraints to simplify the problem.



UNIT 6: LINEAR AND QUADRATIC FUNCTIONS

Key unit competence: Apply graphical behaviour and key features of linear and quadratic functions in solving real-world problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Explain system of simultaneous linear equations and inequalities in two unknowns.
- Identify the solution set of a system from its graphical representation.
- Explain the meaning of the point of intersection and the shaded region in the context of a problem.

Skills:

- Solve systems of linear equations graphically and algebraically (substitution, elimination).
- Graph the solution set for a system of linear inequalities.
- Translate word problems into systems of equations or inequalities.

- Interpret the solution of a system in the context of the original real-world problem.

Attitudes and values:

- Appreciate the power of graphical and algebraic models for making informed decisions.
- Develop logical reasoning and systematic problem-solving strategies.
- Collaborate effectively to analyse problems and justify solutions.

Introduction

Linear and quadratic functions are widely used in real-life situations to describe and solve practical problems. Linear functions apply to situations with a constant rate of change, such as calculating costs, predicting profits, converting units, or determining distance over time at a steady speed. Quadratic functions are used when relationships involve curvature or change in rates, such as modelling the path of a projectile, calculating the maximum area of a fenced region, analysing acceleration in physics, or determining revenue and profit in business when there is an optimal point. Together, these functions help in decision-making, forecasting, engineering, economics, and many daily-life problem-solving contexts.



Introductory activity

- 1) (a) Draw the graph of the quadratic function $y = 2x^2 + 5x - 9$ and $y = 2x - 5$ for the values of x from -4 to 2 .
b) Use the graphs to locate coordinates for intersecting points of the line and the curve.
- 2) Solve the equation $2x^2 + 5x - 9 = 0$. Are solutions the same as the intersecting points of x -axis and the graph of the function $y = 2x^2 + 5x - 9$?
- 3) State the equation of the line of symmetry for the curve of $y = 2x^2 + 5x - 9$.
- 4) Use a graph to solve the equation $2x^2 + 3x - 4 = 0$.

6.1. Finding equation of a straight line given gradient and one point on the line



Activity 6.1:

You are given the straight line that passes through the point $(3, -1)$ and its gradient is 2 .

- 1) Taking (x, y) as any other general point through the line, form an expression for the gradient using the two points $(3, -1)$ and (x, y) .
- 2) Find the equation of the line by equating an expression you obtained in step 1 to the gradient $m = 2$. Simplify the equation in form $y = mx + k$.



Content summary

If a straight line has gradient m and passes through a point (x_1, y_1) , its equation is:

$$y - y_1 = m(x - x_1)$$

This formula is called the **point-slope form** and is used when one point and the gradient are known.

Example 1

A straight line with gradient 3 passes through the point A (3, -4). Find the equation of the line.

Solution

The straight line is given by:

$$y - (-4) = 3(x - 3) \Rightarrow y = 3x - 13$$

Alternatively, let the point B (x, y) represent any general point on the line.

Using points, A and B, we have,

$$\text{gradient} = \frac{y - (-4)}{x - 3}. \text{ Since } \text{gradient} = 3,$$

$$\frac{y - (-4)}{x - 3} = 3$$

$$\Rightarrow y - (-4) = 3(x - 3) \Rightarrow y = 3x - 13$$



Application activity 6.1:

- In each of the following cases, the gradient of a line and a point on the line are given. Find the equation of the line.

$$(a) 3; (3, 1) \quad (b) \frac{1}{2}; (4, 5) \quad (c) -1; (-2, 3) \quad (d) -\frac{1}{5}; (5, 5) \quad (e) 0; (-4, 3)$$

$$(f) \text{Undefined}; (-4, 3) \quad (g) \frac{2}{3}; (-3, 0) \quad (h) -\frac{2}{3}; (0, 2)$$

- In each of the following, find the equation of a line whose gradient and a point through which it passes are:

$$(a) -3; (0, 0) \quad (b) \frac{2}{3}; (12, 13) \quad (c) -\frac{5}{2}; (4, 0) \quad (d) 0; (4, 3)$$

- Find the equations of lines described below.

(a) A line whose gradient is -2 and x-intercept is 1

(b) A line whose gradient is $-\frac{1}{5}$ and y-intercept is 4

- The gradients of two lines l_1 and l_2 are $-\frac{1}{2}$ and 3 respectively. Find their equations if they meet at the point (2, 3).

- Line L_1 has a gradient of -1 and passes through the point (3, 0). Line L_2 has a gradient $\frac{2}{3}$ and passes through the point (4, 4). Draw the two lines on the same pair of axes and state their point of intersection.

6.2 Equation of a straight line joining two points



Activity 6.2:

You are provided with the points A(-1, 1) and B(3, 2) along the straight line.

1. Find the gradient of the line joining the two points A and B.
2. Taking the point C (x, y) as the general point on the line and the point A(-1, 1), find an expression for the gradient of line AC.
3. Equate gradient obtained in step 1 to the gradient obtained in step 3, then Simplify the result.
4. Repeat step (2) and (3) using (3, 2) and (x,y)
5. Compare the results. What do you notice?



Content summary

A straight line is completely determined if **two distinct points** on the line are known. From these two points, the **gradient (slope)** of the line can be calculated, and then the equation of the line can be found. Let the two points on the line be: A(x₁,y₁) and B(x₂,y₂). The gradient is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} \text{ or } \frac{y_1 - y_2}{x_1 - x_2}$$

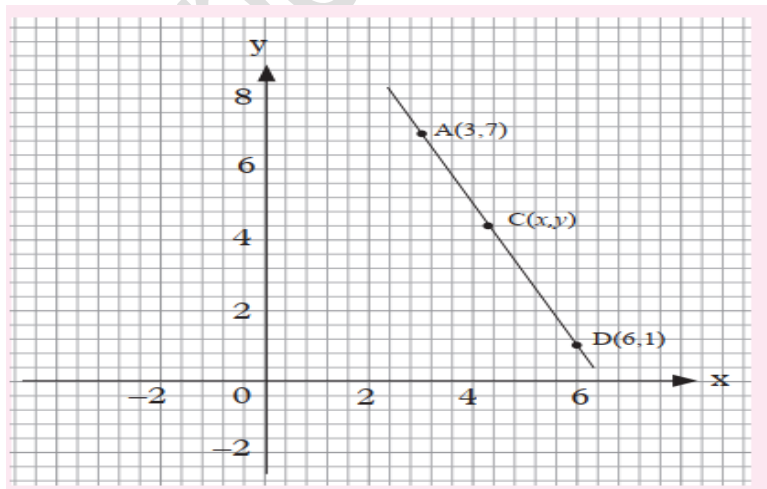
Once the gradient m is known, substitute it and one of the given points into the **point-slope form**: $y - y_1 = m(x - x_1)$ or $y - y_2 = m(x - x_2)$.

Example 1

Find the equation of the straight line which passes through points A(3, 7) and B(6, 1).

Solution

This graph is a sketch of the line.



Using points A and B, we have $\text{Gradient} = \frac{1-7}{6-3} = \frac{-6}{3} = -2$

Point C(x, y) is any general point on line AB.

Using points A and C, we get $\text{gradient} = \frac{y-7}{x-3}$. (We could equally well use points B and C.

Since ACB is a straight line, the two values of the gradient are equal.)

$$\therefore \frac{y-7}{x-3} = -2$$

$$\Leftrightarrow -2(x-3) = y-7 \Leftrightarrow -2x+6 = y-7 \Leftrightarrow 2x+y=13 \text{ (This is the equation of the line).}$$

In general, the equation of a straight line which passes through points (a, b) and (c, d) is given by,

$$\frac{y-b}{x-a} = \frac{d-b}{c-a} \text{ or } \frac{y-b}{x-a} = \frac{b-d}{a-c}.$$

Example 2

Find the equation of the line passing through the points (4, 5) and (8, 7).

Solution

$$\text{Gradient} = \frac{7-5}{8-4} = \frac{2}{4} = \frac{1}{2}.$$

Let (x, y) be a point on the same line. Using one of the points, say (4, 5),

$\text{Gradient} = \frac{y-5}{x-4}$ and the two gradients are equal.

$$\text{Therefore, } \frac{y-5}{x-4} = \frac{1}{2}$$

$$y-5 = \frac{1}{2}(x-4)$$

$$y-5 = \frac{1}{2}x - 2$$

$$y = \frac{1}{2}x + 3$$

Example 3

Find the equation of a line with x-intercept -4 and y-intercept is 3.

Solution

At the x -intercept, $y = 0$ so we have $(-4, 0)$.

At the y -intercept, $x = 0$ so we have $(0, 3)$

We therefore find the equation of the line by joining the points $(-4, 0)$ and $(0, 3)$

$$\text{Gradient} = \frac{\text{change in } y\text{-coordinate}}{\text{change in } x\text{-coordinate}}$$

$$\text{Gradient} = \frac{0 - 3}{-4 - 0} = \frac{-3}{-4} = \frac{3}{4}$$

Taking a general point and the point

$$\text{Gradient} = \frac{y - 3}{x - 0}$$

Equating the two gradients, we get

$$\frac{3}{4} = \frac{y - 3}{x - 0}$$

$$\text{So, } 3x = 4y - 12 \text{ we get } y = \frac{3}{4}x + 3$$



Application activity 6.2:

- Find the equation of the line that passes through the points
(a) $(-1, 1)$, $(3, 2)$ (b) $(7, 2)$ $(4, 3)$ (c) $(2, 5)$, $(0, 5)$ (d) $(5, -2)$, $(6, 2)$ (e) $(6, 3)$, $(-6, 2)$
(f) $(2, -5)$, $(2, 3)$ (g) $\left(\frac{1}{4}, \frac{1}{3}\right)$, $\left(\frac{1}{3}, \frac{1}{4}\right)$ (h) $(0.5, 0.3)$, $(-0.2, -0.7)$
- (a) Find the equation of the straight line which passes through the points $(0, 7)$ and $(7, 0)$.
(b) Show that the equation of the straight line which passes through $(0, a)$ and $(a, 0)$ is $x + y = a$.
- A triangle has vertices A $(-2, 0)$, B $(-1, 3)$ and C $(2, 3)$. Find the equations of the sides of the triangle.
- Two lines, l_1 and l_2 , both pass through the point $(4, k)$.
(a) If l_1 passes through the point $(5, -3)$, and has a gradient $-1\frac{1}{3}$, find the value of k
(b) If l_2 passes through $(-14, 0)$, find its equation
- Find the equations of lines described below
(a) A line whose x -intercept is 0 and passing through the point $(-3, 1)$
(b) A line whose y -intercept is -5 and passing through point $(-4, -1)$

6.3 Equations of parallel lines



Activity 6.3:

Consider the following linear equations:

Line 1: $y = 2x + 1$ Line 2: $y = -3x$ Line 3: $y = 2x - 4$ Line 4: $y = -3x + 5$

- 1) Draw one cartesian plane and plot the four lines. Which one are parallel?
2. What do you notice about the gradient of parallel lines?



Content summary

Lines are parallel if they have the same gradient. Consider two lines $y = m_1x + c_1$ and $y = m_2x + c_2$. These lines are parallel only and only if $m_1 = m_2$.

Example 1

Find the equation of a line which passes through the point (3, 5) and is parallel to $2y = 2 - 6x$.

Solution

The equation of the required line is in the form of $y = mx + c$. The gradient of the line $2y = -6x + 2 \Rightarrow y = -3x + 1$ is -3 . Since the two lines are parallel $m = -3$.

Thus, $y = -3x + c$. The fact that the required line passes through (3,5),

$$5 = -3 \times 3 + c$$

$$5 = -9 + c$$

$$c = 14$$

The required equation is $y = -3x + 14$.



Application activity 6.3:

1. Determine the gradients of the following pairs of equations and state whether their lines are parallel without drawing.

(a) $y = 2x - 7$ (b) $y = 4$
 $3y = 6x + 2$ $y = -3$

(c) $y = 2x + 3$
 $y = 4x + 6$

(d) $5y + 3x + 1 = 0$
 $10y + 6x - 1 = 0$

(e) $2y + x = 2$ (f) $2x + y = 3$
 $3y + 2x = 0$ $3x + y = 1$

2. Find the equation of the line that is parallel to another line whose equation is $4y + 5x = 6$ and passes through the point (8, 5).

3. Find the equation of the line that is parallel to another line whose equation is $y = \frac{2}{5}x + 2$ and passes through the point $(-2, -3)$.
4. A line through the points $(-2, 4)$ and $(3, 5)$ is parallel to the line passing through the points $(a, 6)$ and $(-4, 1)$. Find a .
5. Line L is parallel to a line whose equation is $y = 4x - 7$ and passes through the point $(1, -2)$. Find the equation of line L.

6.4 Equations of perpendicular lines



Activity 6.4

Consider the following linear equations:

Line 1: $y = 2x + 1$ **Line 2:** $y = 3x + 2$ **Line 3:** $y = -\frac{1}{2}x - 3$ **Line 4:** $y = -\frac{1}{3}x$

- 1) In one cartesian plane, plot the 4 lines. Observe and state which pairs of lines are perpendicular.
- 2) (i) Calculate the gradients of all the lines, use the results to complete the table below.

Pair	Gradients of lines		Product of gradients ($m_1 \times m_2$)
	m_1	m_2	

- (ii) What do you notice about $m_1 \times m_2 = -1$?



Content summary

Lines are parallel if they have the same gradient. Consider two lines $y = m_1x + c_1$ and $y = m_2x + c_2$. These lines are perpendicular only and only if $m_1 \times m_2 = -1 \Rightarrow m_2 = -\frac{1}{m_1}$.

Example 1

Find the equation of the line through $(5, 2)$ which is perpendicular to the line $5y - 2x = 10$.

Solution

The required line is perpendicular to $5y - 2x = 10$. If its gradient is m , then

$$m \times \frac{2}{5} = -1$$

$$m = -\frac{5}{2}.$$

If (x, y) is a general point on the required line,

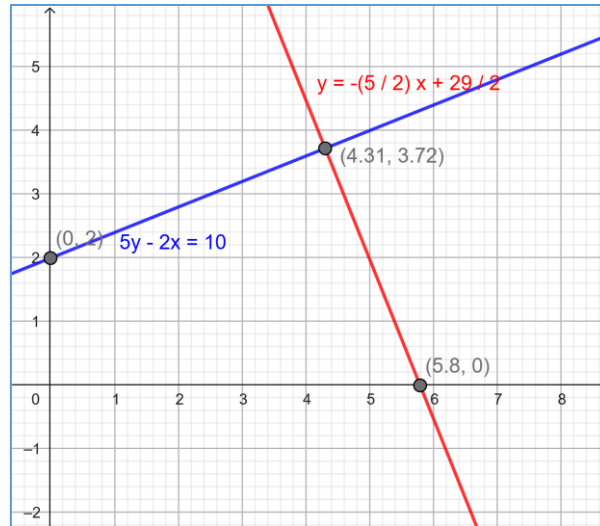
$$\text{then, gradient} = \frac{y-2}{x-5}$$

$$\therefore \frac{y-2}{x-5} = -\frac{5}{2}$$

$$\Leftrightarrow 2y - 4 = -5x + 25$$

$$\Leftrightarrow 2y = -5x + 29$$

$$\Leftrightarrow y = -\frac{5}{2}x + \frac{29}{2} \quad (\text{This is the equation of the required line}).$$



Look at page 95



Application activity 6.4:

- Without drawing, determine which of the following pairs of lines are perpendicular.
 - $y = 2x + 5$ and $2y + x = 3$
 - $2x - y = 7$ and $x + y = 5$
 - $3y = 2x + 1$ and $2y + 3x - 5 = 0$
 - $7x - 2y = -2$ and $14y - 4x = -1$
 - $y = 34x - 2$ and the line through $(8, 10)$ and $(2, 2)$.
 - A line through $(2, 2)$ and $(10, 8)$ and another through $(5, 6)$ and $(8, 2)$.
- Write down the equation of the line perpendicular to:
 - $3x + 4y - 1 = 0$ and passes through $(1, 2)$,
 - $y = 34x + 34$ and passes through the origin,
 - $3x - 2y + 7 = 0$ and passes through $(-1, 0)$,
 - $5y + x + 4 = 0$ and passes through $(3, 5)$.

6.5 Introduction to quadratic functions



Activity 6.5:

Study the following functions by verifying the highest exponent for the variable and decide which of them are quadratic functions. Give reasons to support your answer.

$$(a) y = 3x + 4 \quad (b) y = 3x^2 - 9x - 3 \quad (c) y = x^3 - 4x^2 + 10 \quad (d) y = x^2$$



Content summary

The expression $y = ax^2 + bx + c$, where a , b and c are constants and $a \neq 0$, is called a quadratic function of x or a function of the second degree (highest power of x is two).

Example 1

- (a) $f(x) = x^2 - 9$ (b) $f(x) = 2 - 3x + x^2$
(c) $f(x) = 2x^2 - 3x - 4$ (d) $y = x^2 + 8$



Application activity 6.5:

1) Which of the following functions is *not* a quadratic function?

- A: $f(x) = 3x^2 - 5x + 2$ B: $g(x) = 4 - x^2$
C: $h(x) = (x+1)(x-3)$ D: $p(x) = 2x = 7$ E: $q(x) = -x^2 + 0.5x$

2) A student says the function $f(x) = x(x-1) + 3 - x^2$ is quadratic because it contains x^2 terms. Simplify $f(x)$ and determine whether it is quadratic. Explain your reasoning.

6.6 Tables of values



Activity 6.6:

Given the quadratic function $y = x^2 - x - 6$.

1) Copy and complete the table below.

x	-4	-3	-2	-1	0	1	2	3	4
x^2									
$-x$									
-6	-6	-6	-6	-6	-6	-6	-6	-6	-6
$y = x^2 - x - 6$.									
coordinates (x,y)									

2) State the coordinates from the table in ordered pairs (x, y).



Content summary

Table of values are used to determine the coordinates that are used to draw the graph of a quadratic function. To get the table of values, we need to have the domain (values of an independent variable) and then the domain is replaced in a given quadratic function to find range (values of dependent variables). The values obtained are useful for plotting the graph of a quadratic function. All quadratic function graphs are parabolic in nature.

Example 1

Draw the table of values of $y = x^2$ and $y = -x^2$ for values of x from -5 to $+5$. Plot the graphs.

Solution

Table of $y = x^2$

Make a table of values for (x,y):

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y	25	16	9	4	1	0	1	4	9	16	25

Points obtained are:

$(-5,25)$, $(-4,16)$, $(-3,9)$, $(-2,4)$, $(-1,1)$, $(0,0)$, $(1,1)$, $(2,4)$, $(3,9)$, $(4,16)$, $(5,25)$

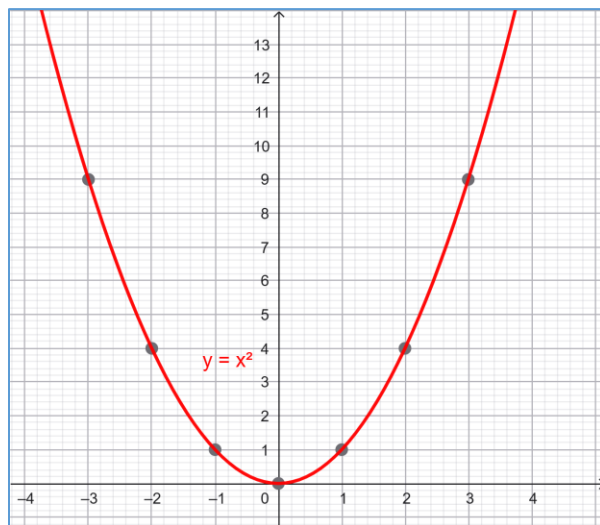
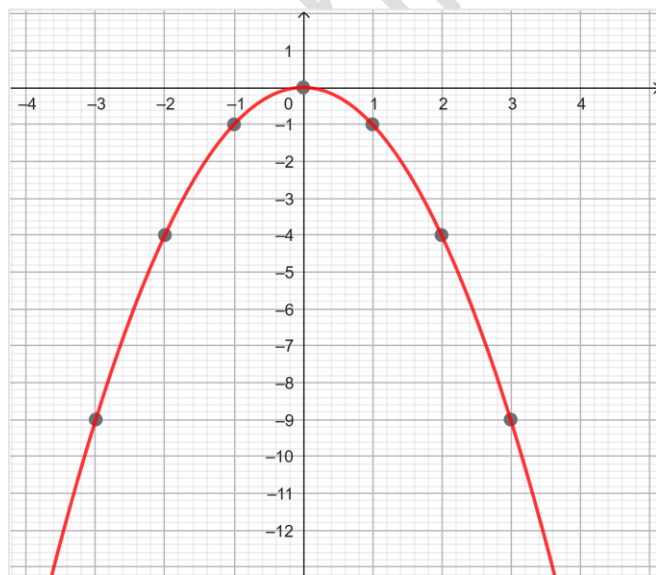


Table of $y = -x^2$

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y	-25	-16	-9	-4	-1	0	-1	-4	-9	-16	-25

The values are $(-5,-25)$, $(-4,-16)$, $(-3,-9)$, $(-2,-4)$, $(-1,-1)$, $(0,0)$, $(1,-1)$, $(2,-4)$, $(3,-9)$, $(4,-16)$, $(5,-25)$



Example 2

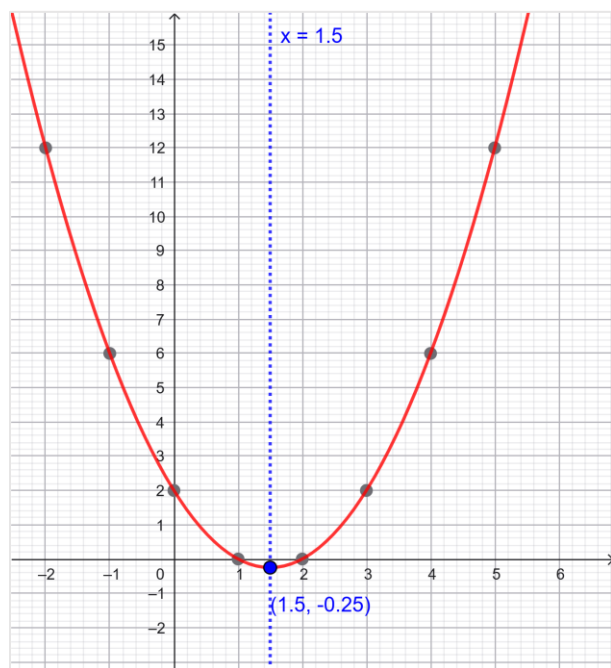
Draw the table of $y = x^2 - 3x + 2$, for values of x between -1 and +4.

Solution

Make a table of values of x and y

x	-1	0	1	1.5	2	3	4
$y = x^2 - 3x + 2$	6	2	0	-0.25	0	2	6

The values are $(-1, 6)$, $(0, 2)$, $(1, 0)$, $(1.5, -0.25)$, $(2, 0)$, $(3, 2)$, $(4, 6)$



Application activity 6.6:

- (a) Draw the table of $y = 1 + x - 2x^2$, taking values of x in the domain $-3 < x < 3$. State the coordinates obtained.
(b) Use the same domain to draw the table of $y = 2x - 5$. State the coordinates obtained.
- Draw the graph of $y = 2x^2 + x - 2$ from $x = -3$ to $x = 2$. Hence, find the appropriate values of the roots of the equation $2x^2 + x - 2 = 0$.
- Copy and complete the following table of values for $y = 6 + 3x - 2x^2$. Plot the graph of the function

x	-2	-1	0	0.5	1	2	3
6							
$3x$							

$-2x^2$							
y	-8	—	—	—	7	4	—

6.7 Vertex of a quadratic function and axis of symmetry



Activity 6.9:

Use internet, AI or dictionary to explain the following terms used for the graph of a quadratic function.

- (a) Line of symmetry
- (b) Maximum point
- (c) Minimum point



Content summary

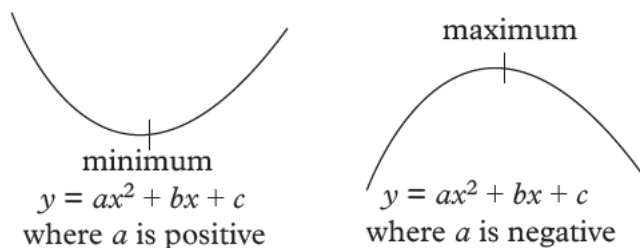
The vertex is the single point where a parabola changes direction. It is either the "peak" (maximum) or the "valley" (minimum) of the function.

For $f(x) = ax^2 + bx + c$, the x-coordinate of the vertex is found using the formula: $x = -\frac{b}{2a}$

Direction: If $a > 0$, the parabola opens upward, and the vertex is the **minimum**. If $a < 0$, it opens downward, and the vertex is the **maximum**.

For any quadratic function $f(x) = ax^2 + bx + c$ whose axis of symmetry is the line $x = h$, the vertex is the point $(h, f(h))$, where $h = -\frac{b}{2a}$.

The following figure shows the vertex: Minimum or maximum.



Page 101

The axis of symmetry is an imaginary vertical line that passes directly through the vertex, dividing the parabola into two identical, mirror-image halves.

Its equation: its equation is always: $x = -\frac{b}{2a}$

Property: For every point on the parabola to the left of this line, there is a corresponding point at the same height to the right.

Example 1

1. Given that $y = x^2 + 2x - 2$ is a quadratic function.

(a) Prepare the table of values for the function $y = x^2 + 2x - 2$ for $-4 \leq x \leq 2$.

(b) Draw the graph.

(c) From the graph, identify;

(i) the value of y when $x = 1.5$

(ii) the value of x when $y = -1$,

(iii) the least value of y .

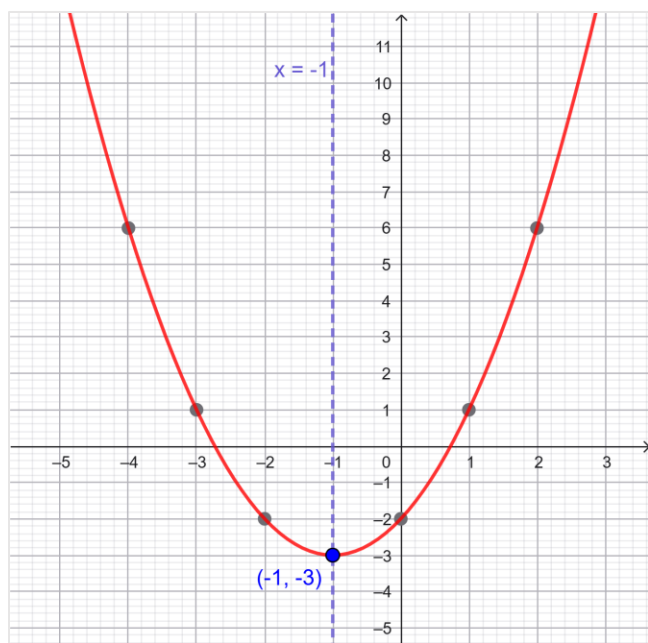
(d) Determine the line of symmetry of the function.

Solution

(a) Table below is the required table of values. Values of y are obtained by adding the values of x^2 , $2x$ and -2 . Note that breaking down the expression into such components makes the working much easier.

x	-4	-3	-2	-1	0	1	2
x^2	16	9	4	1	0	1	4
$2x$	-8	-6	-4	-2	0	2	4
-2	-2	-2	-2	-2	-2	-2	-2
y	6	1	-2	-3	-2	1	6

Table 6.12



- (c) From graph 6.16,
- (i) when $x = 1.5$, $y = 3$,
 - (ii) when $y = -1$, $x = -2.4$ or 0.4 ,
 - (iii) the least value of y is -3 .
- (d) For the graph $y = x^2 + 2x - 2$, the symmetry is line $x = -1$.

Example 2

Draw the graph of $y = 2 + 2x - x^2$ for values of x from -2 to 4 . From the graph, find:

- (a) the maximum value of $2 + 2x - x^2$
- (b) the value of x for which y is greatest
- (c) the range of values of x for which y is positive
- (d) the axis of symmetry.

Solution

The table 6.13 is the required table of values.

x	-2	-1	0	1	2	3	4
2	2	2	2	2	2	2	2
$2x$	-4	-2	0	2	4	6	8
$-x^2$	-4	-1	0	-1	-4	-9	-16
y	-6	-1	2	3	2	-1	-6

Table 6.13

Fig. 6.17 shows the graph of $y = 2 + 2x - x^2$.

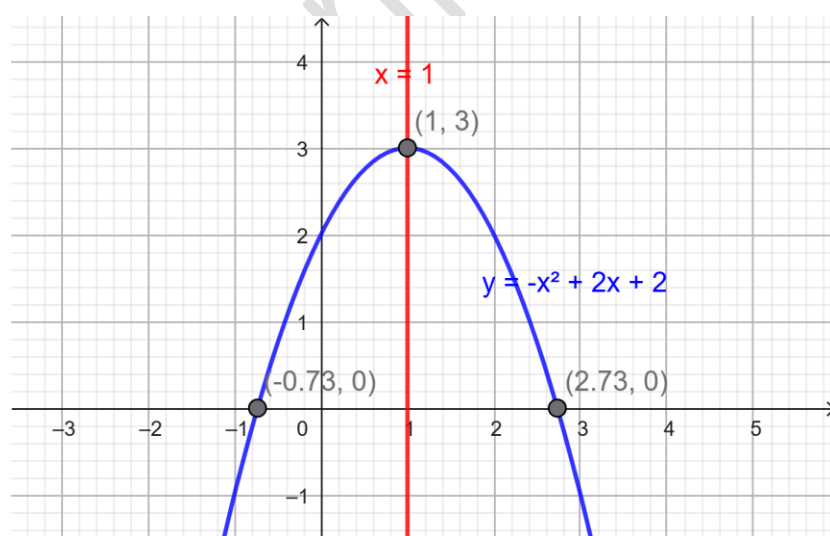
From the graph

- (a) the maximum value of $2 + 2x - x^2$ is 3,
- (b) the value of x for which y is greatest is 1,
- (c) y is positive for all parts of the curve above the x -axis, i.e. when $x > -0.8$ and $x < 2.8$. In short, y is positive over the range $-0.8 < x < 2.8$.

- (d) the axis of symmetry for the graph of $y = 2 + 2x - x^2$ is $x = 1$.

Notice that the graph 6.17 is upside down as compared to graph 6.15.

This is always the case when the coefficient of x^2 is negative.





Application activity 6.7:

1. Plot the graph of $y = x^2 - 4x + 5$ for $-1 < x < 5$. Use the graph to answer the questions below.

- Where does the graph cut x-axis?
- State the axis of symmetry of the graph.
- Find the vertex of the graph.
- Find the value of y when $x=3$.

2. Given that $y = x^2 - x - 2$ complete the following table for values of x and y .

x	x^2	$-x$	-2	y
-2				
-1				
0				
0.5				
1				
2				
3				

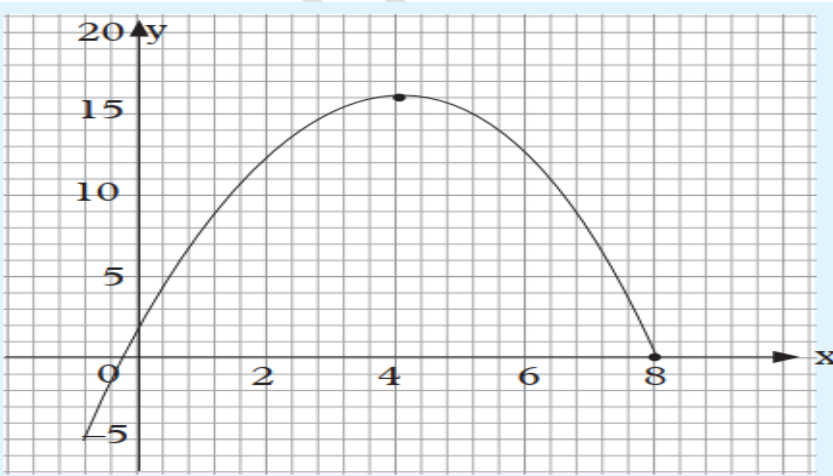
- State the co-ordinates in ordered pairs
- Plot the graph of the function
- Find the axis of symmetry from the graph plotted in (b).
- Find the vertex of the function from the graph plotted in (b).

6.8 Intercepts, vertices and sketching the graph of a quadratic function



Activity 6.8:

Consider the graph in the following graph:

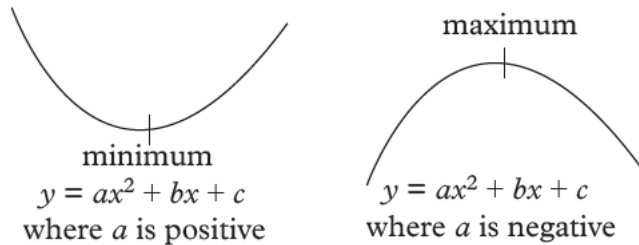


- Read and record the vertex of the graph.
- What is the axis of symmetry from the graph?
- Read and record the points where the graph cuts the axes



Content summary

The vertex of a quadratic function is the point where the function crosses its axis of symmetry. If the coefficient of the x^2 term is positive, the vertex will be the lowest point on the graph, the point at the bottom of the U-shape. If the coefficient of the term x^2 is negative, the vertex will be the highest point on the graph, the point at the top of the $\cap$ -shape. The shapes are as below.



The standard equation of a quadratic function is $y = ax^2 + bx + c$. Since the quadratic expression written as $f(x) = ax^2 + bx + c$, then we can get the y-coordinate of the vertex by substituting the x-coordinate $= -\frac{b}{2a}$.

So, the **vertex** becomes $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$

The axis of symmetry is the x-coordinate of the quadratic function.

Axis of symmetry is therefore calculated from $x = -\frac{b}{2a}$.

The intercepts with axes are the points where a quadratic function cuts the axes.

There are two intercepts i.e. x-intercept and y-intercept. x-intercept for any quadratic expression is calculated by letting $y = 0$ and y intercept is calculated by letting $x = 0$

The graph of a quadratic expression can be sketched without table of values as long as the following are known.

- (a) The vertex (seen in previous lesson)
- (b) The x-intercepts
- (c) The y-intercept.

To find **x-intercepts**, you solve the equation where $f(x) = ax^2 + bx + c$ equals zero as $ax^2 + bx + c = 0$

You can solve for x using the Quadratic Formula: $x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$

On a graph, the **x-intercepts** are written as coordinates: $(x_1, 0)$ and $(x_2, 0)$.

To find **y-intercept**, you set $x=0$ and find $y = f(x) = ax^2 + bx + c$. The required point is $(0, y_0)$ where $y_0 = c$. The y-intercept is always $(0, c)$.

Example 1

Find the vertex of $y = 3x^2 + 12x - 12$. State the axis of symmetry.

Solution

The coefficients are $a = 3, b = 12$ and $c = -12$

The x-coordinate $h = -\frac{b}{2a}, = \frac{-12}{2(3)} = -2$.

Substituting the x-coordinate to get y coordinate, we have

$$y = 3(-2)^2 + 12(-2) - 12 = -24.$$

The vertex is at $(-2, -24)$

The axis of symmetry is the line $x = -2$.

Example 2

Find the intercepts of the graph of the function $y = 2x^2 - 8x + 6$

Solution

$$\text{When } x = 0, y = 2(0)^2 - 8(0) + 6 = 6$$

The y-intercept is $(0, 6)$

$$\text{When } y = 0, 0 = 2x^2 - 8x + 6$$

We therefore solve the quadratic equation for the values of x

$$2x^2 - 8x + 6 = 0.$$

$$2x^2 - 6x - 2x + 6 = 0.$$

$$2x(x - 3) - 2(x - 3) = 0.$$

$$(2x - 2)(x - 3) = 0.$$

$$\text{Either } 2x - 2 = 0 \text{ or } x - 3 = 0.$$

$$x = 1 \text{ or } x = 3.$$

The x-intercepts are $(1, 0)$ or $(3, 0)$

The vertex

The coefficients are $a = 2$, $b = -8$ and $c = 6$

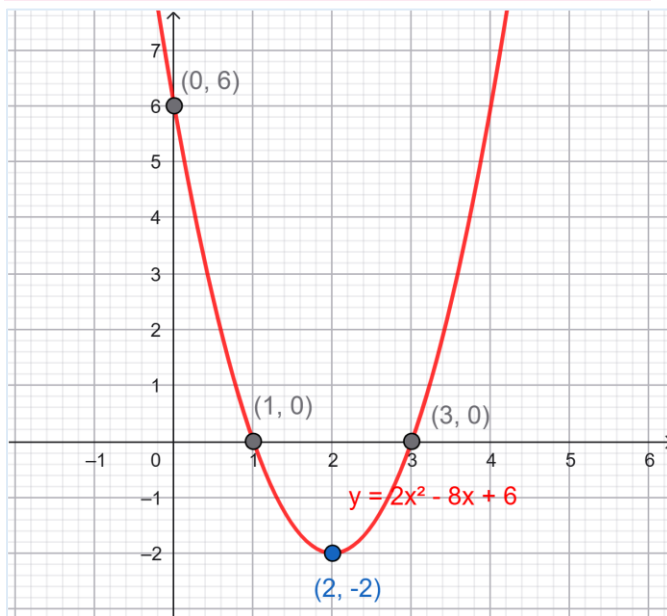
The x -coordinate of the vertex is

$$h = -\frac{b}{2a} = -\left(\frac{-8}{2(2)}\right) = \frac{8}{4} = 2$$

The y -coordinate of the vertex is obtained by substituting the x -coordinate of the vertex to the quadratic expression. We get $y = 2(2)^2 - 8(2) + 6 = -2$

The vertex is $(2, -2)$

The graph is as below.



Application activity 6.8:

1. Sketch the following graphs without tables of values

(a) $y = 2x^2 + 11x + 5$,

(b) $y = 6x^2 + 7x + 2$,

(c) $5x^2 - 7x + 2$

2. Consider the quadratic function $y = 2x^2 + 6x + 3$

- a) Find the intercepts with axes
- b) Find the vertex of the function
- c) Find the axis of symmetry.

- d) Sketch the graph of the function.
3. Without tables of values, state the vertices, intercepts with axes, axes of symmetry, and sketch the graphs.

$$(a) y = 2x^2 + 5x - 1 \quad (b) y = 3x^2 + 8x - 6$$

$$(c) y = 2x^2 - 7x - 15 \quad (d) y = 3 + 4x - 2x^2$$

6.9 Solving quadratic equations using graphical method



Activity 6.9:

1. Complete the following tables of values for;

a) Function $y = 2x^2 + 3x - 2$

X	-3	-2	-1	0	1	2
Y						

b) Function $y = x^2 + 3x + 6$

X	-3	-2	-1	0	1	2
Y						

- State the coordinates in 1 (a) and 1(b) in ordered pairs (x, y).
- Plot the graphs for 1(a) and 1(b) on different Cartesian planes.
- For each graph, read and record the x-coordinate(s) of the point(s) where the graph cuts x-axis.



Content summary

In a graph of the quadratic function $y = f(x) = ax^2 + bx + c$, the x-coordinate of the point where the graph cuts x-axis gives the solution to the quadratic equation $ax^2 + bx + c = 0$ represented by the function.

- When the graph cuts the x-axis at one point, then the equation has one repeated solution.
- When the graph cuts x-axis at two points, then the equation has two different solutions.
- When the graph does not cut x-axis at any point, then the equation has no solution in the field of real numbers.

General form of the graph of the function $y = f(x) = ax^2 + bx + c$

The coefficients a, b, and c in the quadratic function $y = f(x) = ax^2 + bx + c$ each play distinct and important roles in determining the shape, position, and key features of its graph (a parabola).

a) Role of the Leading Coefficient a

- Direction (Concavity):**

- If $a > 0$ (positive), the parabola opens **upwards** (U-shaped).
- If $a < 0$ (negative), the parabola opens **downwards** (∩-shaped).

- **"Steepness" or "Width" of the Parabola:**

- The **absolute value** $|a|$ determines how narrow or wide the parabola is.
- A larger $|a|$ (e.g., $a=4$) → **narrower** parabola (steep curves).
- A smaller $|a|$ (e.g., $a=0.25$) → **wider** parabola (shallow curves).
- If $|a|=1$, the parabola has a "standard" width.
- If $a=0$, the function is no longer quadratic (becomes linear).

b) Role of the Linear Coefficient b

The **linear coefficient b** , together with a , affects the **horizontal position** of the vertex and axis of symmetry.

- **Axis of Symmetry:**

The formula for the axis of symmetry is: $x = -b/(2a)$

So b shifts the axis left or right relative to the y-axis.

- **Interaction with a :**

- The vertex's x-coordinate is $-b/(2a)$.
 - Changing b moves the vertex along a parabolic path even if a and c are fixed.
 - The sign of b relative to a determines whether the vertex is left or right of the y-axis.

- **Effect on turning point location:**

For a given a , increasing b shifts the vertex horizontally.

c) Role of the Constant Term c

The **constant term c** gives the **y-intercept** of the graph.

- **Y-intercept:** $f(0)=c$. So the point $(0,c)$ is where the parabola crosses the y-axis.
- **Vertical Translation:**

Changing c shifts the entire parabola **vertically** without altering its shape or orientation.

d) Combined Effect: Discriminant and Roots

The **discriminant** $D = b^2 - 4ac$ (derived from all three coefficients) determines the **nature of roots (x-intercepts)**:

- If $D > 0$ the function has two distinct real x-intercepts.
- If $D = 0$, the function has one real repeated x-intercept (vertex lies on x-axis).
- If $D < 0$, the function has no real x-intercepts (parabola does not cross x-axis).

Example 1:

(a) Plot the graph of $y = x^2 + 3x - 4$ for values of x from -5 to $+2$.

(b) Use the graph to find:

(i) The value of y when $x = 1.5$

(ii) The value of x when $y = 4$

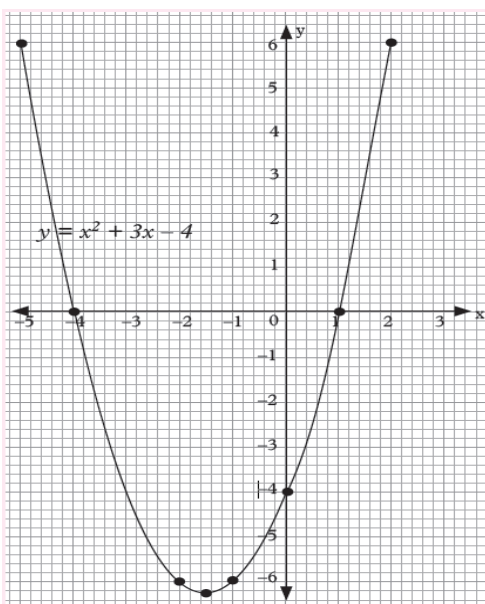
(iii) The value of y when $x = -2.5$

(c) Use the graph to solve the quadratic equation $x^2 + 3x - 4 = 0$.

Solution

(a) Table of values

x	-5	-4	-3	-2	-1	0	1	2
$y = x^2 + 3x - 4$	6	0	-4	-6	-6	-4	0	6



(a) See the graph above,

(b) (i) $y = 2.7$

(iii) $x = 1.7$ or -4.7

(iv) $y = -5.2$

(c) The values of x when $x^2 + 3x - 4 = 0$ are obtained at points where the curve meets the x -axis. i.e.

$x = -4$ and $x = 1$

As $a=1$, the parabola turns upwards.

Example 2

(a) Plot the graph of $y = x^2 - 3x + 2$, for values of x between -1 and $+4$ (include 1.5 in your table of values).

(b) Find the discriminant of $x^2 - 3x + 2$ and deduce the number of roots for $x^2 - 3x + 2 = 0$.

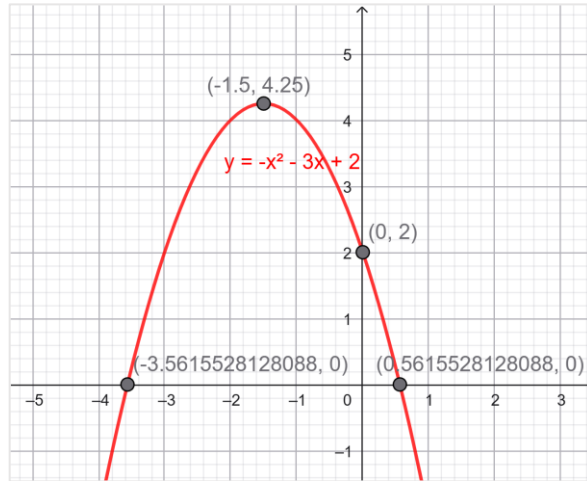
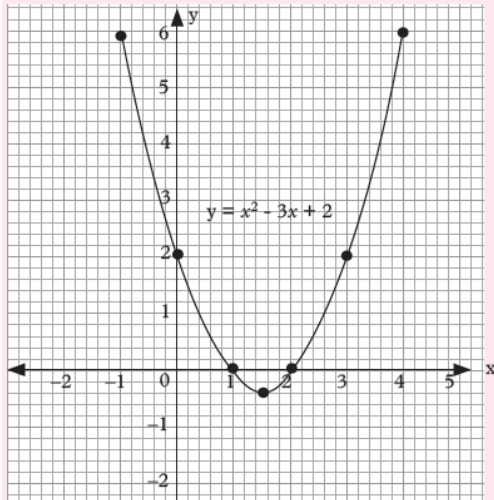
(c) Use the above graph to solve $x^2 - 3x + 2 = 0$

(d) Draw the graph of $y = -x^2 - 3x + 2$ and compare it with the previous one.

Solution

(a) Make a table of values of x and y :

x	-1.5	-1	0	1	1.5	2	3	4
$y = x^2 - 3x + 2$		6	2	0	-0.25	0	2	6
$y = -x^2 - 3x + 2$	4.25	4	2	-2	-4.25	-8	-16	-26



The first figure is the graph of function $y = x^2 - 3x + 2$. It crosses the x-axis at points where $x = 1$ and $x = 2$, i.e.

b) $f(x) = ax^2 + bx + c = x^2 - 3x + 2$, $a=1$, $b=-3$ and $c=2$. $D = b^2 - 4ac = 1$. Therefore, the equation $x^2 - 3x + 2 = 0$ has two roots.

b) Basing on the curve that intersects with x-axis on $x=1$ and $x=2$, the solution of $x^2 - 3x + 2 = 0$ are the points on the curve for which $y = 0$. Therefore, $S = \{1, 2\}$.

Therefore $x = 1$ and $x = 2$ are the solutions of $x^2 - 3x + 2 = 0$.

c) The second graph shows the graph of $y = -x^2 - 3x + 2$. The parabola opens **downwards** ($\cap$ -shaped) which is opposite of the one for $y = x^2 - 3x + 2$ where the parabola opens **upwards** ($\cup$ -shaped). The difference is that the coefficients of x^2 have different signs.



Application activity 6.9:

Given the function $y = f(x) = x^2 - 6x + 5$

- Plot the graph of the function $y = x^2 - 6x + 5$ for $-1 < x < 7$.
- Calculate the Discriminant of $y = x^2 - 6x + 5$ and say the number of roots the equation has.
- Use your graph to solve the equation $x^2 - 6x + 5 = 0$. Is the result the same as your findings in (b)?
- Complete by **True** or **False**:
 - The graph is a parabola that opens downwards
 - The graph intersects the y axis at the point $P(0, 5)$
 - The graph of $y = f(x) = x^2 - 6x + 5$ is the same as the graph of $y = g(x) = x^2 - 6x - 5$



END OF UNIT ASSESSMENT

Section A: Assessing Knowledge and Understanding

1. Which of the following is the correct form of the Cartesian equation of a straight line?

- a) $y = ax^2 + bx + c$
- b) $Ax + By + C = 0$
- c) $\frac{1}{x}$
- d) $y = mx^2 + k$

2. Choose the correct answer. To determine if two lines are parallel, we check the following:

- a) Their gradients multiply to -1 .
- b) Their gradients are equal.
- c) Their y-intercepts are equal.
- d) Their x-intercepts are equal.

3. Choose the correct answer: On the graph of the function $y = f(x) = ax^2 + bx + c$, the constant of x^2 :

- a) It shows the direction of concavity of the graph if it turns down or if it turns up.
- b) It shows the parallelism
- c) It shows the x-intercept.
- d) It shows the y-intercept.

Section B: Assessing skills

1. Find the equation of the line passing through the points $(-2, 1)$ and $((4, 7))$.

2. Given the quadratic function $y = f(x) = x^2 - 4x + 3$, find:

- a) axis of symmetry for its graph
- b) its vertex
- c) Find the discriminant D and solve $x^2 - 4x + 3 = 0$
- d) The graph of the function on $0 \leq x \leq 4$

3. Determine whether the lines $y = 3x - 2$ and $y = -\frac{1}{3}x + 4$ are parallel, perpendicular, or neither.

C: Assessing attitudes and Values

1. Why is it important to learn about linear and quadratic functions?

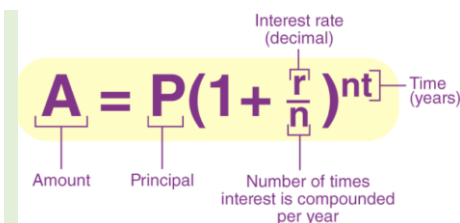
- a) They are only useful for passing exams.
- b) They help in modelling real-world situations like cost, motion, and area optimization.
- c) They are the simplest functions and have no real applications.
- d) They are only used in advanced mathematics.

2. When solving graphically problem involving simultaneous equations, what does the point of intersection represent in a real-world context?

- a) The highest point on the graph.
- b) The common solution to equations, such as break-even points in business or where cost equals revenue.
- c) The gradient of the line.
- d) The y-intercept of the function.

3. What does the vertex of a quadratic graph represent in practical situations?

- a) The starting point of the event.
- b) The maximum or minimum value, such as highest profit or lowest cost.
- c) The x-intercept of the function.
- d) The rate of change of the function.



$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Labels: A (Amount), P (Principal), r (Interest rate (decimal)), n (Number of times interest is compounded per year), nt (Time (years)).

UNIT 7: COMPOUND INTEREST, REVERSE PERCENTAGE AND COMPOUND PROPORTIONAL

Key Unit Competence:

Solve real-world problems involving compound interest, reverse percentages, and proportional change.

Learning objectives:

At the end of this unit, you should be able to:

Knowledge and understanding:

- Explain compound interest, reverse percentage, and compound proportion using multipliers.
- Explain the difference between simple and compound interest and the impact of compounding periods.
- Describe a situation where reverse percentage or compound proportion is applied.

Skills:

- Calculate the final amount, principal, interest rate, or time in compound interest problems using the formula.
- Determine the original value given a percentage increase or decrease (reverse percentage).
- Solve real-world problems involving compound proportional change.
- Compare financial options (e.g., different savings accounts and loan offers) by analysing their long-term costs and benefits.

Attitudes and values:

- Value the importance of saving and investing early due to the power of compound interest.

- Demonstrate critical thinking when evaluating financial claims and advertisements.
- Cultivate a sense of fiscal responsibility and ethical awareness in monetary transactions.

Introduction

In this unit, we will explore how money grows, or debts increase when interest is calculated not only on the initial amount, but also on accumulated interest over time, which is a powerful concept that underpins **savings, investments, and loans** in the real world. Beyond finance, these proportional change principles also apply to **inflation and depreciation of assets**, making this topic essential for informed decision-making in personal finance, business, and everyday life. By mastering percentages, multipliers, and compound formulas, you'll gain tools to compare savings accounts, evaluate loan options, understand investment growth, and analyse any situation where change happens repeatedly over time.



Introductory activity

Your grandparents offer you two choices for your birthday gift:

- **Option A:** 20,000FRW today to put in a savings account that earns 10% interest per year, paid annually.
- **Option B:** 25,000FRW, but you only receive it after three years, with no interest until then.

You choose **Option A** and plan to leave the money untouched for three years.

- 1) Which option seems better at first glance? Why?
- 2) If the bank paid *simple interest* at 10% per year,
 - a) How much interest would you earn each year?
 - b) Calculate the total amount after 3 years with simple interest.
 - c) With simple interest, which option gives more money after 3 years?
3. If interest is "paid annually,"
 - a) what happens to the interest after it is added to your account?
 - b) Year 1: Start with 20,000Frw. Calculate 10% interest. What's the total at the end of Year 1?
 - c) Year 2: Use the *new total* as the starting amount for Year 2. Calculate 10% of that. What's the total at the end of Year 2?
 - d) Year 3: Repeat the process. What's the final amount at the end of Year 3?
 - e) Compare this final amount with Option B (25,000Frw). Which option is better?
4. How much *extra* money did you earn with compound interest compared to simple interest over 3 years?
5. Can you see a quicker way to calculate this without doing three separate steps? (Hint: Think about multipliers.

7.1 Reverse percentage



Activity 7.1

Consider an iron box that costs 450 FRW after a 25% increase in its original price.

- Explain how you can determine the original price of the iron box. Then determine that price.
- Compare the current price and the original price of the iron box. Which is higher? Why?



Content summary

Before talking about reverse percentage, let us have a review on important concepts that will be used:

A. Percentages

A percentage means "per hundred".

- To find $X\%$ of a quantity use $(X/100) \times \text{Amount}$
- Percentage increase by $r\%$: $\text{New} = \text{Original} \times (1 + r/100)$
- Percentage decrease by $r\%$: $\text{New} = \text{Original} \times (1 - r/100)$.

B. Key financial terms

- Savings:** Money set aside, often in a bank, which may earn **interest** (money paid by the bank for using your money).
- Loan:** Money borrowed that must be repaid with **interest** (extra money paid to the lender).
- Principal (P):** The initial amount of money saved or borrowed.
- Interest Rate (r):** The percentage charged or paid per time period (usually per year).
- Time (t):** The duration for which money is saved or borrowed.

C. Simple Interest

- Interest calculated **only on the original principal**. Interest amount is the same every year.
- Formula: Simple Interest $(SI) = P \times r/100 \times t$
- Total Amount $(A) = \text{Principal} + \text{Simple Interest} = P + SI$

Example 1:

Maya saves 15,000Frw. Her bank pays 4% interest per year. How much interest will she earn after 1 year?

Solution:

$$\text{Interest} = 4\% \text{ of } 15,000\text{Frw} = (4/100) \times 15,000\text{Frw} = 0.04 \times 15,000\text{Frw} = 600\text{Frw}$$

$$\text{After 1 year: Total} = 15,000\text{Frw} + 600\text{Frw} = \mathbf{15,600\text{Frw}}.$$

Example 2:

David borrows 50,000Frw at 8% simple interest per year for 3 years to buy a goat.

- How much interest will he pay in total?
- What is the total repayment amount?

Solution:

$$P = 50,000, r = 8\%, t = 3 \text{ years}$$

$$\text{a) SI} = P \times r/100 \times t = 50,000 \times (8/100) \times 3 = 50,000 \times 0.08 \times 3 = 12,000\text{Frw}$$

$$\text{b) Total repayment} = \text{Principal} + \text{Interest} = 50,000 + 12,000 = 62,000\text{Frw}.$$

Reverse percentage:

Reverse percentage (or inverse percentage) is the process of finding the **original amount** when you know the **final amount** and the **percentage change** that has been applied. Essentially, you work backward from the result to find the starting value.

This method is applied when given a quantity after a percentage increase or decrease and one is required to find the original quantity.

When we say something increased by 20%, we multiply the original by **1.20** (the multiplier).

To reverse this, we **divide** the final amount by the same multiplier.

If you know: **Final amount** (after the change) and **Percentage change** (r%):

a) Case of a percentage increase:

$$\text{original} = \frac{\text{Final amount}}{1 + \frac{r}{100}}$$

$$\text{Or using multiplier } m = 1 + \frac{r}{100}, \quad \text{original} = \frac{\text{Final}}{m}$$

b) Case of a percentage decrease:

$$\text{original} = \frac{\text{Final amount}}{1 - \frac{r}{100}}$$

$$\text{Or using multiplier } m = 1 - \frac{r}{100}, \quad \text{original} = \frac{\text{Final}}{m}$$

Example 1:

You see a jacket in a store with a price tag that says 8,400Frw and a sign saying "30% OFF Original Price". You want to know what the original price was before the discount.

Solution:

Understand the change: "30% OFF" means the price was decreased by 30%.

Calculate the multiplier:

$$\text{For a 30\% decrease: multiplier } m = 1 - \frac{30}{100} = 1 - 0.30 = 0.70$$

$$\text{Apply the reverse percentage formula: Original Price} = \frac{\text{Sale price}}{m} = \frac{8400}{0.70} = 12,000\text{Frw}$$

The jacket's original price was **12,000Frw**.

Example 2:

A radio is sold at 6200 FRW after a 40% increase in the price. Find the original price.

Solution

What is required here is to reverse the process of raising the price by the given percentage.

Let the original; price be x

The new price = (100 + 40)% of x = 140% of x

140% of $x = 6200 \text{ FRW}$.

$$\frac{140}{100}x = 6200 \text{ Frw}$$

$$x = \frac{6200 \text{ Frw} \times 100}{140} = 4428.6 \text{ Frw} \approx 4429 \text{ Frw}$$

Example 3:

For purposes of sales promotion, the price of a book has been reduced by 20% to 3 600 FRW. What was the price before the reduction?

Solution

We are required to use the reverse process of decreasing the price by the given percentage.

Let the old price be y .

Then, the new price = $(100-20)\%$ of $y = 80\%$ of y .

80% of $y = 3600 \text{ frw}$

$$y = \frac{3,600 \text{ Frw} \times 100}{80} = 4,500 \text{ Frw}$$

Alternative Method

Old price = $y \text{ FRW}$ which is 100%

New price = 3 600 FRW which is 80%

$y : 3\ 600 = 100 : 80$

$$\frac{y}{3600} = \frac{100}{80} \text{ (a proportion)}$$

$$y = \frac{100}{80} \times 3600 \text{ Frw} = 4500 \text{ Frw} . \text{ The old price is } 4.500 \text{ Frw} .$$



Application activity 7.1:

- 1) A man's daily wage was increased by 25% to 500 FRW. Find how much it was before the increase.
- 2) The price of an article is decreased by 5% to 1 900 FRW. What was the price before the decrease?
- 3) A company produced 23 000 shirts in September. This was 8% less than the August production. How many shirts did the company produce in August?
- 4) A new car falls in value by 30% a year. After a year, it is worth 84 000 FRW. Find the price of the car when it was new.

7.2 Compound interest – step by step method



Activity 7.2

Find out from reference books or the internet the:

1. definition of interest and some of the area of its application.
2. ways of calculating compound interest.

3. difference between compound interest and simple interest.
- 4) Using a step-by-step method, determine the amount accumulated by a principal of 500 000 FRW invested at 8% compound interest for 4 years. What is the total interest realized?



Content summary

When money is borrowed from or deposited in a financial institution, it earns an interest at the end of each interest period as specified in the terms of investment, for example at the end of each year, half year etc. Instead of paying the interest to the owner, it is added to the principal at the end of each period i.e. compounded with the principal. The resulting amount is then taken to be the principal for the next interest period. The interest earned in this period is higher than in the previous one. The interest so earned is called compound interest.

Therefore, compound interest defines the interest calculated on the initial principal and also on the accumulated interest of previous periods of a deposit or loan.

There are two ways of calculating compound interest:

- (a) the step-by-step method
- (b) the compound interest formula.

Step by Step Method

Compound interest can be calculated step by step through compounding the interest generated with the principal. This means that the interest is calculated on the principal plus previously earned interest: After each year the interest I is added to the account P and next year's interest is calculated on the new total $P+I$ by considering that

$$I = \text{Principal} \times \frac{\text{rate}}{100} \times \text{time} = \frac{P \times r \times t}{100}$$

Example 1:

Find the total amount of money accumulated after 3 years if 10 000 FRW is invested at 10% p.a. compound interest.

Solution

Since the principal changes at the beginning of every year, we calculate the interest at the end of each year.

Interest for year 1

$$= (10,000 \times 10 \times 1) / 100 = 1,000 \text{ FRW}$$

Amount after year 1 = $P + I = 10,000 + 1,000 = 11,000 \text{ FRW}$.

Now 11,000 FRW becomes the principal for the 2nd year.

Interest for year 2 = $(11,000 \times 10 \times 1) / 100 = 1,100 \text{ FRW}$

Amount after year 2 = $P + I = 11,000 \text{ FRW} + 1,100 \text{ FRW} = 12,100 \text{ FRW}$

12,100 FRW becomes our principal for the 3rd year.

Interest for year 3 = $(12,100 \times 10 \times 1) / 100 = 1,210 \text{ FRW}$

Amount after year 3 = $P + I = 12,100 \text{ FRW} + 1,210 \text{ FRW} = 13,310 \text{ FRW}$

Example 2:

Find the compound interest earned on 90 000 FRW for 3 years at 7% p.a.

Solution:

Solution

$$\begin{aligned}\text{Interest for year 1} &= \frac{90\,000 \times 7 \times 1}{100} \text{ FRW} \\ &= 6\,300 \text{ FRW}\end{aligned}$$

$$\begin{aligned}\text{Amount after year 1} &= 90\,000 + 6\,300 \\ &= 96\,300 \text{ FRW}\end{aligned}$$

$$\begin{aligned}\text{Interest for year 2} &= \left(\frac{96\,300 \times 7 \times 1}{100} \right) \text{ FRW} \\ &= 6\,741 \text{ FRW}\end{aligned}$$

$$\begin{aligned}\text{Amount after year 2} &= 96\,300 + 6\,741 \\ &= 103\,041 \text{ FRW}\end{aligned}$$

$$\begin{aligned}\text{Interest for year 3} &= \frac{103\,041 \times 7 \times 1}{100} \\ &= 7\,212.87 \text{ FRW}\end{aligned}$$

$$\begin{aligned}\text{Amount after year 3} &= 103\,041 + 7\,212.87 \\ &= 110\,253.87 \text{ FRW}\end{aligned}$$

$$\begin{aligned}\text{Compound interest} &= (110\,253.87 - 90\,000) \text{ FRW} \\ &= 20\,253.87 \text{ FRW}\end{aligned}$$

Page

108

Example 3:

Gatesi borrows 3 800 FRW from Jane at 10% p.a. compound interest. At the end of each year, she pays back 910 FRW. How much does Gatesi owe Jane at the beginning of the third year?

Solution

$$\text{Interest for year 1} = (3,800 \times 10 \times 1) / 100 = 380 \text{ FRW}$$

$$\text{Amount after year 1} = 3,800 + 380 = 4,180 \text{ FRW}$$

$$\text{Less payment of } 4,180 \text{ FRW} - 910 \text{ FRW made after year 1} = 3,270 \text{ FRW}$$

Interest for year 2

$$= (3,270 \times 10 \times 1) / 100 = 327 \text{ FRW}$$

$$\text{Amount after year 2} = 3,270 + 327 = 3,597 \text{ FRW}$$

$$\text{Less payment of } 3,597 \text{ FRW} - 910 \text{ FRW made after year 2} = 2,687 \text{ FRW}$$

Therefore, Jane is owed **2,687 FRW** at the beginning of the third year.

Example 4:

Find the accumulated amount of money after (1.5) years, for 10 500 FRW, invested at the rate of 8% p.a. compounded semi-annually.

Solution

$$P = 10,500 \text{ FRW}, R = 8\% \text{ p.a.}, T = 1\frac{1}{2} \text{ years}$$

There are 3 half years in one and a half years.

The rate of interest for each half year is 4% (i.e., $8\% \div 2$), while the interest period is each $\frac{1}{2}$ year.

$$\text{First half-year interest} = (10,500 \times 4 \times 1) / 100 = 420 \text{ FRW}$$

$$\text{Amount after 1st half} = 10,500 + 420 = 10,920 \text{ FRW}$$

Interest after 2nd half-year = $(10,920 \times 4 \times 1) / 100 = 436.80$ FRW

Amount after 1 year = $10,920 + 436.80 = 11,356.80$ FRW

Interest after 3rd half-year = $(11,356.80 \times 4 \times 1) / 100 = 454.27$ FRW

Amount after $1\frac{1}{2}$ years = $11,356.80 + 454.27 = 11,811.07$ FRW



Application activity 7.2

1) Find the amount and the compound interest for each of the following correct to the nearest FRW:

8 000 FRW invested for 3 years at 4% p.a. compound interest.

2) Determine the difference between the simple interest and compound interest earned on 15 000 FRW for 2 years at 3% p.a.

3) Kampire invested P FRW, which amounted to 8 420 FRW in 3 years at a rate of 4.5% p.a. compound interest. Find the value of P.

4) Determine the compound interest earned on 45 000 FRW after 3 years at the rate of 6% p.a.

7.3 The compound interest formula



Activity 7.3

Consider the case in which 10 000 FRW is invested in a bank for 3 years at the rate of 5% p.a. compound interest.

Using the multiplier as a factor, calculate the amount of money accumulated after 3 years.



Content summary

Assuming 5 000 FRW is the principal amount compounded at 6% p.a. The amount of money accumulated after every year is calculated as follows:

Amount after 1st year.

$$= 5\,000 \text{ FRW} + \frac{5\,000 \times 6 \times 1}{100}$$

$$= 5\,000 \text{ FRW} + 5\,000 \times \frac{6}{100}$$

$$= 5\,000 \text{ FRW} \left(1 + \frac{6}{100}\right)$$

Amount after 2nd year

$$= 5\,000 \left(1 + \frac{6}{100}\right) + 5\,000 \left(1 + \frac{6}{100}\right) \times \frac{6}{100} \times 1$$

$$= 5\,000 \left(1 + \frac{6}{100}\right)^2$$

Amount after 3rd year

$$= 5\,000\left(1 + \frac{6}{100}\right)^2 + 5\,000\left(1 + \frac{6}{100}\right) \times \frac{6}{100} \times 1$$

$$= 5\,000 \left(1 + \frac{6}{100}\right)^3$$

Amount after 4th year

$$= 5\,000\left(1 + \frac{6}{100}\right)^3 + 5\,000\left(1 + \frac{6}{100}\right) \times \frac{6}{100}$$

$$= 5\,000 \left(1 + \frac{6}{100}\right)^4$$

Amount after n years

$$= 5\,000 \left(1 + \frac{6}{100}\right)^n \text{ FRW}$$

Considering the case in which P is invested in a bank for t -interest periods at the rate of $r\%$ p.a. The accumulated amount (A) after the given time is given by:

$$A = P \left(1 + \frac{R}{100}\right)^t$$

Where

- A = Amount after t years
- P = Principal amount
- R = Annual interest rate (in percentage)
- t = Time in years

This is called the compound interest formula. It is conveniently used in solving problems of compound interest especially those involving long periods of investments or payment.

In this method, the accrued compound interest is obtained by subtracting the original principal from the final amount.

Thus, **Compound interest** = **Accumulated amount(A)** – **Principal(P)**

Note that the principal and the interest earned increased after each interest period. We can also deduce that;

$$\text{Compound interest} = \text{Accumulated amount} - \text{Principal amount} \quad I = A - P$$

$$\text{Accumulated amount} = \text{Principal amount} + \text{Compound interest: } A = P + I$$

Importance of Studying Compound Interest in Real Life:

- **Empowers informed financial decisions:** Understanding compound interest helps individuals evaluate loans, savings, investments, and credit options more critically, preventing debt traps and promoting wise borrowing.
- **Encourages long-term saving and investing:** Knowing that money grows very quickly over time motivates early and consistent saving, essential for goals like retirement, education, or buying a home.

- **Highlights the cost of debt:** Demonstrates how quickly borrowed money can accumulate, encouraging responsible credit use and awareness of total repayment amounts beyond monthly payments.
- **Improves financial planning skills:** Enables individuals to calculate future values of savings and compare investment returns, supporting better budgeting and goal-setting.
- **Promotes financial independence and security:** Builds a foundation for wealth accumulation over time, helping individuals build assets and reduce reliance on high-interest loans.
- **Teaches patience and delayed gratification:** Illustrates the power of time in growing money, reinforcing the value of long-term planning over short-term spending.
- **Supports ethical financial awareness:** Helps consumers recognize misleading offers (e.g., “low payments” with high compound rates), promoting fairness and transparency in financial dealings.

Example 1:

A trader deposited 63 000 FRW in a fixed deposit account with a local bank which attracted an interest of 8% p.a. compound interest. Find:

- (a) the total amount after 4 years.
(b) compound interest.

Solution:

$$P = 63\,000 \text{ FRW } n = 4, r = 8\% \text{ p.a. and } A = P \left(1 + \frac{r}{100} \right)^n$$

$$(a) \quad A = 63\,000 \left(1 + \frac{8}{100} \right)^4 = 63\,000 (1.08)^4 = 63\,000 \times 1.36048896 = 85\,710.80 \text{ FRW}$$

$$(b) \quad I = (85\,710.80 - 63\,000) \text{ FRW} = 22\,710.80 \text{ FRW}$$

The compound interest formula when interest is compounded n times per year:

$$A = P \left(1 + \frac{R}{n \times 100} \right)^{n \times t}$$

Where:

- A = Total amount after t years
- P = Principal amount
- R = Annual interest rate (%)
- n = Number of compounding periods per year
- t = Time in years

The **compound interest** is then: $CI = A - P$.

Important notes:

1. **Annually (n=1):** $A = P \left(1 + \frac{R}{100} \right)^t$

2. **Semi-annually (n=2):** $A = P \left(1 + \frac{R}{2 \times 100} \right)^{2t}$

3. **Quarterly (n=4):** $A = P \left(1 + \frac{R}{4 \times 100} \right)^{4t}$

4. **Monthly (n=12):** $A = P \left(1 + \frac{R}{12 \times 100} \right)^{12t}$

5. **Daily (n=365):** $A = P \left(1 + \frac{R}{365 \times 100} \right)^{365t}$

Example 2:

If $P = 5000\text{Frw}$, $R = 6\%$ and $t = 2$ years compounded quarterly,

$$A = P \left(1 + \frac{R}{4 \times 100} \right)^{4t} = 5000 \left(1 + \frac{6}{4 \times 100} \right)^{4 \times 2} \approx 5632.45\text{Frw}$$

Example 3:

Find the accumulated amount that 30 000 FRW will yield if deposited for 2 years at 6% p.a. compounded semi-annually.

Solution

When compounded **semi-annually**, you can use this formula: $A = P \left(1 + \frac{R}{2 \times 100} \right)^{2t}$

Where $P = 30,000\text{Frw}$, $R = 6$, $t = 2$.

$$A = P \left(1 + \frac{R}{2 \times 100} \right)^{2t} = 30000 \left(1 + \frac{6}{2 \times 100} \right)^{2 \times 2} = 33,765\text{Frw}$$

Example 4

Find the compound interest earned on 15 000Frw invested for 3 years, at 20% p.a. compounded quarterly.

Solution: Compounded quarterly means: $A = P \left(1 + \frac{R}{4 \times 100} \right)^{4t}$

Where $P = 15000\text{Frw}$, $R = 20$, $t = 3$.

$$A = P \left(1 + \frac{R}{4 \times 100} \right)^{4t} = 26\,937.84\text{FRW}$$

$$I = 26\,937.84\text{FRW} - 15\,000\text{FRW} = 11\,937.84\text{FRW} = 11\,938\text{FRW}$$



Application activity 7.3

- 1) Determine the accumulated amount for each of the following:
 - (a) 19 000 FRW invested for 2 years at 3% p.a. compound interest.
 - (b) 8 300 FRW invested for 3 years at 4% p.a. compound interest.
- 2) A businesswoman invested 4 500 FRW for 2 years in a savings account. She was paid 4% per annual compound interest. How much did she have in her savings account after 2 years?
- 3) Adams invests 4 500 FRW at a compound interest rate of 5% per annum. At the end of t years, the investment has grown to 5 469.78 FRW. Find the value of t .

7.4 The compound proportional change



Activity 7.4

Consider that 3 people working at the same rate can plough 2 acres of land in 3 days. What do you think will happen if the working days are increased to five working at the same rate? Discuss.



Content summary

Compound proportional change refers to a situation where a quantity changes by a certain percentage or proportion repeatedly over multiple periods. The change in each period is applied to the new amount (not the original), leading to **compounding effects**.

Common Applications:

- Compound interest (growth)
- Annual depreciation (decrease)
- Population growth/decline
- Inflation adjustments
- Investment returns

If a quantity changes by a proportion r (can be positive for increase, negative for decrease) per period, after n periods:

$$A = P \times (1 + r)^n$$

Where:

- A = Final amount
- P = Initial amount
- r = Proportional change per period (as a decimal)
- n = Number of periods

Percentage Increase (Growth)

If increase is $R\%$ per period: $A = P \left(1 + \frac{R}{100} \right)^n$

Percentage Decrease (Depreciation/Decay)

If decrease is $R\%$ per period: $A = P\left(1 - \frac{R}{100}\right)^n$

Example 1: Appreciation

Initial value $P=1000$, annual increase 5% , $n=3$ years:

$$A = 1000 \times (1 + 0.05)^3 = 1000 \times 1.157625 = 1157.63$$

Example 2: Depreciation

Car value $P=20000$, annual depreciation 10% , $n=4$ years:

$$A = 20000 \times (1 - 0.10)^4 = 20000 \times 0.6561 = 13122.$$

Sometimes, a quantity may be proportional to two or more other quantities. In such a case, the quantities are said to be in **proportion**.

Example 3:

Eighteen labourers dig a ditch 80 metres long in 5 days. How long will it take 24 labourers to dig 64 metres long? What assumptions have you made?

Solution

Notice that the number of days depends on the number of labourers as well as on the length of the ditch. Thus, this is a problem in compound proportion.

1. Ratio method

Length of ditch decrease from 80 m to 64 m, i.e. in the ratio 64 : 80.

A shorter ditch takes a shorter time.

$\therefore$ multiply days by $\frac{64}{80}$.

Number of labourers increases from 18 to 24, i.e. in the ratio 24 : 18.

More labourers take a shorter time.

$\therefore$ multiply days by $\frac{18}{24}$.

80 metres of ditch are dug by 18 labourers in 5 days.

64 metres of ditch are dug by 24 labourers in

$$5 \times \frac{64}{80} \times \frac{18}{24} \text{ days} = 3 \text{ days.}$$

2. Unitary method

18 labourers dig 80 metres of ditch in 5 days.

$\therefore$ 1 labourer digs 80 metres of ditch in (5×18) days = 90 days.

$\therefore$ 1 labourer digs 1 metre of ditch in $(90 \div 80)$ days = $1\frac{1}{8}$ days.

$\therefore$ 1 labourer digs 64 metres of ditch in $(1\frac{1}{8} \times 64)$ days = 72 days.

$\therefore$ 24 labourers dig 64 metres of ditch in $(72 \div 24)$ days = 3 days.

The assumptions made in this calculation are:

1. The width and depth of the ditch is uniform.
2. The hardness of the ground is the same throughout.
3. The labourers work at the same rate.

Page 113



Application activity 7.4

- 1) 16 men dig a trench 92 m long in 9 days. What length of trench can 12 men dig in 15 days?
- 2) To unclog a silted drain 85 m long, 15 workers take 10 days. Find how many workers are required to unclog a similarly silted drain 51 m long drain in 5 days.
- 3 Six people pay 40 320 FRW for a 7 day stay at a hotel. How much would eight people pay for a 3 day stay?

4. A car hire company with 24 cars uses 2 940 litres of petrol in 5 days. How long would 4 116 litres of petrol last if the company had 28 cars and the consumption rate does not change?

Consider that 3 people working at the same rate can plough 2 acres of land in 3 days.

7.5 Continued proportions using multipliers



Activity 7.5

Imagine this:

You see a pair of shoes originally priced at **80,000Frw**.

First, there's a **20% discount** for a sale.

Then, the store adds a **10% service charge** on the discounted price.

Finally, at the checkout, you get an **extra 5% off** for using your membership card.

How much will you actually pay?

Is the final discount more or less than just adding/subtracting percentages?



Content summary

What Are Continued Proportions?

When multiple percentage changes happen **one after another**, each acting on the new amount, it's called **continued proportions**.

Why Use Multipliers?

Instead of adding/subtracting percentages (which can be misleading), we use **multipliers** to calculate each change accurately.

Multiplier for a Percentage Change

For a change of $R\%$:

$m = 1 + \frac{R}{100}$. When we have increase, R is positive. When it is decrease, R is negative (or use

$$m = 1 - \frac{|R|}{100}$$

General Formula for Continued Proportions

If initial value = P , and multipliers = $m_1, m_2, \dots, m_n$:

Final Value = $P \times m_1 \times m_2 \times \dots \times m_n$

Finding Overall Percentage Change

Overall % Change = $(m_{\text{total}} - 1) \times 100\%$

where $m_{\text{total}} = m_1 \times m_2 \times \dots \times m_n$

Example 1:

Example 1: Price Changes

A book costs **50**. Price increases by **10%**, then decreases by **15%**, then increases by **5%**. Find final price.

Solution:

Multipliers:

$$m_1=1.10, m_2=0.85, m_3=1.05$$

$$m_{\text{total}}=1.10 \times 0.85 \times 1.05=0.98175$$

$$\text{Final price}=50 \times 0.98175=49.09$$

Overall change: $(0.98175-1) \times 100\% = -1.825\%$ (small decrease)

Example 2: Population Growth

Town population: **10,000**.

Year 1: **+4%**, Year 2: **+3%**, Year 3: **-1%**. Find population after 3 years.

Solution:

Multipliers: $m_1=1.04, m_2=1.03, m_3=0.99$

$$m_{\text{total}}=1.04 \times 1.03 \times 0.99 \approx 1.060788$$

$$\text{Final population} = 10000 \times 1.060788 \approx 10,608.$$

Example 3: Investment Return

Investment: **2000**.

Return: Year 1: **+8%**, Year 2: **-5%**, Year 3: **+12%**. Find final value.

Solution:

Multipliers: $m_1=1.08, m_2=0.95, m_3=1.12$

$$m_{\text{total}}=1.08 \times 0.95 \times 1.12=1.14912$$

$$\text{Final value}=2000 \times 1.14912=\$2298.24.$$



Application activity 7.5:

Q1: A smartphone costs **600**. First discount: **15% off**, Then tax: **8% added**, Then loyalty discount: **5% off**. Find final price in the following way:

- List multipliers: $m_1=?$, $m_2=?$, $m_3=?$
- Multiply them.
- Multiply by initial price.

Q2: Bacteria culture: **5000 cells**.

Hour 1: **+20%**, Hour 2: **+15%**, Hour 3: **-10%**. Find final count in the following way:

- Write multipliers for each hour.
- Find total multiplier.
- Apply to initial count.

Q3. Car value: **\$25,000**.

Depreciates: Year 1: **-12%**, Year 2: **-10%**, Year 3: **-8%**.

Find value after 3 years.



END OF UNIT ASSESSMENT

Section A: Knowledge and understanding

1. Which statement best describes **compound interest**?

- A) Interest calculated only on the original principal.
- B) Interest calculated on the principal plus previously earned interest.
- C) Interest that decreases over time.
- D) Interest paid only at the end of the loan period.

2. Explain in your own words the difference between **simple interest** and **compound interest**. Give one real-life situation where each is used.

3. Define **reverse percentage** and provide an example of a real-life situation where you would need to use it.

Section B: Skills *(Show all your work)*

4. Mutesi invests **50,000 FRW** at an annual compound interest rate of **6%** for **4 years**.

- Calculate the final amount using the compound interest formula.
- Find the total compound interest earned.

5. After a **15% increase**, a teacher's salary is **115,000 FRW** per month. What was her salary before the increase?

6. A car valued at **3,000,000 FRW** depreciates by **12%** in the first year, **10%** in the second year, and **8%** in the third year.

- Find its value at the end of 3 years using multipliers.

Section C: Attitudes and values

7. A friend tells you about a loan offer with “very low monthly payments” but doesn't mention the interest rate or loan term. What is the MOST important question to ask before deciding?

- a) How long do I have to repay the loan?
- b) What is the annual compound interest rate?
- c) Can I pay late without a penalty?
- d) Is the lender trustworthy?

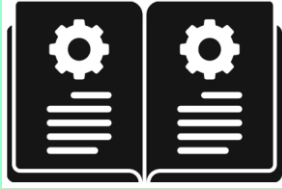
8. Choose the correct answer. Why is it important to read the fine print (or contract) in a loan or savings agreement?

- a) It helps you avoid hidden fees, unfair terms, and long-term debt traps.
- b) It makes the document look more official.
- c) It is only necessary if you don't trust the lender.
- d) It is a legal requirement but doesn't affect your decision.

9. How can understanding compound interest promote financial responsibility? Choose the correct answer.

- a) It shows that borrowing money is always risky.
- b) It encourages early saving and long-term planning because money grows faster over time.

- c) It helps you calculate discounts while shopping.
- d) It explains why loans should be avoided completely.



PROJECT: The loan Challenge - *A Financial Mathematics project for analyzing loan costs and making smart decisions*

1. Project Overview

Big Idea: Students become critical financial analysts investigating how loan duration affects total interest paid. They'll discover that "affordable monthly payments" can come with a hidden massive cost in interest, and learn to make mathematically-informed borrowing decisions.

Learning objective: To calculate monthly installments, total repayment, and total interest for different loan terms using compound interest formulas. To critically compare results and determine optimal borrowing strategies to minimize bank payments.

Critical question: "How can I borrow money without giving too much of my hard-earned cash to the bank as interest?"

Final product: A strategic financial analysis report with clear recommendations supported by mathematical evidence.

2. The investigation scenario

Client's Need: Borrow 10,000,000 FRW for business expansion

Bank offer: 18% annual compound interest,

The client has two options:

- **Option A:** 5-year repayment (60 monthly installments).
- **Option B:** 10-year repayment (120 monthly installments).

Core investigation questions:

1. What will the monthly payment be for each option?
2. How much total interest goes to the bank in each case?
3. Which option gives the bank less money?
4. When might someone choose the more expensive option anyway?
5. What strategies can minimize interest payments?

3. Mathematical Toolkit

With the compound interest, the total amount A after t years, when interest is compounded n times per year is:

$$A = P \left(1 + \frac{R}{n \times 100} \right)^{n \times t}$$

$P = 10,000,000$ FRW, number of $n = 60$ or 120 formula for P .

The **compound interest** is then: $CI=A-P$.

4. Step-by-step investigation guide

Phase 1: Calculate both options

Students work in pairs to ensure accuracy

Option A (5 Years):

Option B (10 Years):

Expected Results:

- Option A: Monthly $\approx$? Total Interest $\approx$?
- Option B: Monthly $\approx$? Total Interest $\approx$?

Phase 2: The Shocking comparison: Create an "Interest exposure analysis table":

Analysis Point	Option A (5 years)	Option B (10 years)	What this Means
Monthly Payment	?	?	Option B is easier monthly
Total to Bank	?	?	
Bank's Interest	?	?	Option B gives ____FRW MORE to bank
Interest as % of Loan	?	?	With Option B, you pay more in interest than you borrowed!
Daily Interest Cost	?	?	Option B costs more per day
What You Could Buy With Interest	A good car	A small house	The interest difference could buy a motorcycle

Phase 3: Deep Analysis - "Who wins?"

Critical Thinking Exercise:

a) Bank's Perspective: Which option does the bank prefer? Why? Calculate: Bank's extra profit

b) Client's Dilemma: Scenario: Client earns 500,000 FRW/month

- With Option A: ?% of income
- With Option B: ?% of income

c) The Breakeven Analysis:

- Extra monthly cash in Option B = ?
- Could this be invested to earn more than the extra interest?
- Investment needed: To make it in 10 years, you'd need about ?% return annually (same as loan rate!)

Phase 4: Smart borrower strategies

Investigate alternative approaches:

1. **The "Best of Both" Strategy:**

- Take Option B (lower payments)
- BUT pay Option A amount (253,900 FRW) when possible
- Result: Pay off in ~5 years but have flexibility in tough months

2. **The "Split Loan" Analysis:**

- What if you borrowed 5M at 5 years and 5M at 10 years?

Phase 5: Final recommendations and presentation

Create a "Smart Borrower's Guide" Poster:

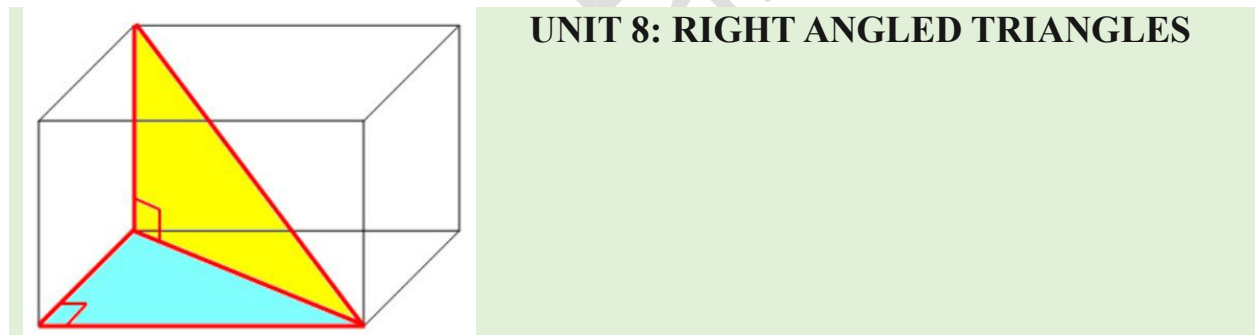
Section 1: The numbers Don't Lie

- Clear comparison charts showing interest disparity
- Visual: "The difference" bar graph

Section 2: Rules for borrowing:

1. Shortest Term Possible: Always choose the shortest term you can afford
2. Interest Awareness: You're not just borrowing X, you're paying back $X + ? \%$
3. The 30% Rule: Never let loan payments exceed 30% of your income

Section 3: Recommendation: Choose one option and say why.



Key unit competence: Apply the Pythagoras' Theorem and trigonometric ratios in finding unknown sides and angles in right-angled triangles and solve real-world problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and Understanding

- Define and relate the sides of a right triangle (hypotenuse, opposite, adjacent) to a given acute angle.
- Express the trigonometric ratios (sine, cosine, tangent) using triangle sides and explain Pythagoras' Theorem.

Skills

- Use Pythagoras' Theorem and trigonometric ratios (SOH-CAH-TOA) to find unknown sides and angles.
- Model real-world problems such as measuring heights or distances with right triangles.

Attitudes and Values

- Value trigonometry as a practical tool for measurement across different fields.
- Develop precision in calculation and collaboration when solving applied challenges.

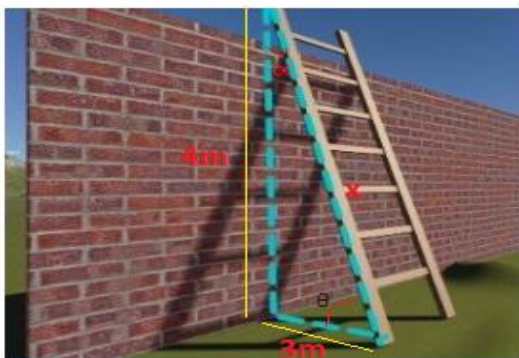
Introduction:

A right-angle triangle is a three-sided shape with one perfect corner, like the corner of a book or a piece of paper. This special 90-degree angle creates simple rules for its sides. We see and use them everywhere in real life! For example, builders use them to make sure walls are perfectly straight and corners are square. When you climb a ladder, the triangle formed by the wall, the ground, and the ladder helps us know if it's safe. So, from building a treehouse to designing a football field, this simple triangle is a hidden but very useful tool.



Introductory activity

Look at the image.



- What shape does the ladder form on the wall?
- What measurements are given?
- What is missing?
- Why might the angle of the ladder matter in real life?
- Through research, give two ways in which x can be calculated.
- Is it possible to find the missing angles in the figure? Explain.

8.1 Median theorem of a right-angled triangle



Activity 8.1

Refer to the figure below (**a triangular face of a pyramid**) and answer to the following questions:

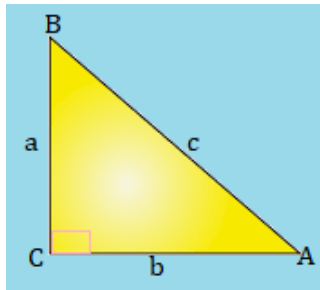


- The **right angle** is marked on the diagram, identify the **hypotenuse** of the triangular face.
- Let the right-angled triangle be $\triangle ABC$, right-angled at A. If BC is the hypotenuse and M is the midpoint of BC. What is the name given to the line segment AM?
- What is the relationship between the line AM and BC in the triangle ABC?
- If the length of the hypotenuse of the triangular face is **20 m**, discuss how to find the length of the AM.



Content summary

Let us start by recalling the **Pythagoras theorem**: it states that the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides.

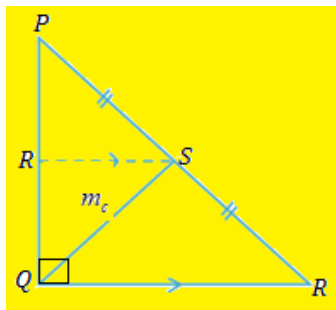


Pythagoras theorem: $c^2 = a^2 + b^2$

The median theorem of a right-angled triangle states that: the median from the right-angled vertex to the hypotenuse is equal to half the length of the hypotenuse.

Median = Half the Hypotenuse

As such, that median subdivides the right-angled triangle into two similar isosceles triangles

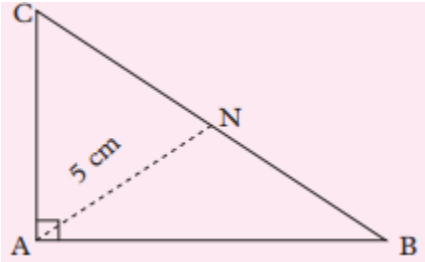


$m_c = \frac{c}{2}$, where m_c median to hypotenuse, c = hypotenuse length.

This means: $QS = \frac{PR}{2}$. Where PR-hypotenuse and QS-Median to hypotenuse.

Example 1

In a right-angled triangle ABC, line BC is the hypotenuse and AN is 5 cm long. What is the length of the hypotenuse?



Solution

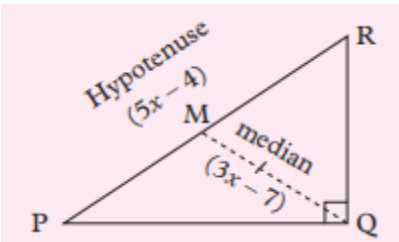
$$\text{Median } AN = \frac{1}{2}BC \text{ (hypotenuse)}$$

$$5 \text{ cm} = \frac{1}{2}BC$$

$$BC = 5 \text{ cm} \times 2 = 10 \text{ cm}$$

Example 2

In a right-angled triangle, the median to the hypotenuse has a length of $(3x - 7)$ cm. the hypotenuse is $(5x - 4)$ long. Find the value x , hence find the length of the hypotenuse.



Solution



$$QM = \frac{1}{2}PR$$

$$3x - 7 = \frac{5x - 4}{2}$$

By cross multiplication

$$2(3x - 7) = 5x - 4$$

$$6x - 14 = 5x - 4$$

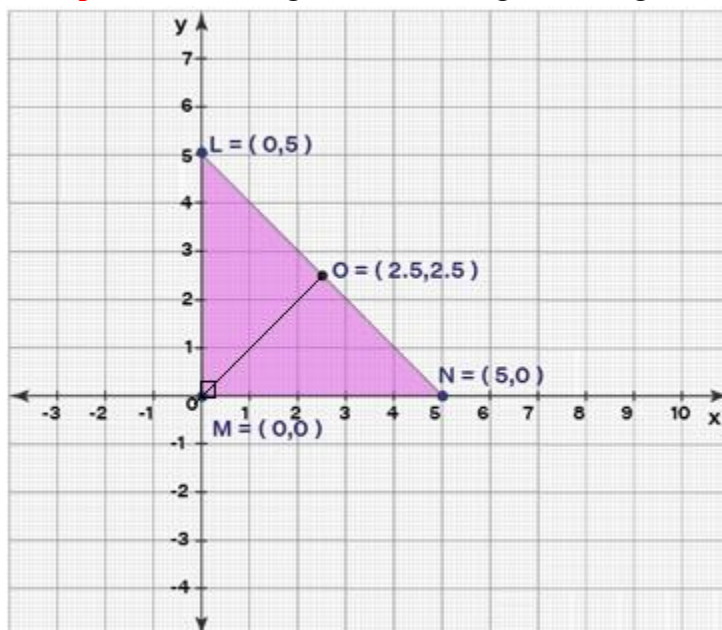
$$x = 10 \text{ cm}$$

Hence, the length of the hypotenuse is

$$\begin{aligned}(5x - 4) \text{ cm} &= (5 \times 10 - 4) \text{ cm} \\ &= 46 \text{ cm}\end{aligned}$$

Median creates two isosceles triangles, useful in coordinate geometry and proof problems.

Example 3: Look at figure. Relate the given triangle to the **Median–Hypotenuse Theorem**



Solution

Step 1. Identify the triangle

From the diagram, the vertices are M(0,0), L(0,5), N(5,0)

Step 2: Check the sides

We have already shown that: ML is vertical with length 5. MN is horizontal with length 5.

So, $ML = MN = 5$

Step 3: Check the angle at M

ML lies on the **y-axis**

MN lies on the **x-axis**

The x-axis is perpendicular to the y-axis, so: $\angle LMN = 90^\circ$

Conclusion

The triangle $\triangle LMN$ is a **right-angled isosceles triangle**, right-angled at M

Right-angled at M(0,0)

Isosceles because two sides are equal (ML=MN)

Step 4: Identify the hypotenuse

Since the right angle is at M, the side opposite it is the **hypotenuse**:

Hypotenuse=LN

$$\text{Length of LN: } LN = \sqrt{(5-0)^2 + (0-5)^2} = \sqrt{25+25} = 5\sqrt{2}$$

Step 5: Find the midpoint of the hypotenuse

Let O be the midpoint of LN.

$$\text{Using the midpoint formula: } O\left(\frac{0+5}{2}, \frac{5+0}{2}\right) = (2.5, 2.5)$$

Step 6: Length of the median to the hypotenuse

The median to the hypotenuse is MO.

$$\text{Using the distance formula: } MO = \sqrt{(2.5-0)^2 + (2.5-0)^2} = \sqrt{6.25+6.25} = \sqrt{12.5} = \frac{5\sqrt{2}}{2}$$

Step 7: Verify the Median–Hypotenuse Theorem

The theorem states: **In a right-angled triangle, the median drawn from the right angle to the hypotenuse is equal to half the hypotenuse.**

Half of the hypotenuse:

$$\frac{1}{2}LN = \frac{1}{2}(5\sqrt{2}) = \frac{5\sqrt{2}}{2}$$

$$\therefore MO = \frac{1}{2}LN$$

Step 8: Conclusion: MO is the **median to the hypotenuse**, $MO = \frac{1}{2}LN$ and Point O is **equidistant from L, M and N**.

Example 4: Consider triangle $\triangle ABC$ with vertices A(0,0), B(6,0), and C(0,8).

- Show that $\triangle ABC$ is right-angled at A.
- Find the coordinates of the midpoint M of the hypotenuse BC.
- Calculate the lengths AM, BM, and CM.
- What do you observe about these three lengths?

Solution

Given: A(0,0), B(6,0), C(0,8)

(a) Show that $\triangle ABC$ is right-angled at A

- Line AB lies on the x-axis.
- Line AC lies on the y-axis.
- The x-axis is perpendicular to the y-axis.

$$\therefore \angle A = 90^\circ$$

(b) Find the midpoint M of the hypotenuse BC

Using the midpoint formula: $M\left(\frac{6+0}{2}, \frac{0+8}{2}\right) = (3, 4)$

(c) Find AM, BM, CM

Using the distance formula:

$$AM = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{9+16} = 5$$

$$BM = \sqrt{(3-6)^2 + (4-0)^2} = \sqrt{9+16} = 5$$

$$CM = \sqrt{(3-0)^2 + (4-8)^2} = \sqrt{9+16} = 5$$

(d) Observation: $AM = BM = CM$. So, the midpoint of the hypotenuse is equidistant from all three vertices.



Application activity 8.1

1) In a right-angled triangle, the median drawn to the hypotenuse is 4.5 cm. The length of the hypotenuse is:

- (A) 9 cm (B) 4.5 cm (C) 13.5 cm (D) 6 cm

2) One leg of a right triangle is 12 cm. The median to the hypotenuse is 7.5 cm. The length of the hypotenuse is:

- (A) 15 cm (B) 9 cm (C) 10.5 cm (D) 12 cm

3) The legs of a right triangle are 4.5 cm and 6 cm. The length of the median to the hypotenuse is:

- (A) 5.25 cm (B) 3.75 cm (C) 7.5 cm (D) 4.5 cm

4) A right-angled triangle has vertices A(-2,0), B(4,0), and C(-2,6).

a) Identify the hypotenuse.

b) Show that the length of this median is half the length of the hypotenuse

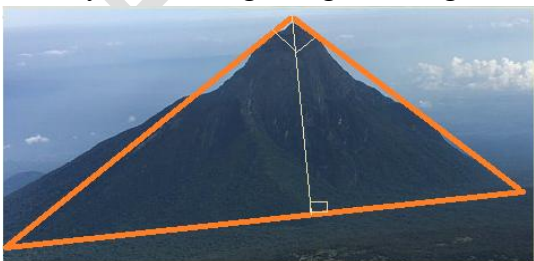
In a right-angled triangle EFG, the hypotenuse is $(3x + 8)$ cm long. The median to the hypotenuse is $(5x - 10)$ long. Find the value of x hence find the length of the median and hypotenuse.

8.2 Altitude (Height) theorem of a right-angled triangle



Activity 8.2

A surveyor uses a right angled triangle to know a Mountain dimensions.



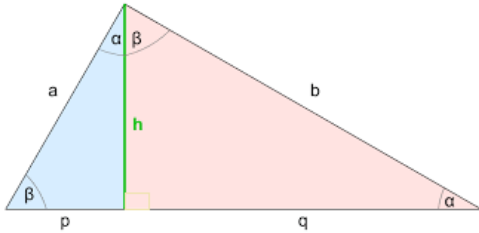
- If the mountain height is 60 m and the hypotenuse is divided in the ratio 1:4, find the horizontal distances from the foot of the mountain to each end.
- Explain how surveyors could use this triangle model to calculate the height of the mountain without climbing it.



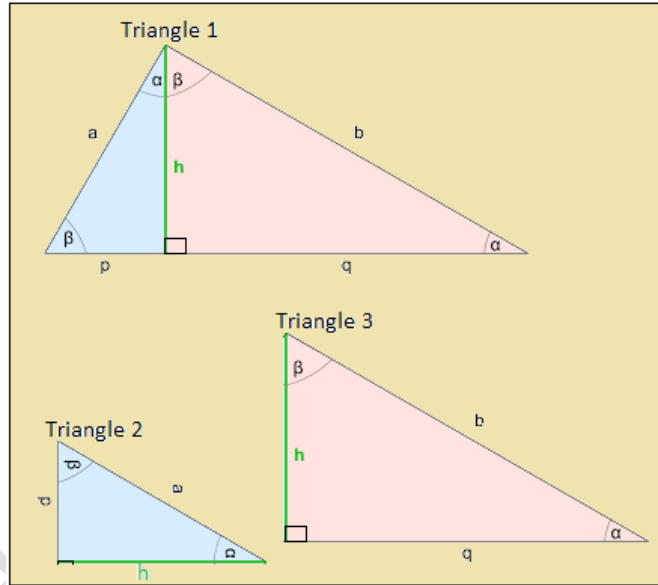
Content summary

The height/altitude to the hypotenuse of a right-angled triangle is the mean proportional between the segments into which it divides the hypotenuse.

As you can see in the picture:



$$h^2 = p \times q$$

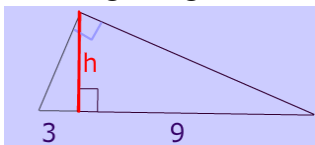


As you can see in the picture, triangle 1 involves other two inner similar triangles which have proportional sides and the altitude /height is the long leg of triangle 2 and short leg of triangle 3. So, from these similar triangles (Triangle 1 and Triangle 2).

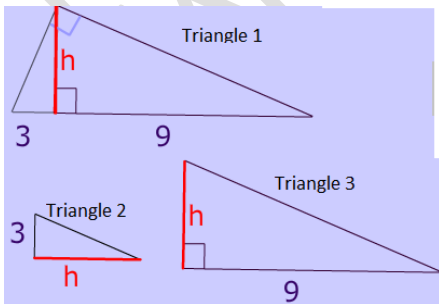
$$\frac{p}{h} = \frac{h}{q} \Rightarrow h^2 = p \times q$$

Example 1

In the right angled below h is the altitude. Find h.



Solution

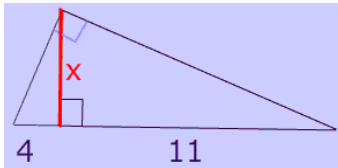


$$\frac{3}{h} = \frac{h}{9} \Rightarrow h^2 = 27$$

$$\Rightarrow h = \sqrt{27} = 3\sqrt{3}$$

Example 2

What is the length of the altitude marked x



Solution

$$x^2 = 4 \times 11$$

$$\Rightarrow x = \sqrt{44} = 2\sqrt{11}$$

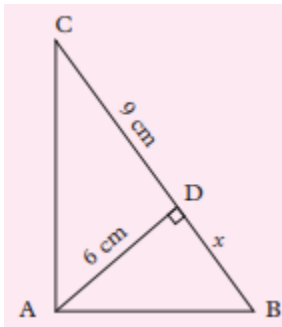
Example 3

In a right-angled triangle ABC, right-angled at A, AD is the altitude from A to hypotenuse BC. If AD = 6 cm and DC = 9 cm, what is the length of segment BD?

- A) 3 cm
- B) 4 cm
- C) 5 cm
- D) 6 cm

Solution

Step 1: draw the figure to represent the situation



Step 2: Solve for the missing or unknown length.

$$6^2 = 9 \times x$$

$$36 = 9x$$

$$\therefore x = 4$$

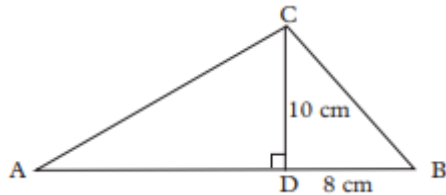
Step 3: Choose the correct answer.

The correct answer is **B) 4cm**.



Application activity 8.2

1. In a right-angled triangle ABC, right-angled at A, AD is the altitude to hypotenuse BC. If $BD = 8$ cm and $DC = 2$ cm, what is the length of AD?
A) 2 cm B) 4 cm C) 6 cm D) 16 cm
2. In right triangle ABC, $\angle A = 90^\circ$, altitude AD meets hypotenuse BC at D. If $AD = 12$ cm and $BD = 9$ cm, what is the length of AC?
A) 13 cm B) 15 cm C) 18 cm D) 20 cm
3. The altitude to the hypotenuse of a right-angled triangle is 8 cm long. If the hypotenuse is 20 cm long, what are the lengths of the two segments of the hypotenuse?
A) 4 cm and 16 cm B) 6 cm and 14 cm C) 8 cm and 12 cm D) 10 cm and 10 cm
4. Find the length AD in triangle ABC below, given that $DC = 10$ cm and $DB = 8$ cm.

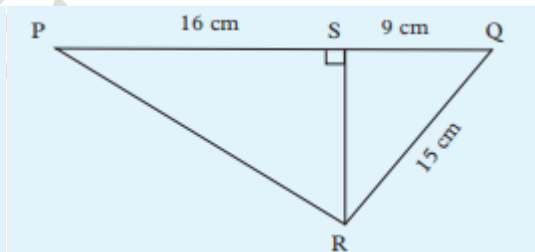
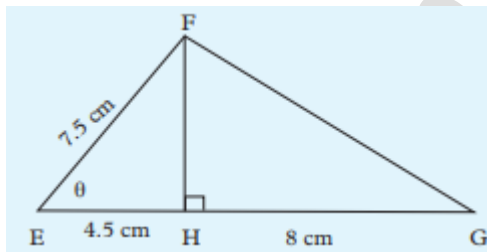


8.3 Theorem of a right-angled triangle



Activity 8.3

Look at the right-angled triangles EFG and PQR in Figures below



(a) Work out and compare the ratios:

(i) $\frac{EF}{EG}$ and $\frac{EH}{EF}$

(ii) $\frac{QR}{PQ}$ and $\frac{QS}{QR}$

(iii) What do you notice about each pair of ratios.

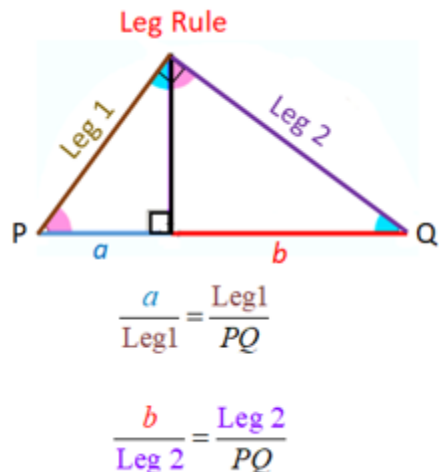
(b) Summarize your observations into a theorem on right-angled triangles.



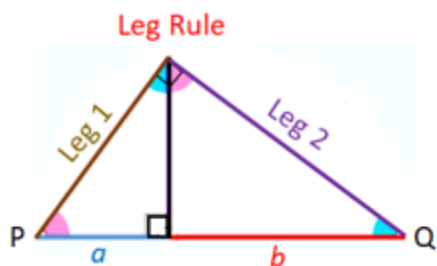
Content summary

The **Leg Rule** (or **Leg Geometric Mean Theorem**) states:

Each leg is the **geometric mean** between the hypotenuse and the segment of the hypotenuse (the projection of the leg on the hypotenuse) adjacent to that leg.



Similarly,

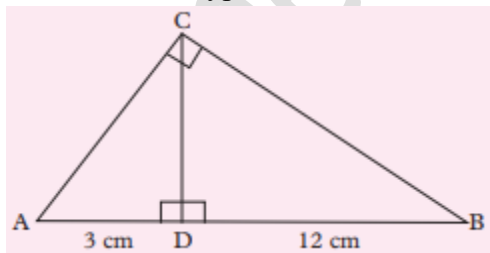


$$(\text{leg 1})^2 = PQ \times a$$

$$(\text{leg 2})^2 = PQ \times b$$

Example 1

The Figure below shows a right angled triangle in which $AB = 12$ cm, $AD = 3$ cm and CD is the attitude to the hypotenuse. Find the length AC .



Solution

For leg AC (adjacent to segment AD):

$$AC^2 = AD \times AB$$

For leg BC (adjacent to segment DB):

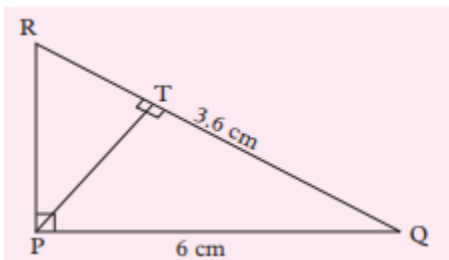
$$BC^2 = DB \times AB$$

So, $AC^2 = 3 \times 15 \text{ cm}$

$$AC = 3\sqrt{5} \text{ cm}$$

Example 2

The figure shows a right-angled triangle PQR in which PQ = 6 cm. QT = 3.6 cm. Find the length of RT. Hence or otherwise find the length of QR.



Solution

Solution

QR is a hypotenuse of $\triangle PQR$

PQ is a leg,

QT is the projection of leg PQ on the hypotenuse.

Applying the leg theorem

$$\frac{\text{leg}}{\text{hypotenuse}} = \frac{\text{projection}}{\text{leg}}$$

$$\frac{PQ}{QR} = \frac{QT}{PQ}$$

Let $RT = x$ hence $QR = x + 3.6 \text{ cm}$

$$\frac{6 \text{ cm}}{(x + 3.6) \text{ cm}} = \frac{3.6 \text{ cm}}{6 \text{ cm}}$$

$$\Rightarrow 3.6 (x + 3.6) \text{ cm}^2 = 36 \text{ cm}^2$$

$$x + 3.6 = \frac{3.6 \text{ cm}}{6 \text{ cm}}$$

$$\Rightarrow x + 3.6 = 10$$

$$x = (10 - 3.6) \text{ cm} = 7.4 \text{ cm}$$

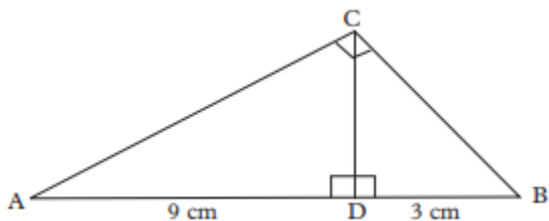
$$RT = 7.4 \text{ cm}$$

$$\Rightarrow QR = 7.4 \text{ cm} + 3.6 \text{ cm} = 10 \text{ cm}$$



Application activity 8.3

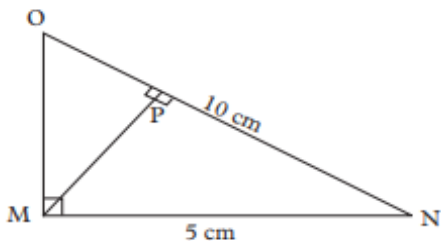
1) In the right-angled triangle ABC, the right angle is at C. The altitude CD meets the hypotenuse AB at D, where AD=9 cm and DB=3 cm.



Find the length of AC and choose the correct answer:

A. $3\sqrt{3} \text{ cm}$ B. $6\sqrt{3} \text{ cm}$ C. $9\sqrt{3} \text{ cm}$ D. 12 cm

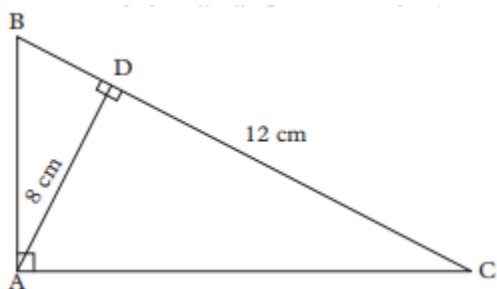
2) In triangle MNO, the right angle is at M. The altitude MP is drawn to the hypotenuse NO.



Given that $MN=5$ cm and $NO=10$ cm, find the length of NP and select the correct answer:
 A) 2.5cm B) 5 cm C) 7.5cm D) 8.5cm.

3) In triangle **ABC**, the right angle is at **A**.

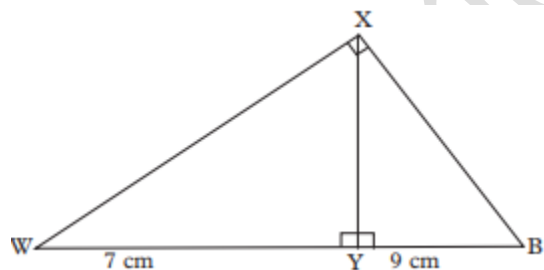
The perpendicular **AD** is drawn to the hypotenuse **BC**, such that $AD=8$ cm and $DC=12$ cm.



Find the lengths of **BD** and **AC** and choose the correct answer:

- A) $BD = 4$ cm, $AC = 10$ cm B) $BD = 6$ cm, $AC = 12$ cm
 C) $BD = \frac{16}{3}$ cm, $AC = 4\sqrt{13}$ cm D) $BD = 8$ cm, $AC = 20$ cm

4) Find the length of **WX** in triangle **WXY** below, given that $WY = 7$ cm and $YB = 9$ cm.



The hypotenuse of a right-angled triangle is 5 cm long and the longer leg of the triangle is 4 cm long. What is the length of the projection of the shorter leg on the hypotenuse?

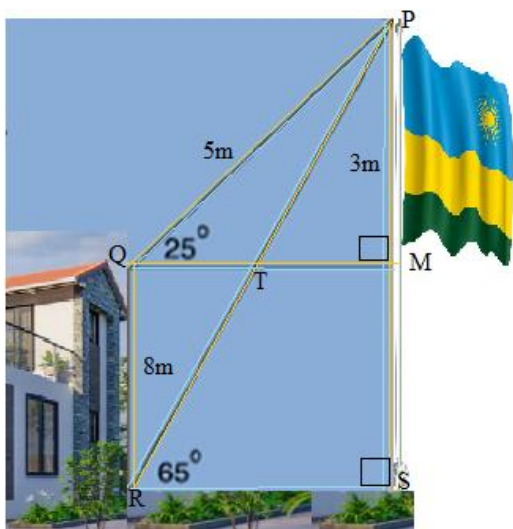
8.4 Introduction to trigonometry

8.4.1 Sine and cosine of an acute angle



Activity 8.4.1

In the figure below, the distance from the top of flag to the top of house PQ is 5m, the distance PM is 3m, the height of the house is 8m, the angle PQM is 25° , the angle PRS is 65° . Find



- The angles QTR and TPM,
- The length of MS,
- The distance from the base of the house to the foot of the flagpole,
- The distance from the base of the house to the top of the flagpole,
- The length MQ can be found using Pythagoras theorem where $QM = \sqrt{PQ^2 - PM^2}$. Is there any other method that can help you to get the same result? Explain.

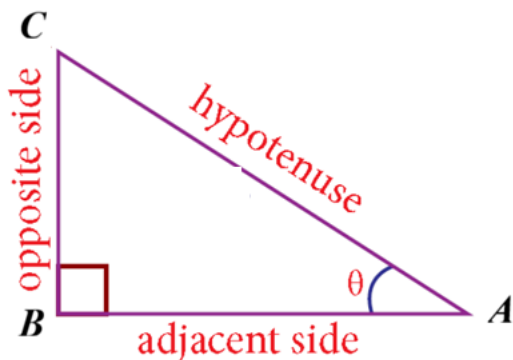


Content summary

Trigonometry is the branch of mathematics concerned with the relationships between the sides and the angles of triangles.

In this lesson, we shall study the relationships between the acute angles and the sides of a right angled triangle

The sides of such a triangle are named, with reference to specific acute angle (*angles are most commonly presented using Greek letters such as α = alpha; β = beta; γ = gamma; δ = delta; θ = theta; ϕ = phi; ω = omega*)



From the figure, BC is named the **opposite side of angle θ** , AB is the **adjacent side of angle θ** and AC is the hypotenuse.

Therefore, **Basic trigonometric ratios** such as **sine**, and **cosine** can be developed through the relationships between the acute angles and the sides

The ratio $\frac{BC}{AC} = \frac{\text{Opposite side to } \theta}{\text{hypotenuse}}$ is called **the sine of angle θ** (write: **sin θ**).

The ratio $\frac{AB}{AC} = \frac{\text{Adjacent side to } \theta}{\text{hypotenuse}}$ is called **the cosine of angle θ** (write: **cos θ**).

Note: The above Basic trigonometric ratios can be summarised as:

The ratio $\sin \theta = \frac{\text{Opposite side to } \theta}{\text{hypotenuse}}$

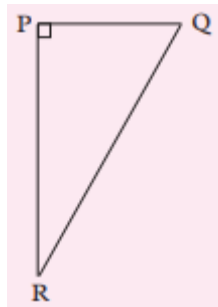
The ratio $\cos \theta = \frac{\text{Adjacent side to } \theta}{\text{hypotenuse}}$

Memory Aid: SOH-CAH-TOA (SOH = Sine = Opposite/Hypotenuse, CAH = Cosine = Adjacent/Hypotenuse)

Example 1

Consider triangle PQR below, Identify:

- (a) the hypotenuse
- (b) identify the:
 - (i) side opposite to $\angle R$
 - (ii) side adjacent to $\angle R$
 - (iii) side opposite to $\angle Q$
 - (iv) side adjacent to $\angle Q$



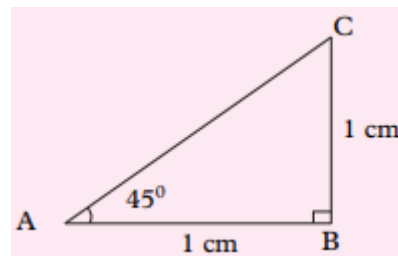
Solution

- (a) hypotenuse is QR
- (b) (i) PQ (ii) PR (iii) PR (iv) PQ

Example 2

Consider the triangle ABC with $AB = 1 \text{ cm}$, $AC = 1 \text{ cm}$.

- (a) Find the length of side AC. Leave result in surds
- (b) Find from the triangle:
 - (i) $\cos 45^\circ$
 - (ii) $\sin 45^\circ$ Leave the answer in surds.



Solution

- (a) Opposite side to $45^\circ = 1 \text{ cm}$; Adjacent side to $45^\circ = 1 \text{ cm}$

By Pythagoras theorem;

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 1^2 + 1^2 = 2$$

$$AC = \sqrt{2} \text{ cm; (hypotenuse)}$$

$$(b) \text{ i) } \cos 45^\circ = \frac{A}{H} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \quad \text{ii) } \sin 45^\circ = \frac{O}{H} = \frac{1}{\sqrt{2}}$$

Using a Calculator to Find Sine, Cosine, and Inverse Functions

Important: Before beginning, ensure your calculator is in **DEGREE** mode (not RADIAN or GRAD).

Finding the Sine or Cosine of an Angle

Follow these steps:

- **To find sine:**

Press **sin**, then enter **the angle**, then press **=**.

*(On some calculators, you may enter the angle first, then press **sin**.)*

- **To find cosine:**

Press **cos**, then enter **the angle**, then press **=**.

*(On some calculators, you may enter the angle first, then press **cos**.)*



Example 3

a) To find **sin 30°**: Press **sin**, then Press **30**, **=**.

Result: 0.5.

b) To find **cos 45°**: Press **cos**, then Press **45**, **=**.

Result: 0.7071.

Finding the Angle from a Sine or Cosine Value (Inverse Functions)

To find an angle when you know its sine or cosine, follow these steps:

Enter the value, then press **2ndF** (or **Shift**), then press **sin** or **cos**, then press **=**.

Example 4

To find the angle θ where $\sin \theta = 0.5$:

Press **0.5**, then **2ndF** (or **Shift**), then **sin**, then **=**.

Result: $\theta = 30^\circ$.

Important Note: Common Values

It is useful to know these common sine and cosine values:

Angle	Sin	Cos
0°	0	1
30°	0.5	0.8660
45°	0.7071	0.7071
60°	0.8660	0.5
90°	1	0

General Trend: As angles increase from 0° to 90°:

(i) Their **sines increase** from 0 to 1.

(ii) Their **cosines decrease** from 1 to 0.

Example 5:

Using a Calculator to find: (i) $\sin 37^\circ$ (ii) $\cos 18^\circ$

Solution

(i) Press **sin**, **37**, **=**. Result: $\sin 37^\circ \approx 0.6018$.

(ii) Press **cos**, **18**, **=**. Result: $\cos 18^\circ \approx 0.9511$.

Example 6:

Find an angle θ , if $\sin \theta = 0.7660$.

Solution

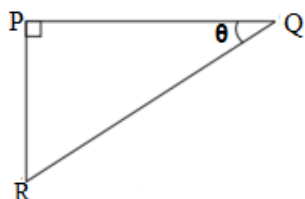
Press **0.7660**, **2ndF** (or **Shift**), **sin**, **=**.

Result: $\theta \approx 50^\circ$.



Application activity 8.4.1

1. Label the hypotenuse, opposite, and adjacent sides for angle θ in a given right triangle diagram.

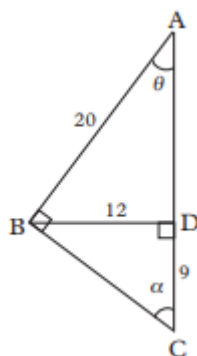


2. In a right triangle, $\sin \theta$ equals:

- a) Adjacent/Hypotenuse
- b) Opposite/Hypotenuse
- c) Opposite/Adjacent
- d) Adjacent/Opposite

3. In the figure, name and give the lengths of two sides which are:

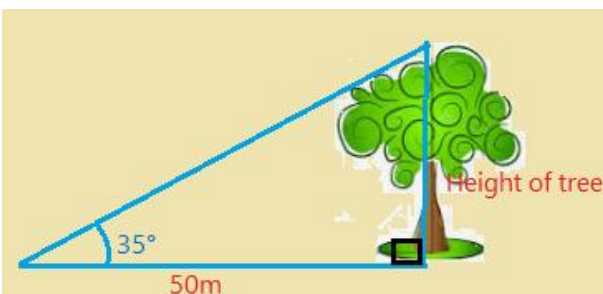
- a) adjacent to angle θ
- b) hypotenuse relative to angle α
- c) Opposite to to angle θ
- d) Write a simplified ratio
 - i) $\sin \theta$
 - ii) $\cos \alpha$



8.4.2 Solving a right-angled triangle using sines and cosine



Activity 8.4.2



Surveyor needs to determine the height of a tree without climbing it. She measures 50m from base and use a clinometer to measure angle of elevation to top as 35° .

- Find the straight-line distance from the surveyor to the top of the tree
- How tall is the tree?
- What other measurements could She determine from this setup?



Content summary

Trigonometric ratios **sine** and **cosine** are useful to find unknown sides and angles. Actually, we know that trigonometric ratios sine and cosine are defined as

$$\sin \theta = \frac{\text{Opposite side to } \theta}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent side to } \theta}{\text{hypotenuse}}$$

So, through trigonometric ratios, we can develop other formulae for example

Using Sine:

- When we know Angle and Opposite side, then
 $\text{Hypotenuse } (H) = O/\sin \theta$
- When we know Angle and Hypotenuse, then
 $\text{Opposite: } O = H \times \sin \theta$
- When we know Opposite and Hypotenuse then
 $\text{Find angle } (\theta) = \sin^{-1}(O/H)$

Using Cosine:

- When we know Angle and Adjacent side then
 $\text{Hypotenuse } (H) = A/\cos \theta$
- When we know: Angle and Hypotenuse then
 $\text{Adjacent: } A = H \times \cos \theta$
- When we know Adjacent and Hypotenuse then
 $\text{angle: } \theta = \cos^{-1}(A/H)$

Note: In right triangle, two acute angles sum up to 90° .

Example1

Given: $\angle A = 40^\circ$, hypotenuse = 15cm. Find:

- i) Opposite (BC)
- ii) Adjacent (AB)

Solution

- i) Opposite (BC) = $15 \times \sin 40^\circ \approx 15 \times 0.6428 = 9.642\text{cm}$
 ii) Adjacent (AB) = $15 \times \cos 40^\circ \approx 15 \times 0.7660 = 11.49\text{cm}$
 $\angle C = 90^\circ - 40^\circ = 50^\circ$

Example 2

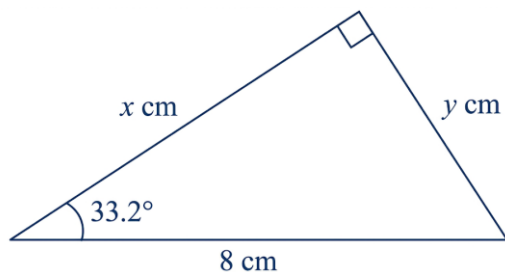
Given: Opposite = 12m, hypotenuse = 20m. Find an adjacent side.

$$\sin \theta = 12/20 = 0.6 \rightarrow \theta = \sin^{-1}(0.6) \approx 36.87^\circ$$

$$\text{Adjacent} = 20 \times \cos 36.87^\circ \approx 20 \times 0.8 = 16\text{m}$$

Example 3

Find the value of x and y in Figure below to 2 s.f.



Solution:

$$\sin 33.2^\circ = \frac{\text{opposite side}}{\text{hypotenuse}} \quad \text{or} \quad \sin 33.2^\circ = \frac{y}{8}$$

$$\Rightarrow y = 8 \sin 33.2^\circ = 8 \times 0.5476 \text{ cm} \\ = 4.3808 \text{ cm} = 4.4 \text{ cm (2 s.f.)}$$

$$\cos 33.2^\circ = \frac{\text{adjacent side}}{\text{hypotenuse}} \quad \text{or} \quad \cos 33.2^\circ = \frac{x}{8}$$

$$\Rightarrow x = 8 \cos 33.2^\circ = 8 \times 0.8368 \text{ cm} \\ = 6.6944 \text{ cm} = 6.7 \text{ cm (2 s.f.)}$$

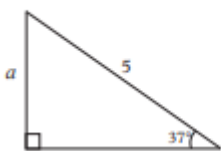
...



Application activity 8.4.2

- Solve triangle: $\angle A = 28^\circ$, hypotenuse = 25cm. Find other sides and $\angle C$.
- In right $\triangle PQR$ (right at Q), $PQ = 7\text{cm}$, $PR = 12\text{cm}$. Find $\angle P$.
- A ladder 6m long leans against wall at 65° . How high up wall does it reach?
- To find hypotenuse when angle and adjacent are known, use:
 - $A = H \times \cos \theta$
 - $H = A / \cos \theta$
 - $O = H \times \sin \theta$
 - $H = O / \sin \theta$
- In Figure, find the lengths marked with letters. The lengths are in cm. (Give your answer correct to 3 decimal places).

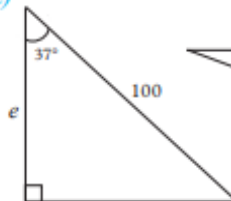
(a)



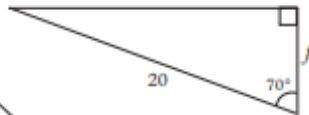
(b)

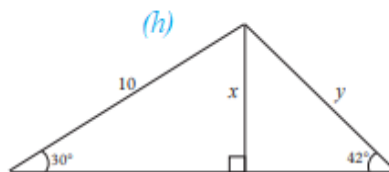
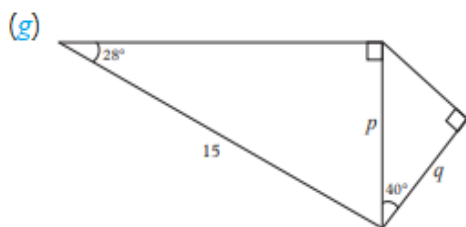


(e)



(f)

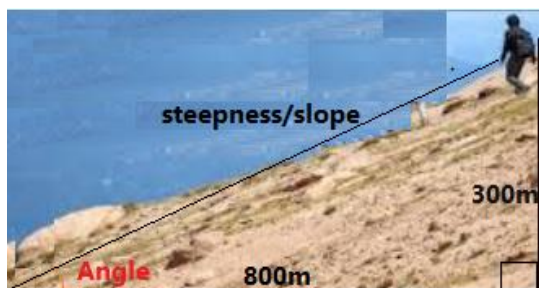




8.4.3 Tangent of the acute angle



Activity: 8.4.3



You are hiking up a mountain trail.
The trail rises 300m vertically over a horizontal distance of 800m.

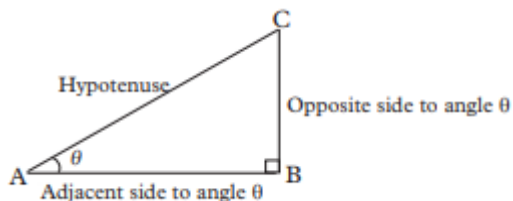
- How steep is the trail?
- What angle does it make with the horizontal?

Hint: This steepness ratio is called the gradient or slope



Content summary

In **Basic trigonometric ratios**, we have also **tangent** that can be developed through the relationships between the acute angle and the sides (Opposite and Adjacent).



The ratio $\frac{BC}{AB} = \frac{\text{Opposite side to } \theta}{\text{Adjacent side to } \theta}$ is called **the tangent of angle θ ($\tan \theta$)**.

Note: The above Basic trigonometric ratio can be summarised as:

The ratio $\tan \theta = \frac{\text{Opposite side to } \theta}{\text{Adjacent side to } \theta}$

Memory Aid: SOH-CAH-TOA (TOA: Tangent = Opposite/Adjacent)

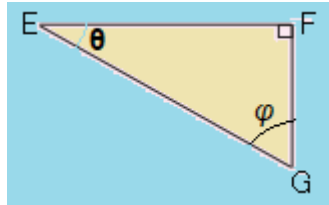
Example 1

Consider triangle EFG below, Identify:

a. Identify:

- (i) side opposite to $\angle \theta$
- (ii) side adjacent to $\angle \theta$

b. Write formula for tan

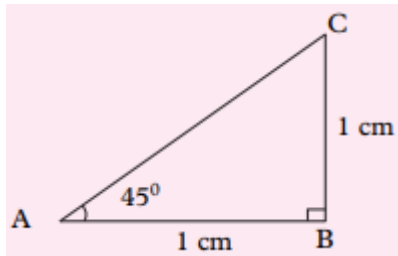


Solution

- (b) (i) PQ (ii) PR (iii) PR (iv) PQ

Example 2

Consider the triangle ABC with AB = 1 cm, AC = 1 cm. Find $\tan 45^\circ$



Solution

- (c) Opposite side to $45^\circ = 1 \text{ cm}$; Adjacent side to $45^\circ = 1 \text{ cm}$

By Pythagoras theorem;

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 1^2 + 1^2 = 2$$

$$AC = \sqrt{2} \text{ cm; (hypotenuse)}$$

(d) $\tan 45^\circ = \frac{O}{A} = \frac{1}{1} = 1$

• Calculator Operations:

Finding $\tan$ of angle: Press **tan** then Press **angle** then Press **=** (or Press angle then Press **tan** depending on calculator)

Finding angle from $\tan$ value: Use **$\tan^{-1}$** or **inv tan** or **arctan**

Press **value** then press **shift** then press **tan** then press **=**

Special Values:

$$\tan 0^\circ = 0$$

$$\tan 30^\circ = 1/\sqrt{3} \approx 0.5774$$

$$\tan 45^\circ = 1$$

$$\tan 60^\circ = \sqrt{3} \approx 1.7321$$

$$\tan 90^\circ = \text{undefined (infinity)}$$

• Mode Check: Always verify calculator is in DEGREE mode

Example

- (a) Find : (i) $\tan 25^\circ$ (ii) $\tan 58^\circ$

- (b) If $\tan \theta = 1.5$, find θ

Solution

- (a) i) On scientific calculator; *press tan, then press 25, then press =*
the result is 0.4663
- (ii) On scientific calculator; *press tan, then press 58, then press =*
the result is 1.6003
- (b) To find θ : Press 1.5 ; Press **shift** ; Press **tan** ; Press **=** , the result is 56.31°



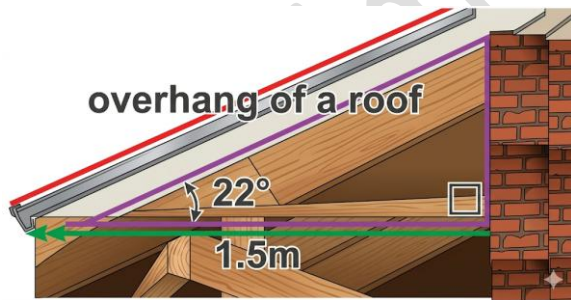
Application activity 8.4.3

1. Define tangent ratio. How does it differ from sine and cosine?
2. In right triangle, opposite = 9, adjacent = 12. Find $\tan \theta$ and θ .
3. If $\theta = 40^\circ$ and adjacent = 15m, find opposite side.
4. Tangent of angle in right triangle is:
 - a) Opposite/Hypotenuse
 - b) Adjacent/Hypotenuse
 - c) Opposite/Adjacent
 - d) Adjacent/Opposite
5. A road rises 4m over 20m horizontal. Find angle of inclination.

8.4.4 Solving a right-angled triangle using tangent of angle



Activity 8.4.4



An architect needs to calculate the exact overhang of a roof with a 22° pitch. The horizontal distance from wall to roof edge is 1.5m. How much does the roof overhang? What calculator steps are needed?



Content summary

- **When to Use Tangent:**
 When we have opposite and adjacent sides (no hypotenuse involved).
 When we have angle and one of opposite or adjacent (not hypotenuse).
- **Problem-Solving Steps:**
Step1: Identify known: sides and angles.
Step2: Choose appropriate ratio (TOA)

The ratio $\tan \theta = \frac{\text{Opposite side to } \theta}{\text{Adjacent side to } \theta}$,

Step 3: Set up equation

$\text{Adjacent side to } \theta \times \tan \theta = \text{Opposite side to } \theta$ or

$$\text{Adjacent side to } \theta = \frac{\text{Opposite side to } \theta}{\tan \theta}$$

$$\theta = \tan^{-1} \left(\frac{\text{Opposite side to } \theta}{\text{Adjacent side to } \theta} \right)$$

Step 4: Solve using calculator

Note: To find remaining elements if needed, you may need:

Pythagoras Theorem for third side and Angle sum for third angle.

Example 1

- a) Given: $\angle A = 35^\circ$, adjacent side = 10cm. Find the Opposite side
 b) Given that the opposite side = 8m, adjacent = 6m. find the angle.

Solution

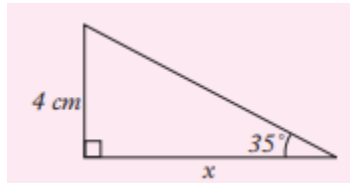
$$(a) \text{ Opposite} = 10 \times \tan 35^\circ \approx 10 \times 0.7002 = 7.002 \text{ cm}$$

$$(b) \tan \theta = \frac{8}{6} = 1.333$$

$$\theta = \tan^{-1}(1.333) \approx 53.13^\circ$$

Example 2

Find the length indicated x in Fig



Solution

The two indicated sides are the opposite and the adjacent sides.

$$\therefore \tan 35^\circ = \frac{4}{x} \Rightarrow x = \frac{4}{\tan 35^\circ} = \frac{4}{0.7002} = 4 \times \frac{1}{0.7002} = 4 \times 1.429$$

$$\therefore x \approx 5.7 \text{ cm (2 s.f.)}$$

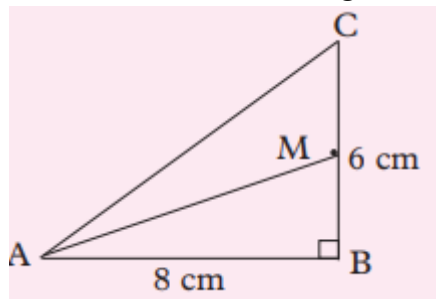
Example 3

ABC is a triangle such that AB = 8cm, $\angle B = 90^\circ$ and BC = 6 cm. M is the midpoint of BC.

Calculate: (a) $\angle BAM$ (b) $\angle MAC$

Solution

Draw a sketch of the triangle ABC, showing all the given information, as in Figure



(a) Join A to M. Using $\triangle ABM$, $BM = 3\text{ cm} \dots$ (midpoint of BC)

$$\tan \hat{BAM} = \frac{BM}{AB} = \frac{3}{8} = 0.375$$

$$\tan^{-1} 0.375 = \angle BAM$$

In a calculator, press $\tan^{-1}$ type 0.75 and press =

$$\therefore \angle BAM = 20.5^\circ$$

$$= \frac{BM}{AB} = \frac{3}{8} = 0.375$$

(b) Using $\triangle ABC$, we can find $\angle BAC$

$$\therefore \tan \angle BAC = \frac{BC}{AB} = \frac{6}{8} = 0.75$$

$$\tan^{-1} 0.75 = \angle BAC$$

In a calculator, press $\tan^{-1}$, type 0.75 and press =

$$\angle BAC = 36.5^\circ$$

$\angle MAC$ is the between $\angle BAC$

and

$\angle BAM$

$$\therefore \angle CAM = \angle BAC - \angle BAM$$

$$= 36.5^\circ - 20.5^\circ$$

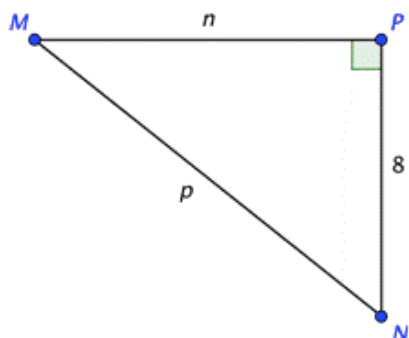
$$= 16^\circ$$

$$\therefore \angle CAM = 16^\circ$$



Application activity 8.4.4

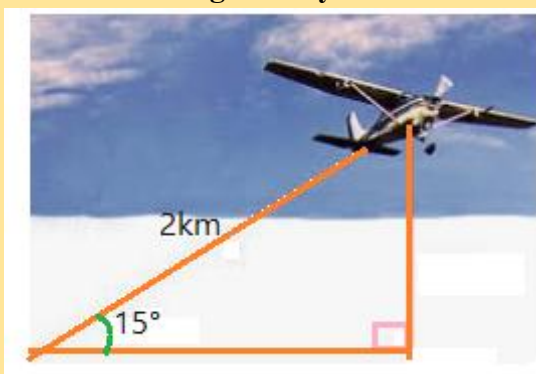
1. Solve triangle: $\angle B = 48^\circ$, adjacent to $\angle B = 12\text{ cm}$.
2. Right triangle has sides: opposite = 15m, adjacent = 8m. Find angles and hypotenuse.
3. (MCQ) When using tangent to solve triangle, you typically have:
 - a) Hypotenuse and angle
 - b) Opposite and hypotenuse
 - c) Opposite and adjacent
 - d) Adjacent and hypotenuse
4. Solve the right triangle shown below. Find the exact side lengths and approximate the angles to the nearest degree.



8.4.5 Application of trigonometric ratios (sine and cosine)



Learning activity 8.4.5



An airplane takes off at a 15° angle. After traveling 2km along its flight path, what altitude has it reached? How far horizontally has it travelled? Air traffic controllers need both values for monitoring.

Content summary

In a right-angled triangle,

- ❖ **Sine ($\sin \theta$):** Ratio of the side **opposite** the angle to the hypotenuse
 $\sin \theta = \text{Opposite/Hypotenuse}$
- ❖ **Cosine ($\cos \theta$):** Ratio of the side **adjacent** to the angle to the hypotenuse
 $\cos \theta = \text{Adjacent/Hypotenuse}$

Note:

Memory Aid: SOH-CAH-TOA (Sine=Opposite/Hypotenuse, Cosine=Adjacent/Hypotenuse)

The **Sine** and **cosine** ratios are essential in different areas of real life. For example:

Application Area	Sine Use Case	Cosine Use Case	Real-World Example
Height and Distance	Finding vertical components	Finding horizontal components	Surveying, architecture
Navigation	Calculating east-west displacement	Calculating north-south displacement	GPS, marine navigation

To solve problem involving sine and cosine the following steps are proceeded:

- **Identify** the right triangle in the problem
- **Label** known sides/angles with O (opposite), A (adjacent), H (hypotenuse)
- **Select** appropriate ratio based on what's known and needed
- **Set up** the equation: $\sin \theta = O/H$ or $\cos \theta = A/H$
- **Solve** for the unknown using algebra and calculator
- **Interpret** the result in the problem's context

Example 1: A 12-meter ladder leans against a wall making a 65° angle with the ground. How far up the wall does it reach?

Solution:

Known: Hypotenuse (ladder) = 12 m, Angle = 65° , Need: Opposite side (height)

Use sine: $\sin(65^\circ) = \text{Opposite} / 12$

$$\text{Opposite} = 12 \times \sin(65^\circ) = 12 \times 0.9063 \approx 10.88 \text{ m}$$

Therefore, the ladder reaches **10.88 meters** up the wall.

Example 2: A ship sails 20 km on a bearing of 060° (60° east of north). How far north and how far east has it traveled?

Solution:

Bearing measured from north: North distance = Adjacent, East distance = Opposite

$$\text{North distance} = 20 \times \cos(60^\circ) = 20 \times 0.5 = 10 \text{ km}$$

$$\text{East distance} = 20 \times \sin(60^\circ) = 20 \times 0.8660 \approx 17.32 \text{ km}$$

Therefore, the ship is **10 km north** and **17.32 km east** of its starting point.

Example 3: A ramp rises 1.5 meters over a horizontal distance of 6 meters. What is the ramp's angle of incline?

Solution:

Known: Opposite = 1.5 m, Adjacent = 6 m, Need: Angle θ

$\cos \theta = \text{Adjacent} / \text{Hypotenuse}$ First find hypotenuse:

$$\sqrt{(1.5^2 + 6^2)} = \sqrt{(2.25 + 36)} = \sqrt{38.25} \approx 6.185$$

$$\cos \theta = 6 / 6.185 \approx 0.97,$$

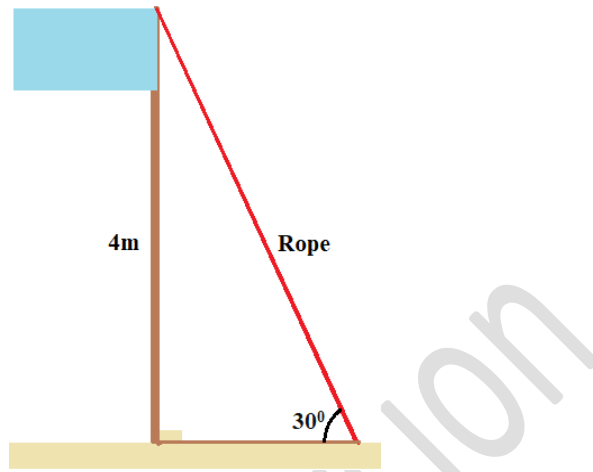
$$\text{so, } \theta = \cos^{-1}(0.97) \approx 14.07^\circ \text{ (same within rounding)}$$

Therefore, the ramp angle is approximately **14°** .

Example 4

A flag pole 4m long, has a rope tied to the top, securing it to the ground.

- How long will the rope need to be?
- How fasten is the rope away from the base of the flag pole.



Solution

$$(a) \sin 30^\circ = \frac{O}{H} = \frac{4}{\text{Rope}}$$

$$\text{Rope} = \frac{4}{\sin 30^\circ} = \frac{4}{0.5} = 8m$$

$$(b) \text{Rope}^2 = h^2 + d^2$$

$$d = \sqrt{8^2 - 4^2} = \sqrt{48} = 4\sqrt{3}m$$

The tangent ratio is the **go-to tool** when the problem involves:

- Height and Distance:** Finding an inaccessible height (like a tree, building) or a distance across a gap.
- Slope and Steepness:** Calculating the grade (%) of a road, ramp, or roof (slope = $\tan(\text{angle}) \times 100\%$).
- Navigation and Mapping:** Basic triangulation.

To solve problems involving application of tangent; we proceed the following steps:

- Identify:** Find the right-angled triangle in the scenario.
- Label:** Mark the given angle, the **Opposite** side, the **Adjacent** side, and the unknown.
- Choose:** Select the correct formula: $\tan(\theta) = O/A$.
- Solve:** Substitute the known values and solve for the unknown.
- Conclude:** State your answer with correct units in the context of the problem.

Example 1 Finding a Height (Basic)

A surveyor stands 50 m from the base of a building. He measures the angle of elevation to the top as 25°. Find the building's height.

Solution

Step 1 - Identify & Label: Angle $\theta = 25^\circ$. Adjacent side (distance) = 50 m. Opposite side (height) = h.

Step 2 - Choose Formula:

$$\tan(\theta) = \text{Opposite} / \text{Adjacent}$$

Step 3 - Solve:

$$\tan(25^\circ) = h / 50$$

$$h = 50 \times \tan(25^\circ) \rightarrow h \approx 50 \times 0.4663 \rightarrow h \approx 23.3 \text{ m}$$

Therefore, the building is approximately **23.3 meters** tall.

Example 2 Finding a Distance

A lifeguard on a 3-meter-high tower spots a swimmer in distress. The angle of depression from the lifeguard to the swimmer is 12° . How far is the swimmer from the base of the tower?

Solution

Step 1 - Identify & Label: Angle of depression = 12° .

The angle inside the triangle is also 12° (alternate angles).

Opposite side (tower height) = 3 m.

Adjacent side (water distance) = d .

Step 2 - Choose Formula:

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 3 - Solve:

$$\tan(12^\circ) = \frac{3}{d}$$

$$d = \frac{3}{\tan(12^\circ)} \approx \frac{3}{0.2126} \approx 14.1 \text{ m}$$

Therefore, the swimmer is about **14.1 meters** from the tower.

Example 3: Finding the Angle (Slope)

A road climbs 15 meters vertically over a horizontal distance of 100 meters. What is the angle of the road's incline?

Solution

Step 1 - Identify & Label:

Opposite side (rise) = 15 m

Adjacent side (run) = 100 m

Angle = θ

Step 2 - Choose Formula:

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 3 - Solve:

$$\tan(\theta) = \frac{15}{100} = 0.15$$

$\theta = \tan^{-1}(0.15)$; This is the inverse tangent function on your calculator

$$\theta \approx 8.5^\circ$$



Application activity 8.4.4

1. A kite string makes 50° with ground. String length = 80m. How high is kite?
2. Road rises uniformly 100m over 500m horizontal. Find road's angle with horizontal.
3. A 10m ladder reaches 8.5m up wall. What angle does it make with ground?
4. For angle of elevation problems, the opposite side represents:
 - a) Distance along slope
 - b) Horizontal distance
 - c) Height/vertical distance
 - d) Straight-line distance
4. An escalator is 20m long and rises 8m vertically. Find its angle with horizontal.
5. During a storm, a 15-meter-tall tree broke at a point 5 meters above the ground. The top portion leaned and touched the ground at a point 8 meters from the tree's base.
 - a) What angle does the broken portion make with the ground?
 - b) How long is the broken portion of the tree?



END OF UNIT ASSESSMENT

Section A: Knowledge and Understanding (6 questions)

1. Define a right-angled triangle and identify the right angle in such a triangle.
2. In a right-angled triangle, explain the meaning of the following sides with respect to a given acute angle:
 - a) Hypotenuse
 - b) Opposite side
 - c) Adjacent side
3. State Pythagoras' Theorem clearly using symbols.
4. Write down the formulas for the following trigonometric ratios:
 - a) Sine of an acute angle
 - b) Cosine of an acute angle
 - c) Tangent of an acute angle
5. What is meant by the median to the hypotenuse in a right-angled triangle? State one property of this median.
6. Explain the height (altitude) theorem in a right-angled triangle in words.

Section B: Skills (9 questions)

7. A right-angled triangle has legs of lengths 6 cm and 8 cm. Calculate the length of the hypotenuse.
8. The hypotenuse of a right-angled triangle is 13 cm and one leg is 5 cm. Find the length of the other leg.

9. In a right-angled triangle, the angle θ has:

opposite side = 7 cm

hypotenuse = 25 cm

Find $\sin \theta$ and $\cos \theta$.

10. Using a calculator, find the value of:

a) $\sin 35^\circ$

b) $\cos 60^\circ$

(Give answers correct to two decimal places.)

11. In a right-angled triangle, $\tan \theta = 3/4$. Find the value of θ using a calculator.

12. A ladder is leaning against a vertical wall. The foot of the ladder is 4 m away from the wall and the ladder makes an angle of 65° with the ground.

Find the length of the ladder.

Section C: Attitudes and Values (3 questions)

13. Explain two real-life situations where trigonometric ratios can be used to measure distances or heights that are difficult to measure directly.

14. While solving trigonometric problems in a group, why is accuracy and clear communication important? Give one example.

15. A student gets different answers when solving a triangle using sine and tangent. What values or habits should the student develop to improve correctness and reliability in calculations?



UNIT 9: CIRCLE AND RELATED THEOREMS

Key unit competence: Use properties of circle and circle theorems in constructing mathematical arguments and solving related problems.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and Understanding

- Identify and define the key parts of a circle, such as the radius, chord, tangent, and sector.
- Explain the eight circle theorems and recall fundamental angle relationships.

Skills

- Apply circle theorems to find unknown angles and lengths in geometric problems.
- Construct logical proofs and solve complex problems by combining multiple theorems.

Attitudes and Values

- Appreciate the logic and structure of geometric proof.

- Develop precision, perseverance, and collaborative reasoning when solving problems.

Introduction

This unit **explores** the essential geometry of the circle, moving from basic terms to mastering its core theorems. You will learn to calculate unknowns, build proofs, and solve complex problems. Ultimately, it aims to **develop** an appreciation for the **beauty and clarity** of geometric reasoning, building precision and teamwork skills that equip you to think and **reason** like a mathematician.



Introductory activity

Look at the image.



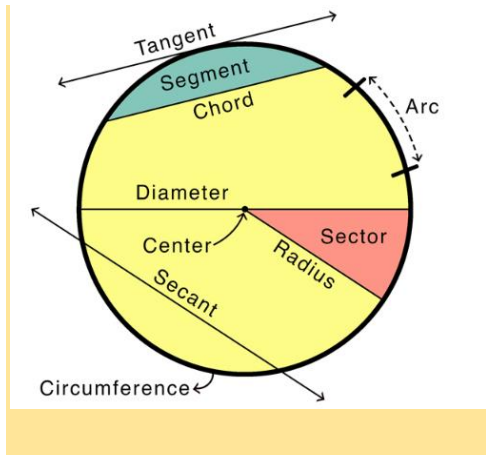
- What point do you think is the **exact centre** of the wheel? Why must this point be special?
- If point **S** is the centre, what do we call the line segment joining **S** to **T, P, R, Q, or U**?
- Which points lie **on the circumference** of the circle?
- Can you identify a line segment that joins two points on the circle but does **not pass through the centre**? What is the name of it?
- Which part of the wheel shows a **sector** of a circle? How can you describe it using angles?
- If a straight road just touches the wheel at one point, what would we call that line?

9.1 Parts or elements of a circle and their properties



Activity 9.1

Observe a bicycle wheel and compare it with parts of the circle:



- What are its main parts?
- How are the spokes arranged relative to the center?
- How would you describe the relationship between the curved rim and the center point?
- Think about the wheel turning what part traces the path?



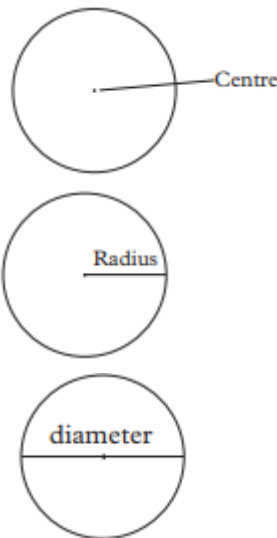
Content summary

The basic elements of a circle are:

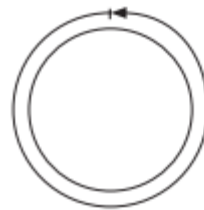
Center (O): Fixed point equidistant from all points on the circle

Radius (r): Distance from center to any point on circle (plural: radii)

Diameter (d): Chord through center ($d = 2r$)

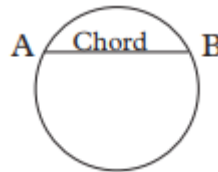


Circumference: Perimeter/distance around the circle

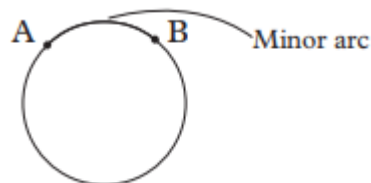


Circumference

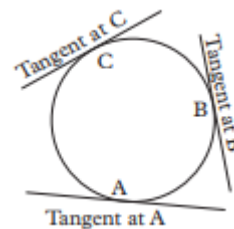
Chord: Line segment joining two points on the circle



Arc: Part of circumference between two points



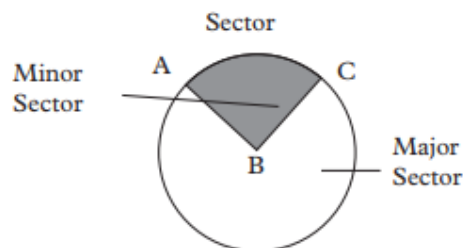
Tangent: Line touching circle at exactly one point



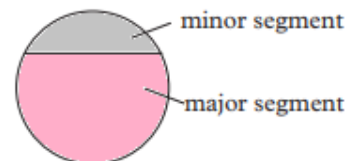
Secant: Line intersecting circle at two points



Sector: Region bounded by two radii and an arc (like pizza slice)



Segment: Region bounded by a chord and an arc



Properties:

- All radii of same circle are equal.

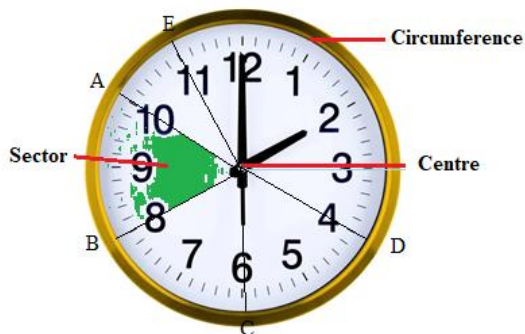
- Diameter is the longest chord of the circle.
- Tangents of the circle are perpendicular to radius at point of contact.

In real life, parts of the circle are used when making different materials:

1. **Bicycle wheel:** Center = hub, Radius = spoke length, Circumference = rim



2. **Clock face:** Center at middle, radii to numbers, circumference is edge



3. **Coin:** Circular shape with all points equidistant from center



4. **Kitchen garden**



Application activity 9.1

1. Explain these terms in your own words: radius, diameter, chord, tangent
2. In a circle with center O , $OA = 5\text{cm}$. How long is the radius? What's the maximum length of any chord?
3. True or False: Every diameter is a chord but not every chord is a diameter.
4. (MCQ) Which statement is ALWAYS true?
 - a) All chords are diameters
 - b) All diameters are chords
 - c) All radii are equal in length to the diameter
 - d) A tangent intersects the circle at two points
5. Draw and label a circle showing: center O , radius OA , diameter BC , chord DE , tangent at point F , and sector OGH .

9.2 Finding arc length and sector area



Activity 9.2

A pizza is cut into 6 equal slices. If the whole pizza has diameter 30cm:



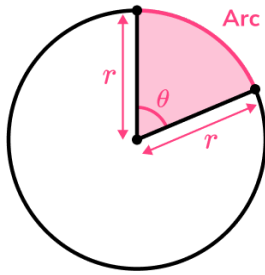
- a) How long is the curved edge of one slice?
- b) What's the area of one slice?
- c) What fraction of the whole pizza is each slice?



Content summary

- **Arc length (L)**

The arc of a circle is part of the circle's circumference depending on the angle at the centre of circle.

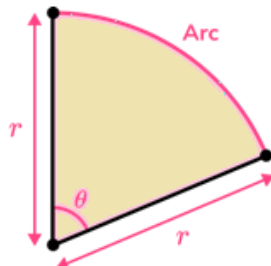


Its length can be found using the formula

$$L = \frac{\theta}{360^\circ} \times C \quad \text{OR} \quad L = \frac{\theta}{360^\circ} \times D\pi \quad \text{OR} \quad L = \frac{\theta}{360^\circ} \times 2r\pi \quad \text{where } \theta \text{ is the sector's central angle.}$$

- **Perimeter of a sector**

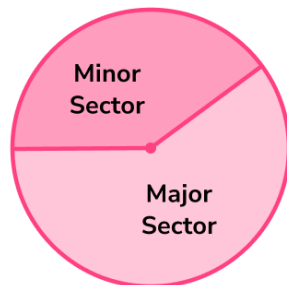
To find the perimeter of a sector you need to find the arc length L and then add it to the two radii (i.e. find the length around the edge of the sector).



$$\text{Perimeter of a sector} = L + 2r$$

Area of a sector

The area of a sector is a part of the area of the full circle.



It can be found by using the formula

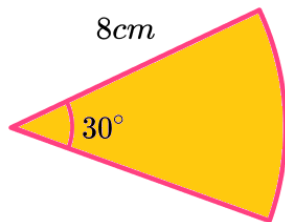
$$A_{\text{Sector}} = \frac{\theta}{360^\circ} \times A \quad \text{OR} \quad A_{\text{Sector}} = \frac{\theta}{360^\circ} \times \pi r^2 \quad \text{where } \theta \text{ is the sector's central angle.}$$

Note:

- **Both Arc Length and sector area are fractions** of whole circle based on central angle
- The unit of **Arc Length** is **linear unit** while the unit of **sector area** is **square unit**.

Example 1

Find sector area of a part of the circle of radius 8cm.



Solution

In this sector, $\theta = 30^\circ$, $r = 8 \text{ cm}$.

The area of the sector can be found using the formula.

$$A_{\text{Sector}} = \frac{30^\circ}{360^\circ} \times \pi \times 8^2 = 16.755 \text{ cm}^2$$

Example 2

Find the length of arc and sector area of a circle of radius 21cm and sector angle 60° . (Take

$$\pi \approx \frac{22}{7})$$

Solution

$$\text{Arc length}(L) = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 2 \times 21 = 22 \text{ cm}$$

$$A_{\text{Sector}} = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 21^2 = 231 \text{ cm}^2$$

Example 3

Pizza with $r = 15 \text{ cm}$ cut into equal 6 slices.

Find the area and perimeter of each slice



Solution

If it has 6 slices, each one has $360 \text{ degrees} \div 6 = 60 \text{ degrees}$.

$$\text{Each slice area} = \frac{60^\circ}{360^\circ} \times 3.14 \times 15^2 = 117.75 \text{ cm}^2$$

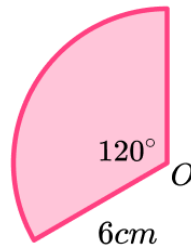
$$\text{Perimeter of a slice} = L + 2r = \left(\frac{60^\circ}{360^\circ} \times 3.14 \times 2 \times 15 \right) + 2 \times 15 = 45.7 \text{ cm}$$



Application activity 9.2

1. Find arc length when $r=10.5\text{cm}$ and $\theta=120^\circ$ ($\pi=22/7$)
2. Calculate area of sector with $r=7\text{cm}$ and $\theta=90^\circ$
3. A sector has area 38.5cm^2 and $r=7\text{cm}$. Find θ .
4. (MCQ) For same radius, if sector angle doubles, what happens to arc length?
a) Halves b) Doubles c) Quadruples d) Remains the same
5. Circle has circumference 88cm . Find arc length for 45° sector.
- 6.

Find the arc length and sector area of the sector shown below. Give your answer to 3 significant figures.



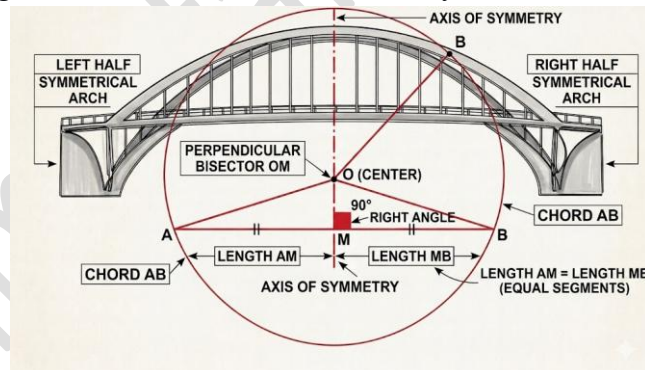
9.3 Properties of chords of a circle



Activity 9.3

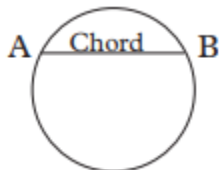
A bridge has circular arches. Engineers need to ensure the arch is symmetrical.

If they draw a perpendicular line from the center of the bridge to the base (chord), what do you notice about where it meets the base?



Content summary

Chord is the line segment joining two points on the circle



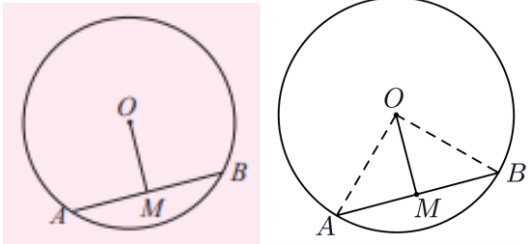
Properties of chords

- ❖ A perpendicular from center to chord bisects the chord (divide the chord into two equal parts).
- ❖ Line from center bisecting chord is perpendicular to chord

- ❖ If two chords of a circle are parallel, then the perpendicular bisector of one is also perpendicular bisector of the other.
- ❖ If two chords of a circle are equal, then they are equidistant from the centre.
- ❖ If two chords of a circle are equidistant from the centre, then their lengths are equal
- ❖ If two chords of a circle are equal, then the angles they subtend at the centre are equal.
- ❖ If two angles at the centre of a circle are equal, then they are subtended by equal chords.

Using right-angled triangles formed by radius, half-chord, and perpendicular distance

Figure shows a circle centre O. M is the midpoint of the chord AB. Show that OM is perpendicular to AB.

 Join OA and OB In Δs OMA and OMB, • $OA = OB$ (radii of the circle) • $AM = MB$ (given that M is midpoint of AB) • OM (common to both triangles) $\therefore \Delta$s OMA and OMB are congruent (sss) $\therefore \angle OMA = \angle OMB$. But these are adjacent angles on a straight line $\therefore \angle OMA$ is a right angle, i.e. OM is perpendicular to AB.	- Perpendicular Bisector Theorem: A line drawn from the center O perpendicular to a chord ($OM \perp AB$) always bisects that chord ($AM = MB$). - Congruence through Radii: By drawing radii from the center to the endpoints of the chord (OA and OB), two congruent right-angled triangles ($\Delta OMA \cong \Delta OMB$) are formed. - Equidistant Chords: Chords that are equal in length are equidistant from the center of the circle. - Right Angles: The intersection of a perpendicular line from the center to a chord always creates a 90° angle at the point of bisection (M).
---	---

Example 1

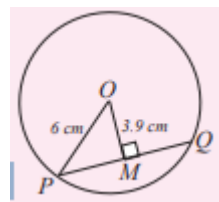
If chord $AB = 16\text{cm}$ and distance from O to $AB = 6\text{cm}$, find radius:

Solution

Half-chord $= 8\text{cm}$, $\text{radius}^2 = 8^2 + 6^2 = 100$, $r = 10\text{cm}$ using Pythagoras theorem.

Example 2

A circle has a radius of 6 cm. A chord of the circle is 3.9 cm from the centre of the circle. Find the length of the chord.



Solution

In Figure,

- is the centre of the circle,
- radius = 6 cm.
- the perpendicular distance from O to the chord OM = 3.9 cm

PQ is the chord whose length is required.

In $\triangle PMO$,

$$OP^2 = OM^2 + PM^2 \text{ (Pythagoras theorem)}$$

$$6^2 = 3.9^2 + PM^2$$

$$\Rightarrow PM^2 = 6^2 - 3.9^2 = 36 - 15.21 = 20.79$$

$$\Rightarrow PM = 4.56 \text{ cm}$$

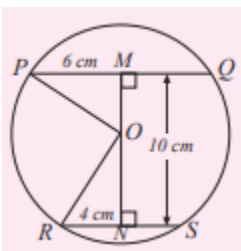
$$\therefore PQ = 2 \times PM = 2 \times 4.56 \text{ cm} = 9.12 \text{ cm}.$$

Example 3

Two parallel chords of a circle are of lengths 8 cm and 12 cm respectively and are 10 cm apart. What is the diameter of the circle?

Solution

Solution Consider Figure below:



In $\triangle PMO$: $PO^2 = PM^2 + MO^2$ (Pythagoras theorem).

In $\triangle RNO$:

$$RO^2 = RN^2 + NO^2 \text{ (Pythagoras theorem).}$$

$$PO = RO = r \text{ (radius)}$$

$$\text{Let } NO = x \Rightarrow MO = 10 - x$$

$$\text{Then } r^2 = 6^2 + (10 - x)^2 \text{ and } r^2 = 4^2 + x^2$$

$$\text{Hence } 16 + x^2 = 36 + (100 - 20x + x^2)$$

$$\Rightarrow 20x = 120$$

$$\text{i.e. } x = 6$$

$$\therefore r^2 = 4^2 + x^2 = 4^2 + 6^2 = 52 \Rightarrow r = 7.211$$

$$\therefore \text{Diameter} = 14.42 \text{ cm}.$$



Application activity 9.3

1. A chord of a circle of radius 9 cm is at a distance of 4 cm from the centre. What is the length of the chord?
2. Two chords are 8cm and 6cm from center. Which is longer? Explain.

3. (MCQ) In a circle, the perpendicular bisector of a chord:
 - a) Always passes through the center
 - b) Sometimes passes through the center
 - c) Never passes through the center
 - d) Is parallel to the chord
4. Prove that equal chords subtend equal angles at the center.
 1. $\angle AOB = \angle COD$

9.4 Theorem 1: Angles at the centre and at the circumference



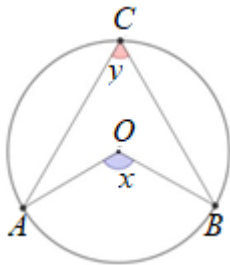
Activity 9.4

Draw a circle and mark an arc AB. Draw angle at center O ($\angle AOB$) and angle at circumference ($\angle ACB$) subtended by same arc AB. Measure them. What relationship do you notice? Try different positions for point C on circumference.



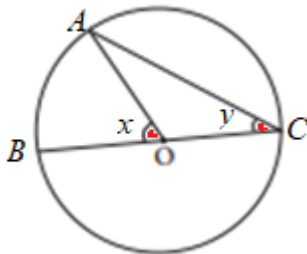
Content summary

In any circle the Angle formed at the centre of a circle is exactly twice the angle formed at the circumference by the same arc.



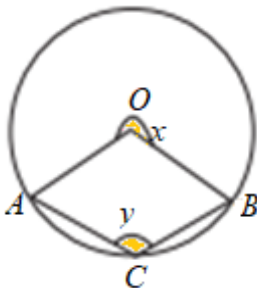
$$\angle AOB = 2\angle ACB$$

$$x = 2y$$



$$\angle AOB = 2\angle ACB$$

$$x = 2y$$

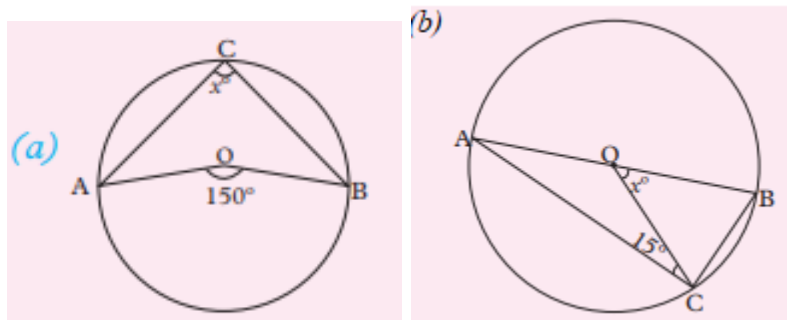


$$\angle AOB = 2\angle ACB$$

$$x = 2y$$

Example 1

Figures below show circles with their centres marked O. The line segments AB, BC and AC are chords of the circles in each case. Determine the size of the angles marked x, y and z.



Solution

(a) In Figure $\angle AOB$ is subtended at the centre while $\angle ACB$ is subtended at the circumference, both by arc AB (or chord AB).

$$\angle ACB = \angle x = \frac{\angle AOB}{2} = \frac{150^\circ}{2} = 75^\circ$$

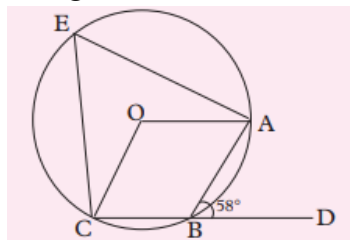
(b) In Figure $\angle COB$ and $\angle CAB$ are subtended by chord CB, at the centre and circumference respectively. OAC is isosceles hence $\angle ACO = \angle CAB = 15^\circ$

$$\angle COB = x = 2\angle CAB$$

$$\therefore \angle COB = 15^\circ \times 2 = 30^\circ$$

Example 2

In Figure, $\angle ABD = 58^\circ$ and CBD is a straight line. Calculate: (a) obtuse $\angle AOC$, (b) $\angle AEC$.



Solution

(a) $\angle ABC = 180^\circ - 58^\circ = 122^\circ$ (Angles on a straight line.)

Now reflex $\angle AOC = 122^\circ \times 2 = 244^\circ$ (Angle subtended at the centre by major arc AC.)

Thus, Obtuse $\angle AOC = 360^\circ - 244^\circ = 116^\circ$ (Angles at a point.)

(b) $\angle AEC$ is subtended at the circumference by the minor arc ABC.

$$\therefore 2 \angle AEC = \text{obtuse } \angle AOC = 116^\circ$$

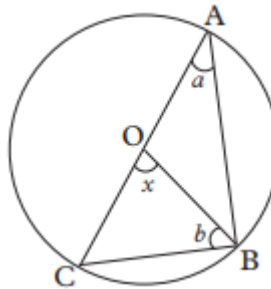
$$\angle AEC = 116^\circ / 2 = 58^\circ.$$



Application activity 9.4

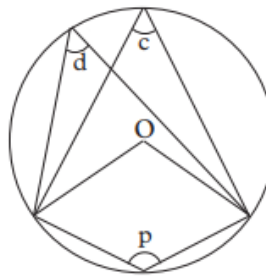
1) In Figure, O is the centre and AC is the diameter.

- (a) If $a = 24^\circ$, find x .
- (b) If $x = 62^\circ$, find b .
- (c) If $a = 52^\circ$, find b .
- (d) If $b = 28^\circ$, find a .
- (e) Express x in terms of a .
- (f) Express b in terms of a .



...

2) In Figure O is the centre of the circle. If $c = 47^\circ$, find d and p .

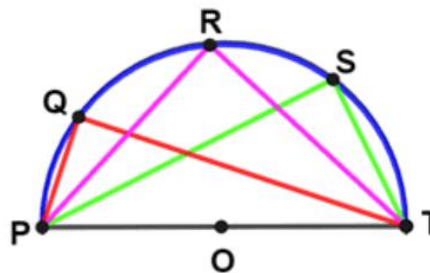


9.5 Theorem 2: Angles in a semicircle



Activity 9.5

Carpenters building arched windows often use the "3-4-5 triangle method" to ensure right angles. Why does this work? Draw semicircle with diameter PT as base. Place the point R anywhere on arc and join to ends of diameter. What type of angle PRT is formed?

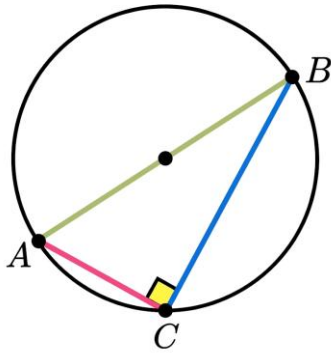


..



Content summary

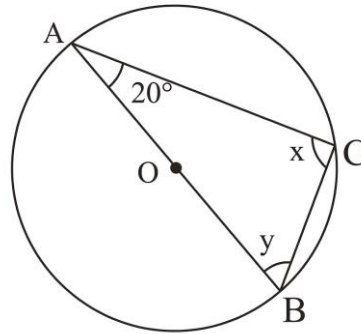
The angle subtended by the diameter at any point on the circumference of a circle is a right angle (90°).



If AB is diameter, and C is any point on circumference, then $\angle ACB = 90^\circ$.

Example 1

In the figure, calculate giving reasons, the angles marked with letters if O is the centre of the circle.



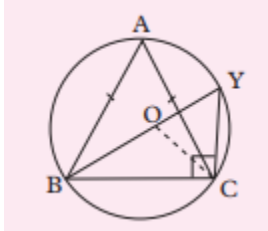
Solution

$\angle x = 90^\circ$, angle subtended by the diameter on the circumference. $\angle y = 180^\circ - (x + 20)$, sum of angles in a triangle add up to 180° . $\angle y = 180^\circ - 110^\circ = 70^\circ$

Example 2

Figure shows a circle centre O, diameter BOY. Triangle ABC is isosceles and $\angle ACB = 63^\circ$. Use the given information to find angles:

- (a) BAC (b) ACY,
(c) CBY (d) ABY (e) BYC



Solution

(a) Using $\triangle ABC$, $\angle ABC = \angle ACB = 63^\circ$

$$\therefore \angle BAC = 180 - (\angle ABC + \angle ACB)$$
$$= 180^\circ - 126^\circ = 54^\circ$$

$$\therefore \angle BAC = 54^\circ$$

(b) Using $\triangle BCY$,

$$\angle BCY = 90^\circ \dots \text{angle on a semicircle}$$

$$\angle ACY = \angle BCY - \angle ACB$$

$$= 90^\circ - 63^\circ = 27^\circ$$

$$\therefore \angle ACY = 27^\circ$$

(c) Join O to C so that BOC is an isosceles triangle.

$$\text{Thus: } \angle BOC = 2\angle BAC = 2 \times 54^\circ$$
$$= 108^\circ$$

$$\therefore \angle CBO = \angle BCO$$

$$= \frac{180^\circ - 108^\circ}{2} = \frac{72^\circ}{2}$$

$$= 36^\circ$$

$$\angle CBO = \angle CBY = 36^\circ$$

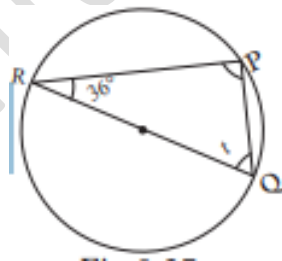
$$(d) \angle ADY = 63^\circ - 36^\circ = 27^\circ$$

$$(e) \angle BYC = 180^\circ - (90^\circ + 36^\circ) = 180^\circ - 126^\circ = 54^\circ$$

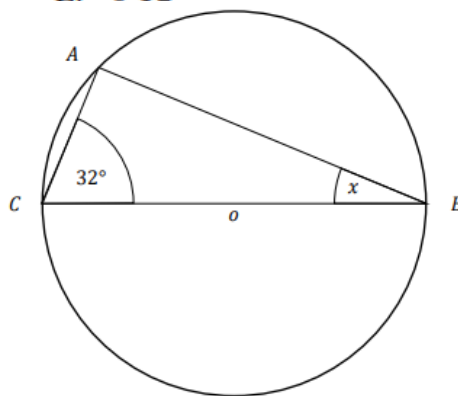


Application activity 9.5

1) Calculate the angle marked t in Figure



2) Points A , B , and C lie on the circumference of a circle. The line BC passes through the centre of the circle, O . Calculate the angle x , giving your reasoning for each step.



9.6 Theorem 3: Angles in the same segment



Activity 9.6

Draw circle with chord AB . Place points C and D on same arc AB (same side of chord). Measure $\angle ACB$ and $\angle ADB$.

What do you notice?

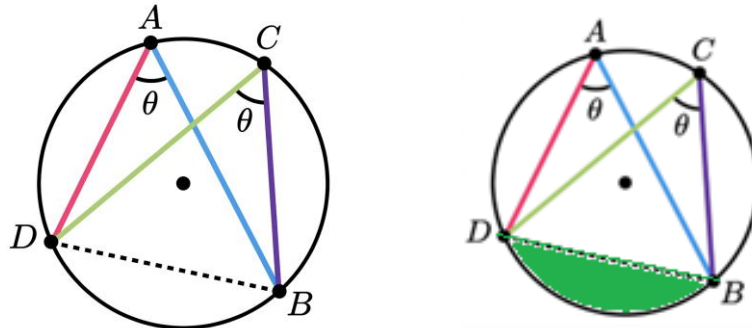
Move C and D to different positions on same arc. Does relationship hold?



Content summary

A **segment of a circle** is a region of the circle's interior bound by an **arc** and a **chord**. The region DB is a segment.

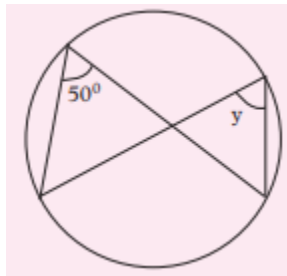
Angles subtended on the circumference by the same chord in the same segment are equal.



$\angle DAB = \angle DCB$ (A and C on same arc DB, on same side of chord DB)

Example 1

In Figure below, find the size of angle y

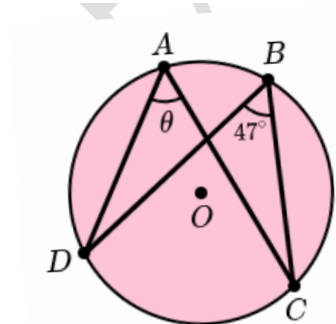


Solution

Angle $y = 50^\circ$ Angle obtained by the same arc at the circumference are equal.

Example 2

Below is a circle with centre O. AC and BD are chords. Calculate the size of angle CAD.



Solution

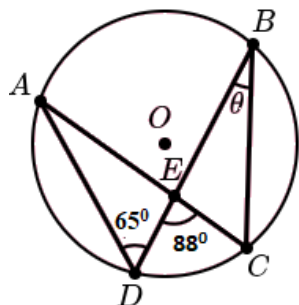
Angle DAC is equal to the angle DBC:

$$\angle DAC = \angle DBC$$

$$\theta = 47^\circ$$

Example 3

A, B, C, and D are points on a circle with centre O. Calculate the size of the angle CBD.



Solution

$\angle DAC = \angle DBC = \theta$; $\angle ADB = \angle ACB = 65^\circ$ angles in the same segment.

$\angle AEB = \angle DEC = 88^\circ$ vertically opposite angles. Similarly, $\angle AED = \angle BEC = x$

Remember angles at centre add up to 360° . Then

$$\angle AEB + \angle DEC + \angle AED + \angle BEC = 360^\circ$$

$$88^\circ + 88^\circ + x + x = 360^\circ$$

$$x = 92^\circ$$

By considering triangle BEC

$$92^\circ + \theta + 65^\circ = 180^\circ$$

$$\theta = 180^\circ - 157^\circ = 23^\circ$$

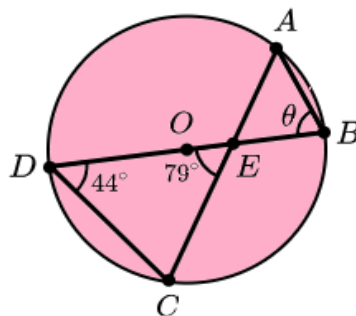


Application activity 9.6

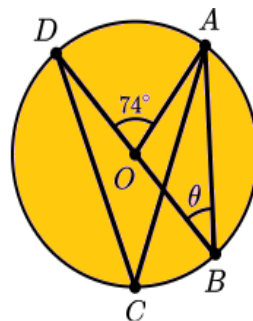
1) (MCQ) If two angles subtended by same chord are equal, the vertices must lie:

- a) On same side of chord b) On opposite sides of chord
- c) Inside the circle d) Outside the circle

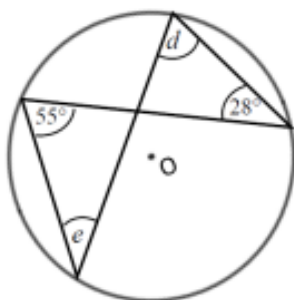
2) The circle with centre O has four points on the circumference. The two chords AC and BD intersect at the point E. Calculate the size of angle ABE.



3) A, B, C, and D are points on the circumference of the circle with centre O. BOD is a straight line. Calculate the size of angle ABD.



4) Work out the size of each angle marked with a letter. Give reasons for your answers



9.7 Theorem 4: Angles in a cyclic quadrilateral



Activity 9.7

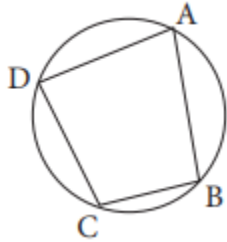
Using instructions given,

- Draw a circle centre O using any convenient radius.
- On the circumference, mark points A, B, C and D in that order and join them to form a quadrilateral.
- Measure angles ABC and ADC. Find their sum.
- Measure angles BAD and BCD. Find their sum.
- What do you notice about the two sums in 3 and 4?
- Are the pairs of angles in 3 and 4 adjacent or opposite?
- Produce side AB of the quadrilateral, and measure the exterior angle so formed. How does the size of this angle compare with that of interior $\angle ADC$?



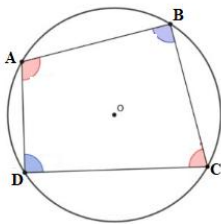
Content summary

A **cyclic quadrilateral** is a quadrilateral whose vertices all lie on a circle.



Theorem states that:

- *Opposite interior angles are supplementary (add up to 180°).*

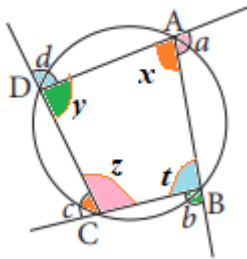


$$\angle A + \angle C = 180^\circ$$

$$\angle B + \angle D = 180^\circ$$

- *Exterior angle = interior opposite angle*

The figure shows exterior angles a , b , c and d of the quadrilateral $ABCD$.



$$a = z$$

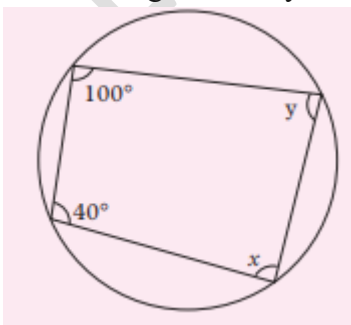
$$c = x$$

$$b = y$$

$$d = t$$

Example 1

Find the angles x and y in the figure below



Solution

Solution

$\angle x = 180^\circ - 100^\circ$ (Opposite angles in a cyclic quadrilateral add up to 180° .)

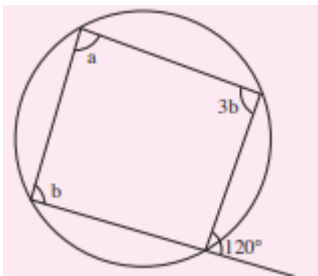
$$x = 80^\circ$$

$y = 180^\circ - 40^\circ$ (Opposite angles in a cyclic quadrilateral add up to 180° .)

$$y = 140^\circ$$

Example 2

Find angles a and b in the Fig 9.49 below.

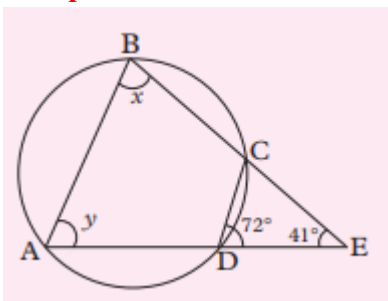


Solution

$a = 120^\circ$ (Exterior angle of a cyclic quadrilateral is equal to opposite interior angle.)

$b + 3b = 180^\circ$, opp. angles of a cyclic quadrilateral. $4b = 180^\circ$ $b = 45^\circ$

Example 3



In the Figure, find the values of x and y

Solution

$\angle ADC = 180^\circ - 72^\circ = 108^\circ$ (Angles on a straight line).

$x + 108^\circ = 180^\circ$ (Opposite angles of a cyclic quadrilateral)

$\therefore x = 180^\circ - 108^\circ = 72^\circ$. Note that $\angle ABC$ is the interior angle opposite to the exterior angle at D.

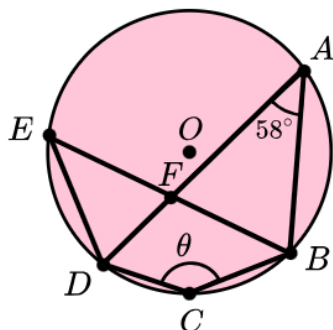
$\angle DCE = 180^\circ - (72^\circ + 41^\circ)$ (angle sum of a triangle) $= 180^\circ - 113^\circ = 67^\circ$

$\therefore y = \angle DCE = 67^\circ$ (exterior angle equals interior opposite angle).

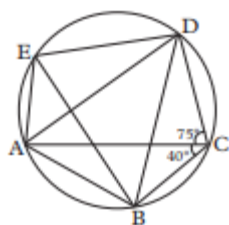


Application activity 9.7

1) A circle with centre O has points A, B, C, D , and E on the circumference. Angle $BAF = 58^\circ$. Calculate the size of the angle BCD .

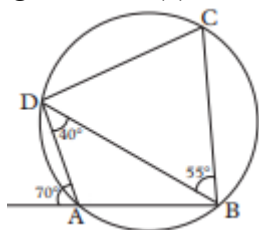


2) A, B, C, D and E are five points, in that order, on the circumference of a circle

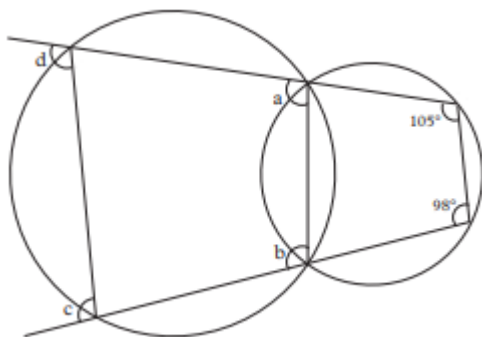


- Write down all angles in the figure equal to $\angle ACB$.
- Write down all angles in the figure supplementary to $\angle BCD$.
- If $\angle ACB = 40^\circ$ and $\angle ACD = 75^\circ$, find the size of $\angle DEB$.

3) In Figure, find: (a) $\angle BCD$ (b) $\angle CDA$.



4) Figure consists of two intersecting circles. Use it to find the angles marked by letters.



9.8 Theorem 5: Tangents to a circle

9.8.1 Meaning of tangent to a circle



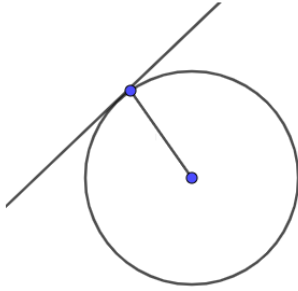
Activity 9.8.1

- Draw a circle, centre O, using any radius.
- Draw a line OB through any point A on the circumference, with B outside the circle.
- At A, construct a line PQ perpendicular to OB.



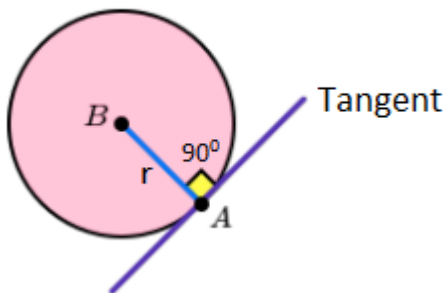
Content summary

A **tangent** of the circle is a line which has one, and only one point in contact with a circle.



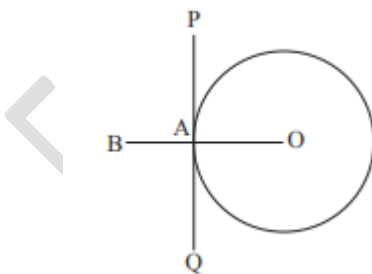
Theorem on Tangent states that

- A tangent to a circle is perpendicular to the radius drawn through the point of contact.
- Conversely stated the perpendicular to a tangent at its point of contact passes through the centre of the circle.



The angle between a tangent and radius is 90 degrees.

To construct a tangent to a circle, we use the fact that a tangent is perpendicular to the circle at the point of contact.

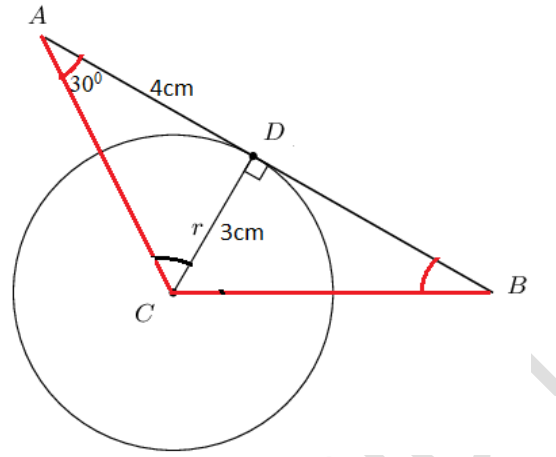


The line PQ is a tangent to the circle at A.

Example 1

In the figure, CD is a radius, AB is tangent to circle of centre C.

- Find the length AC.
- Find the angle ACD.



Solution

Triangle ADC is right angled triangle at point D.

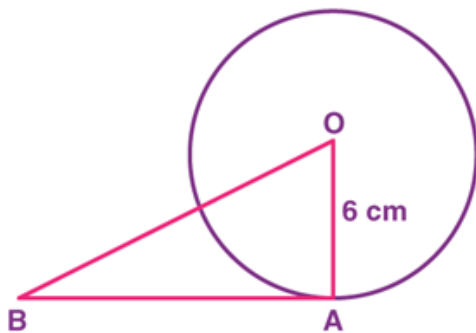
- So, using **Pythagoras theorem:** $AC^2 = AD^2 + CD^2$

$$AC = \sqrt{4^2 + 3^2} \Rightarrow AC = 5\text{ cm}$$

- angle ACD = $180^\circ - (90^\circ + 30^\circ) = 60^\circ$

Example 2

AB is a tangent to a circle with centre O at point A of radius 6 cm. It meets the line OB such that OB = 10 cm. What is the length of AB?



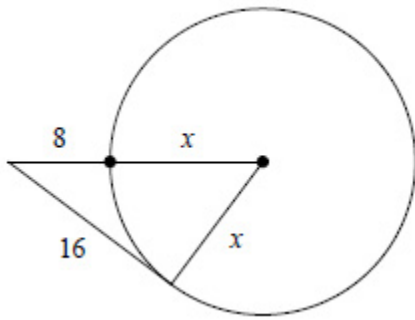
Solution

Since the radius is perpendicular to tangent, then triangle OAB is right angled triangle

Using Pythagoras Theorem: $AB = \sqrt{10^2 - 6^2} \Rightarrow AB = 8\text{ cm}$

Example 3

In the figure; find the value of x.



Solution

Remember, radius is perpendicular to tangent.

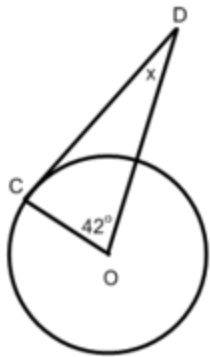
$$(x + 8)^2 = 16^2 + x^2$$

$$x^2 + 16x + 64 = 256 + x^2$$

$$x = 12$$

Example 4

Find the angle at point D.



Solution

Since the radius is perpendicular to tangent, then triangle OAB is right angled triangle at C

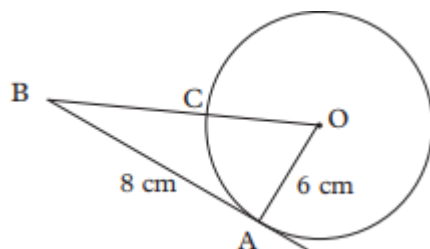
$$\text{So, } 42 + 90 + x = 180^\circ \quad x = 48^\circ$$



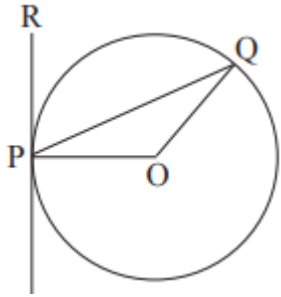
Application activity 9.8.1

1) In the Fig. 9.60 below, O is the centre of the circle while A and C are points on the circumference of the circle. BCO is a straight line and BA is a tangent to the circle.

AB = 8cm and OA = 6cm.

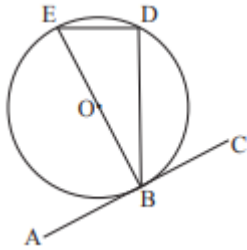


- (a) Explain why $\angle OAB$ is a right angle.
 (b) Find the length BC.
- 2) The Figure shows a circle, centre O. PR is a tangent to the circle at P and PQ is a chord.



Calculate:

- (a) $\angle RPQ$ given that $\angle POQ = 85^\circ$.
 (b) $\angle RPQ$ given that $\angle PQO = 26^\circ$.
 (c) $\angle POQ$ given that $\angle RPQ = 54^\circ$.
 (d) $\angle POQ$ given that $\angle QPO = 17^\circ$.
- 3) In Figure, ABC is a tangent and BE is a diameter to the circle.



Calculate:

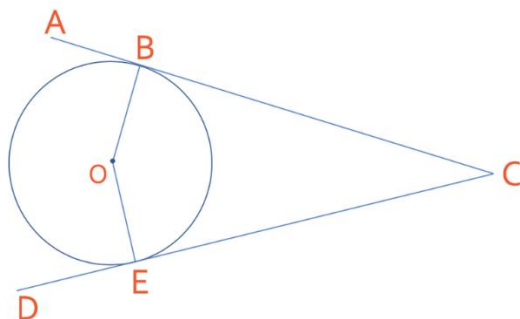
- (a) $\angle EBD$ if $\angle CBD = 33^\circ$.
 (b) $\angle BED$ if $\angle ABD = 150^\circ$.
 (c) $\angle DBC$ if $\angle DEB = 65^\circ$.
 (d) $\angle ABD$ if $\angle BED = 38^\circ$.
- 4) A tangent is drawn from a point 17 cm away from the centre of a circle of radius 8 cm. What is the length of the tangent?

9.8.2 Tangents to a circle from a common point



Activity 9.8.2

- Draw a circle of any radius OB and centre O.



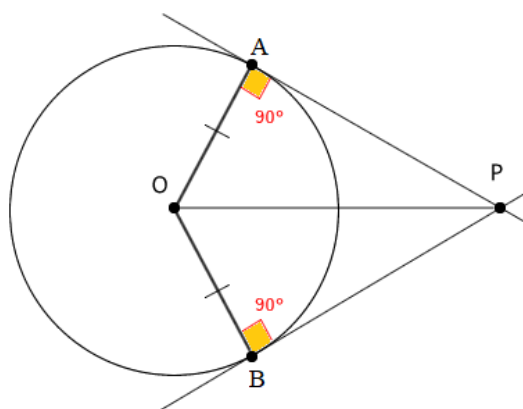
- Mark a point C outside the circle.
- Join OC.
 - a) How are the distance BC and CE?
 - b) Compare angles BCO and ECO.



Content summary

If two tangents are drawn to a circle from a common point:

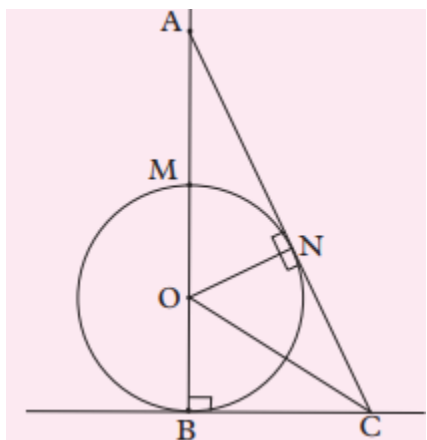
- (a) *The tangents are equal;*
- (b) *The tangents subtend equal angles at the centre;*
- (c) *The line joining the centre to the common point bisects the angles between the tangents.*



$PA = PB$ (where PA, PB tangents from P to points A, B on circle)

Example 1

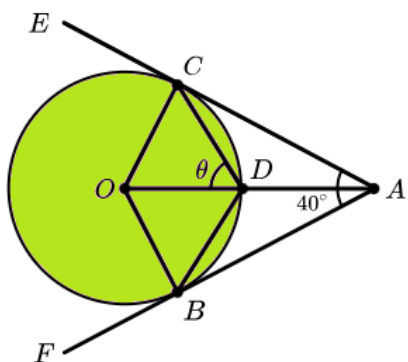
In the figure below, triangle ABC is the tangent to the circle of centre O at two points N and B. Length AM = 6 cm, length BC = 18 cm and the radius of the circle is 9 cm.



Find the following: (a) length AN (b) length AC (c) length OC

Example 2

A circle with centre O has three points on the circumference, B, C and D. The lines AE and AF are tangents to the circle at the points B and C. The tangents meet at the point A. Angle $BAC = 40^\circ$. Calculate the size of the angle ODC.



Solution

In the figure, triangles OAC and OAB are right triangles and equal.

Let consider right triangle OAC, angle $OCA = 90^\circ$, angle $OAC = 20^\circ$ then angle

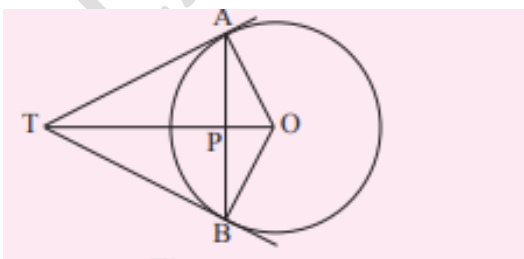
$$\text{Angle } AOC = 180^\circ - (90^\circ + 20^\circ) = 70^\circ$$

Let consider also, isosceles triangle ODC, sides $OC = OD$ and $\angle OCD = \angle ODC = \theta$

$$\text{Therefore } \theta + \theta + 70^\circ = 180^\circ, \text{ and } \theta = \frac{110^\circ}{2} = 55^\circ.$$

Example 3

In Figure, TA and TB are tangents to the circle, centre O. If $\angle ABO = 28^\circ$, what is the size of $\angle ATO$?



Solution

In $\triangle ABO$,

$$\angle ABO = \angle BAO = 28^\circ \text{ (isosceles } \triangle)$$

$$\therefore \angle AOB = 180^\circ - 56^\circ = 124^\circ$$

$$\angle AOT = \frac{1}{2} \angle AOB = 62 \text{ degrees.}$$

Tangents subtend equal angles at the centre of circle.

$$\text{In } \triangle ATO, \angle OAT = 90^\circ$$

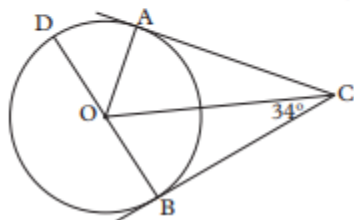
(Tangent is perpendicular to radius.)

$$\therefore \angle ATO = 90^\circ - \angle AOT = 90^\circ - 62^\circ = 28^\circ$$

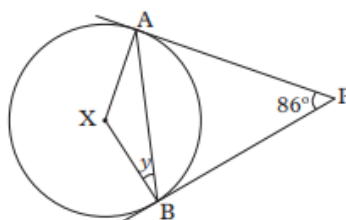


Application activity 9.8.2

1. In Fig 9.67 below, A, B and D are points on the circumference of a circle centre O. BOD is the diameter of the circle while BC and AC are the tangents of the circle. Angle $OCB = 34^\circ$. Work out the size of angle DOA.



2. In Fig 9.68 A and B are points on the circumference of a circle, centre X. PA and PB are tangents to the circle. Angle $APB = 86^\circ$. Find the size of angle y .



9.8.3 Angles in alternate segment



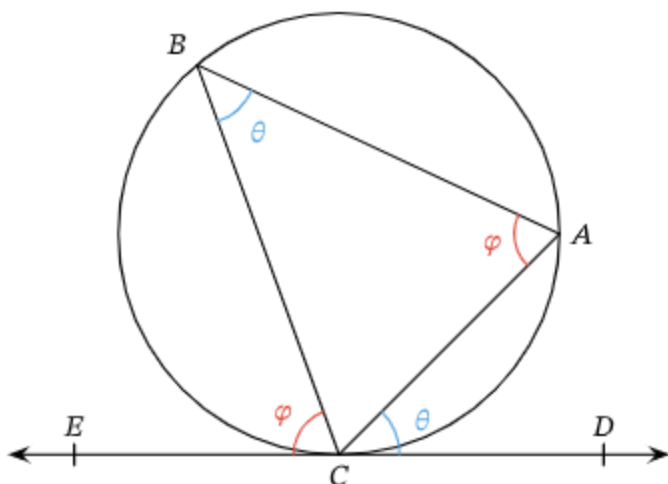
Activity:9.8.3

- 1) Draw a circle of any radius.
- 2) Draw a tangent at any point B.
- 3) Draw a chord BD
- 4) Mark points P, Q, R on the circumference in the same segment. Join BP, BQ, BR, DP, DQ and DR.
- 5) Measure angles BPD, BQD and BRD. What do you notice?



Content summary

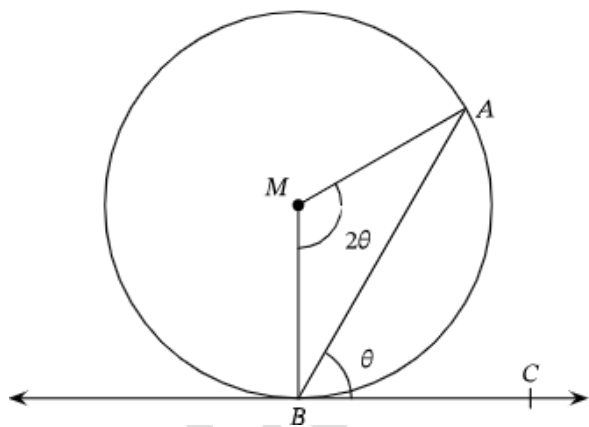
Let A and B be two points on a circle and C be the point where a tangent (passing through E, C, and D) intersects the circle. Then, the angles of tangency $\angle ACD$ and $\angle BCE$ are equal to the angles in the alternate segments $\angle ABC$ and $\angle BAC$ respectively. This is shown below.



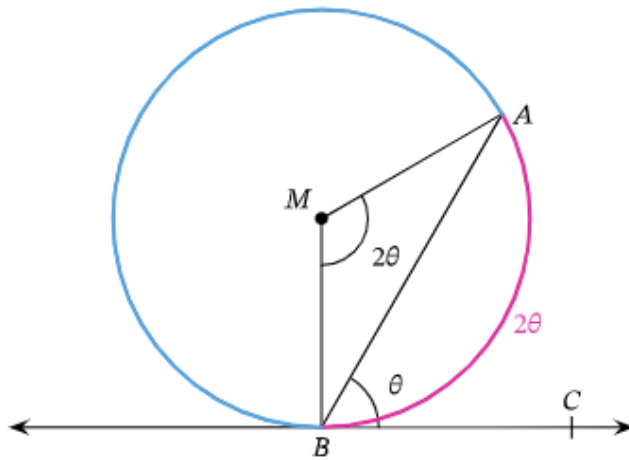
Thus, **theorem** states that the *angle between a tangent and a chord through the point of contact is equal to the angle subtended by the chord in the alternate segment*: $\angle DCA = \angle CBA = \theta$
 Similarly $\angle ECB = \angle CAB = \varphi$.

Note: Angles of Tangency and Central Angles

The angle of tangency $\angle ABC$ is half of the central angle $\angle AMB$. This is shown below.

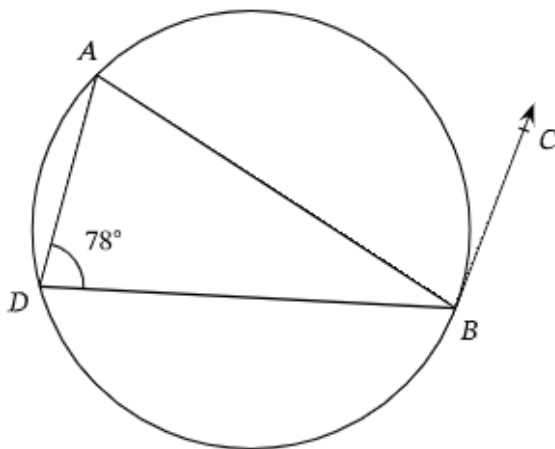


In other words, since both the measure of the central angle and the arc measure are 2θ , they are both twice the angle of tangency.



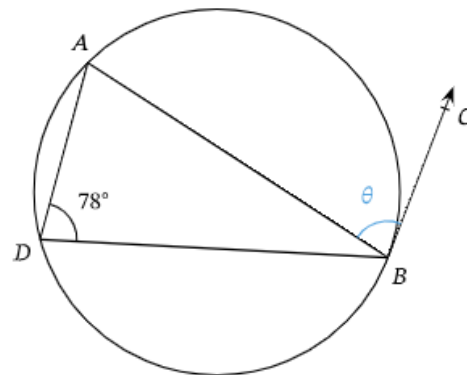
Example 1

Given that BC is a tangent to the circle, find $\angle ABC$.



Solution

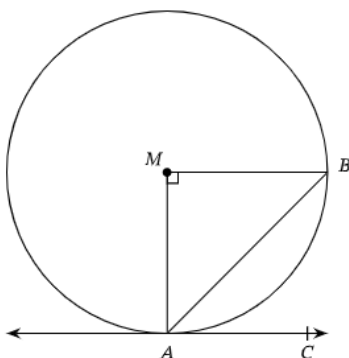
Let us first of all mark the angle we want to find on the diagram with θ .



Since $\angle ADB$ is the angle in the alternate segment to $\angle ABC$, we directly use the theorem to find that $\angle ABC = 78^\circ$.

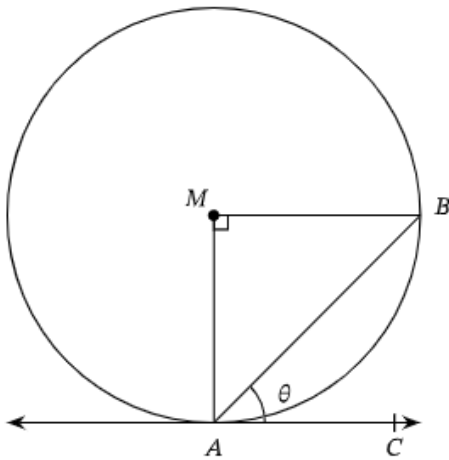
Example 2:

In the figure, find $\angle BAC$.



Solution

Let us begin by marking the angle we have been asked to find on the diagram:



We can see that we have been asked to find the measure of an angle of tangency $\angle BAC$.

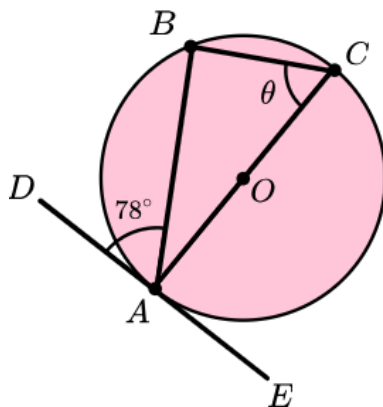
Note that the angle $\angle AMB$ on the diagram is a right angle (90°) because it is marked with a square. We also note that $\angle AMB$ is the central angle subtended by the same arc (i.e., AB) as the angle of tangency.

Recall that the measure of the angle of tangency $\angle BAC$ is equal to half the measure of the central angle subtended by the same arc, which is $\angle AMB$.

Thus, we have $\angle BAC = \frac{1}{2}\angle AMB = \frac{1}{2}(90^\circ) = 45^\circ$

Example 3

The right triangle ABC is inscribed in the circle with centre O . The line DE is a tangent to the circle with the point of contact at A . Angle $BAD = 78^\circ$. Calculate the size of the angle ABC .

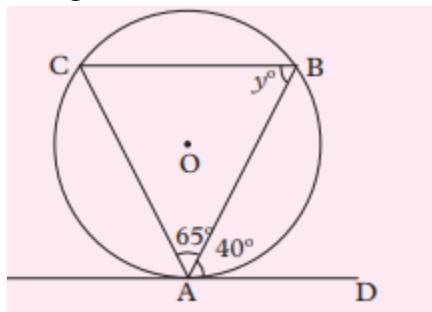


Solution

Since $\angle DAB$ is the angle in the alternate segment to $\angle ACD$, we directly use the theorem to find that $\angle DAB = 78^\circ$.

Example 4

In Figure below, $\angle BAD = 40^\circ$ and $\angle BAC = 65^\circ$. What is y ?



Solution

According to the tangent-chord theorem;

$\angle ACB = \angle BAD = 40^\circ$ But $\angle ACB + \angle BAC + \angle ABC = 180^\circ$ (Angle sum of a triangle.)

$\therefore \angle ABC = 180^\circ - \angle ACB - \angle BAC$

$\angle ABC = 180^\circ - \angle BAD - \angle BAC$ $\angle ABC = 180^\circ - 40^\circ - 65^\circ$

$\angle ABC = 180^\circ - 105^\circ$

$\angle ABC = 75^\circ$.

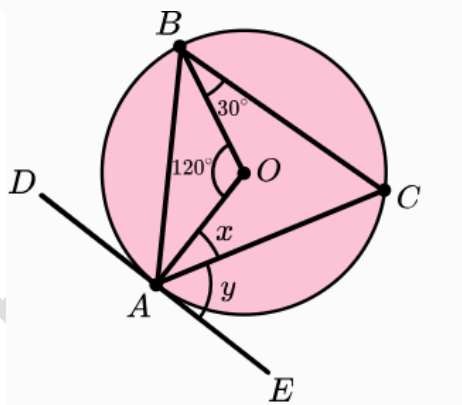
Therefore, $y = 75^\circ$



Application activity 9.8.3

1) (a) ABC is a triangle inscribed in a circle with centre O. The tangent DE passes through the point C. Calculate the size of angle x. Explain your answer.

(b) Hence or otherwise, calculate the value of y. Give a reason for your answer.



2, 3, and 4):

1. If $\angle BAD = 19^\circ$, find $\angle ACB$ in Fig 9.77.

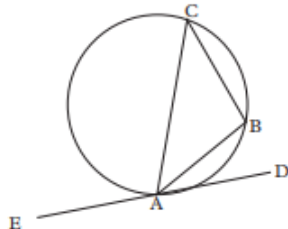
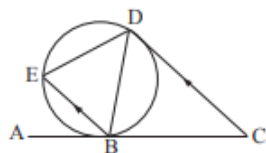


Fig. 9.77

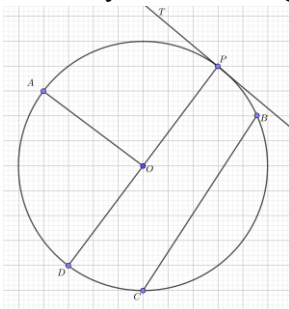
2. In Fig. 9.78, AC is a tangent to the circle and $BE \parallel CD$.



END OF UNIT ASSESSMENT

Section A: Knowledge and Understanding

1. Identify the following parts on the diagram provided



- A chord
- A sector
- A tangent
- The radius

- State the relationship between a radius and a tangent at their point of contact.
- State the Circle Theorem that each of these diagrams represents.

- If $\angle ABE = 42^\circ$, $\angle BDC = 59^\circ$, find $\angle BED$.
- If $\angle DBE = 62^\circ$, $\angle BCD = 56^\circ$, find $\angle BED$.

3. In Fig. 9.79, PR is a tangent to the circle.

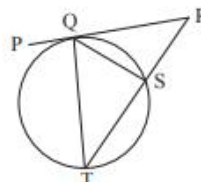
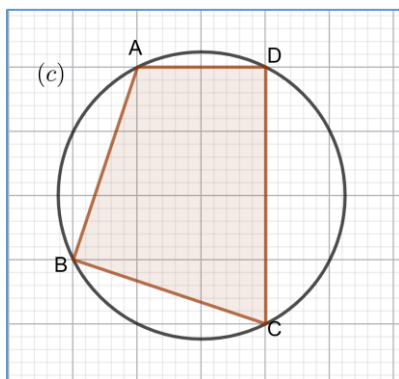
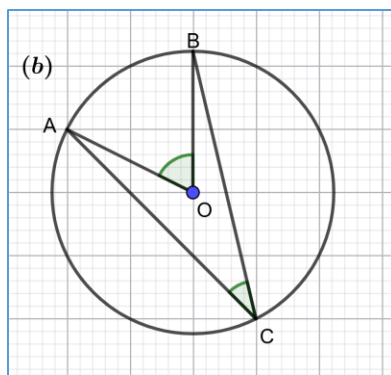
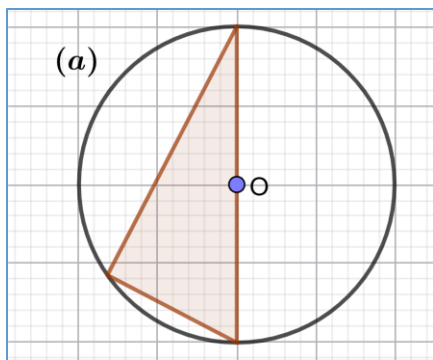


Fig. 9.79

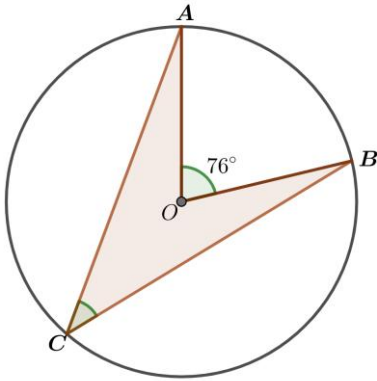
- If $\angle PQT = 66^\circ$, find $\angle QST$.
- If $\angle QTS = 38^\circ$ and $\angle QRS = 30^\circ$, find $\angle QST$.
- If $\angle QTS = 35^\circ$ and $\angle TQS = 58^\circ$, find $\angle QRS$.
- If $\angle PQT = 50^\circ$ and $\angle PRS = 30^\circ$, find $\angle SQT$.



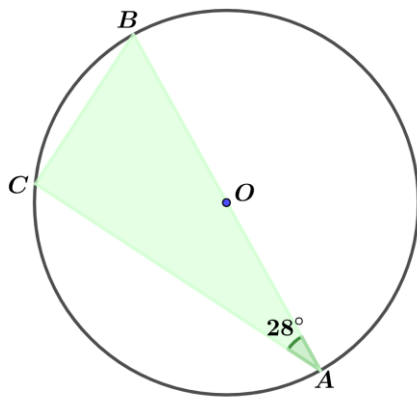
4. The formula for the circumference of a circle is $C=2\pi r$. What does the variable r represent?
5. **True or False:** A line drawn from the centre of a circle perpendicular to a chord will always pass through the midpoint of the chord.
6. What is the measure of the angle in the alternate segment in relation to the angle between the tangent and the chord?

Section B: Skills

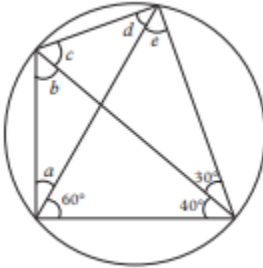
7. A sector of a circle of radius 14 cm has a central angle of 120° . Calculate:
 - a) The length of its arc.
 - b) The area of the sector.
8. In a circle with centre O, a chord AB is 24 cm long. The perpendicular distance from O to AB is 5 cm. What is the length of the radius of the circle?
9. In the diagram, O is the centre. Angle $AOB = 76^\circ$. Find the size of angle ACB.



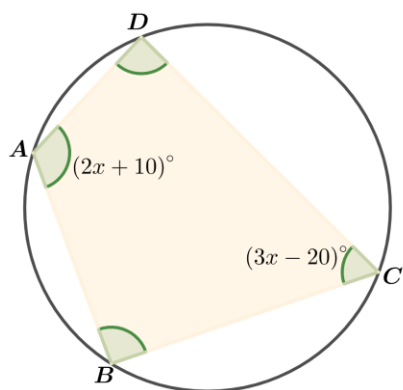
10. In the diagram, AB is a diameter. Angle $BAC = 28^\circ$. Find the size of angle ACB.



11. In Figure, find the angles marked with letters a, b, c, d and e.



12) ABCD is a cyclic quadrilateral. Angle $A = (2x + 10)^\circ$ and angle $C = (3x - 20)^\circ$. Find the value of x and the measure of angle C.



Section C: Attitudes and Values

13. Explain why circle theorems are important in solving geometric problems accurately.
14. While solving a difficult geometry problem involving several circle theorems, what attitudes should a learner develop to succeed?



UNIT 10: COLLINEAR POINTS AND ORTHOGONAL VECTORS IN 2D

Key unit competence: Apply properties of collinearity and orthogonality in solving problems involving vectors.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and Understanding

- Define collinearity for points and orthogonality for vectors and state their mathematical conditions.
- Explain how vectors provide an analytical framework for geometric proofs.

Skills

- Determine collinearity and orthogonality using vector methods, including the dot product.
- Apply these properties to solve problems, such as verifying shapes or finding coordinates.

Attitudes and Values

- Value the clarity and precision that vectors bring to geometric reasoning.
- Collaborate effectively to test and verify geometric conjectures.

Introduction

In the study of two-dimensional geometry, two key concepts provide powerful tools for analysing spatial relationships: collinear points and orthogonal vectors.

Collinear points are sets of points that lie perfectly on a single straight line, sharing a common direction, which allows us to understand alignment and linear paths in a plane.

Orthogonal vectors, on the other hand, describe pairs of vectors that meet at a perfect right angle (90 degrees), representing the idea of perpendicular and independent directions.

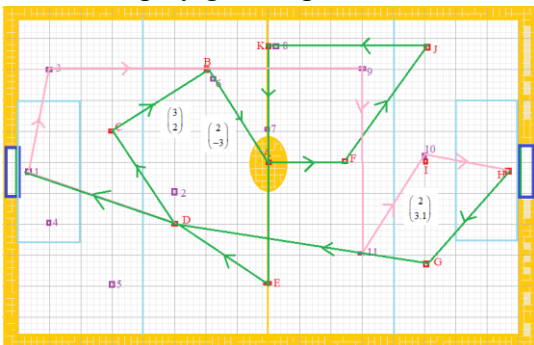
These ideas are deeply connected, as the condition for orthogonality (the dot product of two vectors is zero) can also be used to test if lines formed by points are perpendicular.

Together, these fundamental principles form a cornerstone for solving problems in coordinate geometry, physics, and computer graphics, where determining alignment or calculating projections are essential tasks.



Introductory activity

Look at the play ground picture below:



a) Measure the angle formed by vector CB and BA . What measure do you find? are the vectors perpendicular?

b) If a player I passes the ball to F , then F passes the ball to D ; what do you think on this pathway. Is this straight line or not? Explain.

c) Regardless the team members. Can you identify three players stand on a straight line?

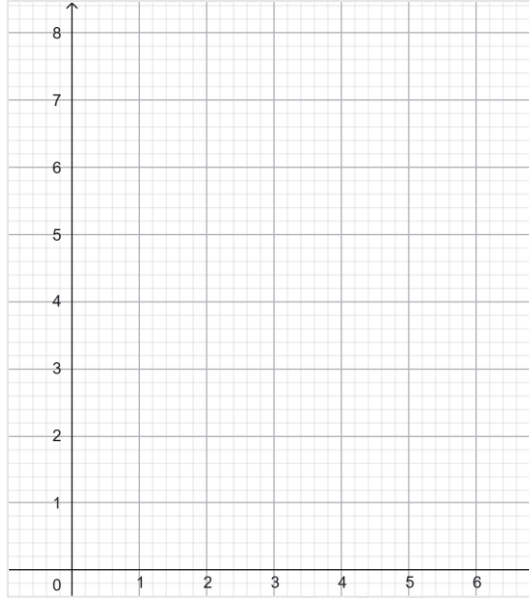
10.1 Collinear points



Activity 10.1

Imagine you are a civil engineer checking the plan for a new water pipeline.

Three houses, located at points A, B, and C on the map, need to be connected by a straight pipeline. You plot their coordinates: House A: (1, 2)
House B: (3, 5)
House C: (5, 8)



Before construction begins, you need to verify whether these three houses lie on the same straight line, so the pipeline can be built without unnecessary turns.

Question: How can you determine mathematically if these three points are perfectly aligned?



Content summary

Collinear points are points that lie on the same straight line.

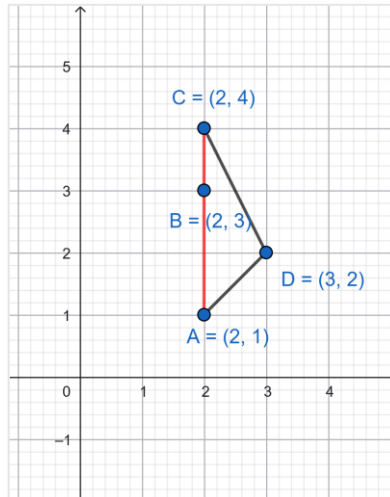
Given points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ they are collinear if:

Method 1: Graphical Method

Plot the points

$A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ on the graph and Verify if they all lie on the same straight line.

On the graph, we see that A, B and C are collinear but A, D and B are not collinear.



Method 2: Slope Method

Slope of AB = Slope of BC = Slope of AC

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}, m_{BC} = \frac{y_3 - y_2}{x_3 - x_2}$$

If $m_{AB} = m_{BC}$, then A, B, C are collinear.

Method 3: Vector method

Points A, B, C are collinear if $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are parallel.

$$\overrightarrow{AB} = k \overrightarrow{BC} \text{ where } k \text{ is a scalar.}$$

Since B is a common point between vectors AB and BC , then A, B and C lie on a straight line, i.e. the points A, B and C are collinear.

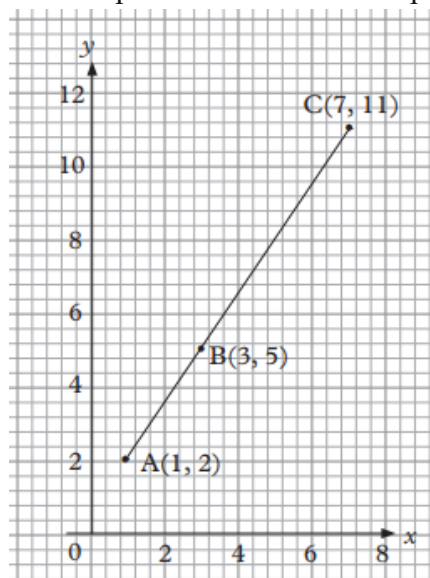
Example 1

Show that the points $A(1, 2)$, $B(3, 5)$ and $C(7, 11)$ are collinear.

Solution

Using graphical method

Plot the points in the Cartesian plane



Since all the points A, B and C lie on straight line, then the points are collinear.

Example 2

Are $P(1, 2)$, $Q(3, 5)$, $R(5, 8)$ collinear?

Solution

By **Slope method**:

$$m_{PQ} = \frac{5-2}{3-1} = \frac{3}{2}$$

$$m_{QR} = \frac{8-5}{5-3} = \frac{3}{2}$$

Since the Slopes are equal then the points P, Q, R are collinear.

Example 3

Verify whether the points $A(-1, 3)$, $B(1, 1)$, and $C(3, -1)$ are collinear.

Solution

Vector method:

$$\vec{AB} = \begin{pmatrix} 1 - (-1) \\ 1 - 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

$$\vec{BC} = \begin{pmatrix} 3 - 1 \\ -1 - 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

Since, $\vec{AB} = \vec{BC} \Rightarrow$ Collinear.

Example 4

Show that the points A (0, -2), B (2, 4) and C (-1, -5) are collinear.

Solution

$$\vec{AB} = \begin{pmatrix} 2 - 0 \\ 4 - (-2) \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

$$\vec{BC} = \begin{pmatrix} -1 - 2 \\ -5 - 4 \end{pmatrix} = \begin{pmatrix} -3 \\ -9 \end{pmatrix}$$

$$\vec{AB} = k\vec{BC}$$

$$\begin{pmatrix} 2 \\ 6 \end{pmatrix} = k \begin{pmatrix} -3 \\ -9 \end{pmatrix}$$

$$\begin{aligned} 2 = -3k \quad (i) \quad k &= \frac{2}{-3} = -\frac{2}{3} \\ 6 = -9k \quad (ii) \quad k &= \frac{6}{-9} = -\frac{2}{3} \end{aligned}$$

Since the value of k is the same in two cases (i) and (ii), it means $\vec{AB} = -\frac{2}{3}\vec{BC}$

Therefore, the points A, B, and C are collinear.



Application activity 10.1

1) **Chose the correct answer:** Points (2, 3), (4, 5), and (6, 7) are:

- a) Collinear
- b) Not collinear
- c) Vertices of a right triangle
- d) None

2) Determine by slope method if (0, 0), (2, 3), (4, 6) are collinear.

3) Use the vector method to check collinearity of (-2, 1), (0, 2), (2, 3)

4) Find t so that (1, 1), (3, t), (5, 9) are collinear.

5) Given P(4, 2), Q(6, 3), R(8, 4), verify collinearity using the area method.

10.2 Orthogonal vectors



Activity 10.2

You are designing a rectangular gate using metal bars. One bar is placed along vector $\vec{u} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$ (horizontal), and the support bar must be perpendicular to it. You propose a bar along $\vec{v} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$ (vertical).

Your supervisor asks: “How can we be sure these bars are perfectly perpendicular using their vector representations?”



Content summary

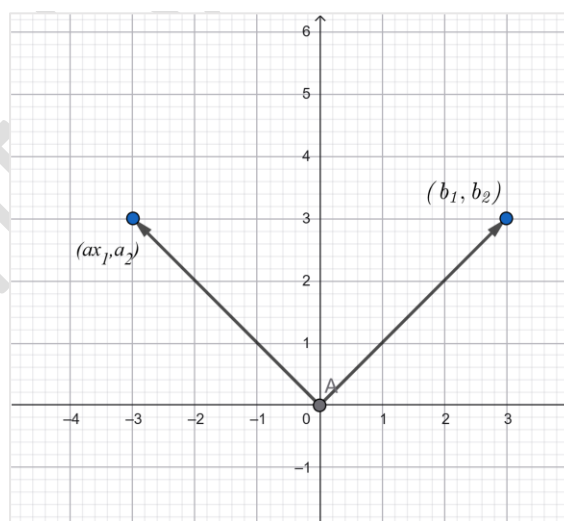
Two vectors are **orthogonal (perpendicular)** if they form a right angle and their dot product is zero.

For vectors $\vec{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$

Then $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2$

If $\vec{a} \cdot \vec{b} = 0$ then $\vec{a}$ and $\vec{b}$ are orthogonal.

They can form a right (90°) angle.



Note: Orthogonal means perpendicular; the zero vector is orthogonal to all vectors but is not considered in geometric problems about orthogonality unless stated.

Example

Show whether the vectors below are orthogonal

(a) $\vec{u} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$

(b) $\vec{a} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

(c) $\vec{p} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and $\vec{q} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$

Solution

(a) $\vec{u} \cdot \vec{v} = 4 \times 0 + 0 \times 3 = 0 \Rightarrow \text{orthogonal}$

(b) $\vec{a} \cdot \vec{b} = -2 \times 5 + 5 \times 2 = 0 \Rightarrow \text{orthogonal}$

(c) $\vec{p} \cdot \vec{q} = 3 \times 2 + -1 \times 6 = 0 \Rightarrow \text{orthogonal}$



Application activity 10.2

1) Vectors $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} -8 \\ 6 \end{pmatrix}$ are:

- a) Parallel
- b) Orthogonal
- c) Neither
- d) Unit vectors

2) Determine if $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$ and $\begin{pmatrix} -14 \\ 2 \end{pmatrix}$ are orthogonal.

3) Find k so that $\begin{pmatrix} k \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$ are orthogonal.

4) Are vectors $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 10 \\ 0 \end{pmatrix}$ orthogonal? Explain.

5) Give an example of a vector orthogonal to $\begin{pmatrix} 7 \\ -2 \end{pmatrix}$

10.3 Problems about points and vectors in two dimensions



Activity 10.3

A surveyor has placed 3 flags at points:

Flag A: (3, 2); Flag B: (7, 6) and Flag C: (11, 10)

A fourth flag D needs to be placed such that:

- (i) A, B, and D are collinear, and
- (ii) CD is perpendicular to AB.

Find the coordinates of D.



Content summary

When working with points and vectors in two dimensions, you learn the following:

1. Find and work with vectors

You already know how to plot a point like $A(2,3)$ or $B(5,7)$. Now we'll connect two points $A(a_1, a_2)$ and $B(b_1, b_2)$, to form a **vector** $\vec{AB}$ which has both length and direction.

Think of it as an arrow from A to B.

- If $A(a_1, a_2)$ and $B(b_1, b_2)$, the vector $\vec{AB} = B - A = (b_1 - a_1, b_2 - a_2)$
- How to add and subtract vectors $\vec{OA}$ and $\vec{OB}$
- How to find the **length** (magnitude) of a vector. If the vector $\vec{AB}$ has coordinates $B - A = (b_1 - a_1, b_2 - a_2)$, then

The magnitude of $\vec{AB} = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2}$

2. Check if points lie in a straight line

Three or more points that lie on the same straight line are called **collinear**. Using vectors, you'll be able to test this easily. For example: Are three football players positioned in a straight line during a free kick setup? Vectors can tell us!

Area of a triangle made by 3 points PQR

where $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ is $= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

If the area is 0. Points are collinear.

3. Find perpendicular vectors

When two lines meet at exactly 90° , we say they are **perpendicular** or **orthogonal**. Using vectors, there's a special check: the **dot product**. If the dot product equals zero, the vectors are perpendicular. This is useful for checking if two paths cross at right angles or if a support beam is truly vertical to the ground.

Example 1

In Fig. 10.5 OCD , $OC = c$ and $OD = d$. C, M and D lie on the same straight line.

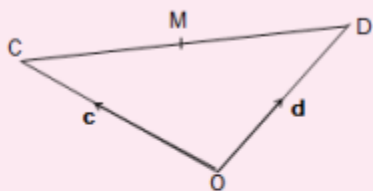


Fig. 10.6

- Express $\vec{CD}$ in terms of vectors $\vec{c}$ and $\vec{d}$.
- M is the midpoint of $\vec{CD}$. What is $\vec{CM}$ in terms of $\vec{c}$ and $\vec{d}$?
- Using your answers to (a) and (b), find $\vec{OM}$ in terms of $\vec{c}$ and $\vec{d}$.

Solution

- $\vec{CD} = \vec{CO} + \vec{OD} = -\vec{OC} + \vec{OD}$
 $= -\vec{c} + \vec{d}$ or $\vec{d} - \vec{c}$
 Therefore, $\vec{CD} = \vec{d} - \vec{c}$.
- $\vec{CM} = \frac{1}{2} \vec{CD}$
 $= \frac{1}{2} (\vec{d} - \vec{c})$
- $\vec{OM} = \vec{OC} + \vec{CM}$
 $= \vec{c} + \frac{1}{2} (\vec{d} - \vec{c})$
 $= \frac{1}{2} \vec{c} + \frac{1}{2} \vec{d}$
 $= \frac{1}{2} (\vec{c} + \vec{d})$

Example 2

In the figure below, A, B and P are collinear points. P divides the line AB in the ratio $AP:PB=7:3$. If $\vec{OA}=\vec{a}$ and $\vec{OB}=\vec{b}$, express $\vec{OP}$ in terms of $\vec{a}$ and $\vec{b}$.

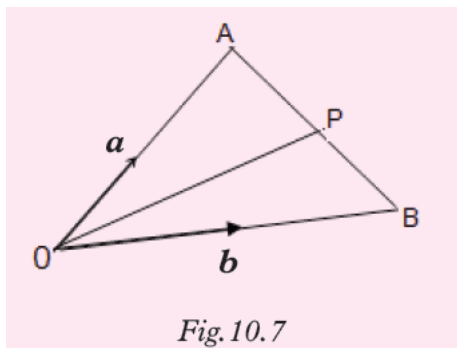


Fig. 10.7

Solution

In triangle OAB , $OA + AB = OB$; $a + AB = b$;

$$AB = b - a.$$

Along AB , $AP = (7/10) AB = (7/10)(b - a)$

$$OP = OA + AP = a + (7/10)(b - a)$$

$$OP = (3/10)a + (7/10)b.$$



Application activity 10.3

1) Chose the correct answer: If the vector $PQ(3, -4)$ and vector $RS(8, 6)$, then vector PQ and vector RS are:

- a) Parallel
- b) Perpendicular
- c) Neither
- d) Collinear

2) $A(0, 0)$, $B(2, 3)$, $C(5, 1)$. Check if the angle ABC is a right angle.

3) Find t if points $(1, t)$, $(3, 2)$, $(5, 4)$ are collinear.

4) $P(1, 2)$; $Q(4, 7)$. Find $R(x, 3)$ such that $PQ \perp QR$.

5) Given $A(2, 1)$, $B(6, 3)$, $C(5, 6)$. Find coordinates of D such that $ABCD$ is a parallelogram with AB perpendicular to BC . Is it a rectangle?



END OF UNIT ASSESSMENT

Part 1: Knowledge and understanding

Multiple Choice questions: Read, work out and chose the correct answer:

1) Three points are collinear if:

- a) They form a triangle with area 10
- b) The area of the triangle formed by them is zero
- c) The distance between them is equal
- d) They form a right triangle

- 2) If vectors $\mathbf{a} = (2, -3)$ and $\mathbf{b} = (6, k)$ are orthogonal, then k equals:
 a) 4 b) -4 c) 9 d) 2
- 3) Points A(2, 3), B(4, 7), and C(6, 11) are:
 a) Vertices of an equilateral triangle
 b) Collinear
 c) Not on the same line
 d) Forming a right angle at B
- 4) The dot product of $\mathbf{u} = \langle 5, -2 \rangle$ and $\mathbf{v} = \langle 3, 4 \rangle$ is:
 a) 23
 b) 7
 c) 14
 d) -7
- 5) Which vector is orthogonal to $\langle 4, -1 \rangle$?
 a) $\langle 1, 4 \rangle$
 b) $\langle 3, 8 \rangle$
 c) $\langle 7, -4 \rangle$
 d) $\langle 1, -4 \rangle$
- 6) The dot product of two vectors is zero. What does this tell you about their geometric relationship?
 a) They are parallel
 b) They have equal magnitude
 c) They are orthogonal (perpendicular)
 d) They are collinear

Part 2: Skills

- 7) Determine whether points P(-1, 2), Q(1, 5), and R(3, 8) are collinear using the area of triangle method.
- 8) Find the value of m such that points A(2, 1), B(3, 4), and C(5, m) are collinear.
- 9) Find k so that $\mathbf{u} = \langle k, 2k \rangle$ is orthogonal to $\mathbf{v} = \langle -6, 3 \rangle$.

Part 3: Attitude and values

- 10) A surveyor is hired to mark the boundary of a rectangular piece of land for a community garden. The intended vertices are: A(2, 1); B(10, 1); C(10, 7); D(2, 7). When checking the actual stakes, point C is found at (11, 7) instead of (10, 7).
- a) To show mathematically that the shape with C at (11, 7) is not a rectangle, which method is **correct**?
 A. Show that all four interior angles are 90° .
 B. Show that the diagonals are perpendicular.
 C. Show that both pairs of opposite sides are parallel.
 D. Show that not all adjacent sides are perpendicular.
- b) What is the area difference between the intended rectangle and the actual quadrilateral?

- A. Intended: 48 sq units; Actual: 54 sq units; Difference: 6 sq units.
 B. Intended: 54 sq units; Actual: 48 sq units; Difference: 6 sq units.
 C. Intended: 42 sq units; Actual: 48 sq units; Difference: 6 sq units.
 D. Intended: 48 sq units; Actual: 56 sq units; Difference: 8 sq units.
 c) Suggest a practical solution that respects both accuracy and the client's concern about costs.
 A. Certify the plot as is but note it's "approximately correct" in the report.
 B. Adjust point B instead of C to make a rectangle without moving C.
 C. Correct the error by moving C to (10,7) now to avoid future disputes.
 D. Accept the landowner's request and keep records as they are.



UNIT 11: ENLARGEMENT AND SIMILARITY IN 2D

Key unit competence: Apply the principles of similarity and enlargement in solving problems involving scale, proportion, and transformation in two dimensions.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and Understanding

- Define enlargement and similarity using scale factor and angle preservation.
- Identify corresponding parts in similar shapes and relate linear, area, and volume scale factors.

Skills

- Perform transformations (enlargement, composite, inverse) and construct images using various scale factors.
- Apply similarity principles, like Thales' Theorem, to calculate unknown lengths, areas, and volumes.

Attitudes and Values

- Appreciate the practical application of scale in fields like design and cartography.
- Develop precise spatial reasoning and collaborative problem-solving skills.

Introduction

This unit explores the mathematics of enlargement and similarity, principles that govern how shapes scale proportionally. You will learn to use the scale factor to perform transformations, calculate unknown dimensions, and understand the relationships between linear, area, and volume scaling. Through application and collaboration, you will develop spatial reasoning and appreciate the role of scale in fields like design, mapping, and technology.



Introductory activity

A Rwandan coffee company packages its product in cartons that are 24 cm long, 12 cm wide, and 16 cm high. For advertising, they plan to build a giant roadside model of this carton, with a specified length of 1.8 meters. To ensure the model maintains the exact same shape (proportions) as the original carton, they have sought your mathematical advice.



- Identify and state two mathematical concepts you would use to determine the correct dimensions for the model.
- Using these concepts, calculate the correct width and height of the advertising model.
- How would you compare your calculated dimensions with those obtained by classmates during a discussion?
- What advice could you offer the company regarding the disposal of used cartons?

11.1 Similarity

11.1.1 Meaning and properties of similarity



Activity 11.1.1:

- Use a mathematics dictionary, reference books, or the internet (e.g., a Google browser or AI) to find the meaning of **similarity** in mathematics.
- Draw a triangle ABC with sides 6 cm, 2 cm, and 5 cm, and measure its angles.
- Draw another triangle PQR whose side lengths are half as long as the corresponding sides of triangle ABC .
- Measure the angles of the two triangles. What do you notice?
- Based on their angles and side lengths, compare the shapes of the two triangles and explain your observations.



Content summary

In mathematics, two figures are **similar** if they have the same shape but not necessarily the same size. This means one figure can be scaled (enlarged or reduced) to exactly match the other.

Formally, two geometric shapes are **similar** if:

- ❖ All corresponding angles are **equal**.

❖ All corresponding sides are in **proportion** (i.e., their ratios are constant). This constant ratio is called the **scale factor** or **ratio of similarity**.

Properties of Similarity

- **Angle Preservation:** Corresponding angles in similar figures are **congruent** (equal in measure).

For Example: If $\triangle ABC$ is similar to $\triangle PQR$, then: $\angle A = \angle P, \angle B = \angle Q, \angle C = \angle R$

- **Proportional Sides:** Corresponding sides are in the same proportion.

For Example: If $\triangle ABC$ is similar to $\triangle PQR$, then:

$$\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = \text{constant}$$

- **Symbol:** The symbol for similarity is $\sim$. For example, $\triangle ABC \sim \triangle DEF$.

Examples



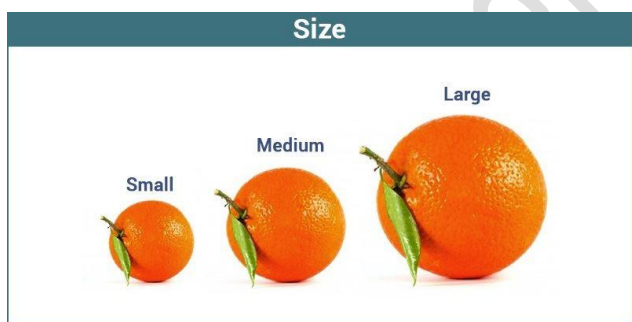
Similar Giraffes



Similar Butterfly



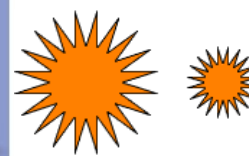
Similar Baskets



Similar Oranges



Similar Bottles



Similar suns



Application activity 11.1.1

1. Define similar figures in your own words.
2. True or False: All rectangles are similar. Justify your answer.
3. Which condition is necessary for two polygons to be similar?
 - a) Same area
 - b) Equal corresponding sides

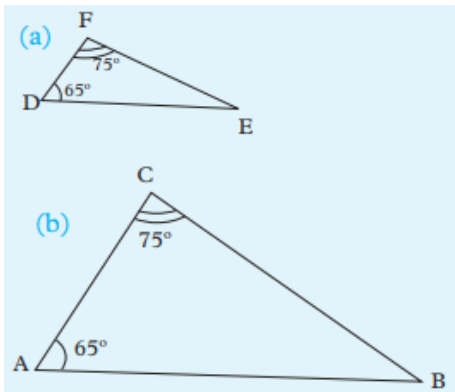
- c) Equal corresponding angles and proportional sides
 d) Same perimeter
4. Construct two triangles ABC and PQR with sides 3 by 3 by 5 cm and 12 by 12 by 20 cm respectively. Measure all the angles. Are triangles ABC and PQR similar? Are all isosceles triangles similar? Are all equilateral triangles similar?
5. The vertices of three right-angled triangles are given below:
 Tr1: A (3, 3), B (4, 5), C (3, 5);
 Tr2: P (1, 3), Q (1, 5), R (2, 4);
 Tr3: X (-2, 3), Y (1, -1), Z (-2, -1).
 Which two triangles are similar?

11.1.2 Similar triangles and polygons



Activity: 11.1.2

1. (a) Observe triangles ABC and DEF (see Figure) and draw them accurately such that $\angle A = \angle D$, $\angle C = \angle F$



- (b) Measure the remaining one angle in each triangle and compare.
 (c) What do you notice?
2. (a) Draw a rectangle PQRS measuring 2 cm by 4 cm.
 (b) Draw another rectangle UVWX measuring 5 cm by 10 cm.
 (c) Find the ratio of corresponding sides of the rectangles.
 (d) What do you notice?



Content summary

Mathematically,

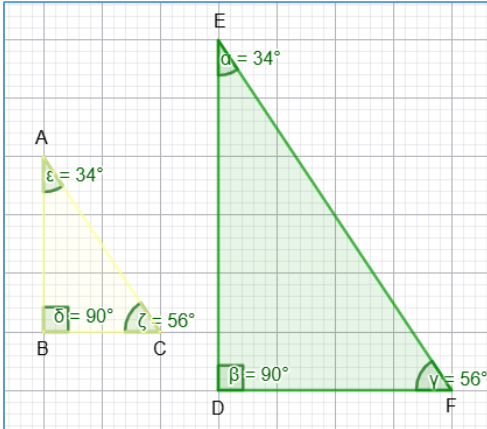
Two triangles are said to be similar, if one of the following three criteria hold:

- AAA or AA criterion: Two triangles are similar if either all the three corresponding angles are equal or any two corresponding angles are equal. AAA and AA criteria are same

because if two corresponding angles of two triangles are equal, then third corresponding angle will definitely be equal.

- SSS criterion: Two triangles are said to be similar, if all the corresponding sides are in the same proportion.
- SAS criterion: Two triangles are similar if their two corresponding sides are in the same proportion and the corresponding angles between these sides are equal.

Let us consider two similar triangles triangle ABC and triangle DEF

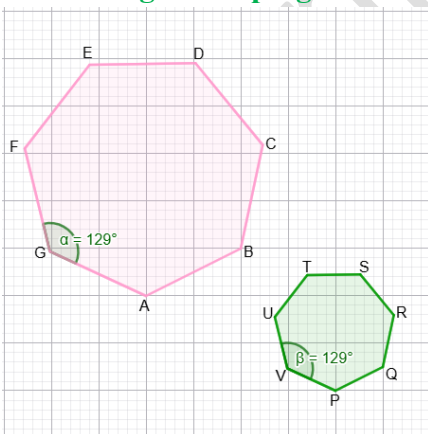


By measurement,

- Corresponding angles are equal : $\angle A = \angle E$, $\angle B = \angle D$, and $\angle C = \angle F$
- Corresponding sides are in the same proportion: $\frac{AB}{ED} = \frac{BC}{DF} = \frac{CA}{FE} = c$

Two or more polygons are similar if the ratio of the corresponding sides is constant and the corresponding angle are equal.

Similar regular heptagons



Similar trapezium

All angles are equal.

Corresponding sides

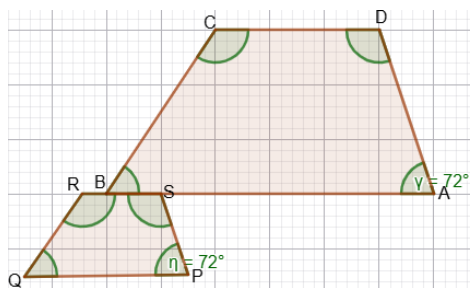
$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CD}{RS} = \frac{DE}{ST} = \frac{EF}{TU} = \frac{FG}{UV} = \frac{GA}{VP} = c$$

Corresponding angles:

$$\angle A = \angle P, \quad \angle B = \angle Q, \quad \angle C = \angle R \quad \text{and}$$

$$\angle D = \angle S$$

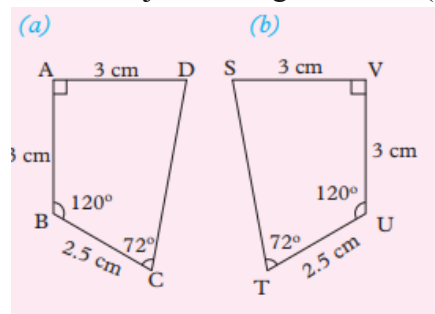
Corresponding sides:



$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CD}{RS} = \frac{DA}{SP} = c$$

Example

Are the objects in Figure below (a) and (b) below similar? Justify your answer.



Solution

Figures (a) and (b) are similar.

Corresponding angles are equal: A corresponds to V = 90°, B corresponds to U = 120°; C corresponds to T = 72°, D corresponds to S = 360° - (90° + 120° + 72°) = 78°

Also,

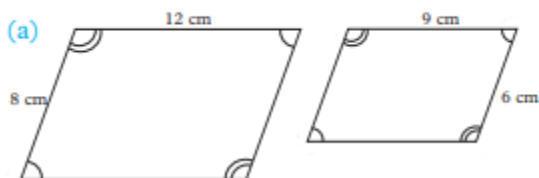
$$\frac{AB}{UV} = \frac{BC}{TU} = \frac{CD}{ST} = \frac{DA}{VS} = 1.$$

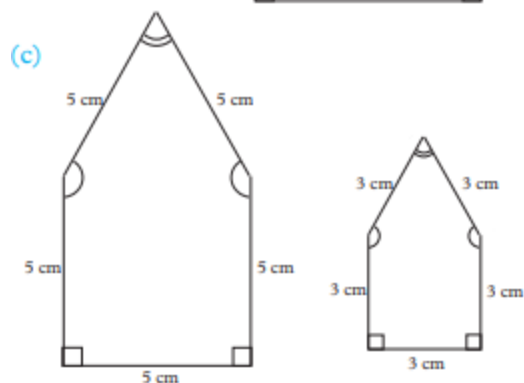
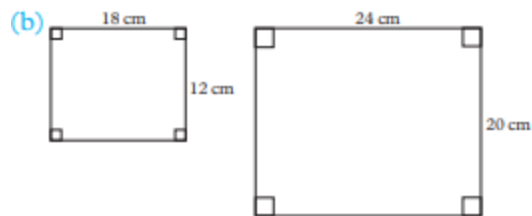
Thus, the two objects are similar.



Application activity 11.1.2

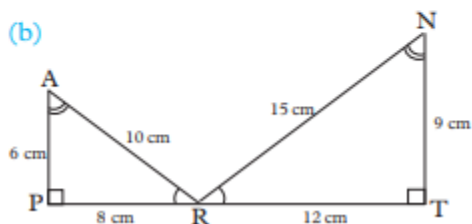
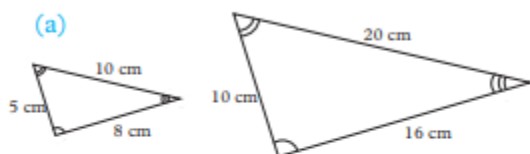
1. State if, and why, the pairs of shapes in Fig. 11.17 are similar.





2.

1. State if, and why, the pairs of shapes in Fig. 11.9 are similar.

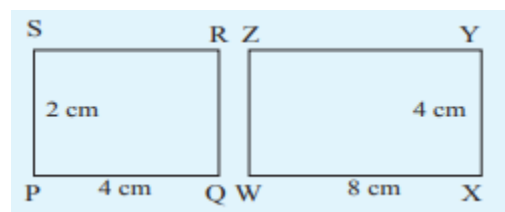


11.1.3 Calculating lengths of sides of similar shapes using similarity and Thales' theorem



Activity: 11.1.3

Consider the rectangles PQRS and WXYZ in Figure below.



(a) Find the values of $\frac{PS}{WZ}$ and $\frac{PQ}{WX}$.

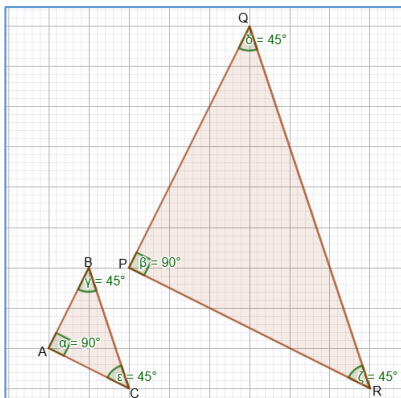
(b) What do you notice between the two results? Make a conclusion.



Content summary

To calculate the lengths of sides of the similar triangles, use the proportionality of sides. E.g.:

Similar triangles ABC and PQR

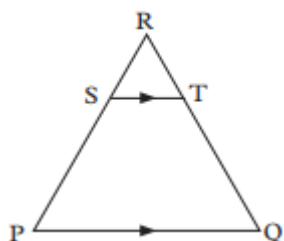


Proportionality of sides is constant:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} = c, \text{ where } c\text{-constant}$$

If two lines are intersected by a set of parallel lines, the parallel lines divide the transversals into proportional segments, use **Thales' Theorem (Intercept Theorem)**:

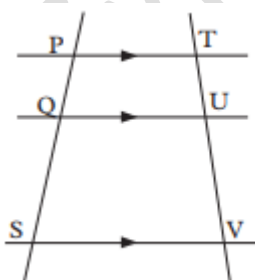
Triangle



Thales' theorem states that:

$$\frac{PS}{SR} = \frac{QT}{TR} \text{ or } \frac{RS}{SP} = \frac{RT}{TQ}$$

Trapezium

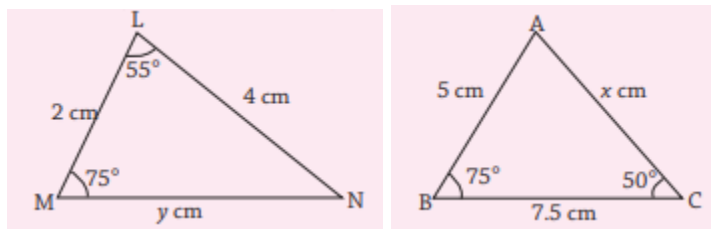


Thales' theorem states that:

$$\frac{PQ}{QS} = \frac{TU}{UV} \text{ or } \frac{SQ}{QP} = \frac{VU}{UT}$$

Example 1

Show that triangle ABC and triangle LMN in Figures below are similar. Hence find the values of x and y.



Solution

Solution

(i) LM corresponds to AB, LN corresponds to AC and MN corresponds to BC.

$$\therefore \frac{AB}{LM} = \frac{AC}{LN} = \frac{BC}{MN} = \text{constant ratio (L.S.F.)}$$

$$\angle LMN = \angle ABC = 75^\circ$$

$$\begin{aligned} \angle MLN &= \angle BAC \\ &= 180^\circ - (75^\circ + 50^\circ) \\ &= 55^\circ \end{aligned}$$

$$\angle LNM = 180 - (55^\circ + 75^\circ) = 50^\circ, \angle ACB = 50^\circ$$

$\therefore \Delta LMN$ and ΔABC are similar.

$$\text{Thus, } \frac{AC}{LN} = \frac{AB}{LM} \Rightarrow \frac{x}{4} = \frac{5}{2}$$

$$2x = 20$$

$$x = 10 \text{ cm}$$

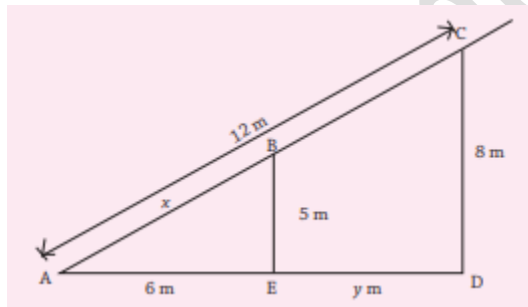
$$\frac{BC}{MN} = \frac{AB}{LM} \Rightarrow \frac{7.5}{y} = \frac{5}{2}$$

$$5y = 15$$

$$y = 3 \text{ cm}$$

Example 2

In Figure, BE is parallel to CD. AC = 12 cm, BE = 5 cm, CD = 8 cm and AE = 6 cm. Find x and y.



Solution

Separating the triangles

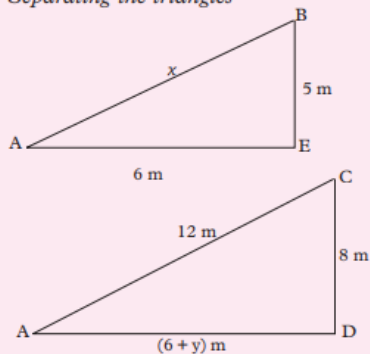


Fig. 11.25

AC corresponds to AB ; AD corresponds to AE and CD corresponds to BE

$\angle AEB = \angle ADC = 90^\circ$ and $\angle CAD = \angle BAE$

$\Rightarrow \triangle AEB$ and $\triangle ADC$ are similar.

$$\frac{AB}{AC} = \frac{BE}{CD} = \frac{AE}{AD}$$

$$\Rightarrow \frac{x}{12} = \frac{5}{8} = \frac{6}{6+y}$$

$$\frac{AB}{AC} = \frac{BE}{CD}$$

$$\therefore \frac{x}{12} = \frac{5}{8}$$

$$8x = 60$$

$$x = 7.5 \text{ m}$$

Also,

$$\frac{AE}{AD} = \frac{BE}{CD}$$

$$\frac{6}{6+y} = \frac{5}{8}$$

$$30 + 5y = 48$$

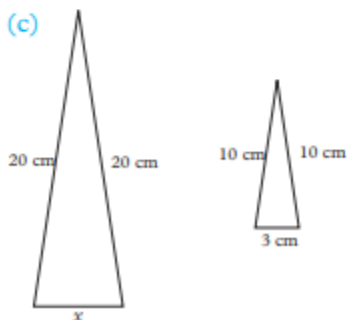
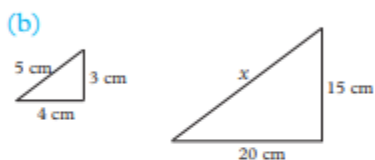
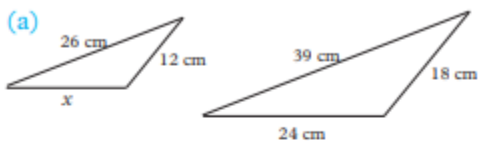
$$5y = 18$$

$$y = 3.6 \text{ m}$$



Application activity 11.1.3

1. The triangles in each pair in Fig. 11.31 are similar. Find x .



11.2 Enlargement

11.2.1 Meaning and properties of enlargement



Activity 12.2.1:

A town council is designing a new playground. The architect has created a small-scale **model** of the playground layout on paper, showing all the equipment, paths, and seating areas.

The model is drawn on a grid where 1 cm represents 1 meter in real life.

The model includes a **triangular climbing frame** with vertices at coordinates (1, 1), (3, 1), and (2, 4) on the grid.

The council loves the design but wants the **actual climbing frame** to be **twice as large** in every dimension (so children have more space to play).

Help the architect enlarge the triangular climbing frame from the model to the real-life size on the blueprint.



Content summary

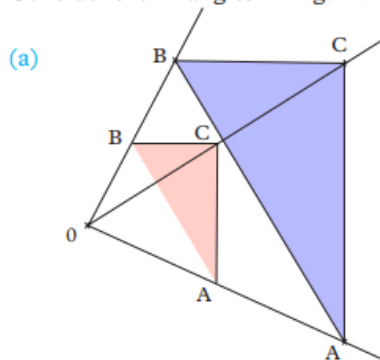
Meaning:

Enlargement is a geometric transformation that changes the size of a shape by a scale factor while preserving its shape and proportions, using a fixed point called the **centre of enlargement**.

Properties:

- ❖ Angles remain the same (the shapes are similar).
- ❖ All lengths are multiplied by the **scale factor** k .
 - If $k > 1$, the image is larger;
 - if $0 < k < 1$, it is smaller;
 - if $k < 0$, the image is inverted relative to the centre.
- ❖ Lines are parallel between object and image.
- ❖ The distance from the centre of enlargement to any image point is $|k|$ times the distance to the corresponding object point.

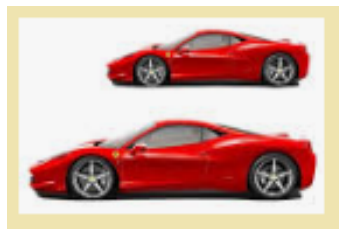
Consider the Triangles in Fig.11.44.



Lines AA', BB' and CC' produced meet at a common point O. The point is called the centre of enlargement. Triangle ABC and A'B'C' are similar.

Note: $\frac{OA'}{OA} = \text{Scale factor}$

Examples of enlarged or reduced size of real objects:



Application activity 11.2.1

Question 1:

- What is meant by *enlargement* in geometry?
- Name the two important elements that define an enlargement.

Question 2:

A triangle is enlarged using a scale factor of 3.

- How does the size of the enlarged triangle compare to the original triangle?
- If one side of the original triangle is 4 cm, find the length of the corresponding side in the enlarged triangle.

Question 3

State **any three properties** of enlargement concerning:

- Shape
- Angles
- Ratio of corresponding sides

Question 4

Triangle ABC has vertices

A(1, 2), B(2, 4), and C(3, 2).

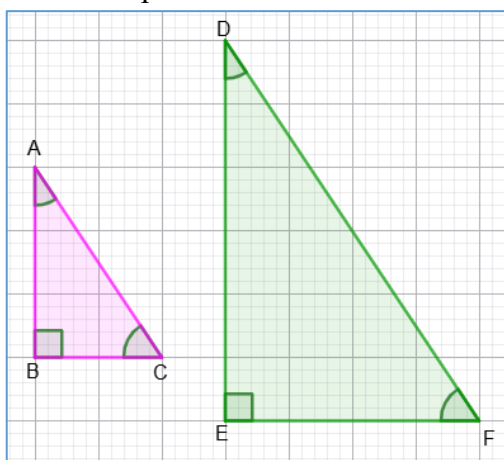
- Find the coordinates of the image of triangle ABC after an enlargement with scale factor 2, using the origin as the centre of enlargement.
- What can you say about the angles of the image compared to the original triangle?

11.2.2 Enlargement with positive scale factor



Activity 11.2.2

Look at the picture below:



- What similarities can you observe between triangle **ABC** and triangle **DEF** in terms of shape and angle sizes
- Which sides of triangle **DEF** correspond to the sides of triangle **ABC**? How can you tell from the diagram?
- By counting grid squares (square units), how many times longer is one side of triangle **DEF** compared to its matching side in triangle **ABC**?
- If triangle **ABC** is enlarged to get triangle **DEF**, what can you say about the direction and sign of the scale factor used?

..



Content summary

Any enlargement requires **scale factor** and **centre of enlargement** to produce an image of the given object.

Enlargement with positive scale factor means that a scale factor k is greater than zero.

Here, we obtain two cases:

- $k > 1$ (gets bigger),
- $0 < k < 1$ (gets smaller, a reduction).

In these two cases, the following steps are proceeded for constructing the image of the object.

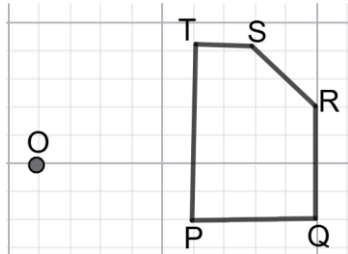
- Draw the object and choose a point O as a center of enlargement.
- Draw lines joins centre with the object points.
- Measure the distance of object point from the centre of enlargement.
- Calculate the corresponding lengths of image points from centre following the rule says that
image distance = Scale factor $\times$ object distance

(e) Mark off the image points and join them to construct the image.

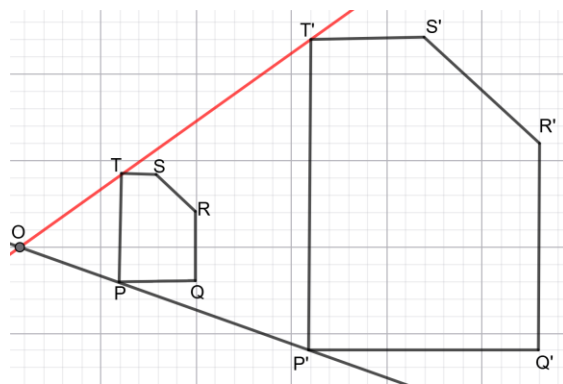
Note: The image is always on the same side as the object.

Example 1

Enlarge the shape PQRST from centre O with scale factor 3



Solution



Description of the image

$$OP' = 3OP$$

$$OQ' = 3OQ$$

$$OR' = 3OR$$

$$OS' = 3OS$$

$$OT' = 3OT$$

$$P'Q' = 3PQ$$

$$Q'R' = 3QR$$

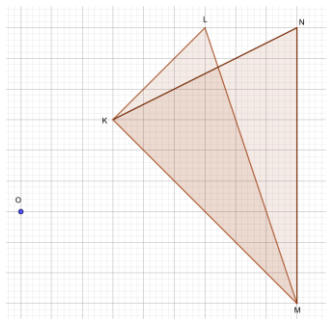
$$R'S' = 3RS$$

$$S'T' = 3ST$$

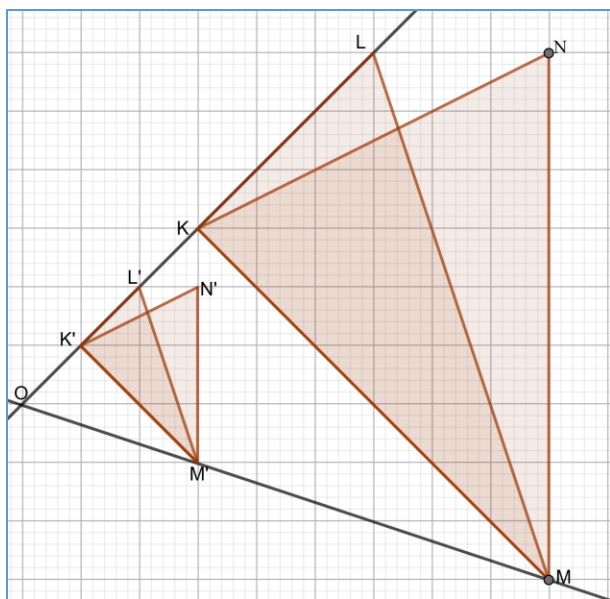
$$T'P' = 3TP$$

Example 2

Reduce the shape KLMN from centre O with scale factor $\frac{1}{3}$

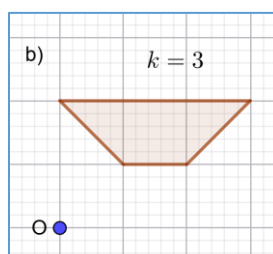
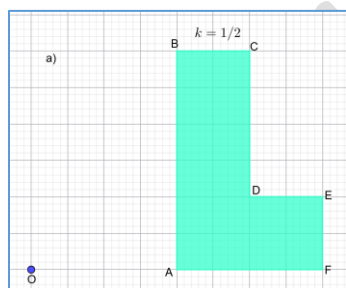


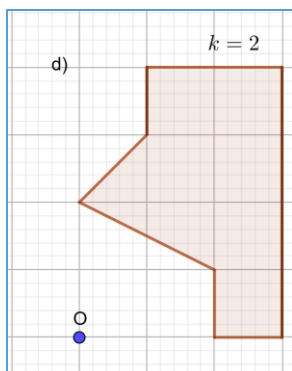
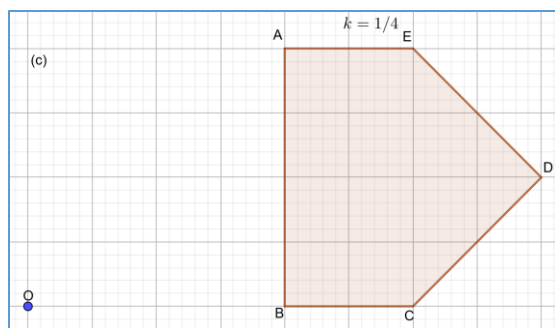
Solution:



Application activity 11.2.2

- 1) Under enlargement of a polygon, what happens to the following if the scale factor is: (i) 3?
(ii) $1/3$?
 - (a) The lengths of corresponding sides
 - (b) Corresponding angles
 - (c) Shape
 - (d) Direction of lines
- 2) Copy and draw the image of each shape under enlargement from centre O with given factor.



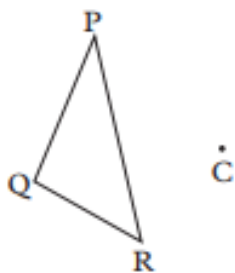


11.2.3 Enlargement with Negative scale factor



Activity 11.2.3

Use scale factor of -4 and centre C to enlarge triangle PQR in figure below about centre C .



Content summary

Enlargement with negative scale factor means that a scale factor k is less than zero.

Here, we have two cases:

- $k < -1$: The image is bigger than the object.
- $-1 < k < 0$: The image is smaller than the object. There is a reduction

In these two cases, the following steps are proceeded for constructing the image of the object.

- Draw the object and choose a point O as a center of enlargement.
- Draw lines joins centre with the object points.
- Measure the distance of object point from the centre of enlargement.
- Calculate the corresponding lengths of image points from centre following the rule says that
image distance = Scale factor $\times$ object distance
- Mark off the image points and join them to construct the image.

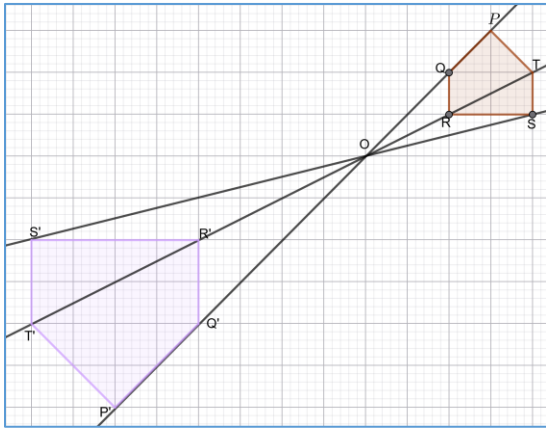
Note:

- The object and the image are on opposite sides of the centre of enlargement,
- Any line on the image is parallel to the corresponding line on the object, but the image is inverted relative to the object.

Example 1

Copy and draw the image of the shape PQRST under enlargement from centre O with scale factor -2

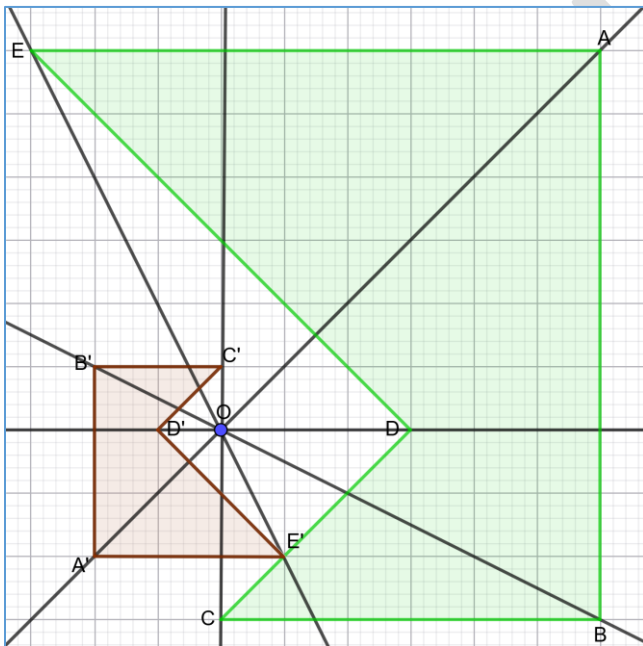
Solution



Example 2

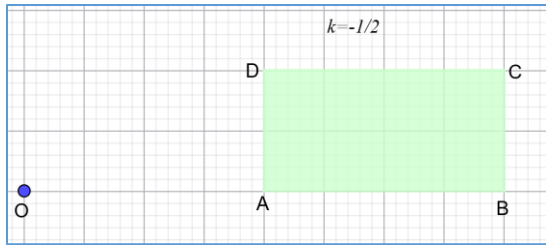
Draw the image of the following shape ABCDE under enlargement from centre O with scale factor $-\frac{1}{3}$

Solution

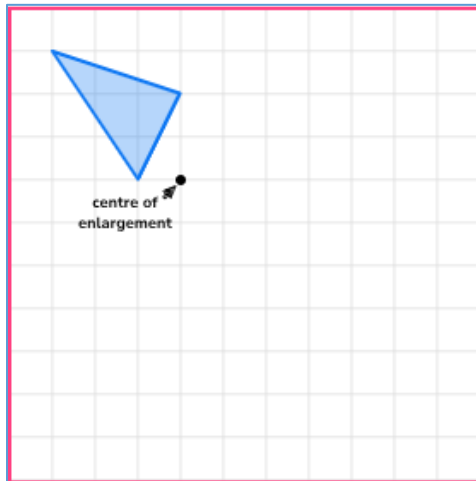


Application activity 11.2.3

1. Copy and draw the image for each figure below, with centre O and scale factor given.



2. Enlarge the shaded shape with scale factor -2 about the point.



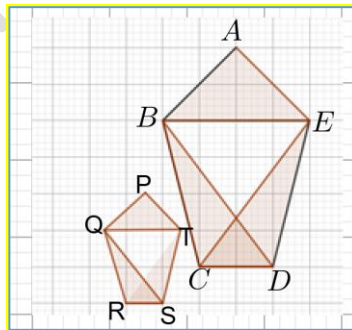
11.2.4 Locating the Centre of enlargement and finding scale factor



Activity 11.2.4

Look at these two pictures,

- Identify pairs of corresponding points
- Join the points that are corresponding,
- Are the lines intersecting?
- How can we call this point?



Content summary

To locate the centre of enlargement,

- Join each object point with its corresponding image point and then do the same on other points.
- Extend the lines til they meet.
- Highlight the point where the lines met as the centre of enlargement.

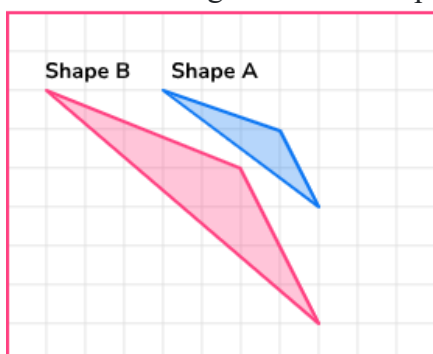
To Calculate scale factor, we use the fact that scale is the ratio of image distance to object distance.

$$k = \text{Image length} / \text{Object length.}$$

Note: Given an object and its image, it is sufficient to construct only two lines in order to obtain the centre of enlargement.

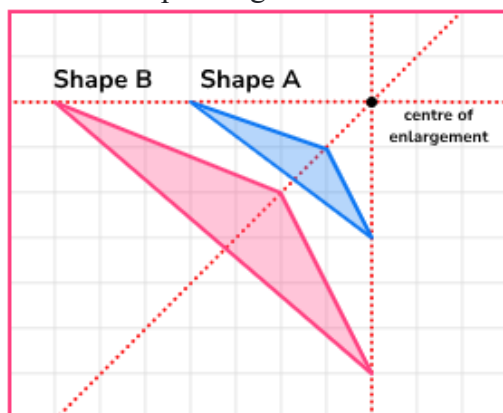
Example 1

Shape A has been enlarged to make shape B. Find the centre of enlargement.



Solution

Find pairs of corresponding vertices and draw ray lines going through the points.

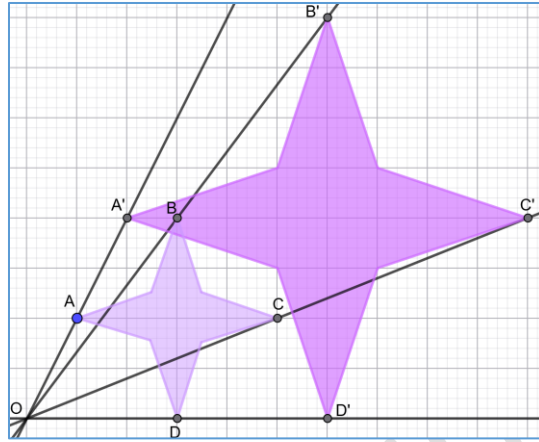
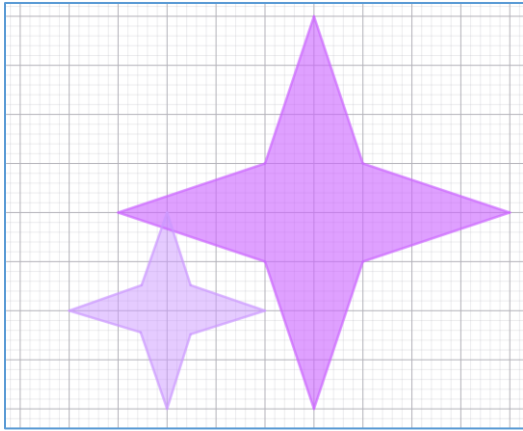


Example 2

A big star is reduced to be smaller one as shown in the figure below. Find the scale factor show the center of enlargement.

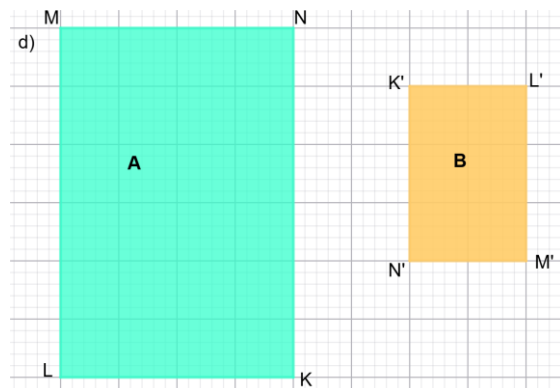
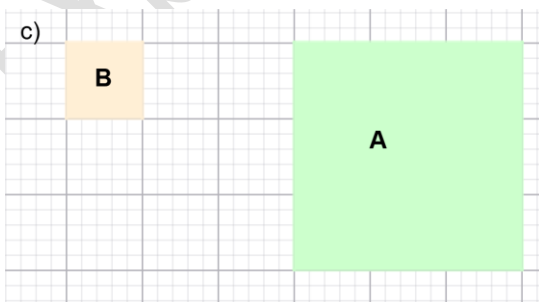
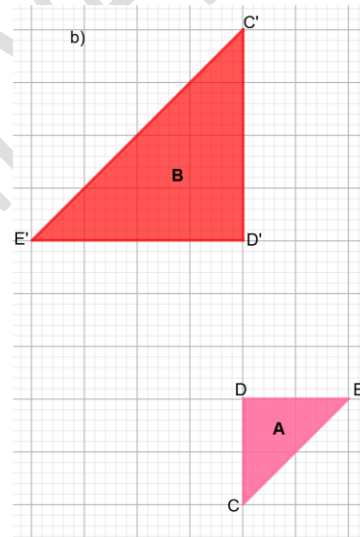
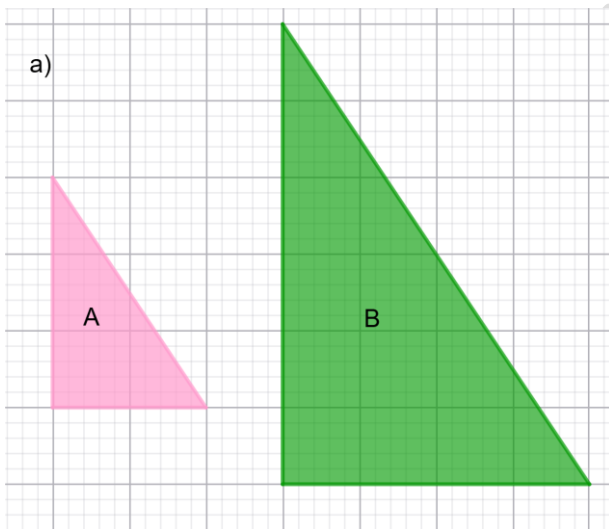
Question

Solution

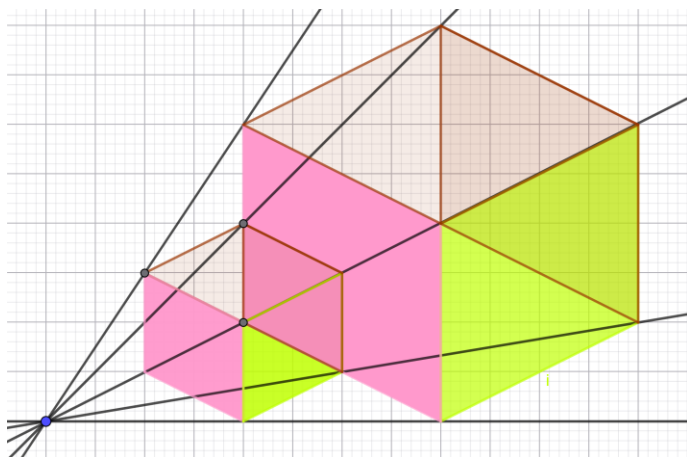


Application activity 11.2.4

1) Find the scale factor when the figure A is transformed to the figure B



2) A small box is magnified to be big one as shown in the figure below. Find the scale factor.



11.2.5 Enlargement in the Cartesian plane



Activity 11.2.5

- On the Cartesian Plane, draw a quadrilateral with vertices $A(0, 3)$, $B(2, 3)$, $C(3, 1)$ and $D(3, -2)$
- In each point, multiply x - and y -coordinate by 3 and write the new points A' , B' , C' , and D' .
- Plot the points A' , B' , C' , and D' in the same graph as (a).
- Join each object point by its corresponding image point.
- Extend the lines until they meet and label the intersection point O .
- Compare the two figures, what do you notice about:
Corresponding sides, Corresponding angles, Shape and Direction of lines?



Content summary

Enlargement with centre is origin $(0,0)$, and scale factor k ; simply the enlargement maps point $P(x,y)$ to $P'(kx, ky)$.

Using a centre $O(a,b)$ and scale factor k then mapping is $(x, y) \rightarrow (a + k(x-a), b + k(y-b))$.

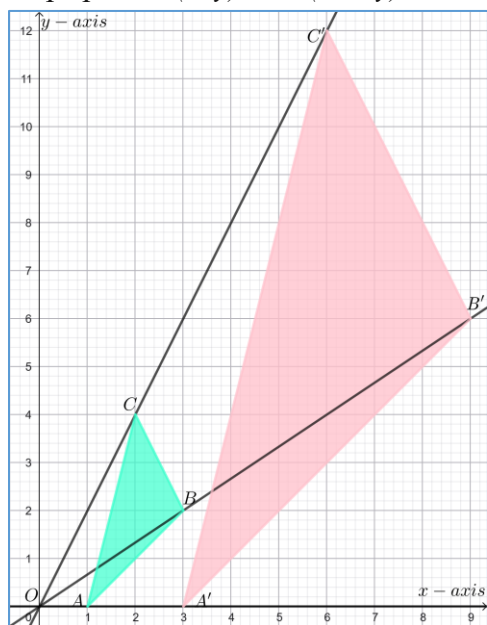
Again, When the centre of enlargement is not the origin, we must carry out complete construction using the steps we learnt earlier

Example 1

In a cartesian plane plot triangle ABC whose coordinates are $A(1, 0)$, $B(3, 2)$ and $(2, 4)$. On the same cartesian plane locate and draw triangle $A'B'C'$ the image of triangle ABC under enlargement scale factor 3 and centre origin.

Solution

Using the steps we learnt earlier and also fact that an enlargement centre origin scale factor K maps point $P(x, y)$ to $P'(kx, ky)$ we obtain $A'B'C'$ as shown in Figure.



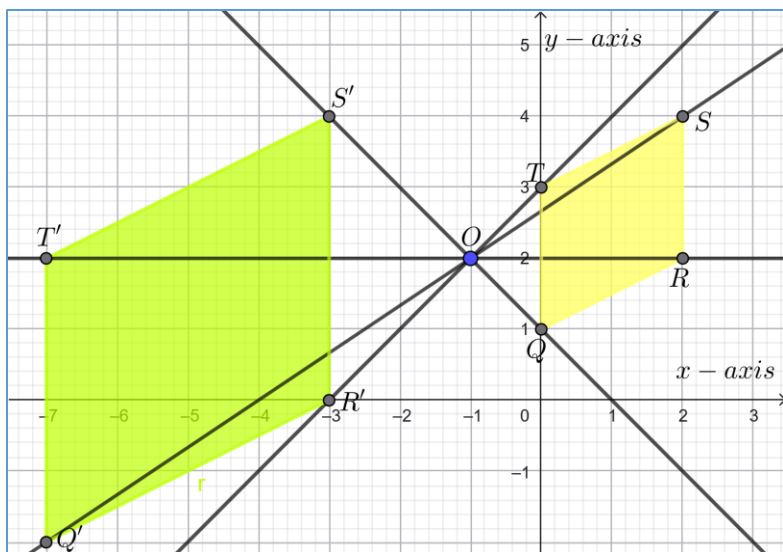
Example 2

The coordinates of a quadrilateral are $Q(0, 1)$, $R(2, 2)$, $S(2, 4)$ and $T(0, 3)$.

- plot the quadrilateral QRST on cartesian plane. Identify the quadrilateral by name?
- By construction, locate and draw the image of quadrilateral QRST on the cartesian plane under enlargement scale factor -2 centre $(-1, 2)$. Name the image as $Q'R'S'T'$.
- Write down the co-ordinates of vertices of the image.

Solution

- The shape is a Rhombus
-



c) Vertices of the image $Q'(-3,4)$, $R'(-7,2)$, $S'(-7,-2)$ and $T'(-3,0)$.



Application activity 11.2.5

1. Points $A(-2, -1)$, $B(1, -1)$, $C(1, -4)$ and $D(-2, -4)$ are vertices of a square. Without drawing, state the coordinates of the vertices of the image square under enlargement with centre origin and scale factor:

(a) -4 (b) -2 (c) -1 (d) $-1/4$

2. Triangle ABC with vertices $A(9, 4)$, $B(9, 1)$ and $C(15, 1)$ is mapped onto $\Delta A'B'C'$ vertices $A'(1, 2)$, $B'(1, 1)$ and $C'(3, 1)$ by an enlargement. Find:

(a) the scale factor of the enlargement.

(b) the co-ordinates of the centre of enlargement.

3. Triangle ABC has vertices $A(1, 2)$, $B(3, 2)$, $C(3, 4)$. Find the co-ordinates of the image $\Delta A'B'C'$ after an enlargement, centre $(0, 0)$ scale factor -3 .

4. The points $A(1, 1)$, $B(3, 2)$, $C(3, 4)$ and $D(1, 3)$ are vertices of a quadrilateral.

(a) Draw the quadrilateral on the Cartesian plane.

(b) Taking $O(4, 0)$ as the centre of enlargement, find the image when the linear scale factor is:

(i) 1.5 (ii) -2.4

11.2.6 Composite enlargement



Activity 11.2.6

A logo is enlarged by 2 for a brochure, and then that image is enlarged by 0.5 for a website icon. What is the overall transformation? What do you think on final image compared to the original logo?

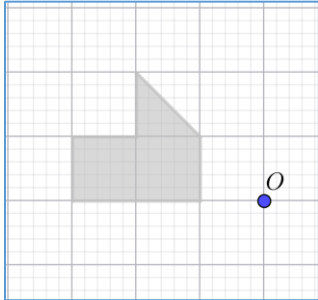


Content summary

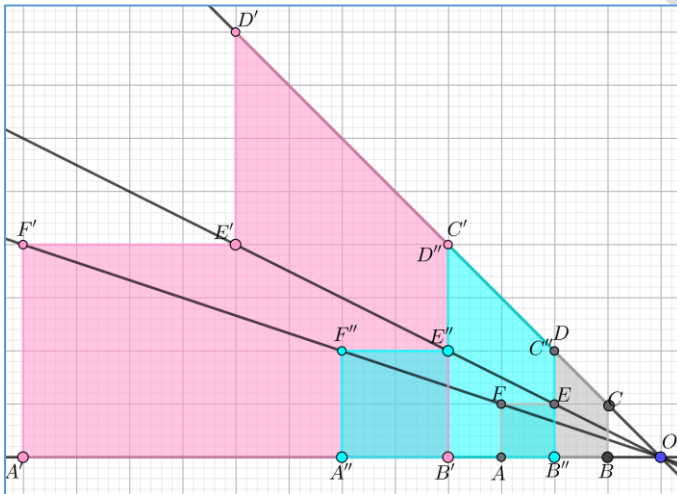
Performing two successive enlargements. The result is a single enlargement. Composite scale factor $k_{\text{total}} = k_1 \times k_2$. For example Enlarge shape with $k=2$, then $k=3$ from the same centre. Overall $k=6$.

Example 1

Draw the image of the figure below after successive enlargement by $k=4$ and $k=1/2$ with centre O .

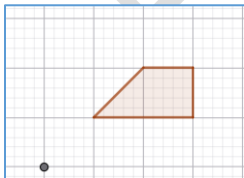


Solution

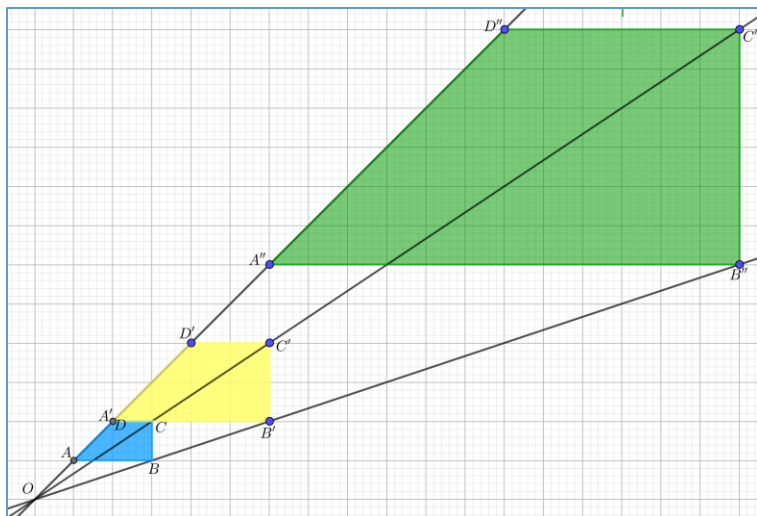


Example 2

The figure below is enlarged successively by $k=2$ and $k=3$ with centre O . Show that the image constructed under successive enlargement is the same as the enlargement of the figure by $k=6$



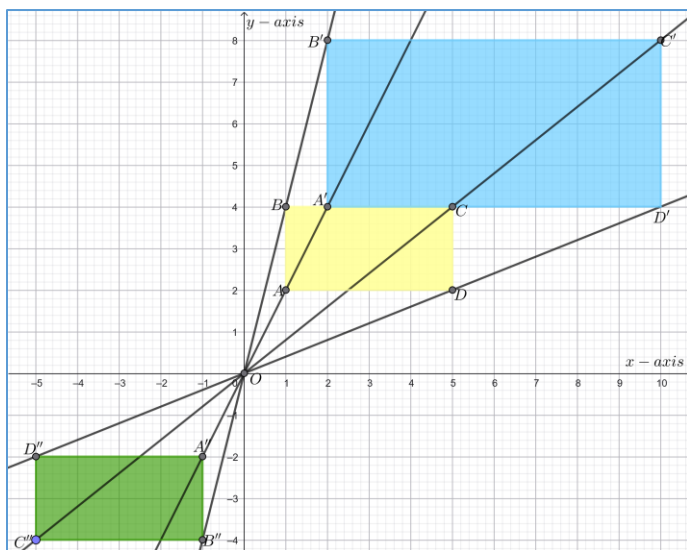
Solution



Example 3

2. Consider a rectangle with vertices at $A(1,2)$, $B(1,4)$, $C(5,4)$, and $D(5,2)$. Apply an enlargement with a scale factor of 2 followed by a scale factor of $-1/2$. Write the coordinates of the final image.

Solution

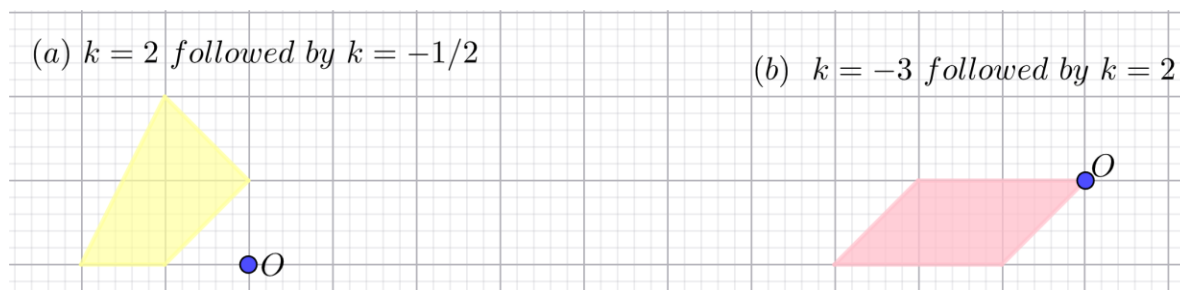


The coordinates of the final image are $A''(-1,-2)$, $B''(-1,-4)$, $C''(-5,-4)$, and $D''(-5,-2)$.

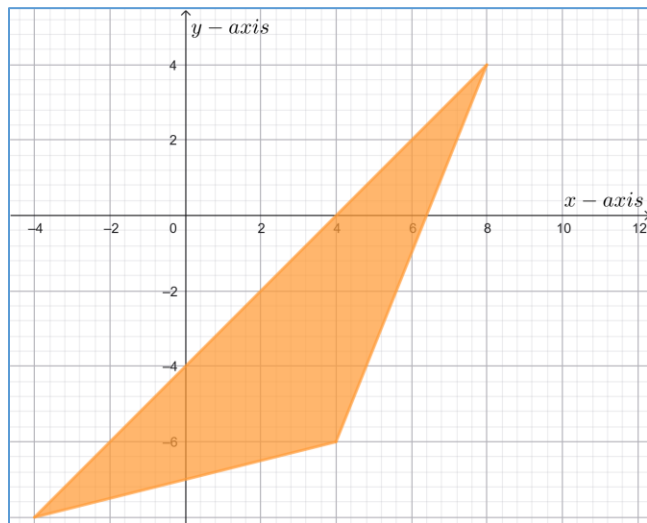


Application activity 11.2.6

1) Draw the image of each figure below after successive enlargement indicated with centre O.



2) Reduce the figure below by a factor of $1/2$ and enlarge the finding by scale factor -3 using the origin as the center of dilation.



11.2.7 Inverse enlargement



Activity 11.2.7

A shape is enlarged with scale factor $k=4$ about a centre point O .

If the image of a point P after the enlargement is at P' , what scale factor would map P' back to P , and what is this transformation called?



Content summary

The inverse transformation reverses an enlargement.

If original enlargement had scale factor k , the inverse has scale factor $1/k$ on same centre.

If original enlargement had scale factor $1/k$, the inverse has scale factor k on same centre.

Example 1

Shape S is enlarged with centre $(0,0)$, scale factor $1/2$ to give shape S' .

Then S' is enlarged with centre $(0,0)$, scale factor $1/4$ to give shape S'' .

What single enlargement maps S directly to S'' ?

What single enlargement maps S'' back to S ?

Example 2

Rectangle has vertices at $A'(2,2)$, $B'(6,2)$, $C'(6,4)$, and $D'(2,4)$.

It was obtained by enlarging rectangle with scale factor $1/3$ about the origin.

Find the coordinates of the vertices of the original rectangle.

Draw them on the cartesian plane.



Application activity 11.2.7

1) Under an enlargement with centre at the origin and scale factor $k=7$, the image of point P is $P'(21,-14)$.

Find the coordinates of P .

2) Point $A'(5,5)$ is the image of point A under an enlargement with centre $C(2,-1)$ and scale factor 4.

Find the coordinates of A .

3) Triangle $\triangle XYZ$ has vertices $X(6,9)$, $Y(12,9)$, $Z(9,15)$.

It is mapped onto $\triangle X'Y'Z'$ by an enlargement with centre $(0,0)$ and scale factor $1/3$.

Find the coordinates of $X'Y'Z'$, then state the scale factor and centre that would map $\triangle X'Y'Z'$ back to $\triangle XYZ$.

11.3. Application of enlargement in real life

11.3.1 Finding lengths of sides of similar shapes using scale factor



Activity 11.3.1

Architects often create scale models of buildings using a specific scale factor. If the actual building is 50 meters tall and the model is constructed at a scale factor of $1:100$, what would be the model's height?



Content summary

To determine the linear scale factor between two similar figures, you can use the ratio of corresponding sides. The formula to calculate the linear scale factor (LSF) is:

$$LSF = \frac{\text{Length of a side in the larger figure}}{\text{Length of corresponding side in the smaller figure}}$$

Recall properties of similarity

- Corresponding Angles: All corresponding angles are equal.
- Proportional Sides: The lengths of corresponding sides are proportional, determined by the linear scale factor.
- Shape Preservation: The overall shape remains unchanged despite differences in size.

Example 1

Given two similar rectangles, if the length of the first rectangle is 8 cm and the corresponding length of the second rectangle is 12 cm, and the width of the first rectangle is 5 cm, find the width of the second rectangle.

Solution

First, calculate the linear scale factor:

$$LSF = \frac{12}{8} = 1.5$$

Then, apply the LSF to the width

The width of the second rectangle is $5\text{ cm} \times 1.5 = 7.5\text{ cm}$

Example 2

Photographers may enlarge images by applying a scale factor to increase the size for printing purposes while maintaining the original proportions. If an original photo measures 10 cm by 15 cm and needs to be enlarged by a scale factor of 3. Find the enlarged photo's measures.

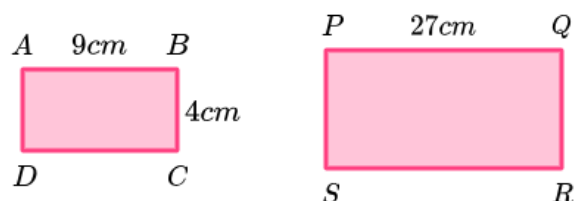
Solution

$$\text{Width} = 10 \times 3 = 30\text{ cm}$$

$$\text{Height} = 15 \times 3 = 45\text{ cm}$$

Example 3

Below are two similar shapes. Find the length QR.



Solution

From the figure,

The sides AB and PQ are a pair of corresponding sides.

The sides BC and QR are a pair of corresponding sides.

The ratio of the lengths AB : PQ is 9 : 27; which simplifies to 1 : 3

This gives a scale factor of enlargement from rectangle ABCD to rectangle PQRS of 3.

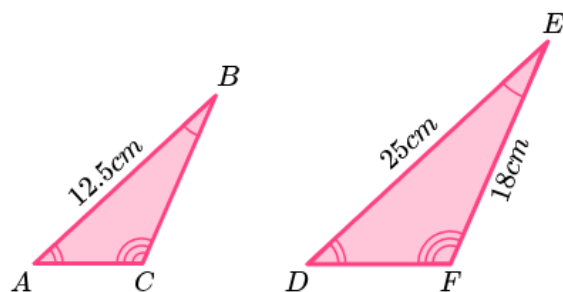
The missing side has been found.

$$QR = 4 \times 3 = 12\text{ cm}$$



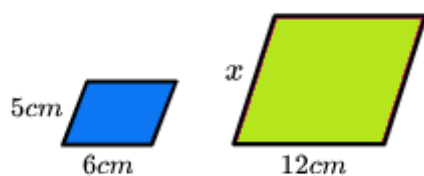
Application activity 11.3.1

1) Here are two similar triangles. Find the length BC.

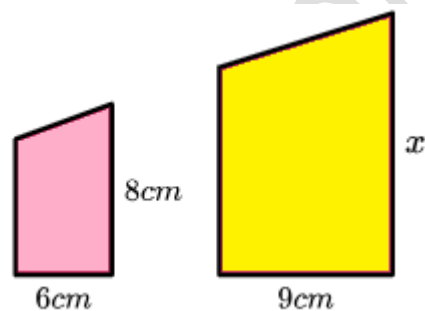


2) Work out the value of x .

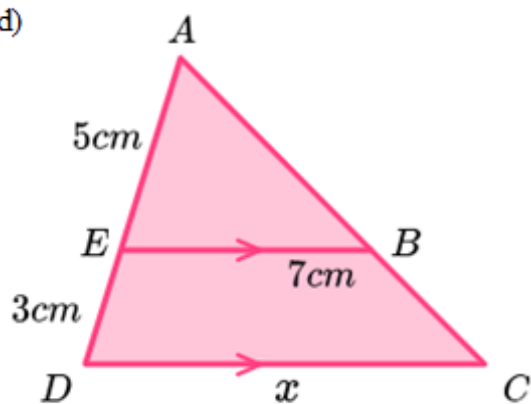
(a)



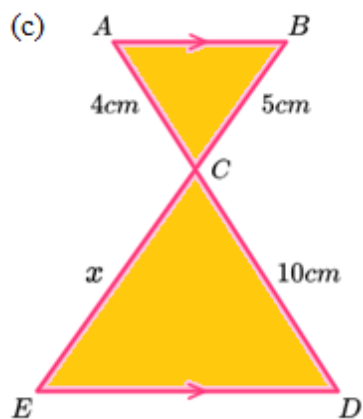
(b)



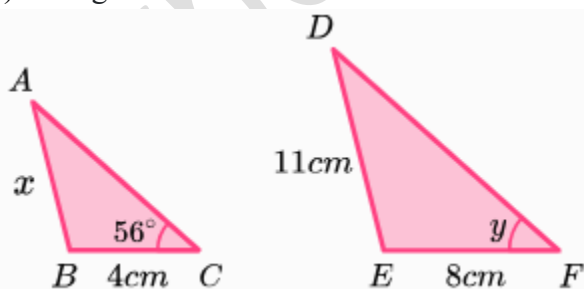
(d)



(c)



3) Triangles ABC and DEF are similar.



(a) Write down the size of angle y

(b) Work out the value of x

4. An enlargement maps line segment XY of length 8 cm to line segment X'Y' ' of length 2 cm
What is the scale factor of the enlargement?
What scale factor would transform X'Y' back to XY?

11.3.2 Finding areas of similar shapes using scale factor



Activity 11.3.2

- Draw a rectangle. Choose a point O anywhere outside the rectangle.
- With O as the centre, enlarge the rectangle with scale factor 2.
- What are the dimensions of the image rectangle?
- Calculate a linear scale factor (L.S.F) of the enlargement.
- Calculate the areas of the two rectangles and their ratio.
- How does this ratio compare with the linear scale factor?



Content summary

We already know that if two shapes are similar their corresponding sides are in the same ratio and their corresponding angles are equal.

When calculating a missing area, we need to calculate the Area Scale Factor(ASF).

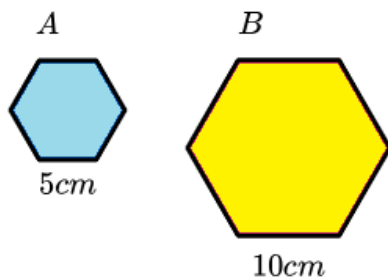
Area Scale Factor is The ratio of the areas of two similar figures. It is equal to the square of the linear scale factor.

$$\text{Area scale factor}(ASF) = (LSF)^2.$$

$$\text{Therefore, } LSF = \sqrt{ASF}$$

Example 1

These two figures are similar. The area of shape A is 60cm^2
Find the area of shape B.



Solution

When writing the ratios, the order is very important.

The ratio of the lengths = length A : length B

The ratio of the lengths = 5:10, which simplifies to 1:2

The scale factor of enlargement from shape A to shape B is 2

As we are finding an area we need to square the scale factor

$$\text{Area Scale Factor} = 2^2 = 4$$

$$\text{The area of shape B} = 4 \times 60\text{cm}^2 = 240\text{cm}^2$$

Alternatively, you can form an equation to solve:

Lengths Areas

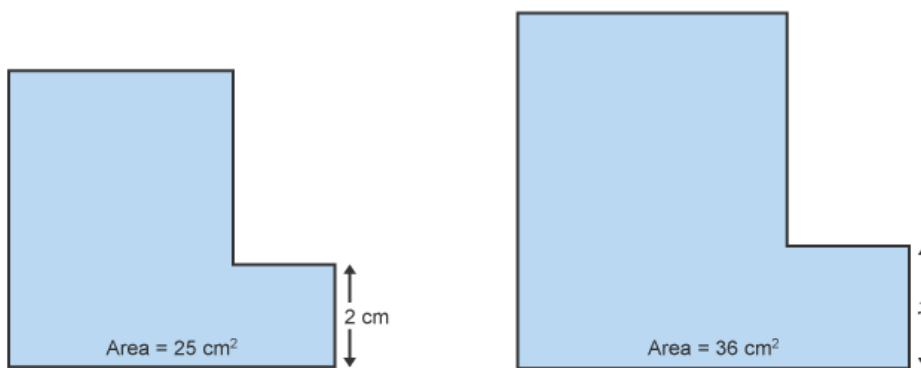
$$\left(\frac{10}{5}\right)^2 = \frac{x}{60}$$

$$240 = x$$

Therefore, The area of shape B is 240cm^2

Example 2

These two shapes are similar. What is the length of x ?



Solution

$$ASF = \frac{\text{Area of bigger}}{\text{Area of smaller}} = \frac{36}{25}$$

$$LSF = \sqrt{ASF} = \sqrt{\frac{36}{25}} = \frac{6}{5}$$

$$x = \frac{6}{5} \times 2 = 2.4\text{cm}$$

Example 3

On a model house, the windows measure $1.2\text{ cm} \times 3\text{ cm}$. The area of the actual house is 10 000 times bigger than the area on the model. Find the dimensions of the windows on the actual house.

Solution

$$ASF = 10\ 000$$

$$LSF = \sqrt{ASF} = \sqrt{10\ 000} = 100$$

Dimensions of image :

$$l = 3\text{cm} \times 100 = 300\text{cm} = 3\text{m}$$

$$w = 1.2\text{cm} \times 100 = 120\text{cm} = 1.2\text{m}$$



Application activity 11.3.2

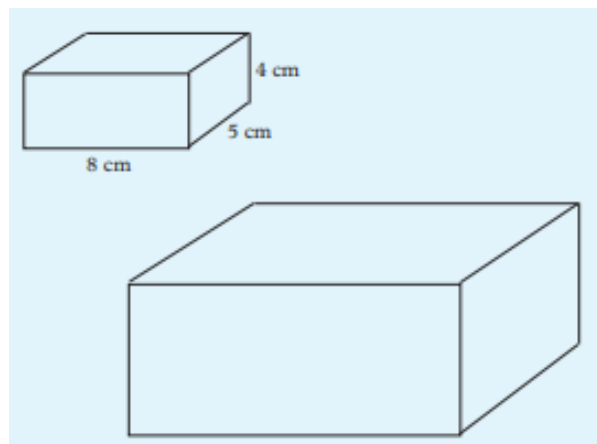
- 1) Two similar figures have corresponding sides 3cm and 5cm. The area of the smaller one is 12 cm^2 . What is the area of the larger one? Give your answer to 1 decimal place.
- 2) Two fridge magnets are mathematically similar.
One magnet is 4cm long and the other is 10cm long.
The area of the smaller magnet is 18 cm^2 .
Calculate the area of the larger magnet.
a) 2.88 cm^2 b) 45 cm^2 c) 112.5 cm^2
- 3) The ratio of the areas of two circles is 16:25.
(a) What is the ratio of their radii?
(b) If the smaller circle has a diameter of 28 cm, find the diameter of the larger circle.

11.3.3 Finding volumes of similar solids using scale factor



Activity 11.3.3

Figure below shows a cuboid and its enlargement with scale factor 2.



- a) What are the dimensions of the image?
- b) Calculate the volumes of the two cuboids and the ratio between them.
- b) How does this ratio compare with the linear scale factor?
- d) If instead the scale factor of enlargement is 3, what are the dimensions and volume of the big cuboid?



Content summary

To find volumes of similar solids, first find the linear scale factor (k) by comparing corresponding lengths, then cube it to get the volume scale factor (k^3) which you use to find the new volume.

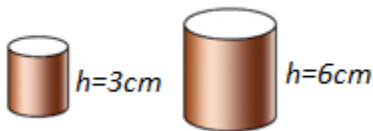
- $New\ Volume = known\ volume \times k^3$
- $\frac{New\ Volume}{known\ volume} = k^3$
- $k = \sqrt[3]{\frac{New\ Volume}{Original\ Volume}}$

Recall Key Relationship

- **Linear Scale Factor (k):** Ratio of corresponding lengths (e.g., side, height, radius).
- **Area Scale Factor (k^2):** ratio of corresponding areas.
- **Volume Scale Factor (k^3):** ratio of corresponding volumes.

Example 1

Two similar cylinders have heights of 3 cm and 6 cm. If the smaller cylinder's volume is $45\pi\text{cm}^3$. Find the volume of larger cylinder.



Solution

Find Linear Scale Factor (k) using heights: $k = \frac{6}{3} = 2$

Find Volume Scale Factor (k^3) using LSF: $k^3 = 2^3 = 8$

Volume of larger cylinder: $V_L = k^3 V_S = 8 \times 45\pi\text{cm}^3 = 360\pi\text{cm}^3$

Therefore, the Volume of larger cylinder is $360\pi\text{cm}^3$

Example 2

Two similar spheres have the volumes given in the figure. Find a linear scale factor



Solution

Equation of finding volume

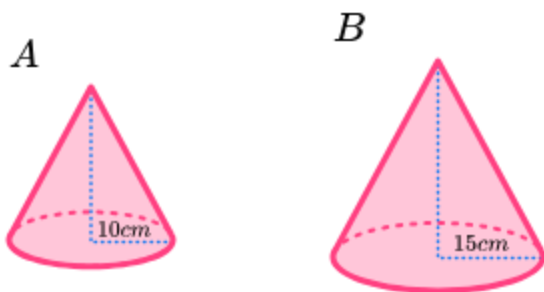
$$V_L = k^3 \times V_S$$

$$k = \sqrt[3]{\frac{27}{8}} = \frac{\sqrt[3]{27}}{\sqrt[3]{8}} = \frac{\sqrt[3]{3^3}}{\sqrt[3]{2^3}} = \frac{3}{2}$$

Example 3

These two shapes are similar. The volume of shape A is 400cm^3 .

Find the volume of shape B:



Solution

Use the given information to write a ratio and work out the scale factor.

When writing the ratios, the order is very important.

The ratio of the lengths is length A : length B

The ratio of the lengths 10 : 15 which simplifies to 1 : 1.5

The linear scale factor of enlargement from shape A to shape B is 1.5

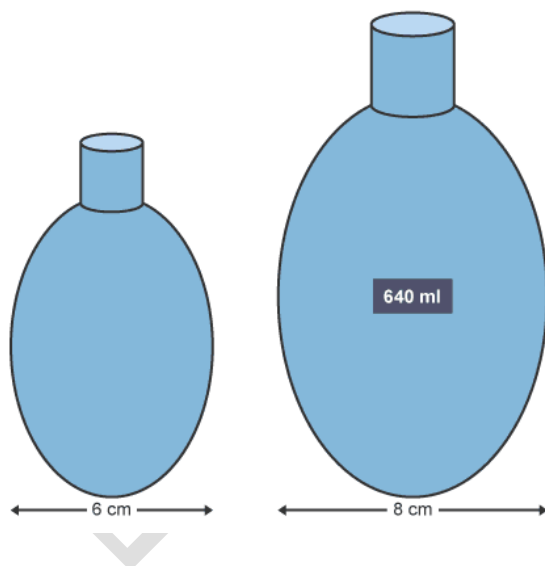
The volume scale factor = $(1.5)^3 = 3.375$

The volume of shape B = $400\text{cm}^3 \times 3.375 = 1350\text{cm}^3$



Application activity 11.3.3

1) The vases below are similar. Calculate the missing volume.



2) Two similar objects have corresponding sides 4 cm and 7 cm. The volume of the larger one is 230 cm^3 . What is the volume of the smaller one? Give your answer to 1 decimal place.

3) Two similar jugs have capacities of 2 litres and 3 litres. If the height of the larger jug is 30 cm, find the height of the smaller one.

4) Two similar cones made of the same wood have masses 4 kg and 0.5 kg respectively. If the base area of the smaller cone is 38.5 cm^2 , find the base area of the larger one?



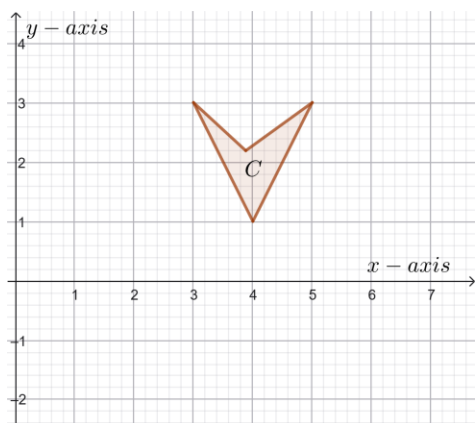
END OF UNIT ASSESSMENT

Section A: Knowledge and Understanding

1. What are the **two key geometric properties** that define two shapes as being mathematically **similar**?
2. In an enlargement, what is the name given to the fixed point from which the shape is scaled, and what does the **scale factor (k)** tell us?
3. State the relationship between the **linear scale factor (k)**, the **area scale factor**, and the **volume scale factor** for similar shapes.
4. **True or False:** If a shape is enlarged with a **scale factor of $k = -2$** , the image is on the opposite side of the centre of enlargement and is twice the linear size of the object.
5. Explain the difference between **enlargement with a positive scale factor** and **enlargement with a negative scale factor**.

Section B: Skills

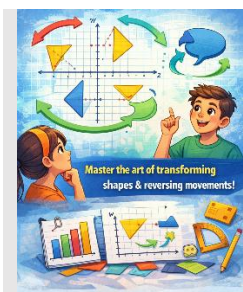
6. Triangles ABC and DEF are similar. $\angle A = 50^\circ$, $\angle B = 60^\circ$, and side $AB = 8$ cm corresponds to side $DE = 12$ cm.
 - a) What is the measure of $\angle F$?
 - b) What is the linear scale factor from triangle ABC to triangle DEF?
7. In triangle PQR, line ST is parallel to QR, with S on PQ and T on PR. Given $PS = 4$ cm, $SQ = 6$ cm, and $PT = 5$ cm, find the length of TR.
8. Two similar rectangles have corresponding sides in the ratio 3:7. If the shorter side of the smaller rectangle is 9 cm, what is the length of the corresponding side on the larger rectangle?
9. Triangle A has vertices at (1, 1), (1, 3), and (3, 1). Using the origin (0,0) as the centre and a scale factor of **$k = 3$** , find the coordinates of the enlarged triangle A'.
10. Shape B is a square with vertices at (2, 2), (2, 4), (4, 4), (4, 2). It is enlarged with centre (3, 3) and scale factor **$k = -1$** . Find the coordinates of the image B'. Describe the single transformation this represents.
11. An object with vertices at (2, 3), (4, 5), and (6, 3) is mapped to an image at (5, 6), (9, 10), and (13, 6). By inspecting corresponding vertices, determine:
 - a) The scale factor of the enlargement.
 - b) The coordinates of the centre of enlargement.
12. Shape C is enlarged with centre (0,0) and scale factor **$k_1 = 2$** . The resulting image is then enlarged with the same centre and scale factor **$k_2 = 0.5$** .



- What single enlargement maps shape C directly to the final image?
- What is this final transformation called?

Section C: Attitudes and Values

- Explain one way in which **enlargement and similarity** are used in real life (for example, maps, architecture, or design).
- Why is **accuracy and precision** important when performing enlargements on the Cartesian plane?
- How can **working collaboratively** help students solve problems involving similarity and enlargement more effectively?



UNIT 12: INVERSE AND COMPOSITE TRANSFORMATIONS IN 2D

Key Unit Competence: Execute combination of transformations in solving geometric problems involving the inverse and composite transformations of shapes.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and Understanding

- Define a composite transformation as a sequence of single movements (translation, reflection, rotation).
- Explain an inverse transformation as the operation that returns a shape to its original position.

Skills

- Perform combined transformations and construct the inverse of a given transformation.

- Apply knowledge of composites and inverses to solve for unknown coordinates or geometric properties.

Attitudes and Values

- Appreciate the relevance of transformations in real-world applications.
- Strengthen spatial reasoning, logical sequencing, and collaborative investigation skills.

Introduction

This unit advances geometry by teaching you to combine and reverse fundamental transformations (translation), reflection, and rotation. You will learn to perform composite sequences and construct precise inverses, skills essential for controlling shape movement in fields like animation and engineering. Your key goal is to apply these transformations to solve geometric problems, such as finding unknown coordinates or proving properties. This work will sharpen your spatial reasoning, logical sequencing, and appreciation for how abstract operations model real-world design, with collaboration deepening your understanding of the structured logic behind moving shapes.



Introductory activity

Read and workout the following task:

- 1) Draw a coordinate plane on your graph paper (x-axis from -6 to 6 , y-axis from -6 to 6).
- 2) Place triangle ABC with vertices: A (1, 1); B (3, 1); C (2, 3).
- 3) First, **rotate triangle ABC 90° counterclockwise** about the origin. You find the shape $A'B'C'$
- 4) Then, **reflect the result $A'B'C'$ over the y-axis**. You get the shape $A''B''C''$.

Let's call this combined move from ABC to the shape $A''B''C''$ as the **transformation M**.

Compare the triangle ABC and the shape $A''B''C''$:

- i) Do they have the same size?
- ii) Do corresponding angles have the same size?
- iii) Do sides have the same size?

12.4 Composite transformations in two dimensions

12.4.1 Introduction to composite central symmetry in two dimensions



Activity 12.4.1

- (a) Plot point A(2,3) and centre O(0,0)
- (b) Reflect A through O to get A' (central symmetry)
- (c) Now take A' and reflect it through a DIFFERENT centre C(4,1)
- (d) Observe the final point A''.
- (e) Repeat with different starting points.

(f) What single transformation could replace these two reflections?



Content summary

A single central symmetry about point C maps P to P' such that C is the midpoint of PP'

In coordinates: If C(h,k), then P'(2h-x, 2k-y) from P(x,y)

Scale factor: -1 (180° rotation about C)

Definition: Two consecutive central symmetries about centres C_1 and C_2

Theorem 1: Composition of two central symmetries is a **translation**

- Proof: Let S_1 be symmetry about C_1 , S_2 about C_2
- For any point P: $S_2(S_1(P)) = P + 2(C_2 - C_1)$
- Translation vector: $\mathbf{v} = 2(\mathbf{C}_2 - \mathbf{C}_1)$

Theorem 2: Composition is **commutative**

- $S_2 \circ S_1 = S_1 \circ S_2$ (both give same translation)
- Because $2(\mathbf{C}_2 - \mathbf{C}_1) = -2(\mathbf{C}_1 - \mathbf{C}_2)$

Theorem 3: Three symmetries about centres C_1, C_2, C_3

- $S_3 \circ S_2 \circ S_1$ is another **central symmetry**
- The new centre is: $\mathbf{C} = \mathbf{C}_1 - \mathbf{C}_2 + \mathbf{C}_3$

Example 1

Find the image of P(1,2) after Symmetry about A(0,0) followed by Symmetry about B(3,1).

Solution:

Step 1: About A(0,0): $P \rightarrow P_1(-1,-2)$

Step 2: About B(3,1): $P_1 \rightarrow P'$

Formula: $P' = (2 \times 3 - (-1), 2 \times 1 - (-2)) = (7, 4)$

Translation vector: $2(\mathbf{B}-\mathbf{A}) = 2(3,1) = (6,2)$

Check: $(1,2) + (6,2) = (7,4) \checkmark$

Example 2

Two symmetries about $C_1(2,3)$ and $C_2(5,7)$. What translation results?

Solution:

$\mathbf{v} = 2(\mathbf{C}_2 - \mathbf{C}_1) = 2(3,4) = (6,8)$

Example 3

A translation by vector (4,-2) results from two central symmetries. If first centre is (1,1), find the second centre.

Solution:

$$v = 2(C_2 - C_1)$$

$$(4, -2) = 2(C_2 - (1, 1))$$

$$(2, -1) = C_2 - (1, 1)$$

$$C_2 = (3, 0)$$

Example 4.

Triangle $\triangle ABC$ has vertices $A(1, 2)$, $B(4, 3)$, $C(3, 5)$

It is first reflected through point $P(0, 0)$ (central symmetry about P), then the image is reflected through point $Q(3, 1)$.

(a) Find the vertices of the final image $\triangle A''B''C''$

(b) What single transformation is equivalent to these two central symmetries?



Application activity 12.4.1

1) Find the image of triangle $\triangle ABC$ with $A(1, 1)$, $B(4, 2)$, $C(3, 5)$ after:

- Symmetry about $D(0, 0)$
 - Followed by symmetry about $E(2, 3)$
- Describe the single transformation.

2) A point P undergoes:

- Symmetry about $F(1, 2)$
- Symmetry about $G(4, 6)$
- Symmetry about $H(2, 1)$

Find the coordinates of the final point if P started at $(0, 0)$.

3) Prove algebraically that for any points C_1 and C_2 :

$$S_2 \circ S_1 = S_1 \circ S_2$$

4) A rectangle has vertices $(0, 0)$, $(4, 0)$, $(4, 2)$, $(0, 2)$. After two symmetries about centres $(1, 1)$ and $(3, 3)$, find:

- a) The image coordinates
- b) Area of final shape
- c) Single transformation

5) A logo design uses composite symmetry:

1. Reflect through centre of circle
2. Reflect through point on circumference

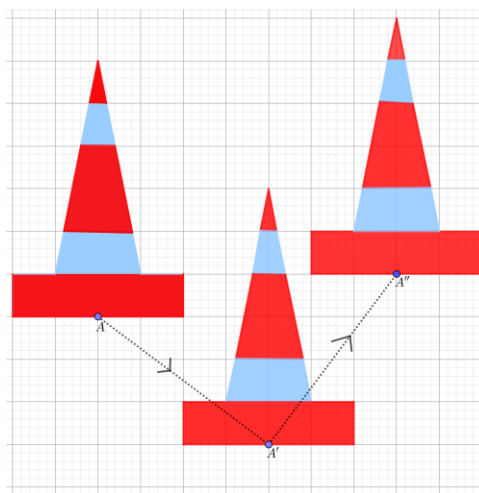
If circle centre is $(0, 0)$, radius 5, and second centre is $(3, 4)$, find where point $(2, 0)$ on logo moves to.

12.4.2 Composite translations in two dimensions



Activity 12.4.2

- Write the translation vector that maps $A \rightarrow A'$.
- Write the translation vector that maps $A' \rightarrow A''$.
- Describe the direction (left/right, up/down) of each translation.
- What single translation would map A **directly** to A'' ?
- Express the **composite translation vector** as the sum of the two individual translation vectors.
- Explain why applying the two translations successively gives the same result as the single composite translation.
- Does the orientation or shape of the object change after the composite translation? Explain why.
- Would the final image change if the order of the two translations were reversed? Justify your answer.
- Give a real-life example that can be modelled by a composite translation similar to the one shown.



Content summary

Composite translations involve applying two or more simple translations (slides) one after another, where the second transformation acts on the result of the first, often combining to form a single, simpler transformation.

The order matters, but the composition of two translations always results in a single, overall translation, found by adding their translation vectors.

Example 1

Triangle ABC is such that A $(-4, 3)$, B $(-1, 1)$ and C $(-2, 5)$.

- Find the co-ordinates of its image after a translation whose vector is $(6, 2)$.
- The image of ABC in (a) above is then translated by the vector $(-2, -5)$. Find the co-ordinates of the new image.
- Plot ABC and its images on the same Cartesian plane.

Solution

$$(a) \quad \mathbf{OA}' = \begin{pmatrix} -4 \\ 3 \end{pmatrix} + \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}, A'(2,5)$$

$$\mathbf{OB}' = \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, B'(5,3)$$

$$\mathbf{OC}' = \begin{pmatrix} -2 \\ 5 \end{pmatrix} + \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}, C'(4,7)$$

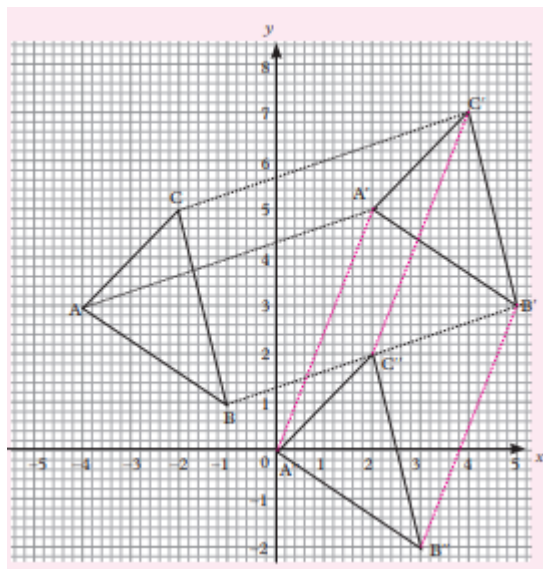
$$(b) \quad \mathbf{OA}'' = \begin{pmatrix} 2 \\ 5 \end{pmatrix} + \begin{pmatrix} -2 \\ -5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, A''(0,0)$$

$$\mathbf{OB}'' = \begin{pmatrix} 5 \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ -5 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}, B''(3,-2)$$

$$\mathbf{OC}'' = \begin{pmatrix} 4 \\ 7 \end{pmatrix} + \begin{pmatrix} -2 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, C''(2,2)$$

- (c) Therefore the co-ordinates of the images are: $A'(2, 5)$, $B'(5, 3)$ and $C'(4, 7)$ and $A''(0, 0)$, $B''(3, -2)$, $C''(2, 2)$.

This can be illustrated on the Cartesian plane as shown in graph



Example 2

Points $A(1,1)$, $B(1, 2)$, $C(4, 2)$ and $D(4, 1)$ represent a rectangle.

(a) Translate $ABCD$ under vector $\begin{pmatrix} -2 \\ -1 \end{pmatrix}$. Write down image co-ordinates A' , B' , C' and D' .

(b) Translate A' , B' , C' and D' under vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$. Write down image co-ordinates for $A''B''C''$ and D'' .

Solution

$$(a) \quad \mathbf{OA}' = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\mathbf{OB}' = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\mathbf{OC}' = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\mathbf{OD}' = \begin{pmatrix} 4 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\therefore A'(-1,0), B'(-1, 1), C'(2, 1) \text{ and } D'(2, 0).$$

$$(b) \quad \mathbf{OA}'' = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\mathbf{OB}'' = \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\mathbf{OC}'' = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$\mathbf{OD}'' = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$\therefore A''(1,2), B''(1,3), C''(4,3) \text{ and } D''(4, 2).$$



Application activity 12.4.2

- (a) Triangle XYZ is such that $X(-2, 1)$, $Y(-3, 2)$ and $Z(-3, 4)$. It is given a translation of vector $(2, 2)$. Find the co-ordinates of its image, $X'Y'Z'$.

(b) The image $X'Y'Z'$ is translated by the vector $(-4, 2)$. Find the new image $X''Y''Z''$.

(c) Plot XYZ and its images $X'Y'Z'$ and $X''Y''Z''$ on the same Cartesian plane.
- The points A (4, 0), B (4, 2), C (0, 2) and D (0, 0) represent a rectangle.

(a) Find the image $A'B'C'D'$ of ABCD under translation represented by vector $(-3, 4)$.

(b) The image of ABCD is further translated by vector $(8, 0)$. Find the co-ordinates of the new image $A''B''C''D''$.

(c) Plot the ABCD and its images on the same Cartesian plane.
- Points A (3, 4), B (3, 7), C (6, 7) and D (9, 4) represent a trapezium.

(a) Find the image $A'B'C'D'$ of trapezium ABCD under a translation represented by vector $(-6, -5)$.

(b) The Image obtained in (a) above is then translated by vector $(3, -2)$. What are the co-ordinates of the new image $A''B''C''D''$?

(c) Plot ABCD and its images on the same Cartesian plane.
- Triangle PQR has co-ordinates $P(0, 0)$, $Q(-3, 2)$ and $R(2, 2)$. (a) What is the image $P'Q'R'$ of PQR under a translation represented by the vector $(-2, -2)$?

(b) The image obtained in (a) above is further translated by the vector $(-4, 5)$. What are the co-ordinates of the new image $P''Q''R''$?

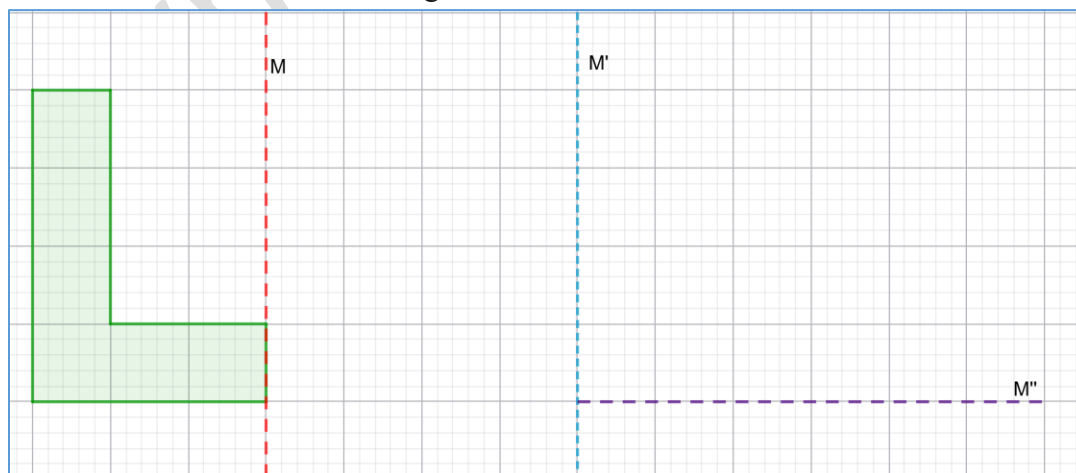
(c) Plot PQR and its two images on the same Cartesian plane.

12.4.3 Composite reflections in two dimensions



Activity 12.4.3

Copy and draw the final image of letter L after reflection on mirror line M, then its image is reflected on M' and then the image is reflected on mirror line M'' .





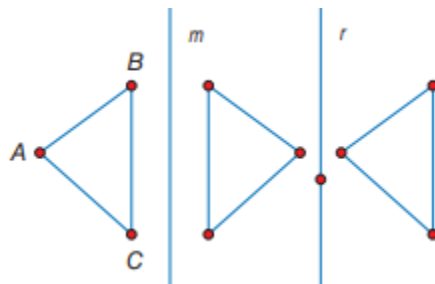
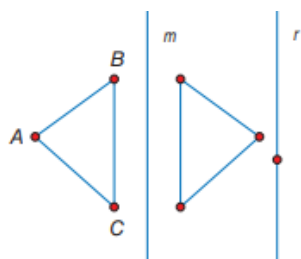
Content summary

A composite reflection is two or more reflection performed one after the other.

Some simple reflections can be performed easily in the co-ordinate plane using the following general rules as we learnt early.

Example 1

Triangle ABC is reflected the mirror line m as shown in the figure. Copy and draw the final image, if the image of triangle ABC is also reflected on the mirror line r .



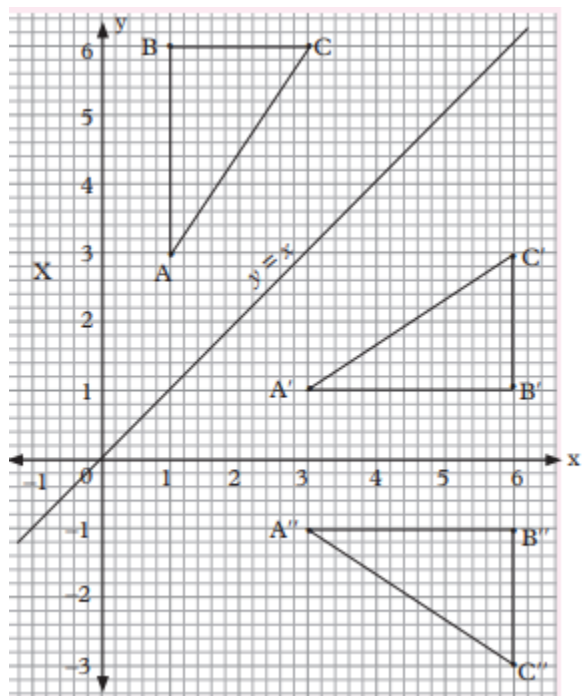
Solution

Example 2

The points A (1, 3), B (1, 6) and C (3, 6) are the vertices of triangle ABC.

- On a Cartesian plane, draw triangle ABC.
- Triangle ABC is reflected in the line $y = x$. Draw the image $A'B'C'$ on the same Cartesian plane and state its co-ordinates.
- Triangles $A'B'C'$ is then reflected in the line $y = 0$ (x-axis). Draw the image $A''B''C''$ of $A'B'C'$ on the same Cartesian plane and state its co-ordinates.

Solution



Co-ordinates of the images. $A'(3,1), B'(6,1), C'(6,3)$ $A''(3,-1), B''(6,-1), C''(6,-3)$

Example 3

A quadrilateral ABCD has vertices $A(2,1), B(4, 1), C(4, 3)$ and $D(2, 3)$.

(a) Find the image of the quadrilateral ABCD under a reflection in the line $x = 0$.

(b) State its co-ordinates under a reflection in the line $y = -x$.

Example 4

The vertices of a quadrilateral are $A(2, 0.5), B(2, 2), C(4, 3.5)$ and $D(3.5, 1)$. Find the image of the quadrilateral under reflection in line $y=0$ followed by reflection in line $y = -x$.

Solution

'Reflection in line $y=0$ followed by reflection in line $y=-x$ means that we first obtain the image under reflection in line $y=0$ and then reflect this image in line $y = -x$.

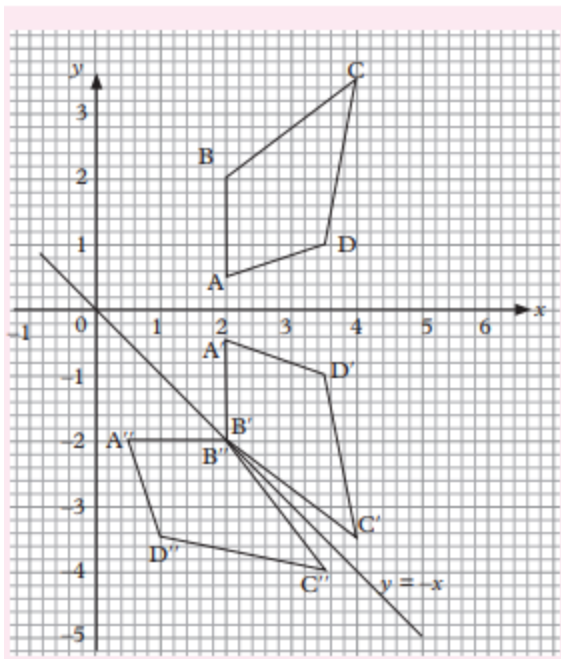
In the graph,

$A'B'C'D'$ is the reflection of ABCD in line $y = 0$.

$A''B''C''D''$ is the reflection of $A'B'C'D'$ in line $y = -x$.

Thus the required image vertices are:

$A(2, -0.5), B(2, -2), C(4, -3.5)$ and $D(3.5, 1)$ $A''(0.5, -2), B''(2, -2), C''(3.5, -4)$ and $D''(1, 3.5)$



Application activity 12.4.3

1) (a) Draw triangle ABC with vertices as A (6, 8), B (2, 8), C (2, 6).

Draw the lines $y = 2$ and $y = x$.

(b) Draw the image of triangle ABC after reflection in;

(i) The y-axis. Label it triangle A'B'C'.

(ii) The x-axis. Label it triangle A''B''C''.

(iii) The line $y=x$. Label it triangle A'''B'''C'''.

(c) Write down the co-ordinates of the image of point A in each case.

2) (a) Plot triangle 1 defined by (3,1), (7,1) (7,3).

(b) Reflect triangle 1 in the line $y = x$ onto triangle 2.

(c) Reflect triangle 2 in the line $y = 0$ (x-axis) onto triangle 3.

(d) Reflect triangle 3 in the line $y = -x$ onto triangle 4.

(e) Reflect triangle 4 in the line $x = 2$ onto triangle 5.

(f) Write down the co-ordinates of triangle 5.

3) (a) Construct triangle 1 at (2, 6), (2, 8), (6, 6).

(b) Reflect triangle 1 in the line $x + y = 6$ onto triangle 2

(c) Reflect triangle 2 in the line $x = 3$ onto triangle 3.

(d) Reflect triangle 3 in the line $x + y = 6$ onto triangle 4

(e) Reflect triangle 4 in the line $y = x - 8$ onto triangle 5.

(f) Write down the co-ordinates of triangle 5.

4) (a) Draw and label the following triangles.

Triangle 1: (3, 3), (3, 6), (1, 6)

Triangle 2: (3,-1), (3, -4), (1, -4)

Triangle 3: (3, 3), (6, 3), (6, 1)

Triangle 4: (-6, -1), (-6, -3), (-3, -3)

Triangle 5: (-6, 5), (-6, 7), (-3, 7)

(b) Find the equation of the mirror line for the reflection;

(i) Triangle 1 onto triangle 2

(ii) Triangle 1 onto triangle 3

(iii) Triangle 1 onto triangle 4

(iv) Triangle 4 onto triangle 5.

12.4.4 Composite rotations in two dimensions



Activity 12.4.4

You are given a line segment with end points A (2, 2) and B (2, 5).

(a) Draw line AB on a Cartesian plane.

(b) Fix point A and rotate point B through 60° in clockwise direction.

(c) Write down the co-ordinate B' the image of B.

(d) Leaving A as a fixed point, rotate point B' 90° in anticlockwise direction.

(e) Write down the co-ordinate B'' the image of B'



Content summary

Composite rotations in 2D combine multiple rotations into a single equivalent rotation, where successive rotations by different angles result in a total rotation by their sum.

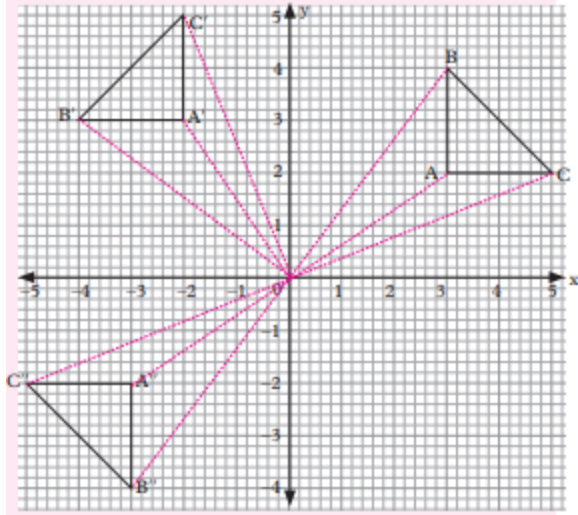
Example 1

Given the vertices of triangle ABC as A (3, 2), B (3, 4) and C (5, 2);

(a) Rotate triangle ABC through $+90^\circ$ about the origin onto A'B'C'. Write down the co-ordinates of A'B'C'

(b) Rotate A'B'C' through $+90^\circ$ about the origin onto . Write down the coordinates of A''B''C''

Solution



Co-ordinates of the Images. $A'(-2, 3)$, $B'(-4, 3)$, $C'(-2, 5)$ $A''(-3, -2)$, $B''(-3, -4)$, $C''(-5, -2)$

Example 2

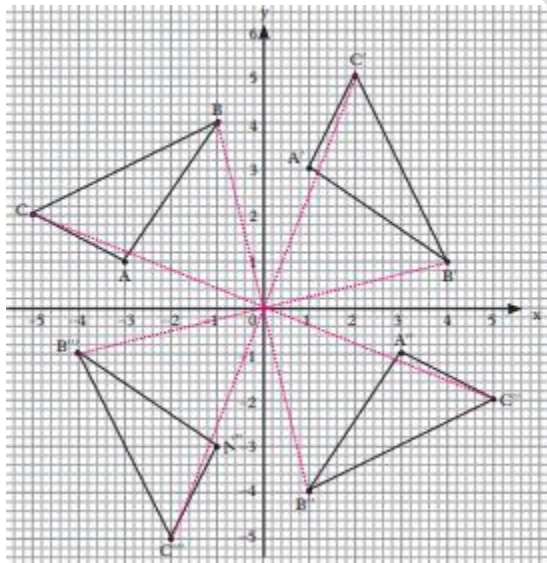
Consider triangle $A(-3, 1)$, $B(-1, 4)$, and $C(-5, 2)$.

(a) Rotate triangle ABC through -90° about the origin onto A', B', C' .

(b) Rotate $A'B'C'$ through -90° about the origin onto triangle $A''B''C''$.

(c) Rotate $A''B''C''$ through -90° about the origin to get triangle $A'''B'''C'''$. Write down the co-ordinates of images $A'B'C'$, $A''B''C''$ and $A'''B'''C'''$.

Solution



(d) $A'(1, 3)$, $B'(4, 1)$, $C'(2, 5)$ $A''(3, -1)$, $B''(1, -4)$ $C''(5, -2)$ $A'''(-1, -3)$, $B'''(-4, -1)$, $A'''(-1, -3)$ $C'''(-2, -5)$,



Application activity 12.4.4

1) (a) Draw triangle one at $A(1, 2)$, $B(1, 6)$, $C(3, 5)$.

(b) Rotate triangle one 90° clockwise centre $(1, 2)$ onto triangle two.

- (c) Rotate triangle two 180° centre $(2, -1)$ onto triangle three.
 (d) Rotate triangle three 90° clockwise, centre $(2, 3)$ onto triangle four (e) Write down the co-ordinates of triangle four.

2) (a) Draw and label the following triangles on a Cartesian plane.

Triangle 1: $(3, 1), (6, 1), (6, 3)$

Triangle 2: $(-1, 3), (-1, 6), (-3, 6)$

Triangle 3: $(1, 1), (-2, 1), (-2, -1)$

Triangle 4: $(-3, 1), (-3, 4), (-5, 4)$

(b) Describe fully the following rotations.

- (i) 1 onto 2
- (ii) 1 onto 3
- (iii) 1 onto 4
- (iv) 3 onto 2

12.5 Mixed transformations in two dimensions



Activity 12.5

Given triangle ABC has vertices $A(0, 2), B(2, 2)$ and $C(2, 5)$.

(a) Plot triangle ABC on a Cartesian plane.

(b) Reflect triangle ABC in the line $x = 3$ to get the image $A'B'C'$. Write the co-ordinates of $A'B'C'$.

(c) Translate triangle $A'B'C'$ by the vector $(2, 2)$ to get the image $A''B''C''$. Write down the co-ordinates of $A'', B'',$ and C'' .



Content summary

Mixed composite transformations in two dimensions is a transformation in which an object can undergo several different transformations, one after the other.

This is done such that the image of the preceding transformation becomes the object of the next transformation.

When an object, A, undergoes a transformation R followed by another transformation N, this is represented as $N[R(A)]$ or $NR(A)$.

Example 1

A is a triangle with vertices $(2, 1), (2, 4)$ and $(4, 1)$. R is a rotation of 90° . Clockwise about the origin and N is a reflection in the y-axis. Find the vertices of the image of A if it undergoes the following transformations:

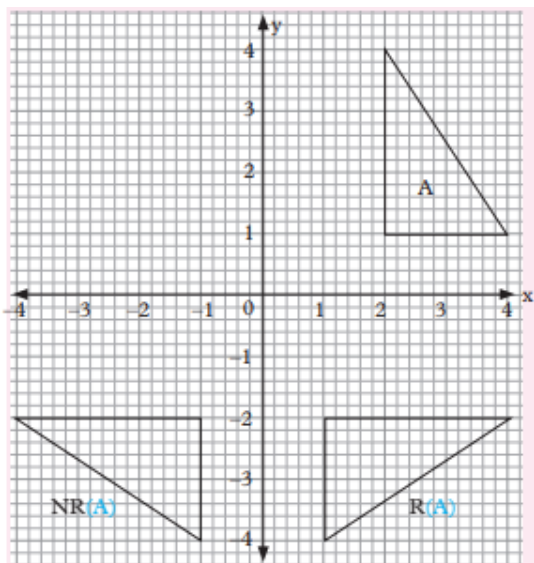
- (a) $NR(A)$ (b) $RN(A)$

Solution

The graph shows triangle A under these transformations and its image $NR(A)$ and $R(A)$.

(a) $(2, 1), (2, 4)$ and $(4, 1)$

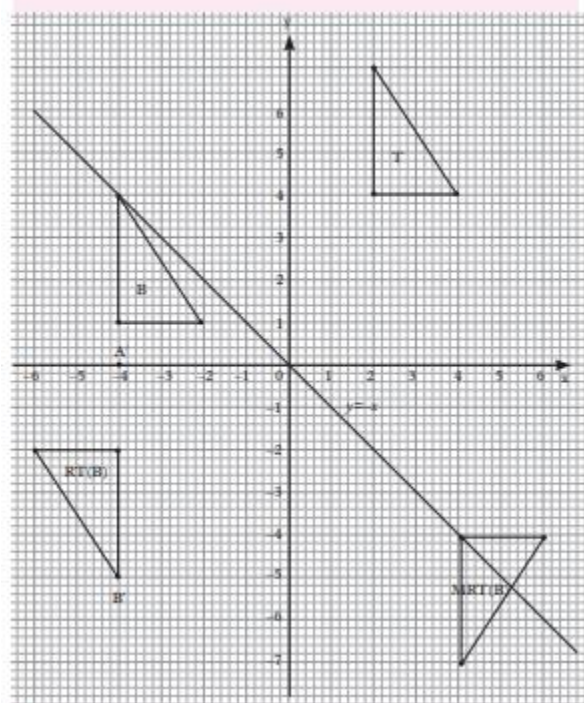
(b) The final image of A has vertices $(-1, -2), (-1, -4)$ and $(-4, -2)$.



Example 2

Triangle B has vertices at $(-2, 1)$, $(-4, 1)$ and $(-4, 4)$. T is a translation represented by the vector $(8, 3)$, R is a reflection in the line $y = -x$ and M is a rotation of 90° anticlockwise about the origin. Find the vertices of MRT(B).

Solution



Vertices of MRT (B) are $(4, -4)$, $(6, -4)$, $(4, -7)$.



Application activity 12.5

1) Draw x- and y-axes with values from -8 to $+8$ and plot the point P $(3, 2)$. R denotes 90° clockwise rotation about $(0, 0)$; X denotes reflection in $x = 0$. H denotes 180° rotation about $(0,$

0); T denotes translation $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$. For each question, write down the co-ordinates of the final image of P.

- (a) $R(P)$ (b) $T(P)$ (c) $TH(P)$ (d) $HX(P)$ (e) $R^2(P)$ (f) $TR(P)$ (g) $RT(P)$
 (h) $XT(P)$ (i) $XX(P)$ (j) $XTH(P)$

12.6 Inverse transformations in two dimensions



Activity 12.6

You are given co-ordinates of figure ABCD as $A(0, 0)$, $B(2, 2)$, $C(-2, 2)$ and $D(-2, 0)$.

- (a) On a Cartesian plane, draw the figure ABCD.
 (b) Rotate figure ABCD through 90° about point $(0, 0)$ in clockwise direction to get the image $A'B'C'D'$.
 (c) Translate the image $A'B'C'D'$ with $T = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ to get image $A''B''C''D''$.
 (d) Translate $A''B''C''D''$ with $T = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ to get the Image $A'''B'''C'''D'''$.
 (e) Plot all the images on the same Cartesian plane.



Content summary

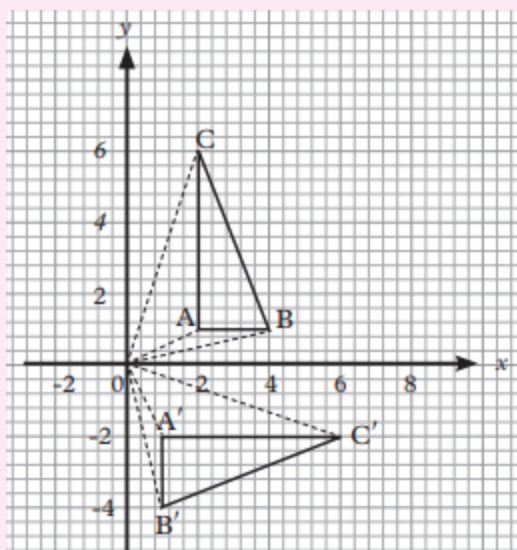
The inverse of a transformation is the transformation which maps the image back to the object. The following are examples of inverse transformations are defined by the following cases.

1. If R denotes θ° clockwise rotation, about $(0, 0)$, then R^{-1} denotes θ° anticlockwise rotation about $(0, 0)$.
2. If translation T has vector, the translation which has the opposite effect has vector. This is written as T^{-1} .
3. For all reflections, the inverse is the same reflection. For example if X is reflection then X^{-1} is also reflection. The symbol T^{-3} means $(T^{-1})^3$ i.e. perform T^{-1} three times.

Example 1

An image of a triangle with vertices $A'(1, -2)$, $B'(1, -4)$ and $C'(6, -1)$ is formed after an object ABC undergoes a clockwise rotation through 90° about the origin. Find by construction the co-ordinates of its object, ABC.

Solution



$A(2, 1)$, $B(4, 1)$ and $C(2, 6)$

Example 2

$P'Q'R'S'$ is the image trapezium whose vertices are $P'(1, 3)$, $Q'(4, 3)$, $R'(1, 1)$ and $S'(6, 1)$. $P'Q'R'S'$ is obtained after reflection in the $x=0$.

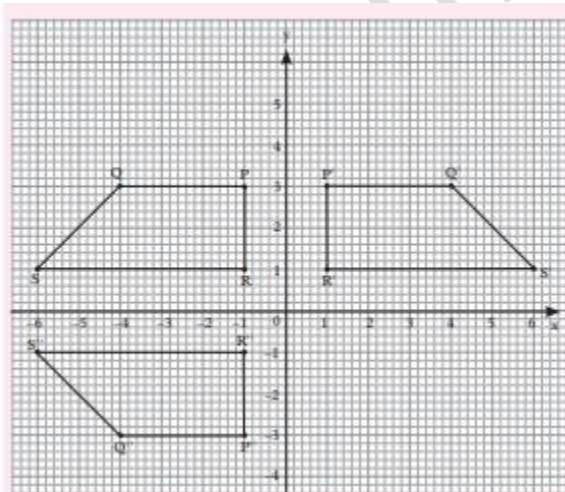
(a) Obtain the co-ordinates of object trapezium PQRS.

(b) Write down the co-ordinates of $P''Q''R''S''$ the image of PQRS when reflected in the line $y=0$.

Solution

All the co-ordinates can be obtained from graph below.

- The line $x=0$ is y -axis.
- The line $y=0$ is x -axis.



$P(-1, 3)$, $Q(-4, 3)$, $R(-1, 1)$ and $S(-6, 1)$. $P''(-1, -3)$, $Q''(-4, -3)$, $R''(-1, -1)$, and $S''(-6, -1)$.



Application activity 12.6

Draw x- and y-axes with values from -8 to $+8$ and plot the point $Q(3, 2)$. R denotes 90° clockwise rotation about $(0, 0)$; X denotes reflection in $x = 0$. H denotes 180° rotation about $(0, 0)$; T denotes translation $(3, 2)$.

Perform each of the following transformation. Write down the co-ordinates of the final image of Q .

- | | |
|-----------------|-------------------|
| 1. $R^{-1}(Q)$ | 6. $T^{-1}R(Q)$ |
| 2. $T^1(Q)$ | 7. $R^{-1}T(Q)$ |
| 3. $(T^3)H(Q)$ | 8. $XT^1(Q)$ |
| 4. $HX^{-1}(Q)$ | 9. $XX^{-1}(Q)$ |
| 5. $RR^{-1}(Q)$ | 10. $XT^{-1}H(Q)$ |



END OF UNIT ASSESSMENT

Section A: Knowledge and Understanding

- What is a **composite transformation**? If transformation T_1 is followed by transformation T_2 , what notation do we use for the combined transformation?
- What is meant by the **inverse** of a geometric transformation T ? What is the result when a transformation is combined with its inverse?
- Based on the lessons, state the final result of the following composites:
 - Two central symmetries (point reflections) about different centres.
 - Two reflections in parallel lines.
 - Two rotations about the same centre.
- True or False:** The order in which you perform transformations **always** matters in a composite transformation. (For example, "Reflect then Rotate" gives the same result as "Rotate then Reflect").
- A shape is translated by the vector $(5, -2)$. Describe in words what the **inverse transformation** would do to the shape.
- Explain how **translation**, **reflection**, and **rotation** differ from one another.
- State the inverse of each of the following transformations:
 - Translation by vector $(3, -2)$
 - Rotation of 90° clockwise about the origin
 - Reflection in the x-axis
- Explain why performing a transformation followed by its inverse returns a shape to its original position.

Section B: Skills (9 questions)

- A point $A(2, 1)$ is translated by the vector $(4, -3)$ and then reflected in the y-axis. Find the final coordinates of the image.

10. Triangle ABC has coordinates A(1,1), B(3,1) and C(2,4)

The triangle is rotated 90° anticlockwise about the origin and then translated by (2,-1).

Find the coordinates of the image of triangle ABC. Hence, draw the object and its image.

11

1. Draw triangle XYZ with vertices at X (-2, 4), Y(-2, 1), Z (-4, 1). Find the image of XYZ under the following combinations of transformations.

- (a) A reflection in the line $x = 0$.
- (b) A rotation through an angle of 180° about (0, 0).

12

4. (a) Plot the triangle ABC at A (4, 6), B (1, 6), C (1, 4). Draw the line $y = 2$ and $y = x$.

(b) Plot the image of triangle ABC after reflection in;

- (i) The y-axis. Label it triangle 1
- (ii) The line $y = 2$. Label it triangle 2.
- (iii) The line $y = x$. Label it triangle 3.

(c) Write down the co-ordinates of the image of A in each case.

13.

2. Plot triangle PQR at P(2, 2), Q(6, 2), R(6, 4). Find the image of PQR under the following combinations of transformations; a translation, under vector $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ followed by a reflection in the line $x=3$. Write down the co-ordinates of the final image of point P.

Section C: Attitudes and Values

14. Give one **real-life application** where composite or inverse transformations are useful.

15. Why is **logical sequencing** important when performing composite transformations?

16. Explain how **collaborative work** can help students correctly perform and verify inverse transformations.



UNIT 13: INTERPRETATION OF STATISTICAL GRAPHS

Key Unit Competence: Student should be able to interpret statistical graphs from both ungrouped and grouped datasets in informing conclusions or decisions.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and understanding:

- Differentiate between raw (ungrouped) and organized (grouped) data.
- Identify and describe common statistical graphs, such as bar charts, pie charts, and histograms.

Skills:

- Extract and interpret information from various graphs to identify trends and patterns.
- Draw reasoned conclusions and make simple predictions based on data analysis.

Attitudes and Values:

- Recognize the importance of using data to inform decisions in real life.
- Develop a critical and accurate approach to interpreting and presenting statistical information.

Introduction:

This unit focuses on the essential skill of interpreting data, moving from raw numbers to meaningful insights. You will learn to distinguish between ungrouped and grouped data, and master visual tools like histograms and bar charts to analyze information. Your goal is to extract trends, draw sound conclusions, and make data-driven decisions, while also developing a critical eye for accuracy and representation. This empowers you to become both an informed consumer and a clear communicator of statistical information in everyday life.



Introductory activity

Our school canteen manager, Rukundo, is confused! Last week, he sold different snacks but can't remember which were most popular. He needs Your Help as **Data Detectives** to figure out which snacks students really like, so he knows what to buy more of next week.

"Mr. Rukundo gave us his sales notes, but they're all mixed up!"

MONDAY: chips, juice, cookies, chips, fruit, chips.

TUESDAY: chocolate, chips, juice, cookies, chips

WEDNESDAY: fruit, juice, chips, chocolate, chips

THURSDAY: cookies, chips, juice, chips, fruit

FRIDAY: chips, chocolate, juice, chips, cookies

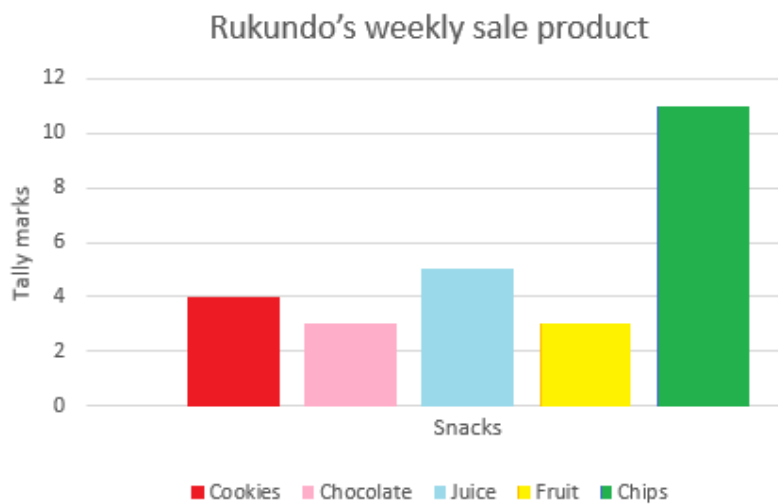
1) Read and answer:

- (a) Which snack was sold most often?
- (b) Which snack was sold least?
- (c) Is this easy or hard to figure out? Why?

2) Copy and complete tally chart below.

Snack	Tally Marks	Total
Chips		
Juice		
Cookies		
Fruit		
Chocolate		

3) If a **Simple Bar Chart** below shows **Rukundo's weekly sale product**.



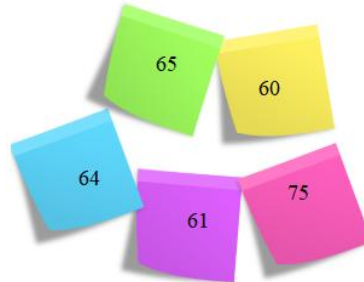
- a) "Which bar is tallest? What does that mean?"
- b) "Which two snacks had the same sales?"
- c) "Is this chart easier to understand than the messy list? Why?"

13.1 Meaning and examples of ungrouped and grouped data



Activity 13.1

Work in groups: each group receives 20 sticky notes with different numbers (ages, test scores, heights in cm)



Task 1: Record each number individually on worksheet (creates ungrouped data)

Task 2: Create categories/ranges (e.g., 10-14 years, 70-79 marks)

Task 3: Sort sticky notes into these categories

Key Questions:

- (a) Which is easier to read: individual numbers or grouped categories?
- (b) When would you use each method?
- (c) What information might you lose when grouping data?



Content summary

Ungrouped Data: Data presented in its original form, listing each observation individually.

Ungrouped Data shows exact values, no categories or intervals and it is useful for small datasets, and preserves original precision.

Grouped Data: Data organized into classes or intervals. It is essential for analyzing large datasets, calculating central tendencies (mean, median, mode) efficiently, and creating frequency distributions or histograms. It deals with the following **key terms**:

Class Interval: Range of values in a group (e.g., 10-19)

Class Limits: Lowest and highest values in interval

Class Boundaries: Exact limits for continuous data

Class Mark/Midpoint: Average of upper and lower limits

Frequency: Number of observations in each class

Note: Key Difference

Ungrouped data keeps every single data point, while grouped data condenses them into summaries (like frequency tables or histograms) for better understanding of overall trends, sacrificing individual precision for clarity on large datasets.

They both play the important role as foundation of statistics topics like mean, median, mode, and graphical representations (histograms, frequency polygons etc).

Examples of Ungrouped Data:

1. Individual test scores: 78, 85, 92, 67, 88, 74
2. Ages of family members: 12, 15, 38, 42, 45
3. Daily temperatures: 21°, 23°, 20°, 22°, 24°, 21°
4. A list of individual ages: 22, 35, 28, 41, 30.
5. Scores for each student on a test: 85, 92, 78, 95, 88.
6. Daily rainfall amounts in millimeters: 0.5, 1.2, 0, 2.1, 0.8.

Examples of Grouped Data:

- a) **Age Groups:** Children (0-12), Teenagers (13-19), Adults (20-64), Seniors (65+).
- b) **Test Scores (by range):** 70-79 (5 students), 80-89 (12 students), 90-100 (8 students).
- c) **Income Levels:** Under 20000FRW, 20000FRW-40000FRW, 40000FRW-60000FRW, etc..

Examples of converting Ungrouped to Grouped Data:

Given: Heights of 20 plants (cm):

12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 38, 40, 42, 45, 48, 50, 52, 55, 58, 60

Step 1: Determine range: $60 - 12 = 48$

Step 2: Decide number of classes (let's use 5)

Step 3: Class width $\approx 48 \div 5 = 9.6 \approx 10$

Step 4: Create intervals:

10-19, 20-29, 30-39, 40-49, 50-60

Step 5: Tally and count:

Height (cm)	Tally	Frequency
10-19		3
20-29		4
30-39		4
40-49		5
50-60		4

13.2 Reading and interpreting statistical graphs or diagrams for ungrouped data



Activity 13.2

A shopkeeper records the **number of customers per day** in one week. The table below shows the records done by a shopkeeper.

Day	Number of Customers
Monday	85
Tuesday	72
Wednesday	98
Thursday	105
Friday	150
Saturday	180
Sunday	90

- Look at the table. Which day had the most customers? Which had the fewest?
- How many more customers came on Friday than on Tuesday?
- What do you notice about the number of customers from Monday to Sunday? Can you describe the pattern?
- Why do you think Saturday had the highest number? Why might Tuesday be lower?
- What was the *total* number of customers for the week? What is the *average* (mean) number of customers per day? (This establishes the need for summarization).
- "Is it easy to see the pattern and compare days at a quick glance using just this table?"
- Use a bar chart to display this information.



Content summary

Ungrouped data (such as daily customer numbers) can be better understood when represented using graphs rather than tables alone. While tables show exact values, graphs make it easier to quickly compare data, identify patterns and trends over time, and recognize the highest and lowest values.

Learners are introduced to the **purpose of graphs**, which is to present data visually in order to:

- Compare values easily
- Observe trends and changes over time
- Identify maximum and minimum values
- Understand proportions within a whole

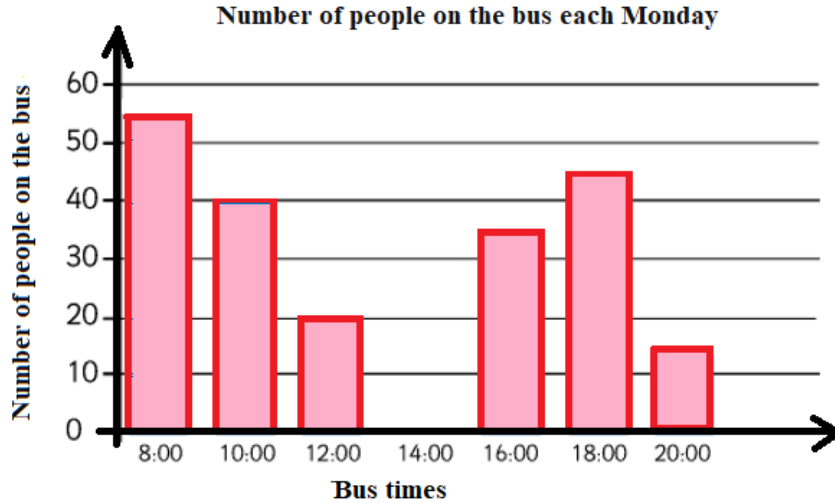
The lesson then explains the **main types of graphs used for ungrouped data**:

- **Bar charts**, which use equal-width bars with equal spacing to compare discrete categories. Proper labeling of axes, equal intervals, and a clear title are essential.
- **Line graphs**, which plot points and join them with straight lines to show how values change over time, making them suitable for identifying trends and patterns.

- **Pie charts**, which represent data as parts of a whole using sectors of a circle, where the total equals 100% or 360°, helping to understand proportions and percentages.

Example 1

This bar chart shows the number of people who get on a bus on Mondays.



- Make a frequency table
- On a Monday which bus is most likely to be full?
- which bus have the fewest people on it? Why?
- How many people used the bus altogether before 13:00?
- How often do the buses run?
- Do you think that the bus time table should change? How and why?
- 25 people were on the 14:00 bus? Draw in the bar.

Solution

Based on the bar chart, the approximate numbers of people on each bus are:

- 08:00 → 55 people
- 10:00 → 40 people
- 12:00 → 20 people
- 16:00 → 35 people
- 18:00 → 45 people
- 20:00 → 15 people

Frequency table

Bus time	Number of people
08:00	55
10:00	40
12:00	20
16:00	35
18:00	45

20:00	15
-------	----

a) The **08:00 bus**, because it has the highest number of people (55). This is likely due to people going to work or school.

b) The **20:00 bus** has the fewest people (15).

This is likely because most people have already returned home by the evening.

c) Before 13:00, the buses are at **08:00, 10:00, and 12:00**.

Total = $55 + 40 + 20 = 115$ $55 + 40 + 20 = 115$

Total = 115 people

d) From the given times, the buses run **every 2 hours**.

e) Yes.

- **More buses** could be added in the **morning (around 08:00)** and **early evening (18:00)** because these times are very busy.

- **Fewer buses** may be needed late in the evening (20:00) since fewer people use the bus.

This would reduce overcrowding and improve efficiency.

f) To draw the bar:

- Add a new bar at **14:00**

- The height of the bar should reach **25** on the vertical axis (halfway between 20 and 30)

Example 2

The bar chart shows the number of students who passed the end of year exam.

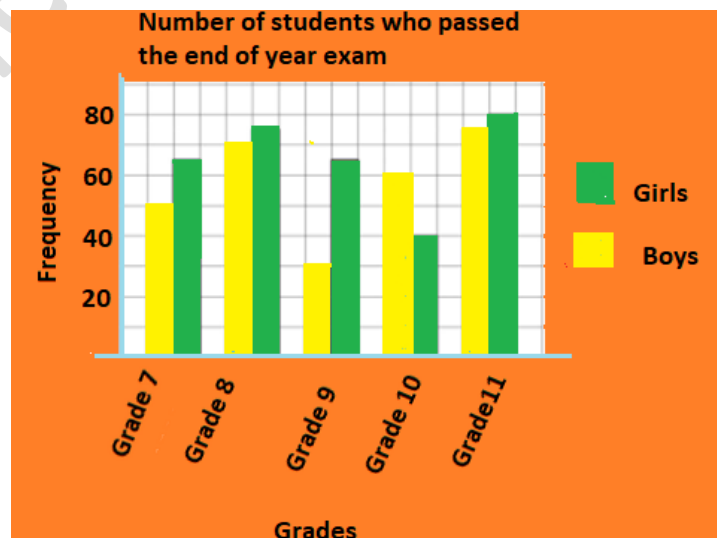
a) How many students in each grade passed the exam?

b) How many students passed the exam at the end of the year in total?

c) How many more girls passed the exam than boys in grade 9?

d) Which two year groups had the same number of girls who passed the exam?

e) Overall, girls did better than boys. Is this statement true or false?



Solution

a.

Grade	Boys	Girls	Total
Grade 7	50	65	115

Grade 8	70	75	145
Grade 9	30	65	95
Grade 10	60	40	100
Grade 11	75	80	155

b. total students passed Exam are : $115+145+95+100+155=610$ students

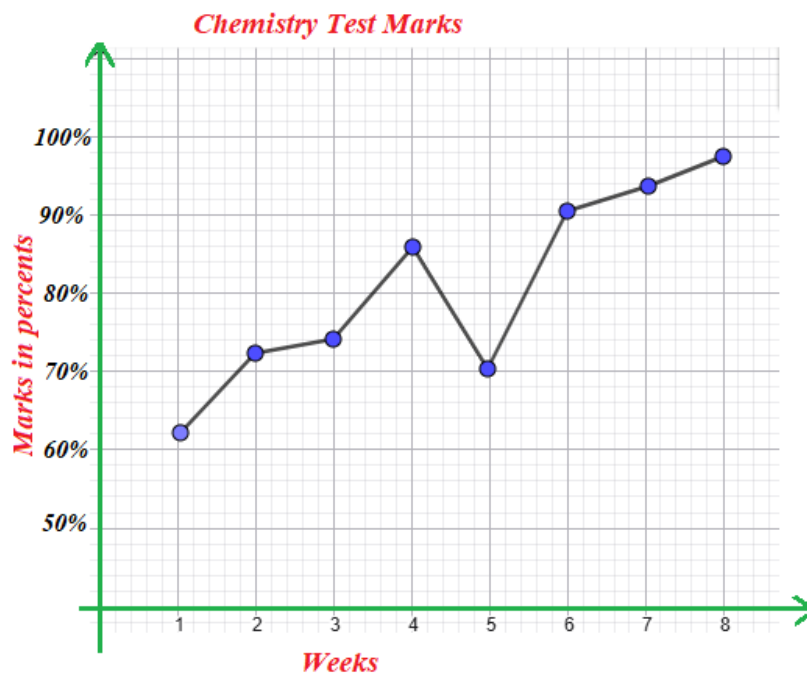
c. in grade 9, the girls passed exam are 35 more than boys ($65-30=35$)

d. Grade 7 and Grade 9 where the number of girls passed exam is 65 each.

e. Yes. Because, in 610 students passed exam, 325 students are girls while the boys are 285 students.

Example 3

The Line Graph below shows a learner's weekly performance progress in Chemistry subject.



- In which week did the learner get the lowest performance?
- In which week did the learner get the highest performance?
- Which way is the line trending?
- Does the line graph indicate any problems? Explain how to solve them.
- Make a frequency table

Solution

a) The learner got the lowest performance in **Week 1** (about **62%**).

b) The learner got the highest performance in **Week 8** (about **98%**).

c) The line is **generally trending upward**, showing improvement over time, although there is a drop in Week 5.

d) Yes. There is a **drop in performance in Week 5** compared to Week 4.

This problem could be solved by:

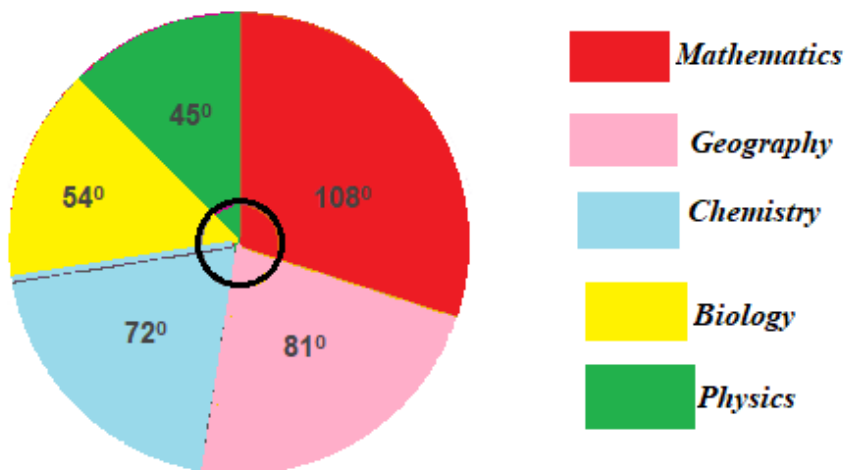
- Revising difficult topics covered that week
- Providing extra practice and remedial lessons
- Improving study habits and assessment feedback

e) Frequency table

Week	Marks (%)
1	62
2	72
3	74
4	86
5	70
6	90
7	94
8	98

Example 4

The pie-chart shows the marks obtained by a student in an examination. If the student secures 440 marks in all, calculate the marks in each of the given subjects.



Solution

Given (from the pie chart)

Total marks = 440

Total angle of a circle = 360°

Subject	Mathematics	Geography	Chemistry	Biology	Physics
Angle	108°	81°	72°	54°	45°

Formula:

$$\text{Marks in a subject} = \frac{\text{Angle in a subject}}{360^\circ} \times 440^\circ$$

$$\text{Marks in Maths} = \frac{108^0}{360^0} \times 440^0 = 132$$

$$\text{Marks in Geo} = \frac{81^0}{360^0} \times 440^0 = 99$$

$$\text{Marks in Chem} = \frac{72^0}{360^0} \times 440^0 = 88$$

$$\text{Marks in Bio} = \frac{54^0}{360^0} \times 440^0 = 66$$

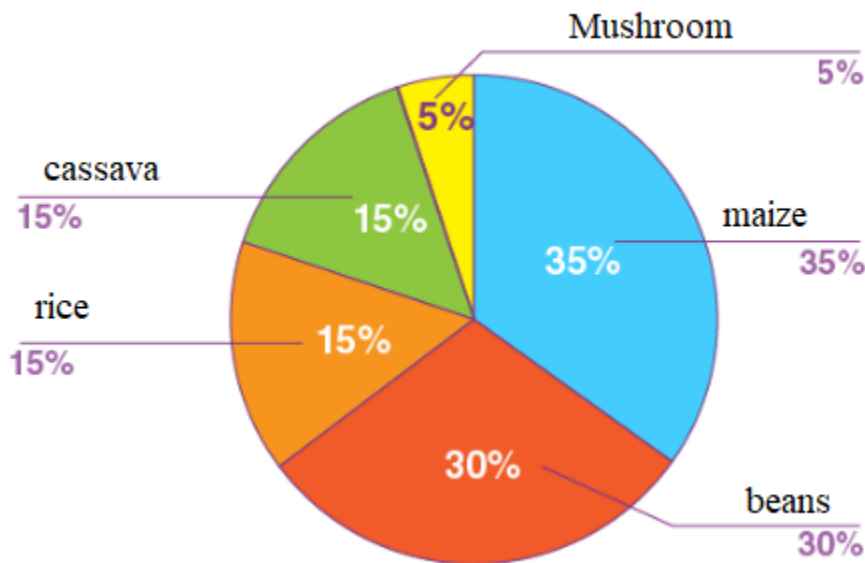
$$\text{Marks in Phys} = \frac{45^0}{360^0} \times 440^0 = 55$$

Final Answer (Marks in each subject)

Subject	Mathematics	Geography	Chemistry	Biology	Physics	Total
Marks	132	99	88	66	55	440

Example 5

The following pie chart shows the percentage of food production of a particular area in Rwanda.



- a) If there are total of 15 tons of food production,
- How many kilograms of maize, beans, cassava, rice and Mushroom are there?
 - Use frequency table to record this information.
- b) Find the central angle of Mushroom sector.

Solution

Using the pie chart information:

- **Maize: 35%**
- **Beans: 30%**
- **Rice: 15%**

- **Cassava: 15%**
- **Mushroom: 5%**

Total food production = **15 tons**

Recall: **1 ton = 1,000 kg**

So, **15 tons = 15,000 kg**

(a)(i) Kilograms of each food type

We calculate each share using:

Amount=Percentage×15,000 kg

Maize (35%)

$$0.35 \times 15,000 = 5,250 \text{ kg}$$

Beans (30%)

$$0.30 \times 15,000 = 4,500 \text{ kg}$$

Rice (15%)

$$0.15 \times 15,000 = 2,250 \text{ kg}$$

Cassava (15%)

$$0.15 \times 15,000 = 2,250 \text{ kg}$$

Mushroom (5%)

$$0.05 \times 15,000 = 750 \text{ kg}$$

(ii) **Frequency table of food production**

Food type	percentage	Quantity (kg)
Maize	35%	5250kg
Beans	30%	4500kg
Rice	15%	2250kg
Casava	15%	2250kg
Mushroom	5%	750kg
Total	100%	15000kg

(b) Central angle of the Mushroom sector

A full circle = **360°**

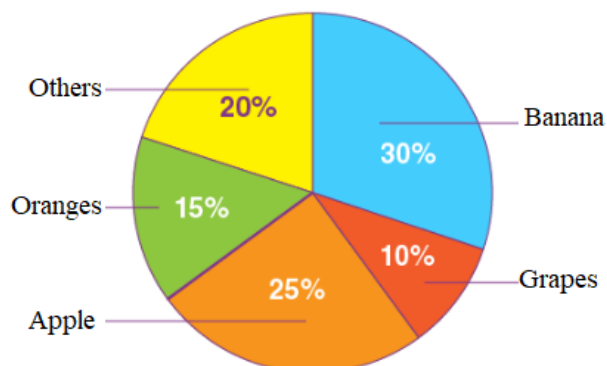
Mushroom = **5%**

$$\frac{5}{100} \times 360^\circ = 18^\circ$$



Application activity 13.2

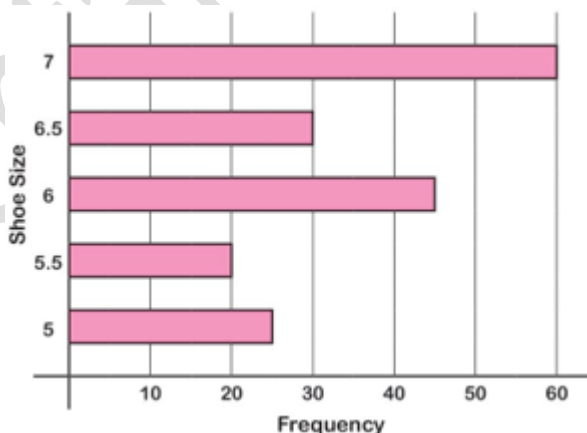
1) The following pie chart shows the sale of different fruits in a day for a shop:



- If a total of 1200 kg of fruits were sold in a day, calculate the amount of bananas sold (in kg)
- Find the difference between sales of grapes and oranges.
- Calculate the central angle for others.

2) The bar chart shows the shoes size of a group of senior 3 students.

- which shoes size is the mode?
- How many more students wear size 7 than wear size 5?
- What is the range in shoe sizes?
- What fraction of students wear size 7?
- How many students wear smallest size? And which size is that?



3. A person spends his time on different activities daily (in hours):

Activity	Office work	Exercise	travelling	Watching shows	Sleeping	Miscellaneous
Number of hours	9	1	2	3	7	2

- Draw a pie chart for this information.
- Draw a line graph

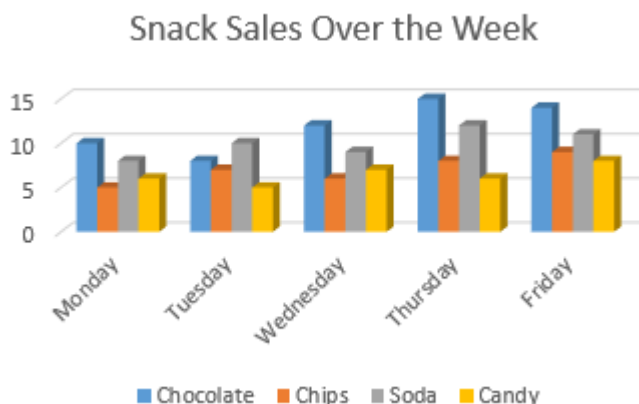
13.3 Problems on interpreting statistical graphs or diagrams for grouped data



Activity 13.3

Look at the Pictures given in **Part A** and **Part B**. Answer the questions that follow.

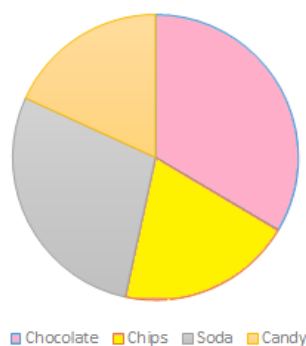
Part A: Bar Chart below shows Snack Sales Over the Week



- (a) Which snack sold the most overall during the week? chocolates
- (b) On which day were the most chocolates sold? Thursday
- (c) How many more sodas than chips were sold on Thursday?
- (d) Which snack had the least variation in sales across the week?
- (e) Describe any trend you notice for any snack (increasing, decreasing, or fluctuating).

Part B: Pie Chart below shows the Weekly Sales of Snacks

Weekly Sales of Snacks



- (a) Which snack occupies the largest slice of the pie?
- (b) Estimate what fraction of the total sales were chocolates.
- (c) If the store wants to promote the least sold snack, which one should it be?
- (d) Compare the pie chart with the bar chart – what extra information does the bar chart show that the pie chart does not?

..



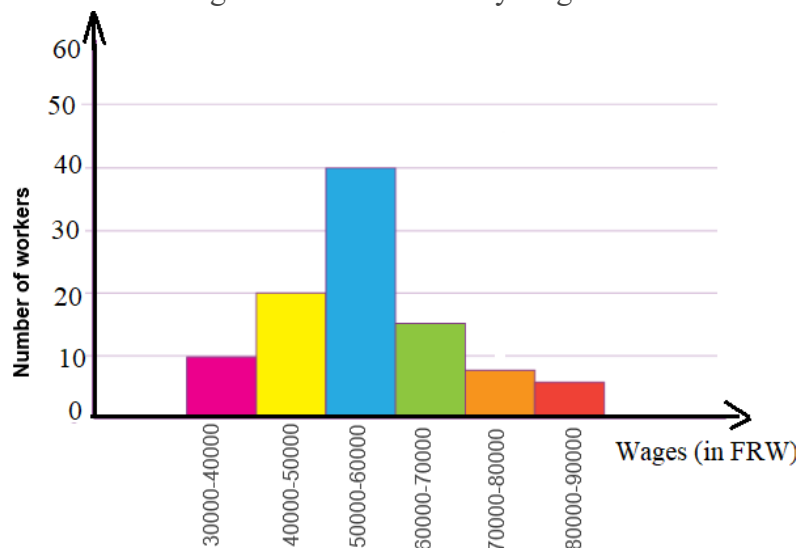
Content summary

Statistics involves collecting and organizing data to understand patterns and trends. When data is **grouped into intervals**, we can use **histograms** to show how often values occur, **frequency**

polygons to connect the midpoints of each group and display trends, and **pie chart** to show central angles and sector percentage for the groups. Interpreting these graphs helps us read frequencies, spot patterns, and compare different groups of data. Together, these tools make it easier to analyze **grouped data** and communicate information clearly.

Example 1

The below histogram shows the weekly wages of workers at a construction site:



Answer the following questions:

- How many workers get wages of 60000-70000(FRW)?
- Construct a frequency distribution table.
- What is the cumulative frequency for the class 50-60?

Solution

- 16 workers
-

Daily wages in FRW	Number of workers
30000-40000	10
40000-50000	20
50000-60000	40
60000-70000	16
70000-80000	8
80000-90000	6

- The cumulative frequency for the class 50-60 = $10 + 20 + 40 = 70$.

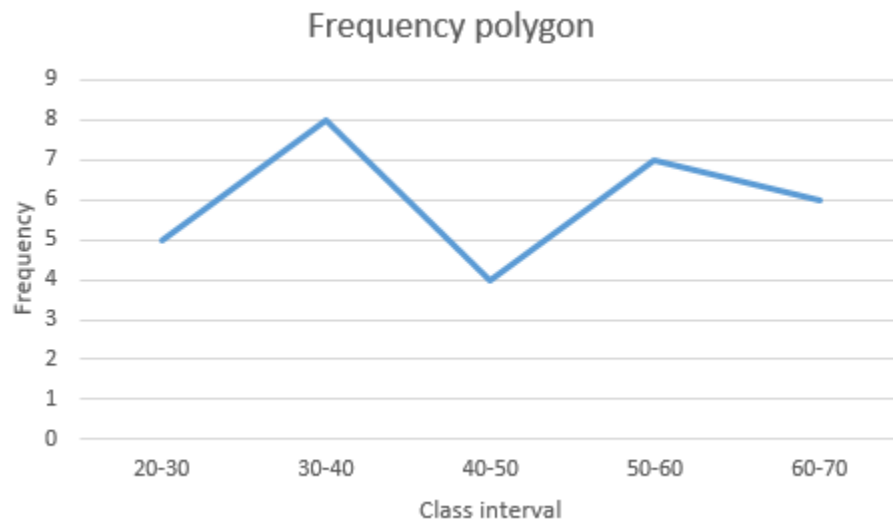
Example 2

Construct frequency polygon and a pie chart for the given frequency distribution below:

Class interval	20-30	30-40	40-50	50-60	60-70
Frequency	5	8	4	7	6

Solution:

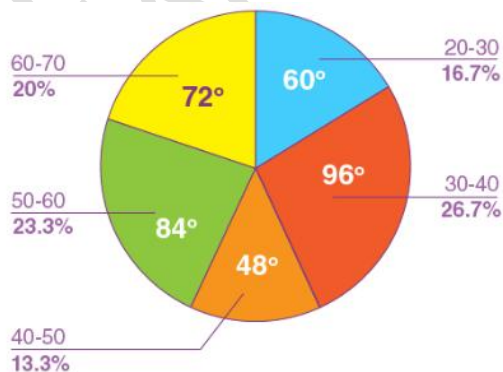
(a)



(b) The central angles are

Class interval	Frequency	Central angle
20-30	5	$\frac{5}{30} \times 360^\circ = 60^\circ$
30-40	8	$\frac{8}{30} \times 360^\circ = 96^\circ$
40-50	4	$\frac{4}{30} \times 360^\circ = 48^\circ$
50-60	7	$\frac{7}{30} \times 360^\circ = 84^\circ$
60-70	6	$\frac{6}{30} \times 360^\circ = 72^\circ$
Total	30	360°

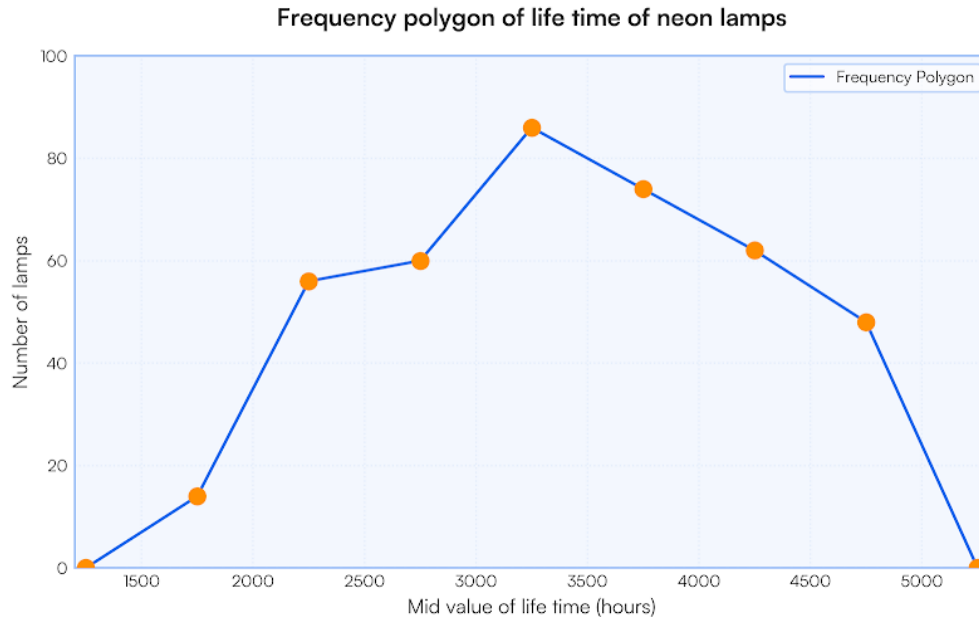
The pie chart is –



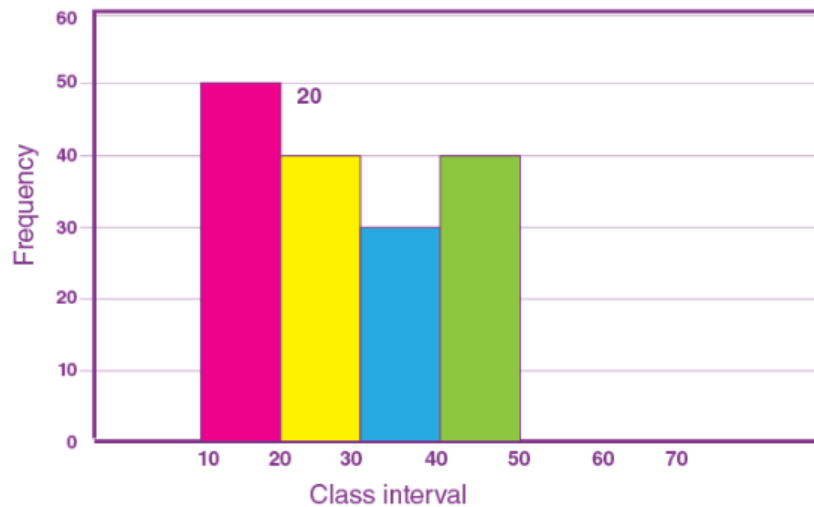


Application activity 13.3

1) The frequency polygon shows information about life time of lamps.



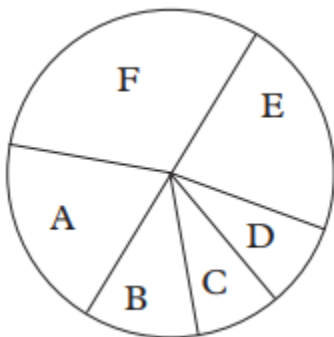
- What is the title of the graph shown above?
 - What does the horizontal axis represent?
 - What does the vertical axis represent?
 - What is meant by the *mid value of lifetime* in this graph?
 - At which mid value of lifetime is the number of neon lamps **highest**?
 - Approximately how many neon lamps have a lifetime around **3,200 hours**?
 - At which mid value of lifetime is the number of lamps **lowest** (excluding zero values)?
 - Describe the general trend of the frequency polygon from 1,500 hours to 5,000 hours.
 - Between which two consecutive mid values does the number of lamps increase most sharply?
 - What can you conclude about the most common lifetime of neon lamps from the graph?
- 2) Study the given histogram:



Answer the following questions:

- Prepare a cumulative frequency distribution table for the above histogram.
- Which class interval has the maximum frequency?
- Which class interval has the minimum frequency?

3. A total of 120 AIDS patients were sampled out and their mass (kg) recorded in classes of 10 beginning with 30 – 39, 40 – 49.... 80 – 89. The data was represented in a pie chart as in figure



The sectors in this pie chart represent the suggested classes.

- By measuring each of the angles at the centre, find the frequencies of the distribution.
- Make a frequency distribution table for the data.
- Calculate:
 - the mean
 - the median
 - the mode of distribution



END OF UNIT ASSESSMENT

Section A: Knowledge and Understanding

1. What is the difference between ungrouped data and grouped data? Provide a simple example of each.

2. **Match** the graph type to its best description:

Bar Chart

Shows parts of a whole as proportional slices

Pie Chart

Shows frequency of continuous data in adjacent intervals

Histogram

Compares discrete categories using bars of different heights

3. Look at the following dataset of test scores: 45, 67, 78, 82, 91, 55, 63, 71, 85, 88.

To create a grouped frequency table, you might create intervals like 40-49, 50-59, etc. What is one piece of specific information that is lost when you convert this raw data into these grouped intervals?

4. **True or False:** In a histogram, the bars must always touch each other, while in a bar chart for favorite colors, there should be gaps between the bars.

5. What is the main purpose of creating a graph (like a bar chart or pie chart) from a table of numbers?

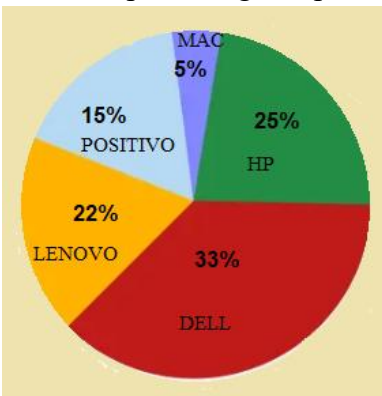
6. Choose the appropriate method to be applied on the scenarios below and explain why:

Scenario 1: Recording daily temperature for one week.
comparison.

Scenario 2: Analyzing exam results for 500 students

Section B: Skills

7) The pie chart below is used to represent people's choice of Laptop's brands. The circle as a whole here is represented by all the people who took part in the survey. Since it is a whole, the sum of all percentages represented in a pie graph must add up to 100%



(a) Which laptop brand is the most preferred by the people who took part in the survey?

Give the percentage that supports your answer.

(b) What percentage of people prefer Lenovo laptops?

How does this compare with those who prefer HP?

(c) If 200 people took part in the survey, how many people chose Dell laptops?

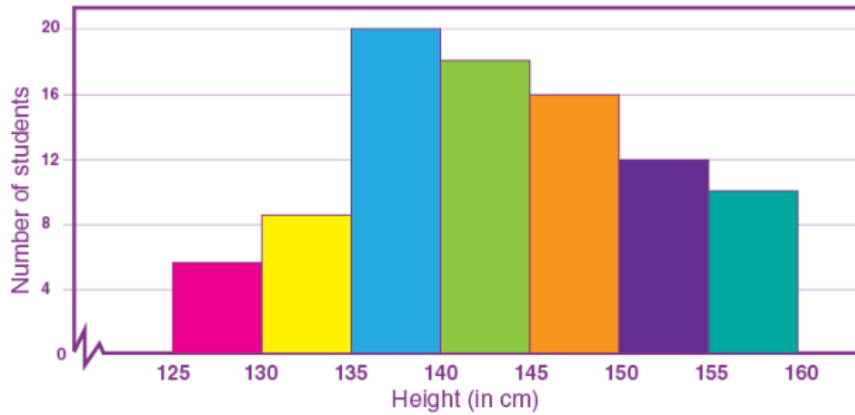
Show your calculation.

(d) Which two laptop brands together account for more than half (50%) of the total choices?

Explain your answer using the percentages shown in the pie chart.

Question 8:

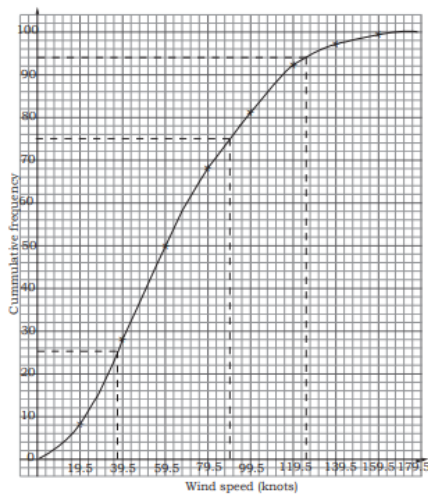
The following histogram shows the height of students within a class:



Answer the following questions:

- How many students have a height of less than 140 cm?
- How many students have a height greater than 140 cm but less than 155 cm?

9) Figure below is a cumulative frequency graph representing some grouped data



from S2SB (unit test question 1)

Use the graph to:

- Make a frequency distribution table
- Calculate the mean of the data
- Calculate the median
- Calculate the mode of the distribution.

Section C: Attitudes and Values

10) Imagine you are the head of the school's Environment Club.

You have two sets of data:

Data A: A pie chart from a survey showing that 70% of students say they are very concerned about plastic waste.

Data B: A bar chart from a waste audit showing that plastic bottles are the 3rd most common item in the school bins, after paper and food scraps.

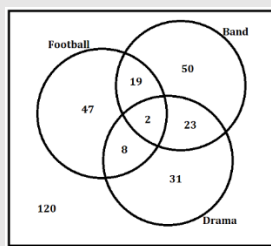
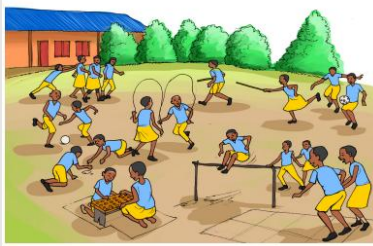
Describe **one specific action** you would propose for the club based on **Data A**, and a **different action** you would propose based on **Data B**. Explain the link between the data and your proposed action.



11. A classmate shows you a project conclusion that states: Our bar chart proves that playing video games causes lower test scores in Mathematics."

a) What is **wrong** with this conclusion based solely on a bar chart showing two sets of averages (high video game use vs. low video game use)?

b) What **other information** or **alternative explanation** should you consider before accepting this cause-and-effect claim? (Name one).



UNIT 14: PROBABILITY WITH VENN DIAGRAMMS

Key Unit Competence: Determine the probability of an event from real life situation using Venn diagrams.

Learning Objectives:

By the end of this unit, you should be able to:

Knowledge and Understanding:

- Define essential probability terms, use set notation, and identify sample spaces and events with Venn diagrams.
- Explain the fundamental probability formula.

Skills:

- Construct and interpret Venn diagrams to analyze combined events.
- Apply the probability formula to solve real-world problems using data from diagrams.

Attitudes and Values:

- Appreciate the clarity that visual tools like Venn diagrams provide for complex scenarios.
- Value collaboration and recognize probability as a practical tool for risk assessment and decision-making.

Introduction

This unit teaches you to solve real-world probability problems using Venn diagrams as a visual tool. You will learn to model overlapping events, use set notation to identify outcomes, and apply the probability formula to calculate chances.

By constructing and interpreting these diagrams, you will sharpen your analytical skills and develop an appreciation for how Venn diagrams can simplify uncertainty, aiding in informed decision-making.



Introductory activity

Imagine your class is planning an end-of-term celebration. The committee has done a quick survey of 30 classmates, asking two simple **Yes** or **No** questions:

Question A: Do you like chips as a party snack?

Question B: Do you like popcorn as a party snack?



Popcorn



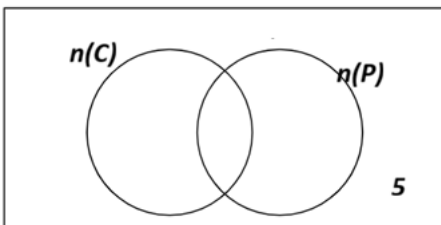
Chips

Suppose the results show: 22 people like chips, 18 people like popcorn and 5 people like **neither** chips nor popcorn.

1) Use the given information to fill in a Venn diagram representing the number of students who like:

(i) Only chips, (ii) Only popcorn, (iii) Both snacks, (iv) Neither snack.

Consider $n(C) = 22$ and $n(P) = 18$.



2) If a student is chosen at random from the class:

a) What is the probability that they like **only popcorn**?

b) What is the probability that they like **both** chips and popcorn?

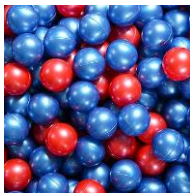
3) The committee decides to bring both snacks to the party. Based on the survey, is bringing both snacks a good idea? Justify your answer using probabilities from your Venn diagram.

14.1 Meaning of Probability of an Event



Activity 14.1

A teacher places **5 red balls and 3 blue balls** in a box.



A **student** picks one ball without looking.

As the class, discuss:

- Is it possible to predict the color of the ball to be picked with certainty?
- Which color is more likely to be picked?

This leads to the idea of **chance** and **probability**.



Content summary

- **Probability** is a measure of how likely an event is to happen.

- An **experiment** is an action with uncertain outcomes (e.g., tossing a coin).
- An **outcome** is a possible result of an experiment.
- An **event** is a set of outcomes.

The **probability of an event** is a measure of how likely that event is to occur, based on a specified random experiment.

If an experiment is repeated a large number of times under similar conditions, the **probability** $P(E)$ of an event E is defined as: **Experimental probability of an event E**

noted as: $P(E) = \frac{\text{Number of trials in which event E occurs}}{\text{Total number of trials}}$

The **experimental** probability is also called **empirical** probability.

Suppose you toss a fair coin 100 times and get “Heads” 48 times.

The experimental probability of Heads is:

$$P(\text{Heads}) = 48/100 = 0.48.$$

If the number of trials is very large, this ratio tends to stabilize near the **theoretical probability** (which for a fair coin is 0.5).

Therefore, the **theoretical probability** $P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$

The probability values lie between **0 and 1**:

- $P(A) = 0$: impossible event
- $P(A) = 1$: certain event

Example 1:

A coin is tossed once. Find the probability of getting a head.

Solution: $P(\text{Head}) = \frac{1}{2}$

Example 2:

A bag contains 4 green and 6 yellow balls. Find the probability of selecting a green ball.

Solution: $P(\text{green}) = \frac{4}{10} = \frac{2}{5}$

Example 3:

A die is thrown. Find the probability of getting an even number.

Solution $P(\text{Even}) = \frac{3}{6} = \frac{1}{2}$



Application activity 14.1

1) A card is drawn from a standard deck of 52 cards.



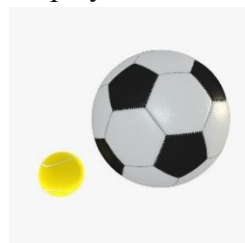
- a) Find the probability of drawing a heart.
 - b) Is this event certain, impossible, or possible?
- 2) A bag contains 7 white and 3 black stones.
Find the probability of selecting a black stone.
- 3) Write down the probability of an impossible event.
- 4) **Chose the correct answer.** The probability of an event that is certain is:
A) 0 B) $\frac{1}{2}$ C) 1 D) Greater than 1

14.2 Set Concepts and Total Number of Outcomes Using Venn Diagrams



Activity 14.2

The pupils of senior three class were asked about the sports they play. 17 of them play football. 14 play tennis. 5 of them play both football and tennis. There are 30 pupils in the class.



- (a) Draw a Venn diagram to show this information.
- (b) How many play football but not tennis?
- (c) How many play neither football nor tennis?



Content summary

A **set** is a collection of distinct objects (elements), described using curly braces like $A = \{1, 2, 3\}$, while a **Venn diagram** visually represents these sets and their logical relationships (overlap, differences) using overlapping circles or ovals within a rectangle (universal set) to show intersections, unions, and complements intuitively. They illustrate concepts like shared elements (intersection) and all elements combined (union).

Basic Venn diagrams can illustrate the interaction of two or three sets. Here are some ways Venn diagrams are used in set theory:

The union of two sets A and B (denoted by $A \cup B$) is represented by all the points in both circles

The intersection of two sets A and B (denoted by $A \cap B$) is represented by the points where the circles and overlap.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - [n(A \cap B) + n(A \cap C) + n(B \cap C)] + n(A \cap B \cap C)$$

The complement of a set A (denoted by A') is represented by all the points outside the circle but within the universal set.

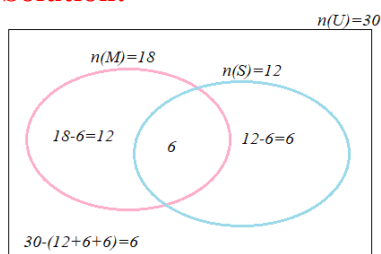
The difference of two sets A and B (denoted by $A-B$) is represented by the points in circle that are not in circle.

Example 1:

In a class of 30 students 18 like Mathematics, 12 like Science and 6 like both. Use Venn diagram to find the number who like:

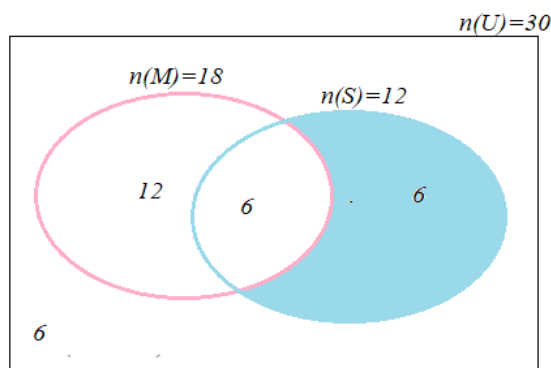
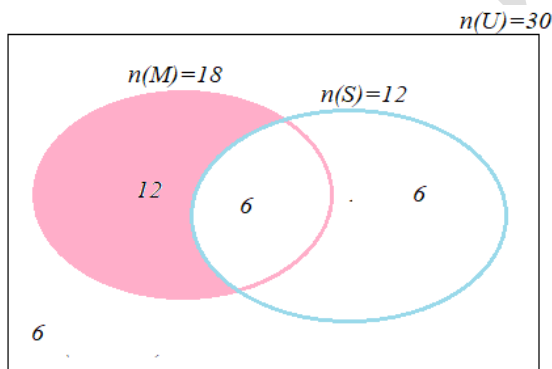
- Only Mathematics
- Only Science
- Neither subject

Solution:

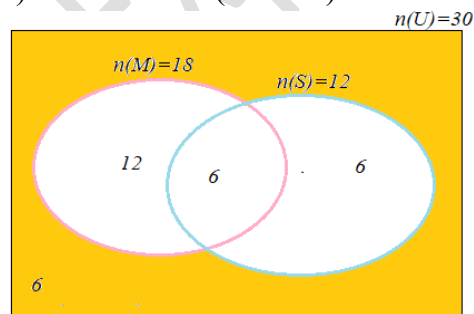


a) Only Mathematics = $18-6=12$

b) Only Science = $12-6=6$



c) Neither = $30-(12+6+6)=6$

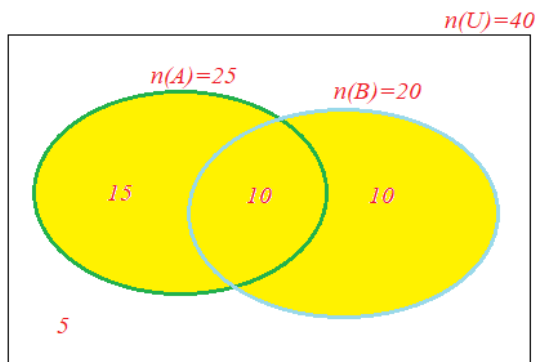


Example 2:

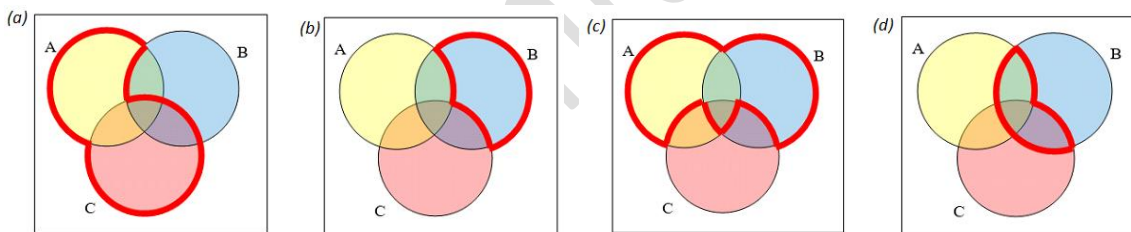
If $n(U)=40$, $n(A)=25$, $n(B)=20$, and $n(A \cap B)=10$, Use Venn diagram to find $n(A \cup B)$.

Solution:

$$n(A \cup B) = 15 + 10 + 10 = 35$$

**Application activity 14.2**

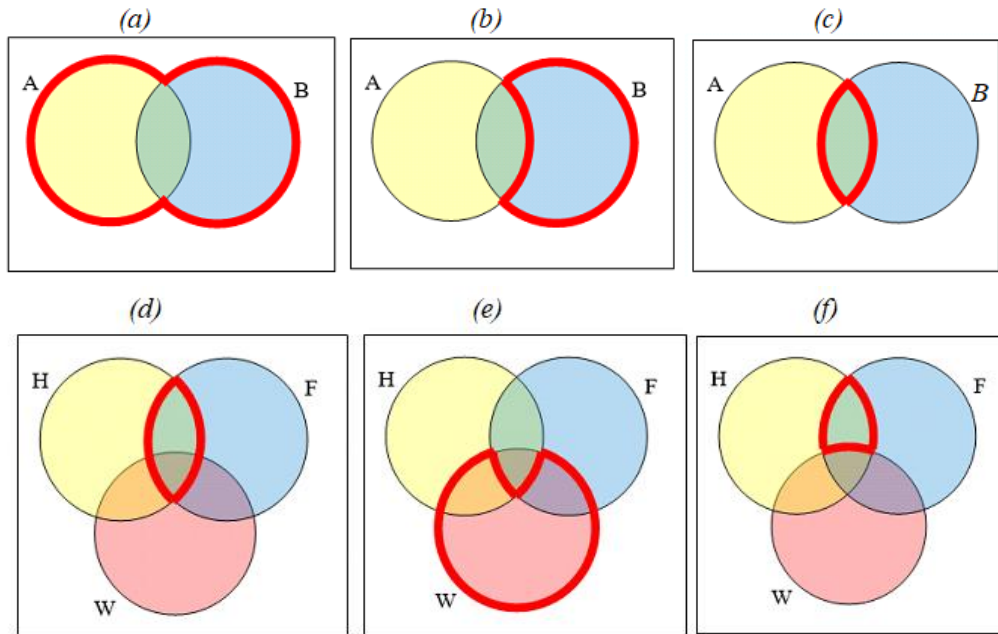
- 1) In a school of 50 learners, 30 study French, 20 study English, and 10 study both. Use Venn diagram to find how many study: a) Only French b) Only English c) Neither language
- 2) The shaded region representing $A \cap B$ contains elements that are:
A) Only in A B) Only in B C) In A or B D) In both A and B.
- 3) If $n(A) = 15$, $n(B) = 12$, and $n(A \cap B) = 5$, find $n(A \cup B)$ and chose the correct answer:
A) 17 B) 22 C) 27 D) 32.
- 4) Write an expression for the shaded region:



- 5) A survey asked buyers whether color, size, or brand influenced their choice of cell phone. The results are the following: 5 only said color, 8 only said size, 16 only said brand, 20 said only color and size, 42 said only color and brand, 53 said only size and brand, 102 said all three, 20 said none of these. How many people were influenced by brand?

14.3 Determining Probability Using Venn Diagrams**Activity 14.3**

Describe the region indicated in each case below and indicate how to find the cardinality of each one.



Content summary

As you interpret the information, remember:

- 1) Probability: $P(A \text{ and } B) = P(A \cap B)$.
- 2) $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- 3) $P(\text{not } A) = 1 - P(A)$
- 4) Since Probability using Venn diagrams:

$$P(A) = \frac{n(A)}{n(U)}$$

$$P(A \cup B) = \frac{n(A \cup B)}{n(U)}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(U)}$$

Venn diagrams help visualize overlapping events.

Example 1:

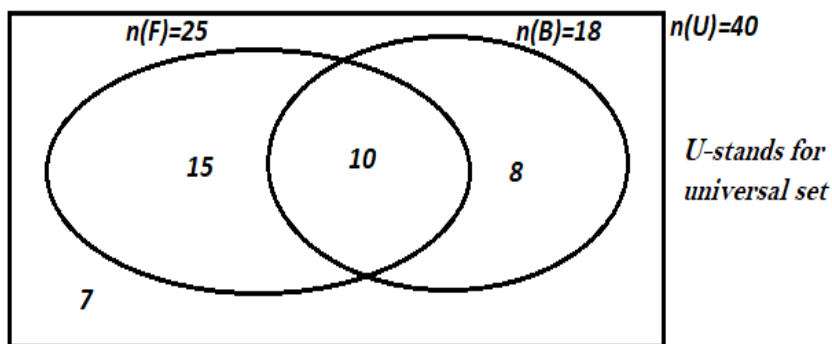
In a class of 40 students: 25 like football; 18 like basketball; 10 like both.

Find the probability that a student likes:

- a) Football
- b) Basketball or football

Solution:

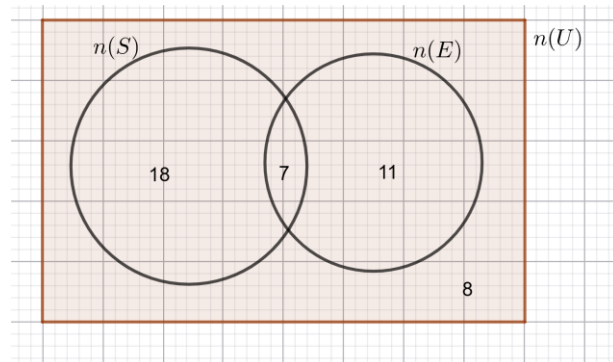
Step1: Draw Venn diagram



$$n(F \cup B) = 25 + 18 - 10 = 33 \quad P(F) = \frac{25}{40} = \frac{5}{8} \quad P(F \cup B) = \frac{33}{40}$$

Example 2:

This Venn diagram represents the people attending a convention on climate change. S represents all of the scientists, and E represents all of the environmentalists.



- How many people are at the convention?
- Find the probability that a people attending a convention is:
 - A Scientist
 - An Environmentalist,
 - both Scientist and Environmentalist,
 - neither Scientists nor Environmentalists
 - Environmentalists but not scientists?

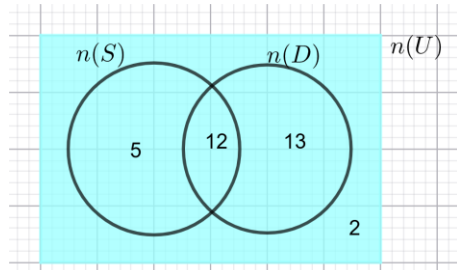
Solution

- $n(U) = 18 + 7 + 11 + 8 = 44$ people.
- $n(S) = 18 + 7 = 25$ people. Therefore, $P(S) = \frac{25}{44} \approx 56.8\%$
 - $n(E) = 8 + 11 = 19$ people. Therefore, $P(E) = \frac{19}{44} \approx 43.18\%$
 - $n(S \cap E) = 7$ people. Therefore, $P(S \cap E) = \frac{7}{44} \approx 15.9\%$
 - $n(S \cup E)' = 8$ people. Therefore, $P(S \cup E)' = \frac{8}{44} = \frac{2}{11} \approx 18.18\%$

v) $n(E - S) = 11 \text{ people}$. Therefore, $P(E - S) = \frac{11}{44} = 25\%$

Example 3

This Venn diagram shows the members of Everlasting Peace Club in presentation. D represents the members who dance and S represents the members who Sing.



- (a) How many members are in the Club?
 (b) What is the probability that a member of Everlasting Peace Club in presentation?
- Sings
 - Dances
 - Dances but not sings,
 - does neither of the presentation,
 - do both.

Solution

(a) $n(U) = 5 + 12 + 13 + 2 = 32 \text{ people}$ students

(b) i) $n(S) = 5 + 12 = 17 \text{ people}$. Therefore, $P(S) = \frac{17}{32}$

ii) $n(D) = 12 + 13 = 25 \text{ people}$. Therefore, $P(D) = \frac{25}{32}$

iii) $n(D - S) = 13 \text{ people}$. Therefore, $P(D - S) = \frac{13}{32}$

iv) $n(S \cup D)' = 2 \text{ people}$. Therefore, $P(S \cup D)' = \frac{2}{32} = \frac{1}{16}$

v) $n(S \cap D) = 12 \text{ people}$. Therefore, $P(S \cap D) = \frac{12}{32} = \frac{3}{8}$

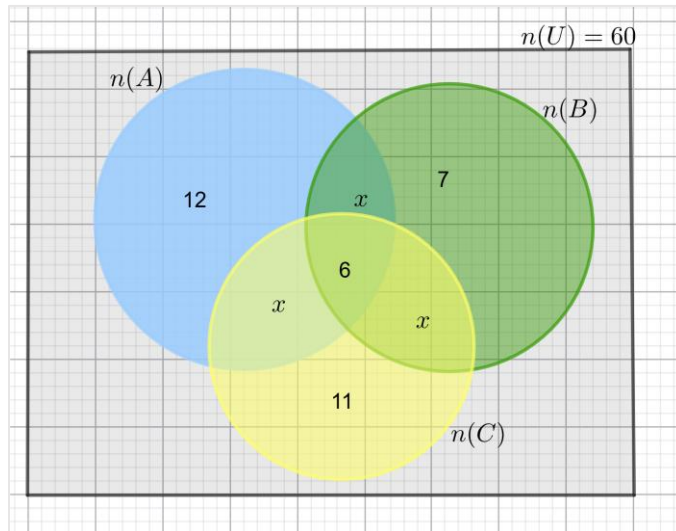
Example 4

All 60 different Vitamin pills, contain at least one of the Vitamins A, B and C. Twelve have A only, 7 have B only, and 11 have C only. If 6 have all three vitamins and there are x having A and B only, B and C only and A and C only.

- How many pills contain vitamin A?
- What is the probability that a randomly selected pill contains **only vitamin A**?
- What is the probability that a randomly selected pill contains **exactly two vitamins**?
- What is the probability that a randomly selected pill contains **vitamin A and vitamin B but not vitamin C**?

- e) What is the probability that a randomly selected pill contains **vitamin C**?
- f) What is the probability that a randomly selected pill contains **at least two vitamins**?

Solution



Solving for x:

Total pills = 60

$$12 + 7 + 11 + 6 + x + x + x = 60$$

$$36 + 3x = 60$$

$$3x = 24$$

$$x = 8$$

Substituting the value of x in the Venn diagram, we found:

B only = 7; C only = 11; A, B, and C = 6; A and B only = 8; B and C only = 8; A and C only = 8

(a) Pills containing vitamin A are:

- A only = 12
- A and B only = 8
- A and C only = 8
- A, B, and C = 6

$$\text{Total containing A} = 12 + 8 + 8 + 6 = 34$$

Pills containing each vitamin

- **Vitamin A**
 $12 + 8 + 8 + 6 = 34$
- **Vitamin B**
 $7 + 8 + 8 + 6 = 29$
- **Vitamin C**
 $11 + 8 + 8 + 6 = 33$
- **At least two vitamins**
 $8 + 8 + 6 = 22$

(b) Probability that a pill contains only vitamin A (A only = 12)

$$P(\text{A only}) = \frac{12}{60} = \frac{1}{5}$$

(c) Probability that a pill contains exactly two vitamins.

(Exactly two vitamins $8+8+8=24$)

$$P(\text{Exactly two}) = \frac{24}{60} = \frac{2}{5}$$

(d) Probability that a pill contains A and B but not C (A and B only = 8)

$$P(A \cap B) = \frac{8}{60} = \frac{2}{15}$$

(e) Probability that a pill contains vitamin C

$$P(C) = \frac{33}{60} = \frac{11}{20}$$

(f) Probability that a pill contains at least two vitamins

$$P(\text{At least two}) = \frac{30}{60} = \frac{1}{2}$$



Application activity 14.3

1) In a class of 60 learners: 20 like Mathematics, 25 like Biology, 15 like both.

Find the probability that a learner likes:

a) Mathematics

b) Biology only

c) Mathematics or Biology

2) A survey of 80 people shows: 45 like tea, 30 like coffee, 20 like both

Find $P(\text{tea})$ and $P(\text{tea or coffee})$.

3) If $n(U)=100$ and $n(A)=40$, $P(A)$ is (select the correct answer):

A) 0.04 B) 0.4 C) 4 D) 40.

4) If $P(A \cup B) = 1/10$ and $n(U)=50$. The number of elements in A union B is: (choose the correct answer): A) 5. B) 10. C) 15. D) 20.



END-OF-UNIT ASSESSMENT

Part A: Assessing knowledge and understanding

Choose the correct answer:

1. The probability of an impossible event is:
A. 1 B. 0 C. $\frac{1}{2}$ D. Greater than 1
- 2) If $n(U)=20$ and $n(A)=5$, then $P(A)$ is:
A. $\frac{1}{5}$ B. $\frac{1}{4}$ C. $(\frac{1}{5})$ D. 4
- 3) The region common to two sets A and B is written as:
A. $A \cup B$ B. A' C. $A \cap B$ D. U
- 4) In a Venn diagram, elements outside all circles but inside the rectangle represent:
A. Intersection B. Union C. Universal set D. Neither set

Work out:

- 5) Write the formula for: a) Probability of an event A. b) Number of elements in $A \cup B$

Part B: Assessing skills

- 6) A bag contains 6 white balls and 4 black balls.
 - a) Write the total number of outcomes.
 - b) Find the probability of picking a black ball.
- 7) A die is thrown once. (4 marks)
 - a) List the sample space.
 - b) Find the probability of getting an odd number.
- 8) In a class of **60 learners**: 32 like **Mathematics**, 28 like **Biology**, 12 like **both subjects**
 - a) Draw a Venn diagram to represent the information.
 - b) Find the probability that a learner likes: i) Mathematics. ii) Biology only.
iii) Mathematics or Biology.

Section C: Assessing Attitudes and Values

- 9) Explain one reason why **Venn diagrams** make probability problems easier to understand.
- 10) Give one real-life example where **probability analysis** can assist in decision-making.
- 11) Explain how **collaboration** can help students accurately solve probability problems with Venn diagrams.

BIBLIOGRAPHY

1. Rwanda Education Board (2015). Mathematics syllabus for Ordinary Level. Rwanda Ministry of Education.
2. E. Johnson, G. Johnson, T. Bokru (1977). Algebra 1 A two part course, Addison-Wesley publishing company, Phillipines
3. Macrae, M. and et al. (2010). New General Mathematics: student book I. Pearson education Ltd.
4. Allen R (2004). Intermediate Algebra for College Students, Pearson Education, Inc, New Jersey.
5. Richard E (1997). Algebra I, Addison – Wesley Publishing Company, Inc, Phillipines.
6. Maina L.et al (2011). Excel and Succed Mathematics Form2, Longhorn Publishers, Nairobi.
7. Maina L et al (2011). Excel and Succed Mathematics Form 3, Longhorn Publishers, Nairobi
8. Engelsohn H., Feit J (1980). Basic Mathematics, Moriah Publishing, New York.